PEARL
Power and Energy Analysis and Research Laboratory

Escuela de Ingeniería Industrial y Aeroespacial Toledo
Universidad de Castilla-La Mancha

UCLM Universidad Castilla-La Mancha

# Tools for the optimal operation and planning of power systems

## Lecture 4: Uncertainty and risk

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# Solutions to the proposed exercises

https://github.com/AlvaroGarciaCerezo/CourseOR/tree/main/Lecture%204



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1. Introduction
2. Stochastic programming (SP)
3. Robust optimization (RO)
4. SP-RO approach

# 1. Introduction

# Uncertainty

**Deterministic** approach

min/max objective function with **known** input data

subject to:
Constraints that involve **known** input data.

# Uncertainty

**Deterministic** approach → Suboptimal solutions

min/max objective function with **known** input data

subject to:
Constraints that involve **known** input data.

# Uncertainty

min/max objective function with **uncertain** input data

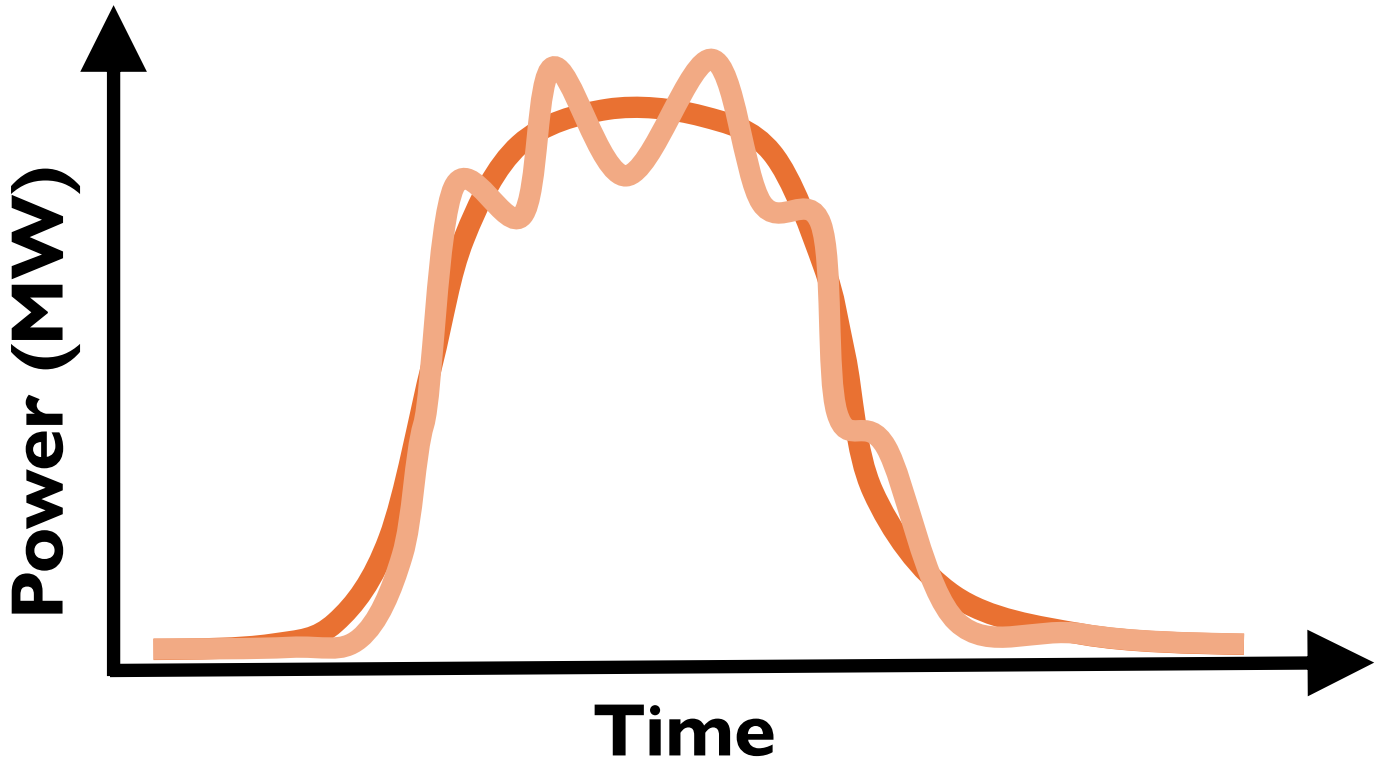
subject to:

Constraints that involve **uncertain** input data.

# Uncertainty

Examples of uncertain input data:

- Renewable Production Availability (RPA).



# Uncertainty

Examples of uncertain input data:

- Renewable Production Availability (RPA).
- Decommission of power plants.
- Offers of the generating units in electricity markets.
- Demand of consumers.
- Regulatory changes.
- Investment cost.

# Two-stage problem

**FIRST STAGE**

**SECOND STAGE**

# Two-stage problem

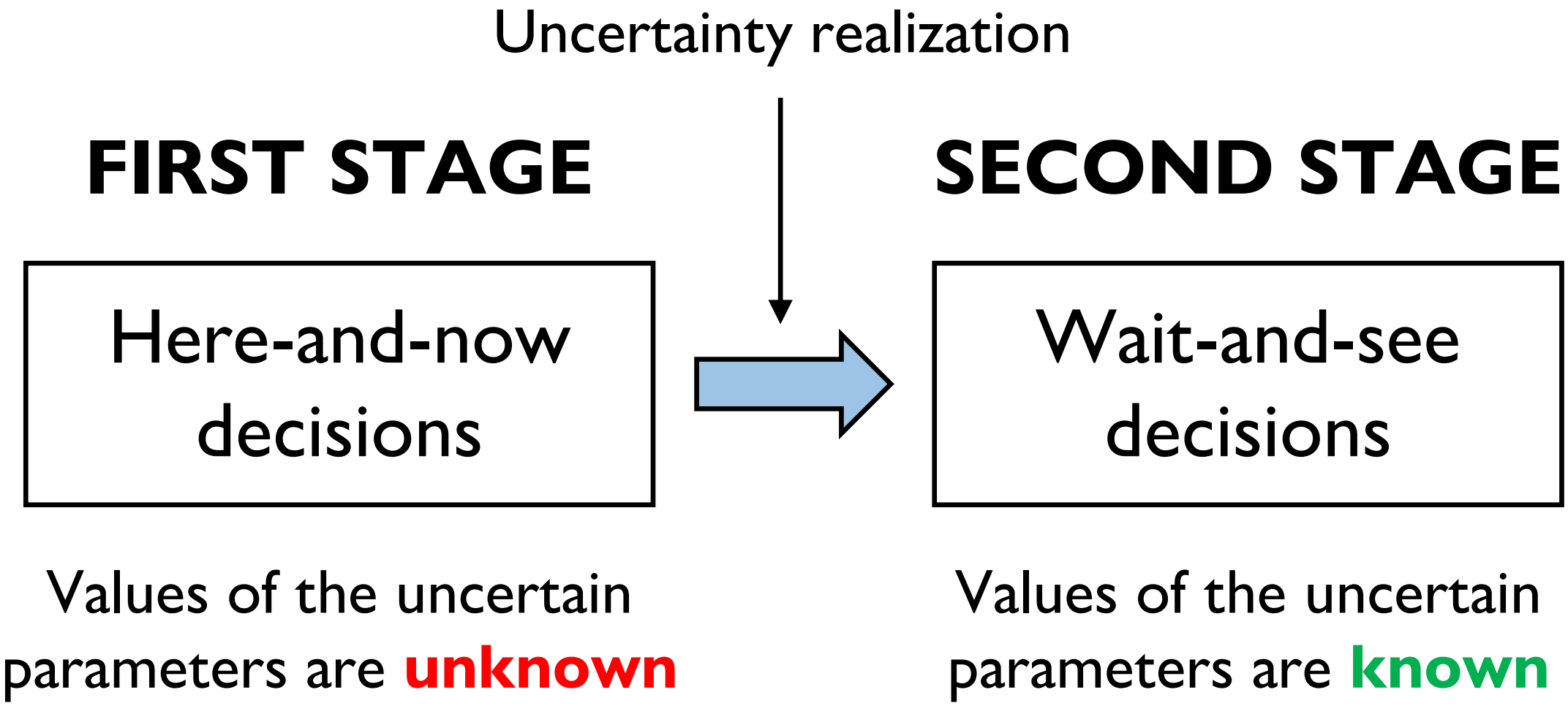
**FIRST STAGE**

**SECOND STAGE**

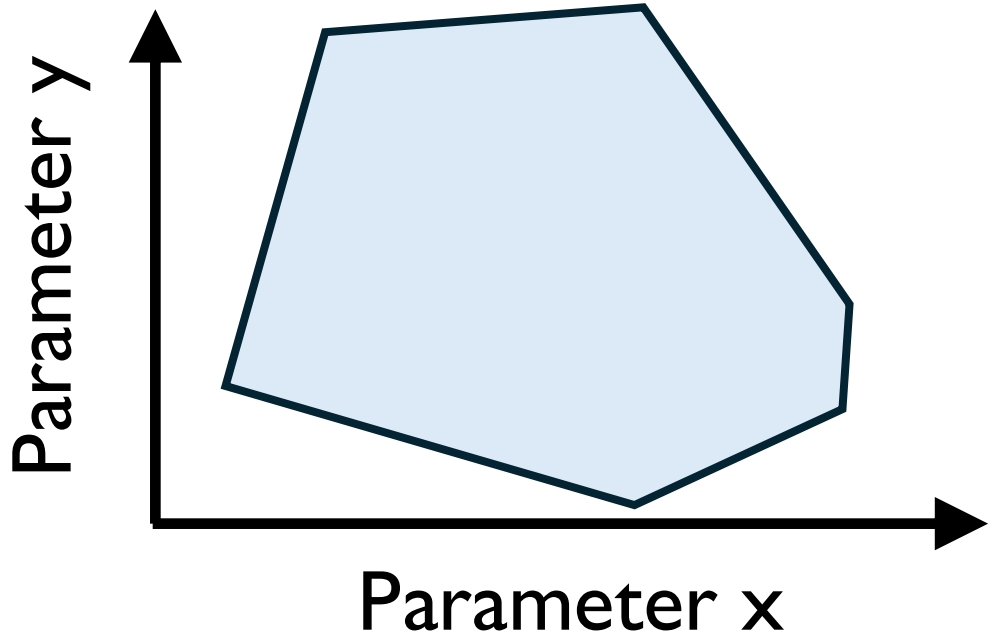
Here-and-now decisions

Values of the uncertain parameters are **unknown**

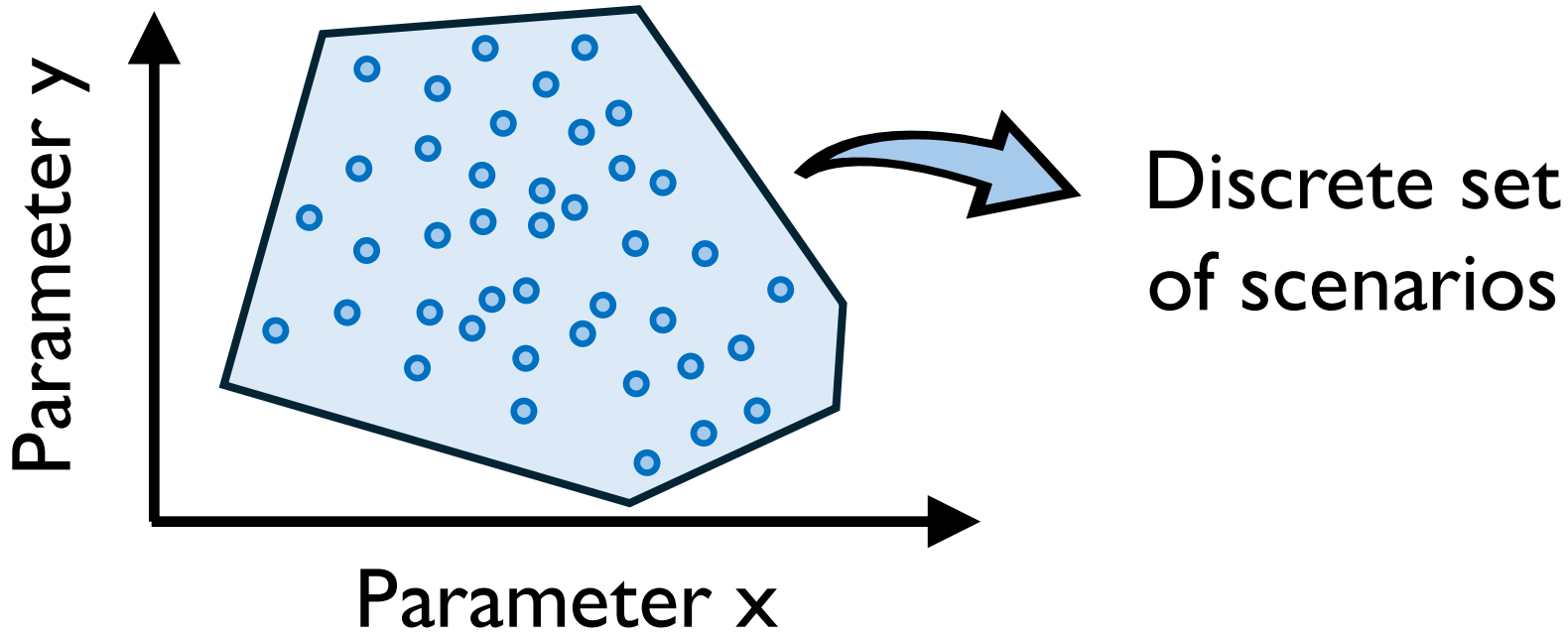
# Two-stage problem



# Stochastic programming

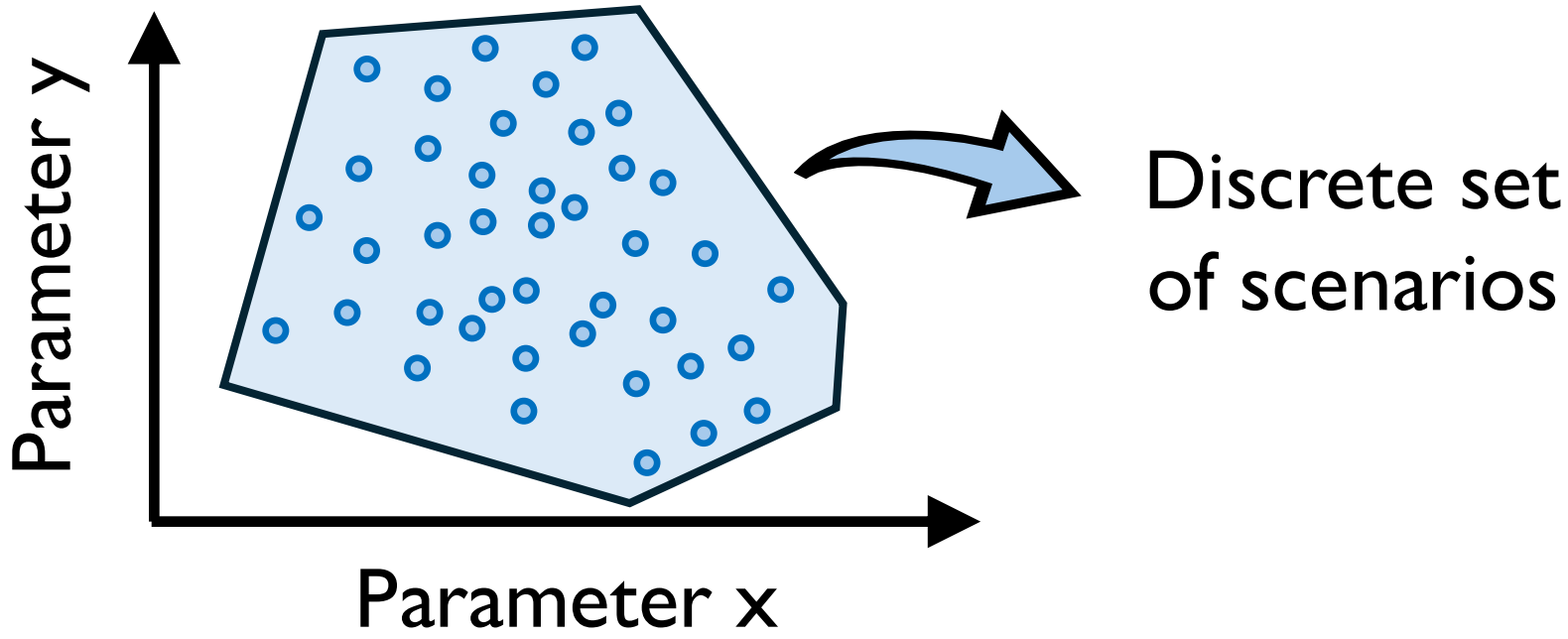


# Stochastic programming




Discrete set
of scenarios

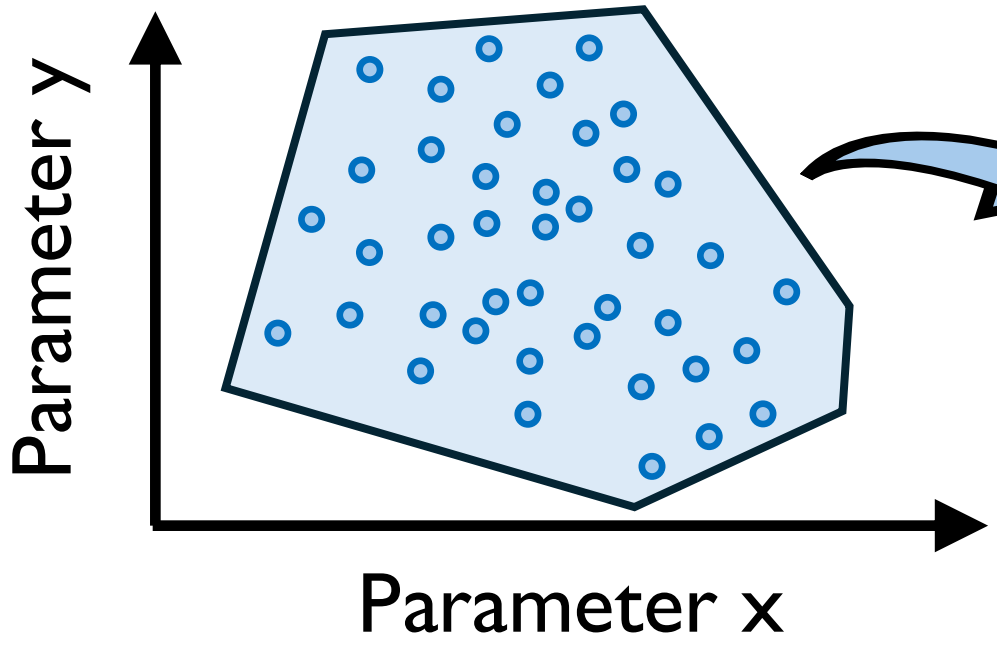
# Stochastic programming



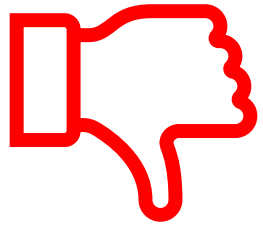

Risk-neutral framework → Expected value

Risk-averse framework → Conditional value-at-risk

# Stochastic programming




Discrete set of scenarios

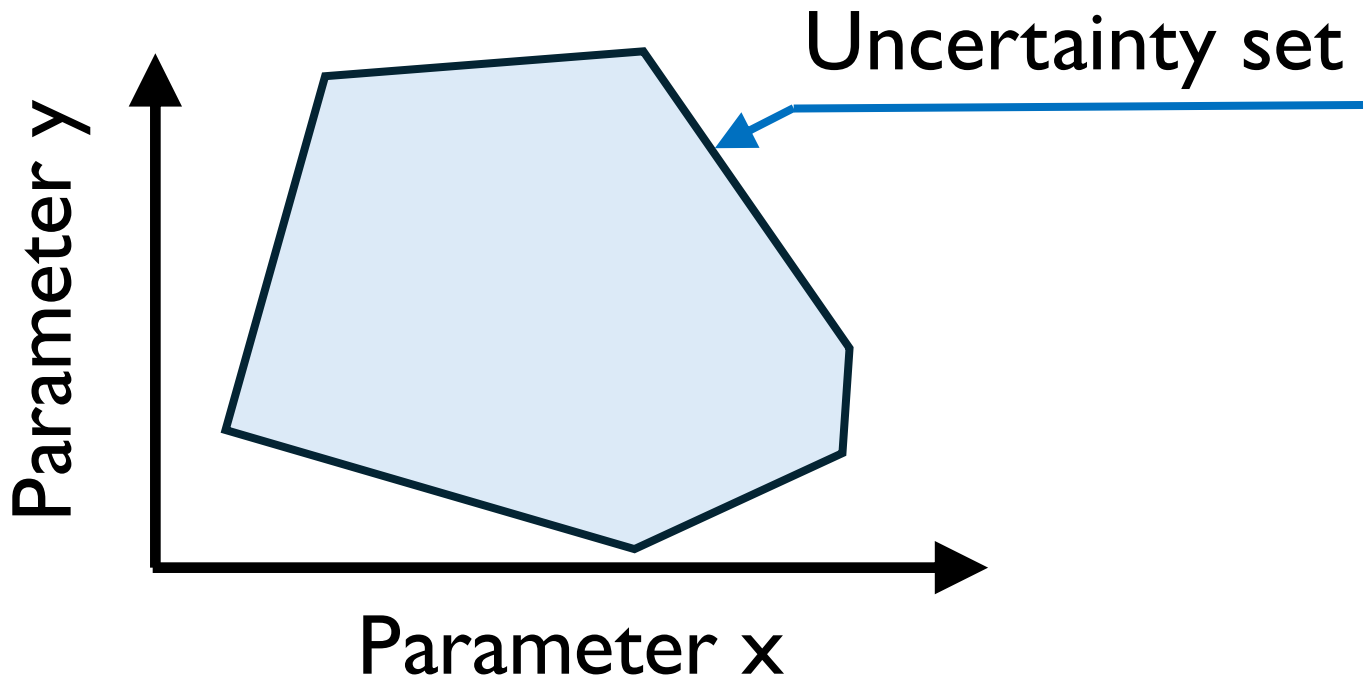


Lack of information about the probability density function

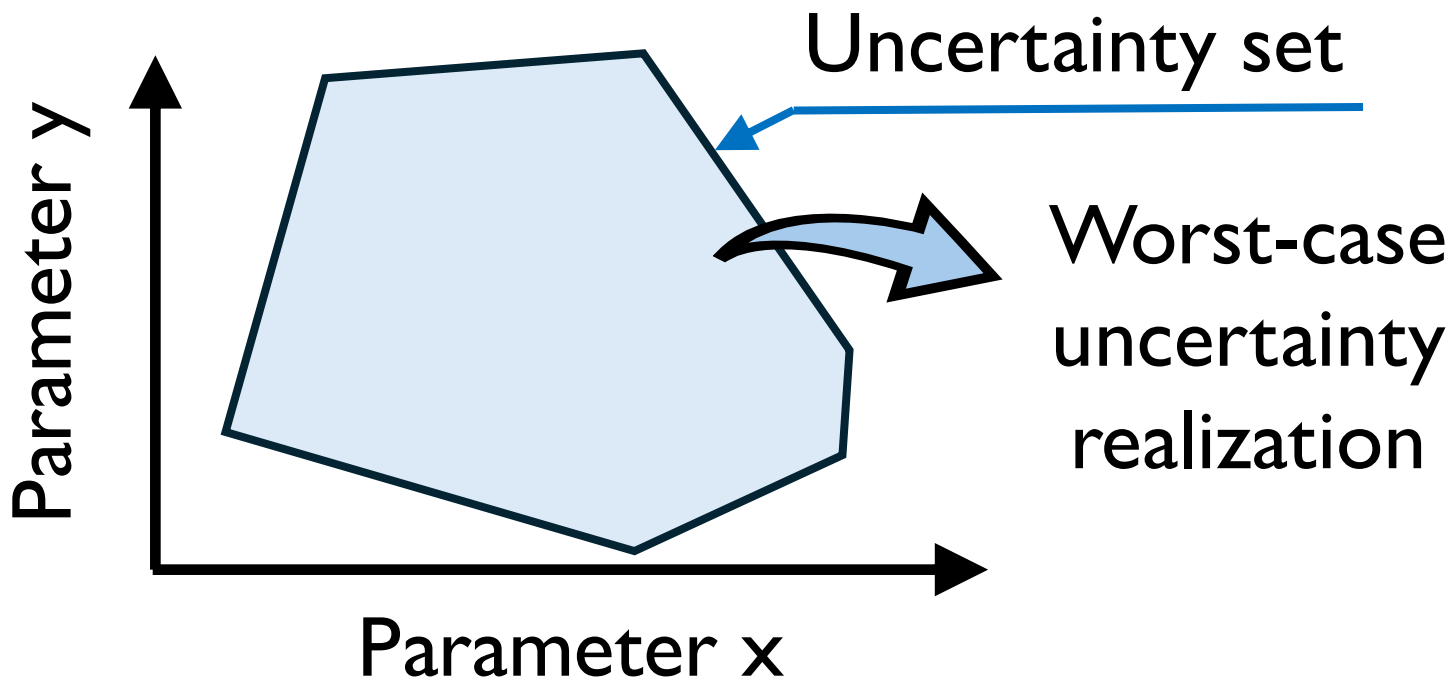
Risk-neutral framework → Expected value

Risk-averse framework → Conditional value-at-risk

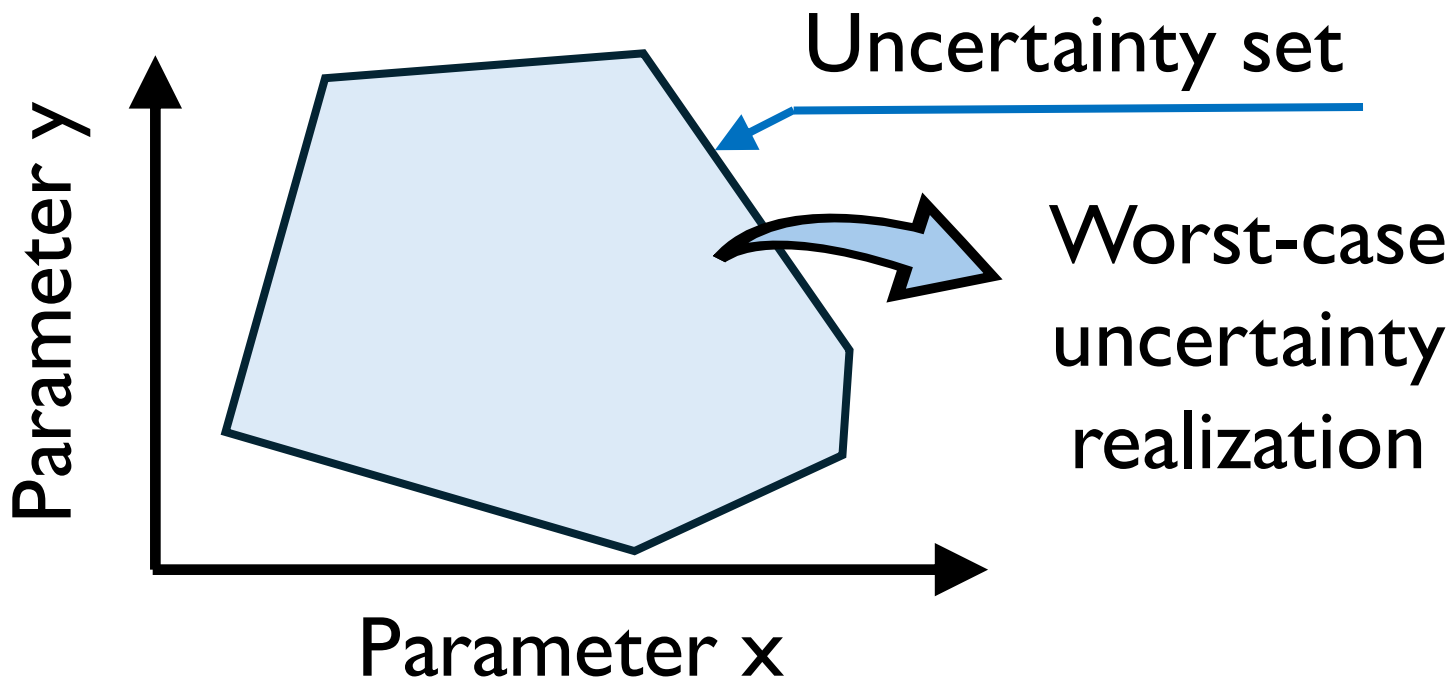
# Robust optimization



# Robust optimization

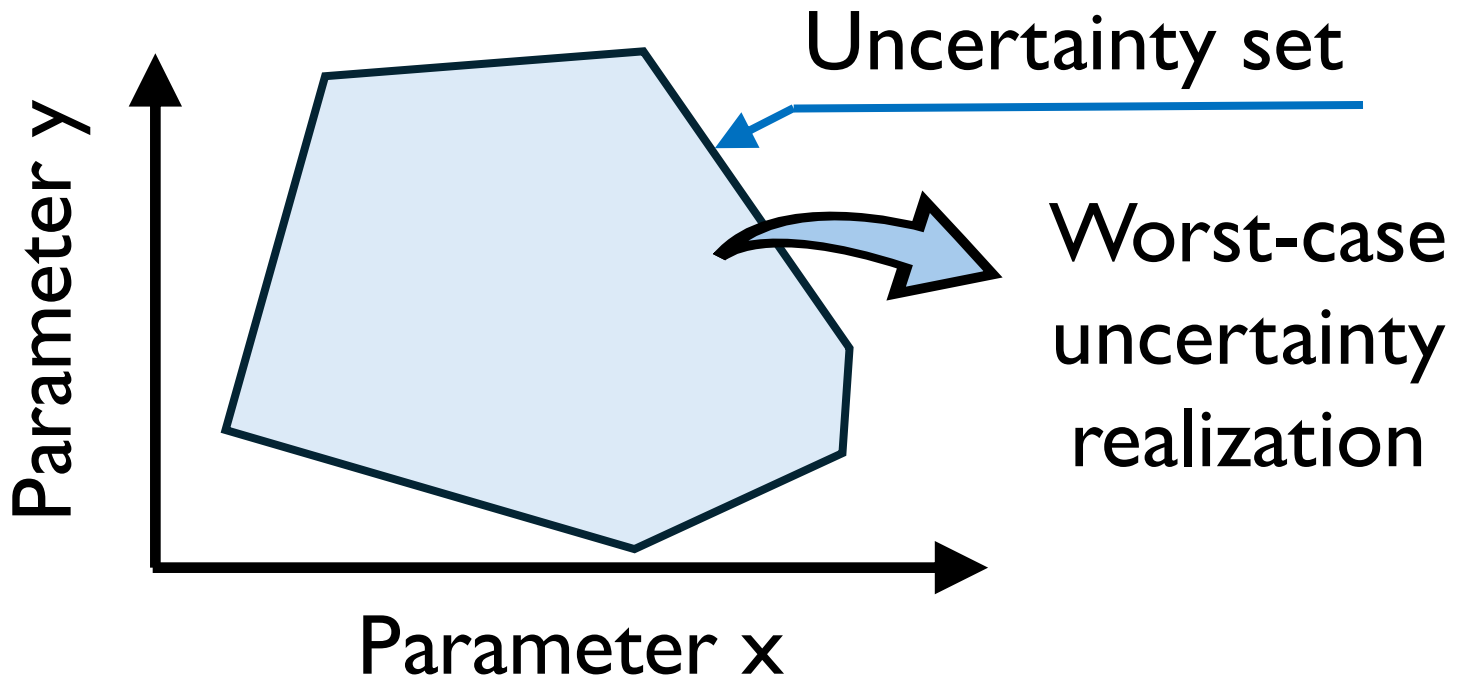


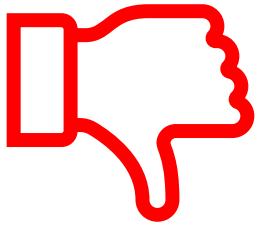
# Robust optimization




Robust optimization without recourse $\rightarrow \min_{x \in X} \max_{u \in \mathcal{U}} f(x, u)$

# Robust optimization

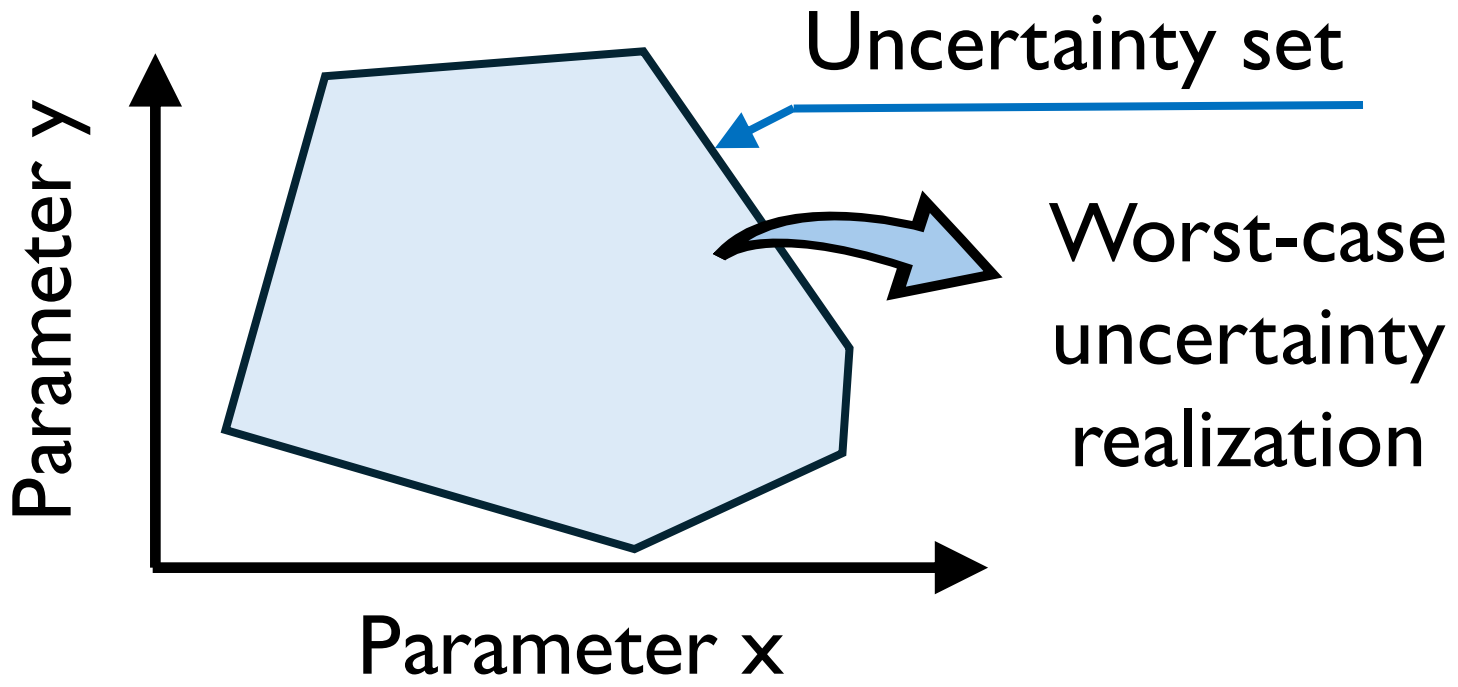


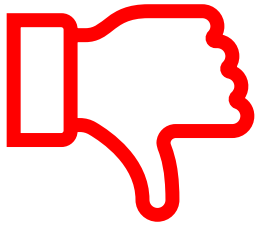



Too conservative solution

Robust optimization without recourse → $\min_{x \in X} \max_{u \in \mathcal{U}} f(x, u)$

# Robust optimization



Too conservative solution

Robust optimization without recourse → $\min_{x \in X} \max_{u \in \mathcal{U}} f(x, u)$

Robust optimization with recourse → $\min_{x_1 \in X_1} \max_{u \in \mathcal{U}} \min_{x_2 \in X_2} f(x_1, u, x_2)$

# What will you learn?

- To solve risk-neutral and risk-averse two-stage stochastic programming problems.
- To carry out out-of-sample analyses.
- To build bidding and offering curves.
- To solve two-stage robust optimization problems.
- To apply decomposition techniques.
- To apply acceleration strategies.

# 2. Stochastic Programming (SP)

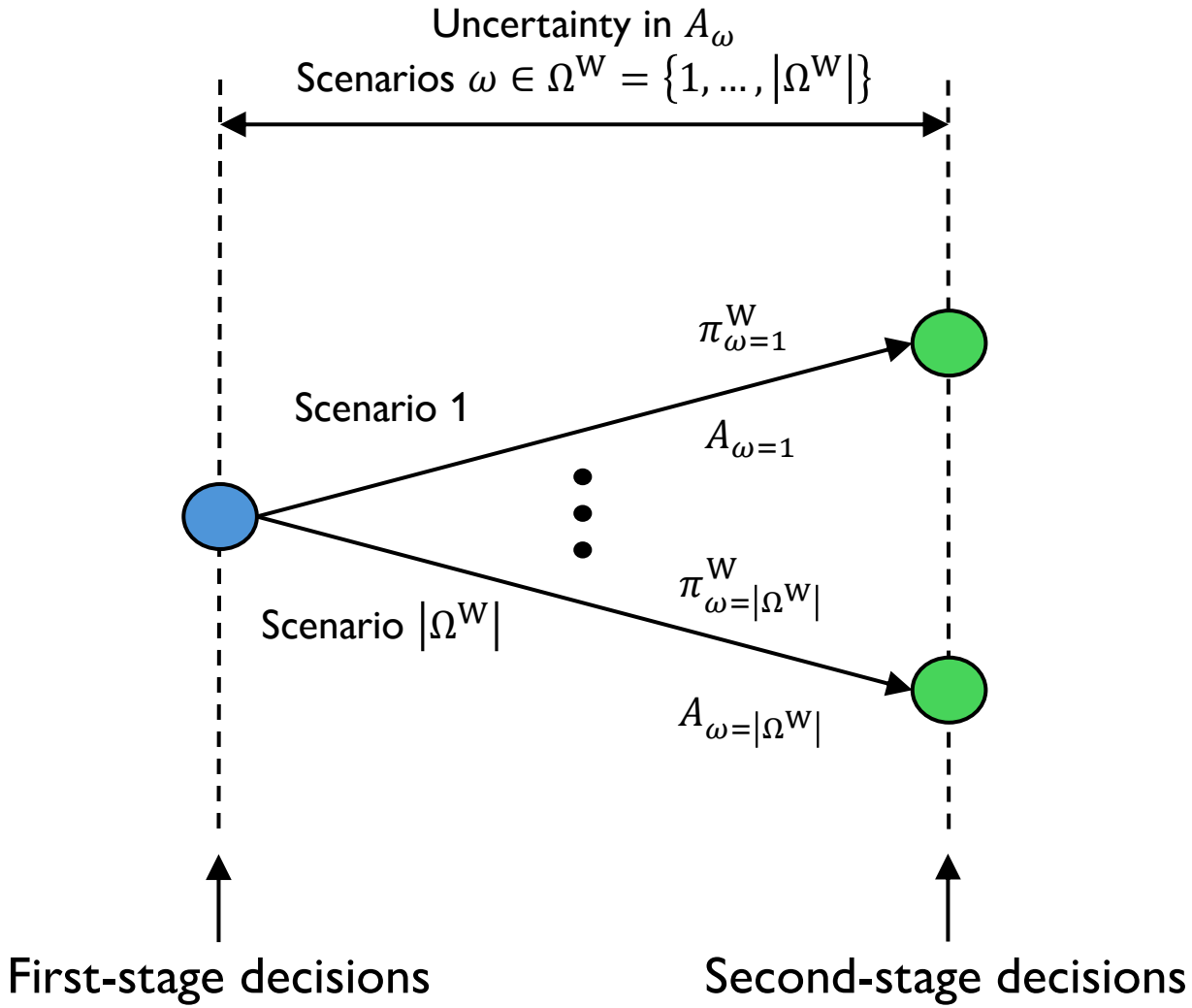
# Stochastic programming

Use of scenarios:

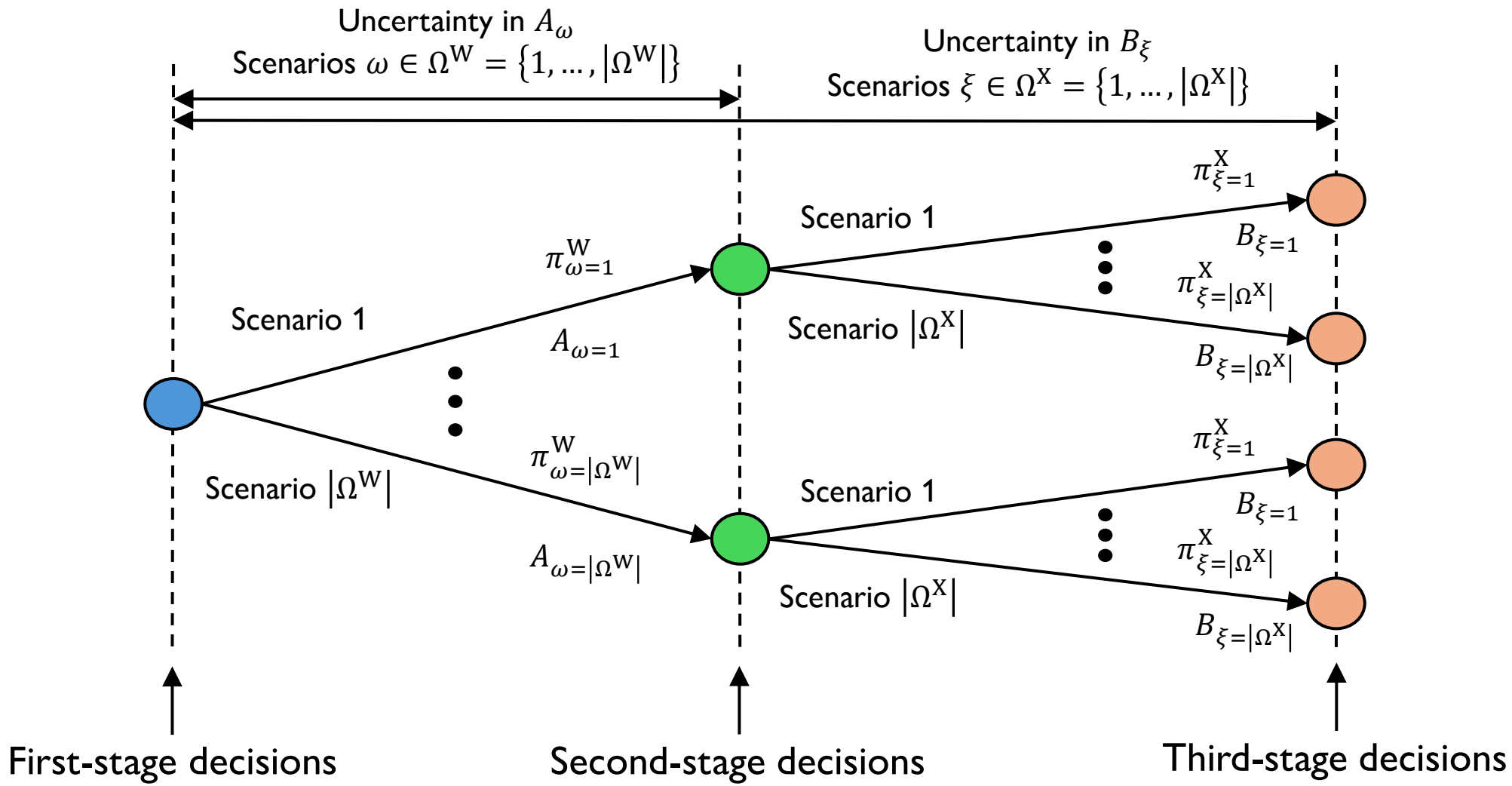
- Each scenario $\omega$ is associated with a value of the probability $\pi_{\omega}^{\mathrm{W}}$ and the uncertain parameters that vary with $\omega$.

- $\sum_{\omega \in \Omega^{\mathrm{W}}} \pi_{\omega}^{\mathrm{W}} = 1.$

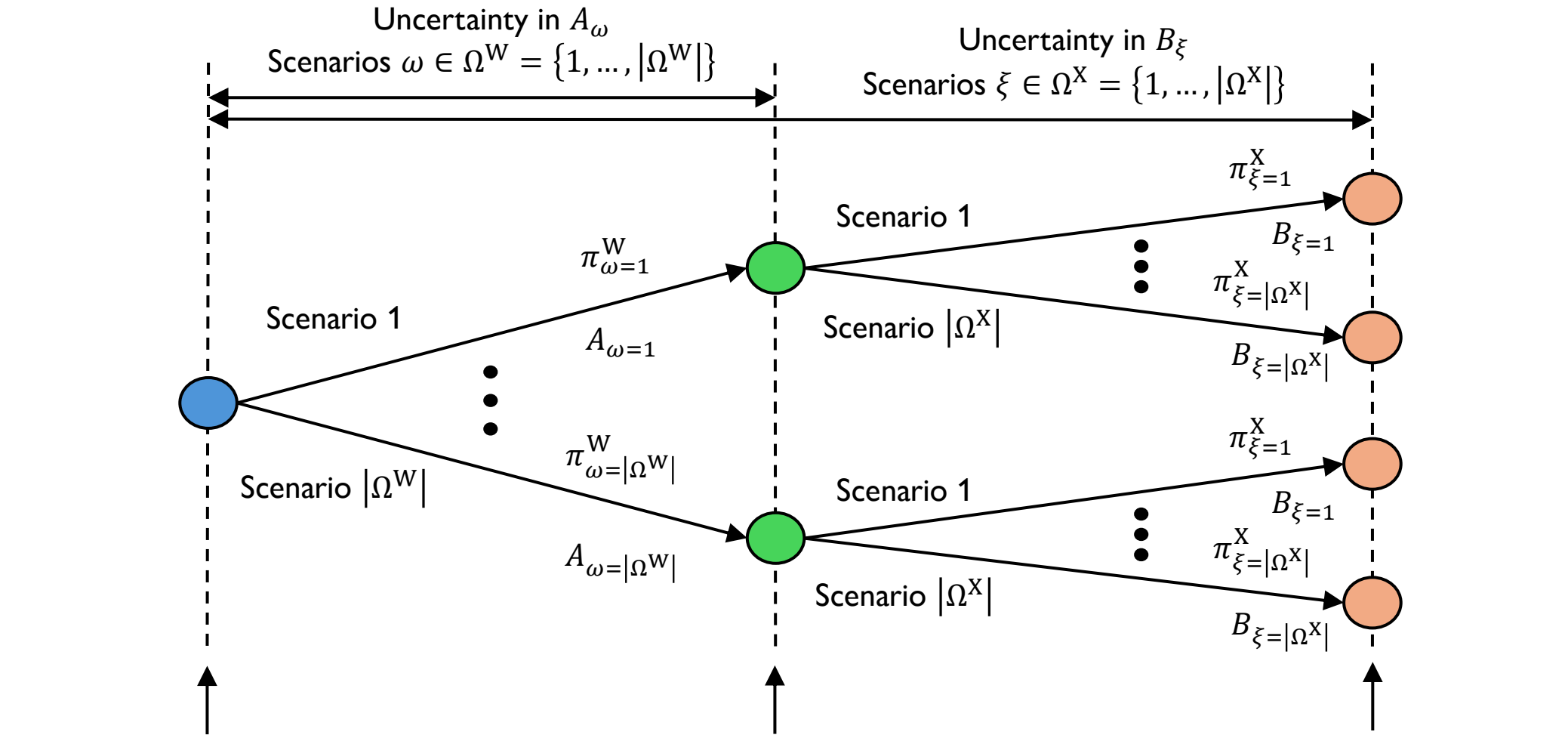
# Stochastic programming



# Stochastic programming



# Stochastic programming




First-stage decisions
(Here-and-now → $\omega$ and $\xi$)

Second-stage decisions
(Wait-and-see → $\omega$
Here-and-now → $\xi$)

Third-stage decisions
(Wait-and-see → $\omega$ and $\xi$)

# Stochastic programming

- Risk-neutral approach.
- Quality metrics.
- Out-of-sample analysis.
- Building bidding and offering curves.
- Risk-averse approach.
- Decomposition techniques.

# Stochastic programming

- **Risk-neutral approach**.
- Quality metrics.
- Out-of-sample analysis.
- Building bidding and offering curves.
- Risk-averse approach.
- Decomposition techniques.

# SP – Risk-neutral approach

$$\min_{\boldsymbol{x}\in\Re^n} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \mathbb{E}[V(\boldsymbol{x},\xi)]$$

**s.t.:**

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$V(x,\xi) = \min_{y\in\Re^k} \quad \{\boldsymbol{q}(\xi)^T\boldsymbol{y} : \boldsymbol{T}(\xi)\boldsymbol{x} + \boldsymbol{R}(\xi)\boldsymbol{y} \leq \boldsymbol{h}(\xi), \boldsymbol{y} \in \boldsymbol{Y}\}$$

# SP – Risk-neutral approach

$$\min_{x \in \Re^n} \quad z = \boldsymbol{c}^T \boldsymbol{x} + \mathbb{E}[V(\boldsymbol{x}, \xi)]$$

Expected value

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$V(x, \xi) = \min_{y \in \Re^k} \quad \{\boldsymbol{q}(\xi)^T \boldsymbol{y} : \boldsymbol{T}(\xi)\boldsymbol{x} + \boldsymbol{R}(\xi)\boldsymbol{y} \leq \boldsymbol{h}(\xi), \boldsymbol{y} \in \boldsymbol{Y}\}$$

# SP – Risk-neutral approach

$$\min_{x;y_\omega,\forall\omega} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \sum_{\omega\in\Omega^W} \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^W$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^W$$

# Exercise 1 – Centralized GEP – Risk-neutral

$$\min_{\Phi} \sum_{\omega \in \Omega^{W}} \pi_{\omega}^{W} \sum_{t \in \Omega^{T}} \rho_{t}^{T} \left( C^{LS} p_{t\omega}^{LS} + \sum_{g \in \Omega^{G}} C_{g}^{G,V} p_{gt\omega}^{G} \right) + \sum_{g \in \Omega^{G,New}} C_{g}^{G,I} CRF_{g} \overline{p}_{g}^{G}$$

**s.t.:**

$$\sum_{g \in \Omega^{G,New}} C_{g}^{G,I} CRF_{g} \overline{p}_{g}^{G} \leq \overline{C}^{I}$$

$$0 \leq \overline{p}_{g}^{G} \leq \overline{P}_{g}^{G,New}, \forall g \in \Omega^{G,New}$$

# Exercise 1 – Centralized GEP – Risk-neutral

$$\sum_{g \in \Omega^{\mathrm{G}}} p^{\mathrm{G}}_{gt\omega} = P^{\mathrm{D}}_{t\omega} - p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega$$

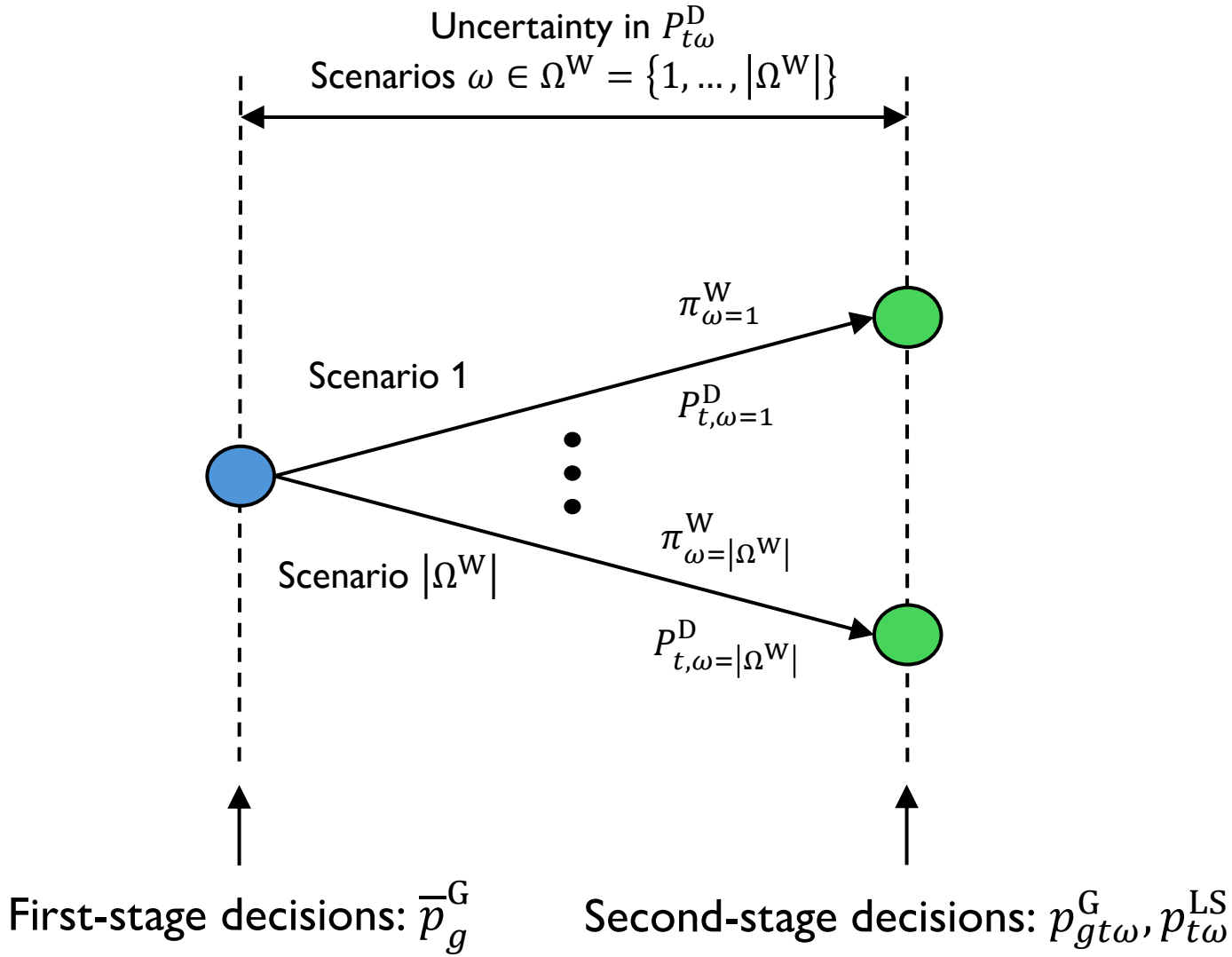
$$0 \leq p^{\mathrm{LS}}_{t\omega} \leq P^{\mathrm{D}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{P}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t, \forall \omega$$

$$\Phi = \left\{ p^{\mathrm{G}}_{gt\omega}, \forall g, \forall t, \forall \omega; \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}; p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega \right\}$$

# Exercise 1 – Centralized GEP – Risk-neutral



# Exercise 1 – Centralized GEP – Risk-neutral

Solve the risk-neutral two-stage SP centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_01.gms

# Exercise 1 – Centralized GEP – Risk-neutral

Deterministic model → Lecture 2, exercise 7.

| Model | Deterministic | Stochastic Programming |
|---|---|---|
| Constraints | 24 | 204 |
| Variables | 16 | 163 |

# Exercise 1 – Centralized GEP – Risk-neutral

| Model | Deterministic | Stochastic Programming |
|---|---|---|
| Objective function value (M€) | 45.28 | 48.77 |
| $\overline{p}^{\mathrm{G}}_{g_2}$ (MW) | 210.232 | 396.899 |
| $\overline{p}^{\mathrm{G}}_{3}$ (MW) | 389.768 | 283.101 |

# Stochastic programming

- Risk-neutral approach.
- **Quality metrics**.
- Out-of-sample analysis.
- Building bidding and offering curves.
- Risk-averse approach.
- Decomposition techniques.

# SP – Quality metrics

What is the advantage of using the stochastic programming model instead of the deterministic model?

Quality metrics:

- Value of the Stochastic Solution (VSS).
- Expected Value of Perfect Information (EVPI).

# SP – Quality metrics – VSS

The Value of the Stochastic Solution quantifies the cost of ignoring uncertainty when making decisions.

# SP – Quality metrics – VSS

The Value of the Stochastic Solution quantifies the cost of ignoring uncertainty when making decisions.

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem $\rightarrow z^S$.

# SP – Quality metrics – VSS

The Value of the Stochastic Solution quantifies the cost of ignoring uncertainty when making decisions.

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem $\rightarrow z^{\mathrm{S}}$.
- Step 2: Solve the deterministic problem considering expected values of the uncertain parameters $\rightarrow \boldsymbol{x}^{\mathrm{D}}$.

# SP – Quality metrics – VSS

The Value of the Stochastic Solution quantifies the cost of ignoring uncertainty when making decisions.

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem $\rightarrow z^{\mathrm{S}}$.
- Step 2: Solve the deterministic problem considering expected values of the uncertain parameters $\rightarrow \boldsymbol{x}^{\mathrm{D}}$.
- Step 3: Solve the SP problem fixing first-stage decision variables to $\boldsymbol{x}^{\mathrm{D}} \rightarrow z^{\mathrm{D}}$.

# SP – Quality metrics – VSS

The Value of the Stochastic Solution quantifies the cost of ignoring uncertainty when making decisions.

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem → $z^{\mathrm{S}}$.
- Step 2: Solve the deterministic problem considering expected values of the uncertain parameters → $\boldsymbol{x}^{\mathrm{D}}$.
- Step 3: Solve the SP problem fixing first-stage decision variables to $\boldsymbol{x}^{\mathrm{D}}$ → $z^{\mathrm{D}}$.
- Step 4: Compute $VSS = z^{\mathrm{D}} - z^{\mathrm{S}}$.

# Exercise 2 – Centralized GEP – VSS

Compute the VSS for the risk-neutral two-stage SP centralized GEP problem presented in exercise 1. Note that the constraints that involve only first-stage decision variables are not needed in Step 3 of the VSS' procedure.

# Exercise 2 – Centralized GEP – VSS

Compute the VSS for the risk-neutral two-stage SP centralized GEP problem presented in exercise 1. Note that the constraints that involve only first-stage decision variables are not needed in Step 3 of the VSS' procedure.

Solution:

High cost due to load shedding

$$VSS = z^{\mathrm{D}} - z^{\mathrm{S}} = 784.57 - 48.77 = \text{M€}735.8$$

# SP – Quality metrics – EVPI

The Expected Value of Perfect Information quantifies the maximum value a decision maker is willing to pay for obtaining perfect information.

# SP – Quality metrics – EVPI

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Procedure:

- Step 1: Solve the SP problem $\rightarrow z^S$.

# SP – Quality metrics – EVPI

The Expected Value of Perfect Information quantifies the maximum value a decision maker is willing to pay for obtaining perfect information.

Procedure:

- Step 1: Solve the SP problem $\rightarrow z^{S}$.
- Step 2: Solve the SP problem with perfect information, i.e., all variables depend on scenario realization so that all variables are wait-and-see decisions $\rightarrow z^{P}$.

# SP – Quality metrics – EVPI

The Expected Value of Perfect Information quantifies the maximum value a decision maker is willing to pay for obtaining perfect information.

Procedure:

- Step 1: Solve the SP problem $\rightarrow z^{\mathrm{S}}$.
- Step 2: Solve the SP problem with perfect information, i.e., all variables depend on scenario realization so that all variables are wait-and-see decisions $\rightarrow z^{\mathrm{P}}$.
- Step 3: Compute $EVPI = z^{\mathrm{S}} - z^{\mathrm{P}}$.

# Exercise 3 – Centralized GEP – EVPI

Compute the EVPI for the risk-neutral two-stage SP centralized GEP problem presented in exercise 1.

# Exercise 3 – Centralized GEP – EVPI

$$\min_{\Phi} \sum_{\omega \in \Omega^{W}} \pi_{\omega}^{W} \left( \sum_{t \in \Omega^{T}} \rho_{t}^{T} \left( C^{LS} p_{t\omega}^{LS} + \sum_{g \in \Omega^{G}} C_{g}^{G,V} p_{gt\omega}^{G} \right) + \sum_{g \in \Omega^{G,New}} C_{g}^{G,I} CRF_{g} \overline{p}_{g\omega}^{G} \right)$$

**s.t.:**

$$\sum_{g \in \Omega^{G,New}} C_{g}^{G,I} CRF_{g} \overline{p}_{g\omega}^{G} \leq \overline{C}^{I}, \forall \omega$$

$$0 \leq \overline{p}_{g\omega}^{G} \leq \overline{P}_{g}^{G,New}, \forall g \in \Omega^{G,New}, \forall \omega$$

Note that this term is multiplied by $\pi_{\omega}^{W}$

# Exercise 3 – Centralized GEP – EVPI

$$\sum_{g \in \Omega^{\mathrm{G}}} p^{\mathrm{G}}_{gt\omega} = P^{\mathrm{D}}_{t\omega} - p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{LS}}_{t\omega} \leq P^{\mathrm{D}}_{t\omega}, \forall t, \forall \omega$$

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$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{p}^{\mathrm{G}}_{g\omega}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t, \forall \omega$$

$$\Phi = \left\{ p^{\mathrm{G}}_{gt\omega}, \forall g, \forall t, \forall \omega; \overline{p}^{\mathrm{G}}_{g\omega}, \forall g \in \Omega^{\mathrm{G,New}}, \forall \omega; p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega \right\}$$

# Exercise 3 – Centralized GEP – EVPI

Solution:

$$EVPI = z^{S} - z^{P} = 48.77 - 46.12 = \text{M€}2.65$$

The decision maker is willing to pay M€2.65 or less for obtaining perfect information.

# Stochastic programming

- Risk-neutral approach.
- Quality metrics.
- **Out-of-sample analysis**.
- Building bidding and offering curves.
- Risk-averse approach.
- Decomposition techniques.

# SP – Out-of-sample analysis

We are solving the SP problem considering a set of in-sample scenarios → In-sample analysis.

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We are solving the SP problem considering a set of in-sample scenarios → In-sample analysis.

What happens if the realized scenario is not considered in the set of in-sample scenarios?

# SP – Out-of-sample analysis

We are solving the SP problem considering a set of in-sample scenarios → In-sample analysis.

What happens if the realized scenario is not considered in the set of in-sample scenarios?

We should perform out-of-sample analyses.

# SP – Out-of-sample analysis

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem considering the set of in-sample scenarios $\rightarrow z^{\mathrm{I}}, x^{\mathrm{I}}$.

# SP – Out-of-sample analysis

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem considering the set of in-sample scenarios $\rightarrow z^{\mathrm{I}}, x^{\mathrm{I}}$.
- Step 2: Solve the SP problem considering the set of out-of-sample scenarios and fixing first-stage decision variables to $\boldsymbol{x}^{\mathrm{I}} \rightarrow z^{\mathrm{O}}$.

# SP – Out-of-sample analysis

Procedure for two-stage SP problems:

- Step 1: Solve the SP problem considering the set of in-sample scenarios $\rightarrow z^{\mathrm{I}}, x^{\mathrm{I}}$.
- Step 2: Solve the SP problem considering the set of out-of-sample scenarios and fixing first-stage decision variables to $\boldsymbol{x}^{\mathrm{I}} \rightarrow z^{\mathrm{O}}$.
- Step 3: Compare $z^{\mathrm{I}}$ with $z^{\mathrm{O}}$.

# SP – Out-of-sample analysis

The out-of-sample analysis measures how robust the first-stage decisions are against other scenarios not considered in the in-sample analysis.

The closer the solution of the out-of-sample analysis to the solution of the in-sample analysis, the better.

# Exercise 4 – Centralized GEP – Out-of-sample analysis

Solve the risk-neutral two-stage SP centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_04.gms

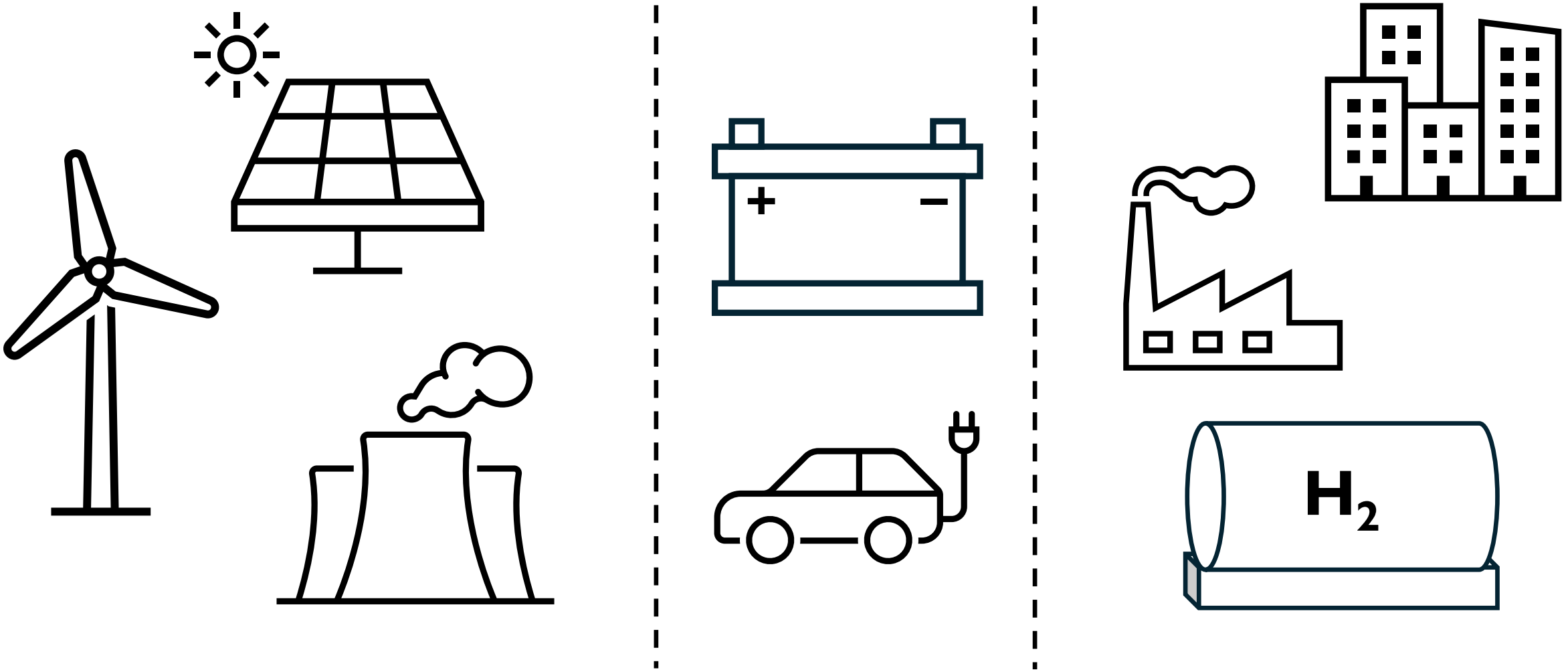
Solution:

- In-sample analysis (exercise 1) $\rightarrow z^{\mathrm{I}} =$ M€48.77.
- Out-of-sample analysis $\rightarrow z^{\mathrm{O}} =$ M€48.82.

# Stochastic programming

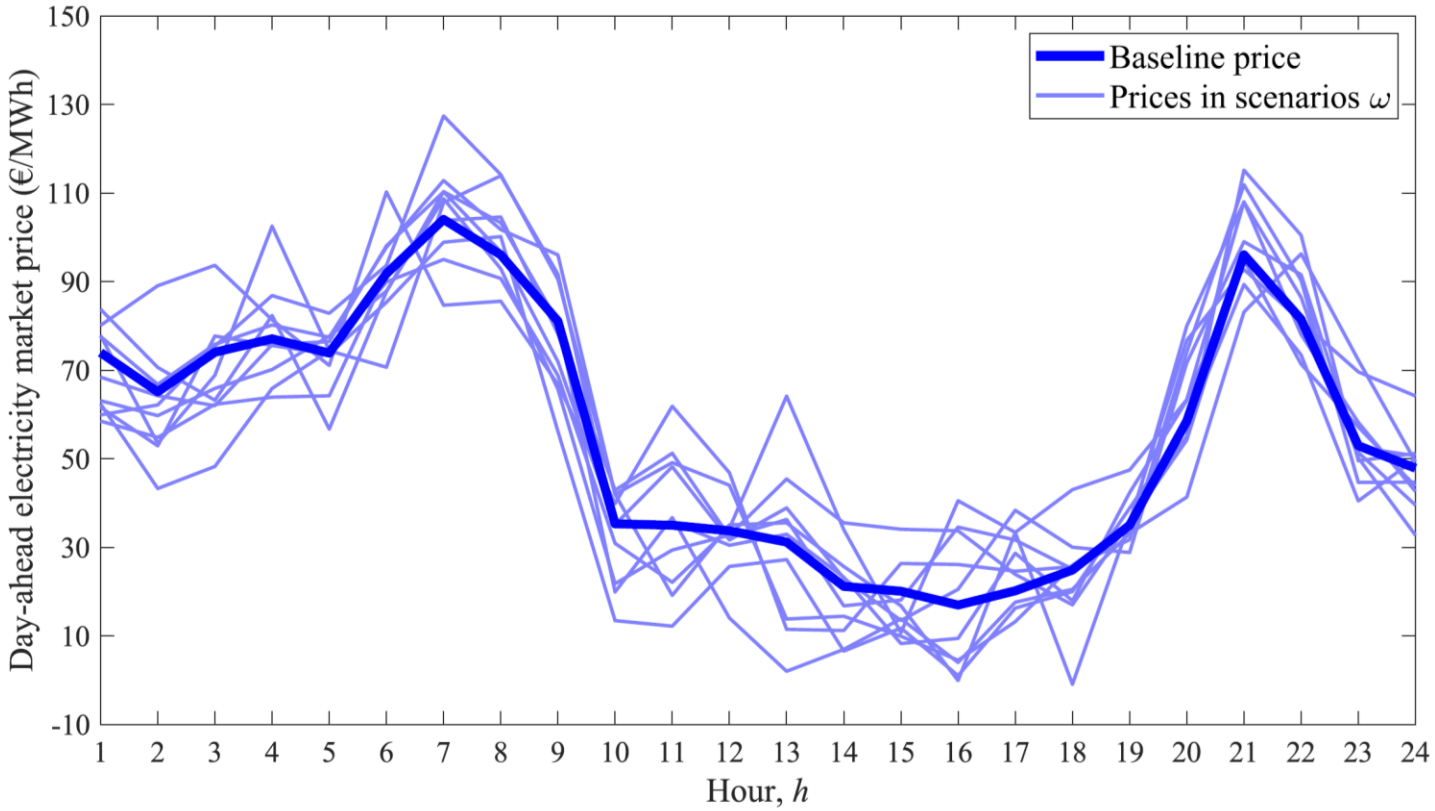
- Risk-neutral approach.
- Quality metrics.
- Out-of-sample analysis.
- **Building bidding and offering curves**.
- Risk-averse approach.
- Decomposition techniques.

# Exercise 5 – VPP' self-scheduling – Risk-neutral



# Exercise 5 – Scenario generation

## DA electricity market price scenarios $\omega$.




A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

# Exercise 5 – Scenario generation

DA electricity market price scenarios $\omega$.

$$\lambda_{t\omega}^{\mathrm{DA}} = \bar{\lambda}_{t}^{\mathrm{DA}} + \varepsilon_t + \phi\left(\lambda_{t-1,\omega}^{\mathrm{DA}} - \bar{\lambda}_{t-1}^{\mathrm{DA}}\right), \forall t \in \Omega^{\mathrm{T}}, \forall \omega \in \Omega^{\mathrm{W}}$$

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

# Exercise 5 – Scenario generation

DA electricity market price scenarios $\omega$.

$$\lambda_{t\omega}^{\mathrm{DA}} = \bar{\lambda}_t^{\mathrm{DA}} + \varepsilon_t + \phi\left(\lambda_{t-1,\omega}^{\mathrm{DA}} - \bar{\lambda}_{t-1}^{\mathrm{DA}}\right), \forall t \in \Omega^{\mathrm{T}}, \forall \omega \in \Omega^{\mathrm{W}}$$

Baseline price

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

# Exercise 5 – Scenario generation

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Random error term

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

# Exercise 5 – Scenario generation

DA electricity market price scenarios $\omega$.

$$\lambda_{t\omega}^{\mathrm{DA}} = \bar{\lambda}_{t}^{\mathrm{DA}} + \varepsilon_t + \phi\left(\lambda_{t-1,\omega}^{\mathrm{DA}} - \bar{\lambda}_{t-1}^{\mathrm{DA}}\right), \forall t \in \Omega^{\mathrm{T}}, \forall \omega \in \Omega^{\mathrm{W}}$$

Scaled factor of the autoregressive term

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

# Exercise 5 – VPP' self-scheduling – Risk-neutral

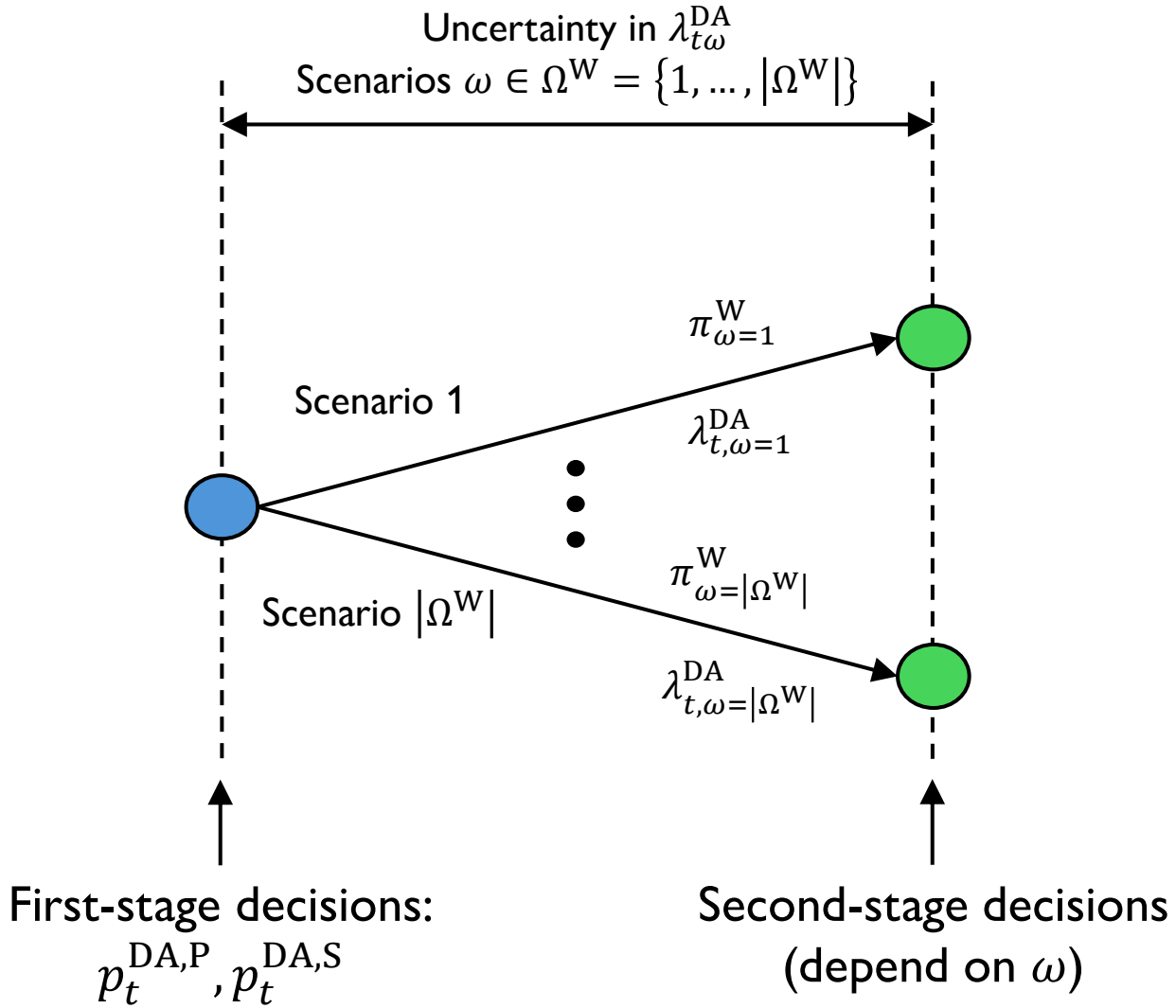
$$\max_{\Phi} \sum_{\omega \in \Omega^{\mathrm{W}}} \pi_{\omega}^{\mathrm{W}} \sum_{t \in \Omega^{\mathrm{T}}} \left( \lambda_{t\omega}^{\mathrm{DA}} p_{t}^{\mathrm{DA,S}} - \left( \lambda_{t\omega}^{\mathrm{DA}} + \lambda_{t\omega}^{\mathrm{C}} \right) p_{t}^{\mathrm{DA,P}} - C^{\mathrm{G,V}} p_{t\omega}^{\mathrm{G}} \right)$$

$\lambda_{t\omega}^{\mathrm{C}}$: Tolls and charges added to $\lambda_{t\omega}^{\mathrm{DA}}$.

Note that $\lambda_{t\omega}^{\mathrm{C}} = 0.3 \left| \lambda_{t\omega}^{\mathrm{DA}} \right|$ in this exercise.

Hence, the uncertain parameter is $\lambda_{t\omega}^{\mathrm{DA}}$.

# Exercise 5 – VPP' self-scheduling – Risk-neutral



# Exercise 5 – VPP' self-scheduling – Risk-neutral

$$\max_{\Phi} \sum_{\omega \in \Omega^{W}} \pi_{\omega}^{W} \sum_{t \in \Omega^{T}} \left( \lambda_{t\omega}^{DA} p_{t}^{DA,S} - \left( \lambda_{t\omega}^{DA} + \lambda_{t\omega}^{C} \right) p_{t}^{DA,P} - C^{G,V} p_{t\omega}^{G} \right)$$

**s.t.:**

$$0 \leq p_{t}^{DA,P} \leq \overline{P}^{DA,P}, \forall t$$

$$0 \leq p_{t}^{DA,S} \leq \overline{P}^{DA,S}, \forall t$$

$$p_{t}^{DA,S} - p_{t}^{DA,P} = p_{t\omega}^{G} + p_{t\omega}^{PV} + p_{t\omega}^{B,D} - p_{t\omega}^{B,C} - p_{t\omega}^{D}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{G} \leq \overline{P}^{G}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{PV} \leq \overline{P}^{PV} A_{t}^{PV}, \forall t, \forall \omega$$

# Exercise 5 – VPP' self-scheduling – Risk-neutral

$$e_{t\omega}^{\mathrm{B}} = e_{t-1,\omega}^{\mathrm{B}} + p_{t\omega}^{\mathrm{B,C}}\eta^{\mathrm{B,C}} - p_{t\omega}^{\mathrm{B,D}}/\eta^{\mathrm{B,D}}, \forall t, \forall \omega$$

$$\underline{E}^{\mathrm{B}} \leq e_{t\omega}^{\mathrm{B}} \leq \overline{E}^{\mathrm{B}}, \forall t, \forall \omega$$

$$e_{t\omega}^{\mathrm{B}} \geq E^{\mathrm{B,F}}, t = \left|\Omega^{\mathrm{T}}\right|, \forall \omega$$

$$u_{t\omega}^{\mathrm{B}} \in \{0,1\}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{B,C}} \leq \overline{P}^{\mathrm{B,C}} u_{t\omega}^{\mathrm{B}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{B,D}} \leq \overline{P}^{\mathrm{B,D}} \left(1 - u_{t\omega}^{\mathrm{B}}\right), \forall t, \forall \omega$$

# Exercise 5 – VPP' self-scheduling – Risk-neutral

$$p_{t\omega}^{\mathrm{D}} = P_t^{\mathrm{D,I}} + p_{t\omega}^{\mathrm{D,F}}, \forall t, \forall \omega$$

$$\underline{P}_t^{\mathrm{D,F}} \leq p_{t\omega}^{\mathrm{D,F}} \leq \overline{P}_t^{\mathrm{D,F}}, \forall t, \forall \omega$$

$$\sum_{t \in \Omega^{\mathrm{T}}} p_{t\omega}^{\mathrm{D,F}} \geq \underline{E}^{\mathrm{D,F}}, \forall \omega$$

$$\Phi = \left\{ p_t^{\mathrm{DA,P}}, p_t^{\mathrm{DA,S}}, \forall t; e_{t\omega}^{\mathrm{B}}, p_{t\omega}^{\mathrm{B,C}}, p_{t\omega}^{\mathrm{B,D}}, p_{t\omega}^{\mathrm{D}}, p_{t\omega}^{\mathrm{G}}, p_{t\omega}^{\mathrm{PV}}, u_{t\omega}^{\mathrm{B}}, \forall t, \forall \omega \right\}$$

# Exercise 5 – VPP' self-scheduling – Risk-neutral

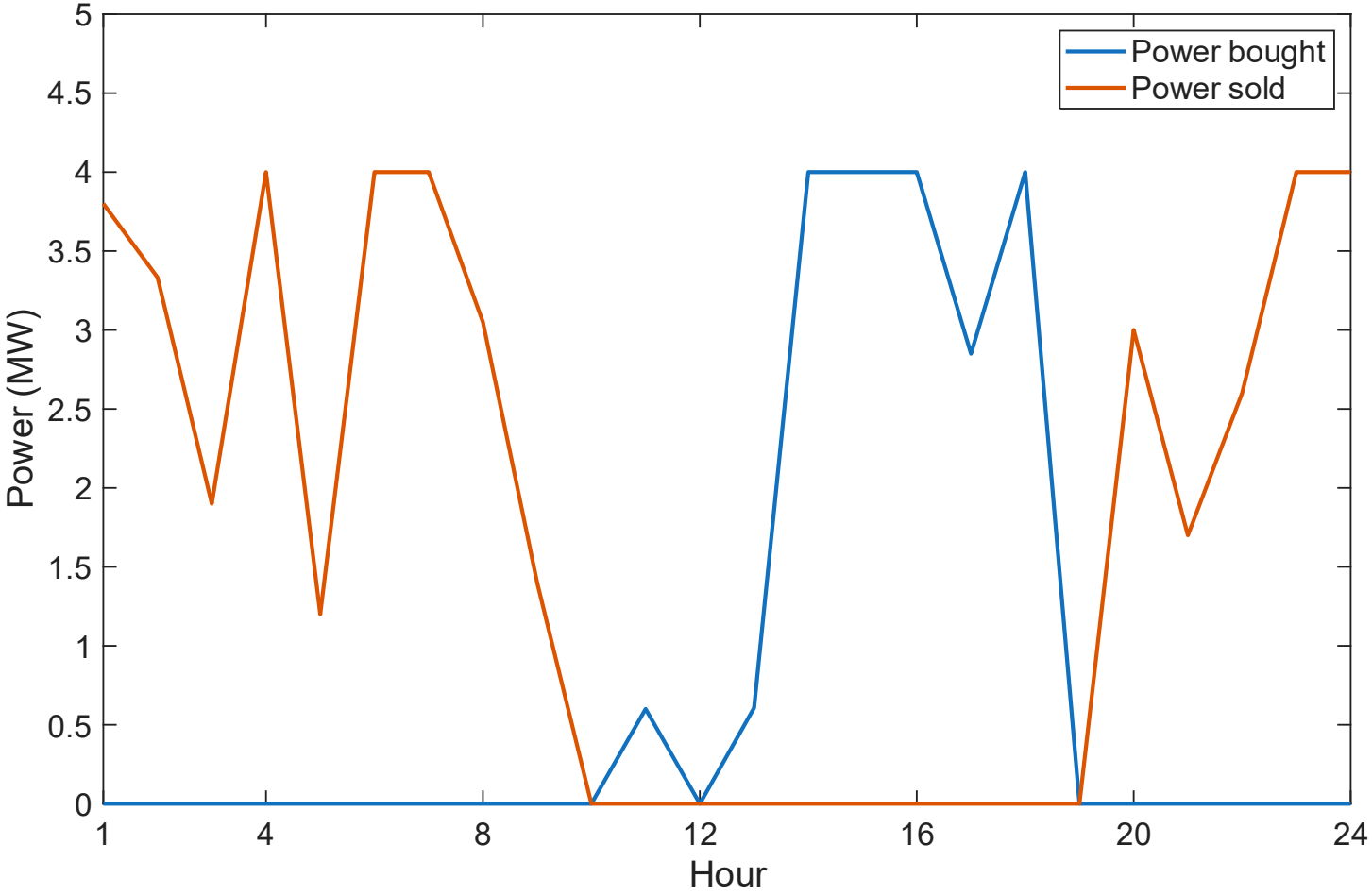
Solve the risk-neutral two-stage SP self-scheduling problem of a price-taker VPP using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_05.gms

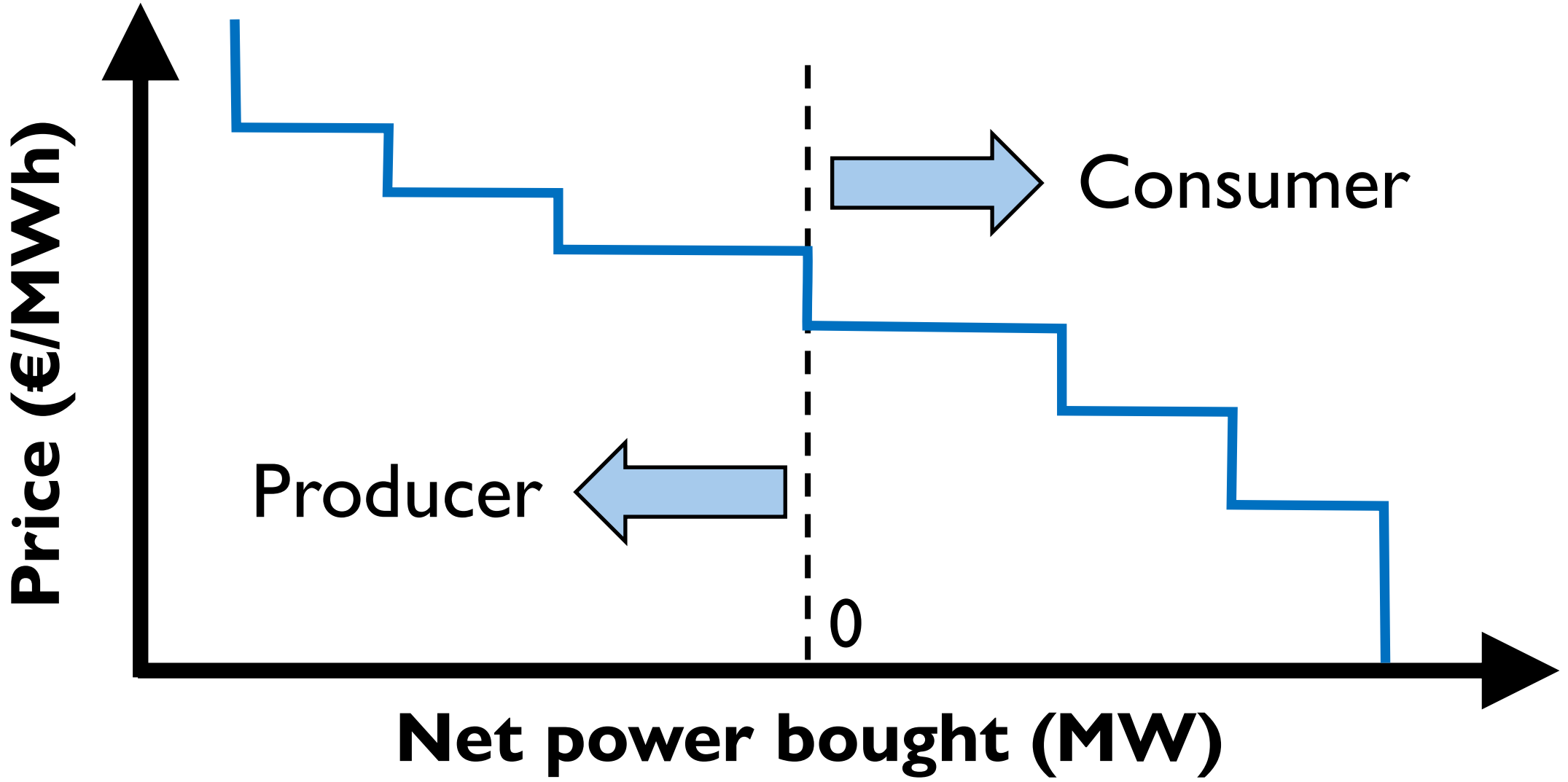
Solution:

Objective function value = − €778.54.

# Exercise 5 – VPP' self-scheduling – Risk-neutral



# SP – Building DA bidding and offering curves



# SP – Building DA bidding and offering curves

Net power bought in the DA electricity market:

$$p_{t\omega}^{\mathrm{DA}} = p_{t\omega}^{\mathrm{DA,P}} - p_{t\omega}^{\mathrm{DA,S}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t, \forall \omega$$

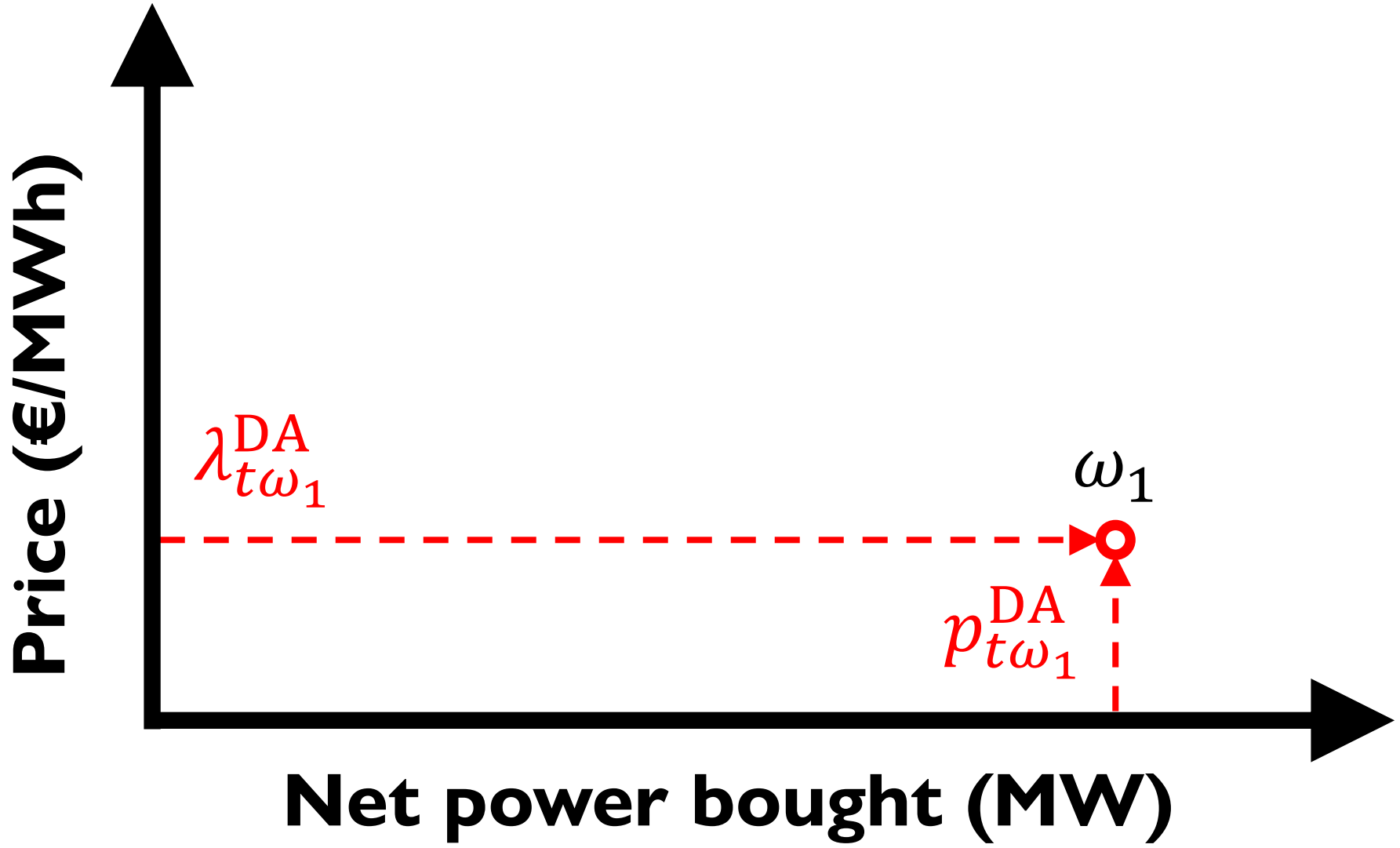
# SP – Building DA bidding and offering curves

Monotonically decreasing curve:

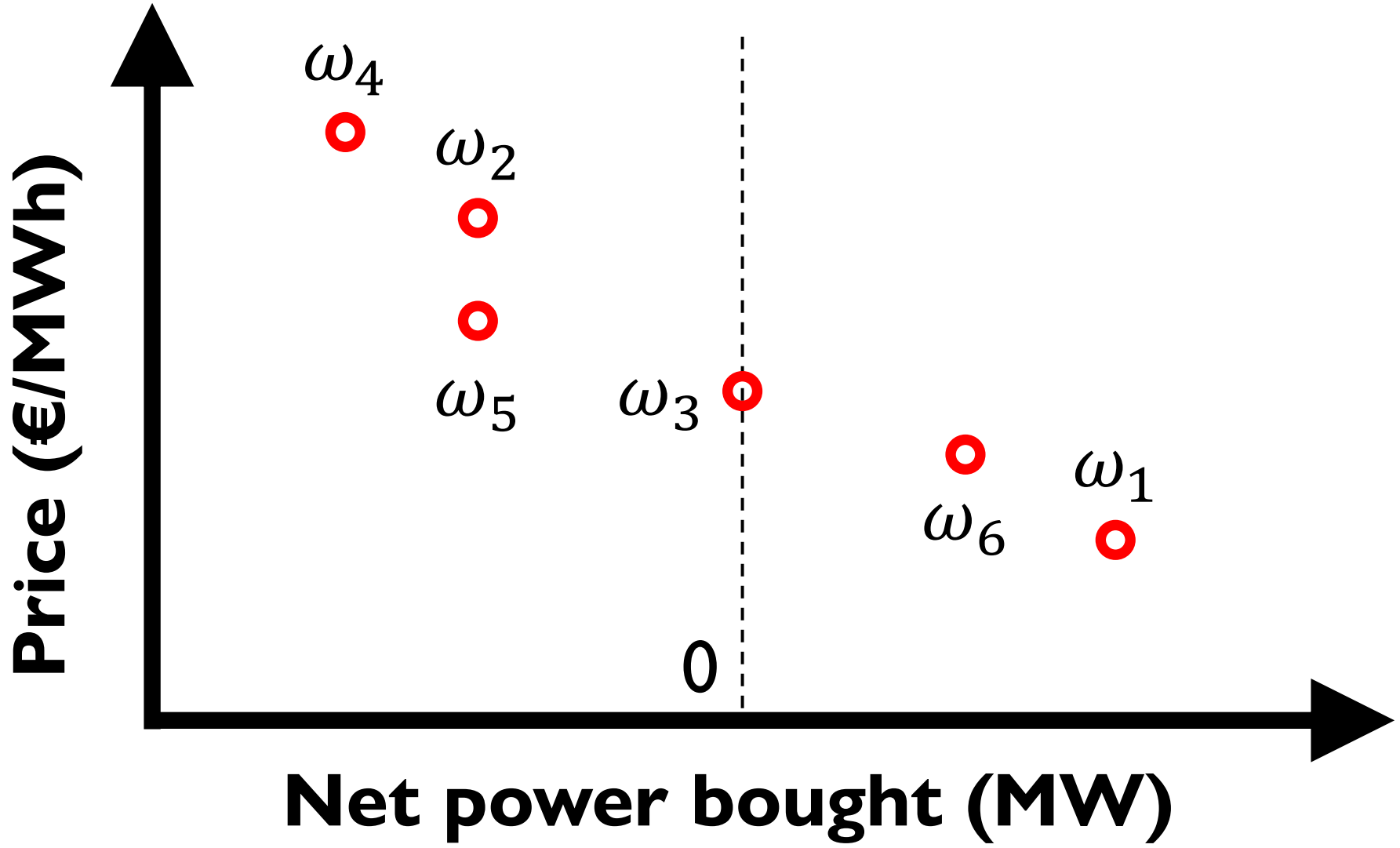
$$p_{t\omega}^{\mathrm{DA}} \geq p_{t\tilde{\omega}}^{\mathrm{DA}} \text{ if } \lambda_{t\omega}^{\mathrm{DA}} \leq \lambda_{t\tilde{\omega}}^{\mathrm{DA}}, \forall h, \forall \omega, \tilde{\omega}$$

If the DA electricity market price in scenario $\omega$ is lower than or equal to the price in scenario $\tilde{\omega}$, then the net power bought in scenario $\omega$ must be higher than or equal to the net power bought in scenario $\tilde{\omega}$.

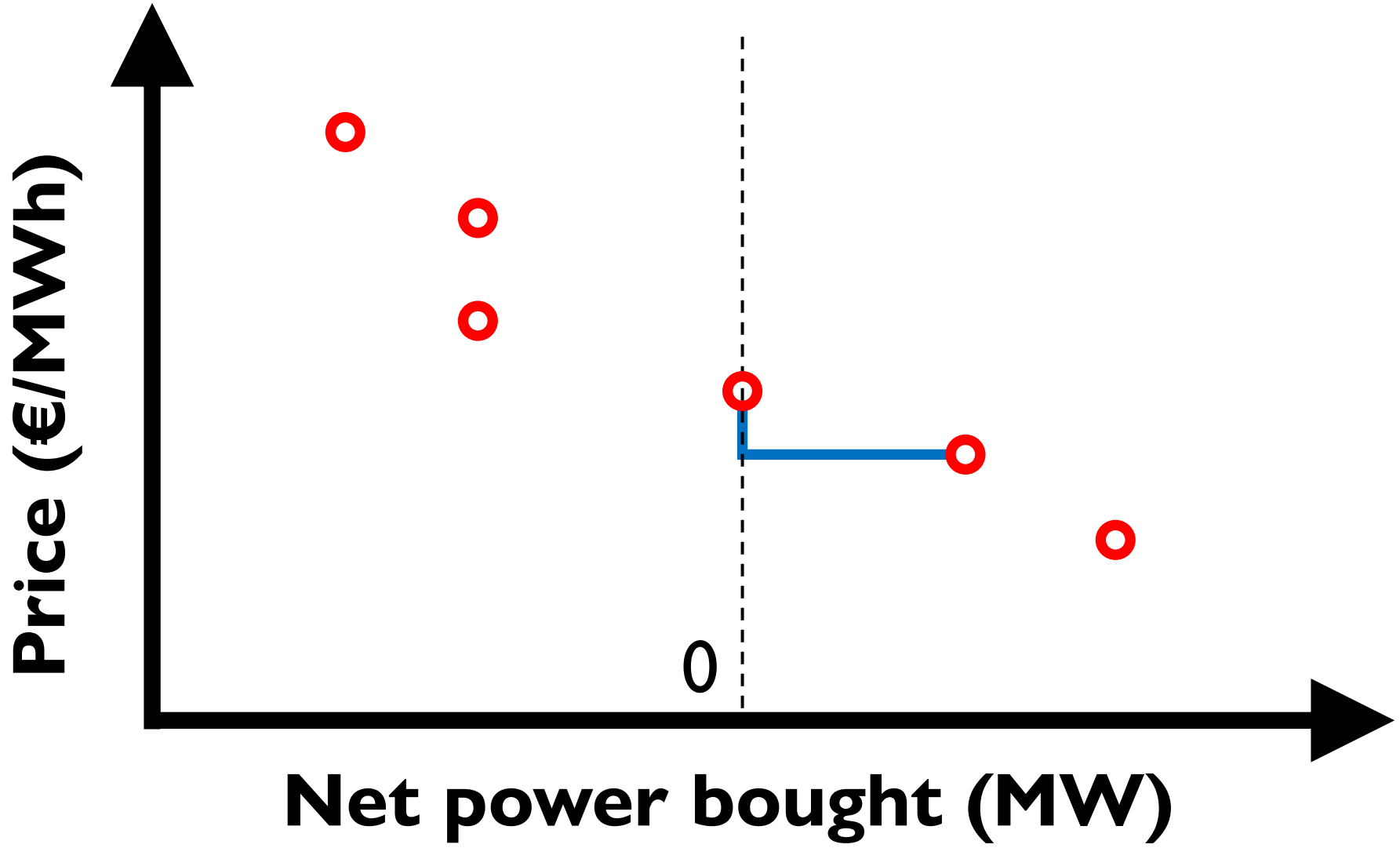
# SP – Building DA bidding and offering curves



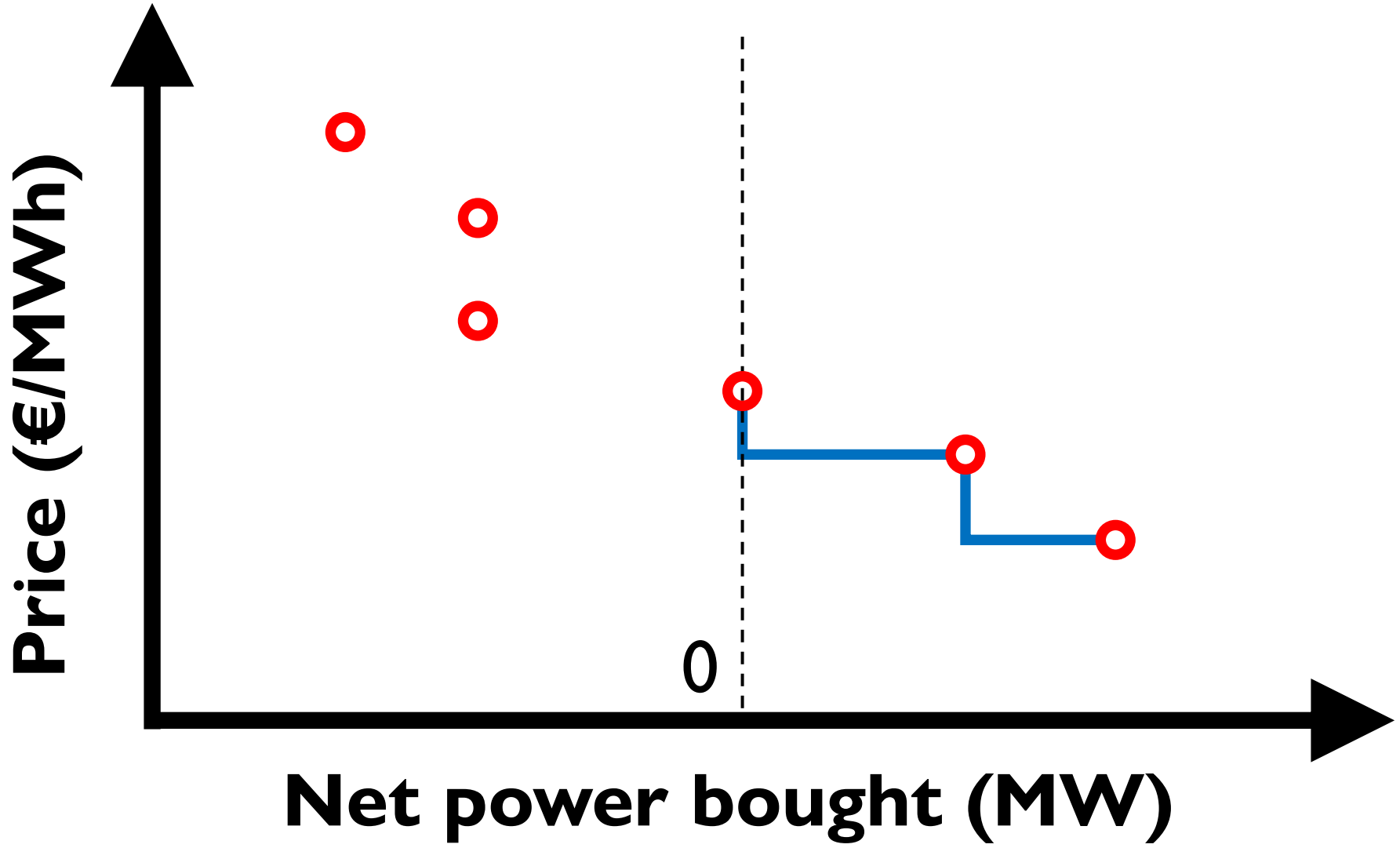
# SP – Building DA bidding and offering curves



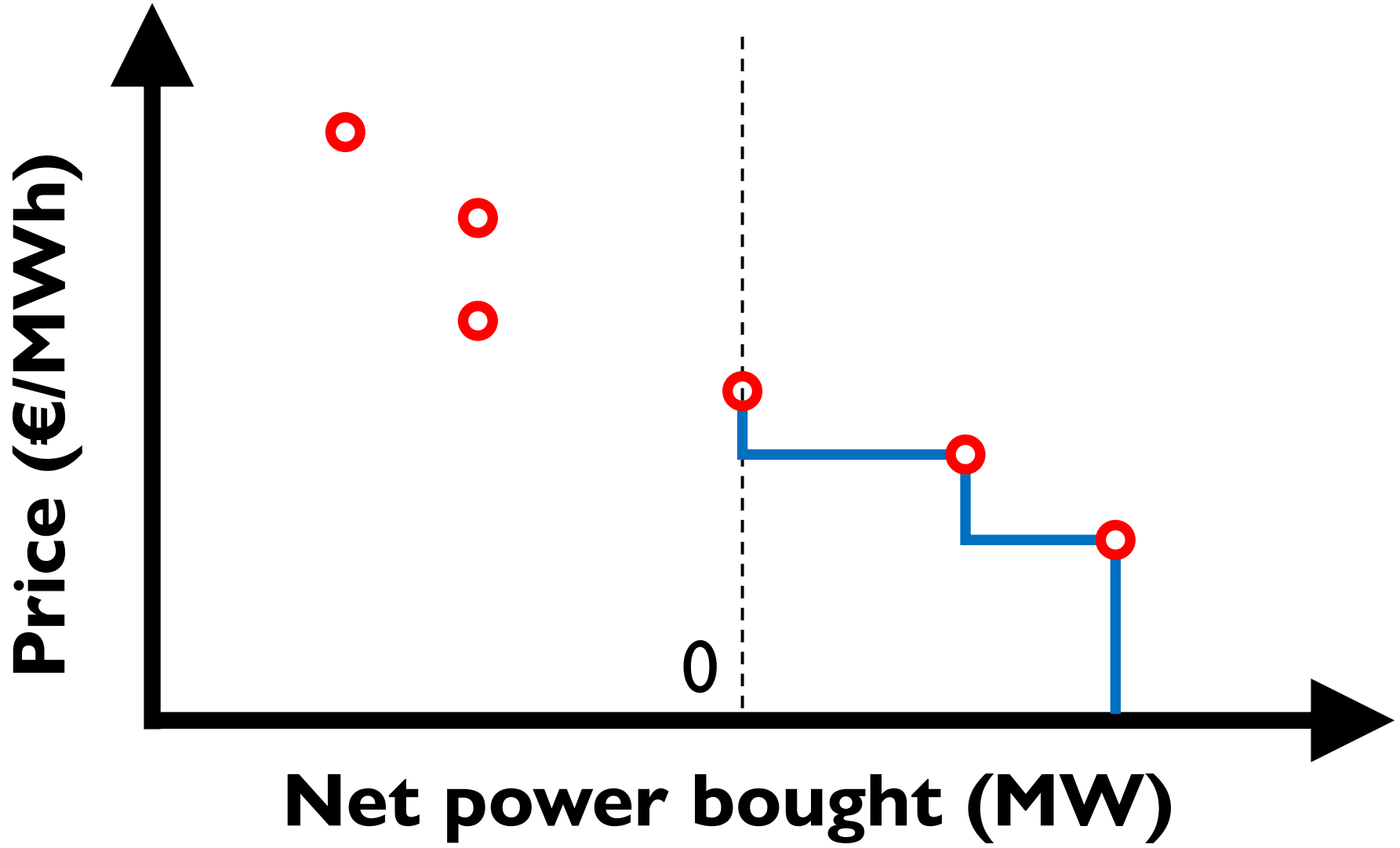
# SP – Building DA bidding and offering curves



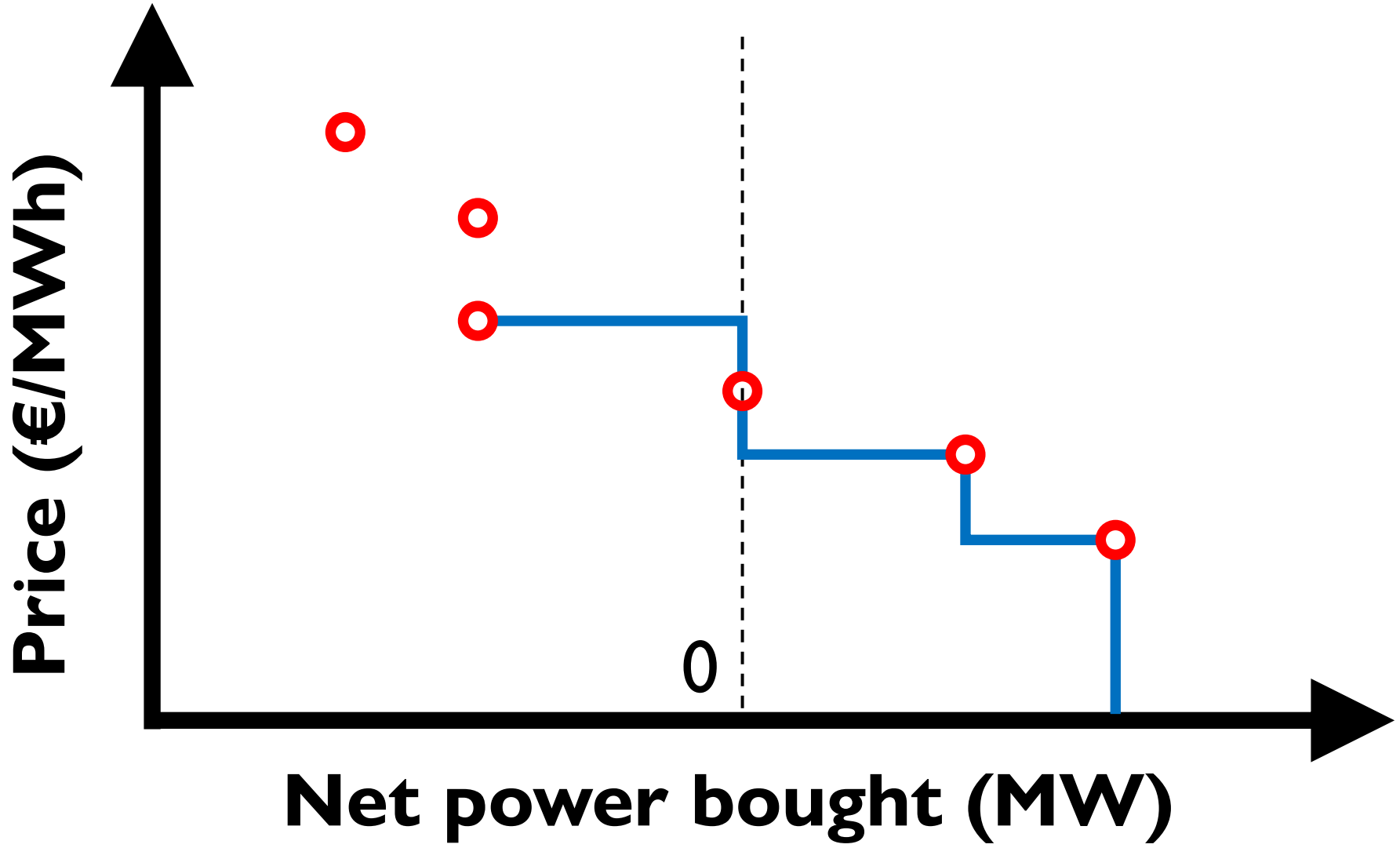
# SP – Building DA bidding and offering curves



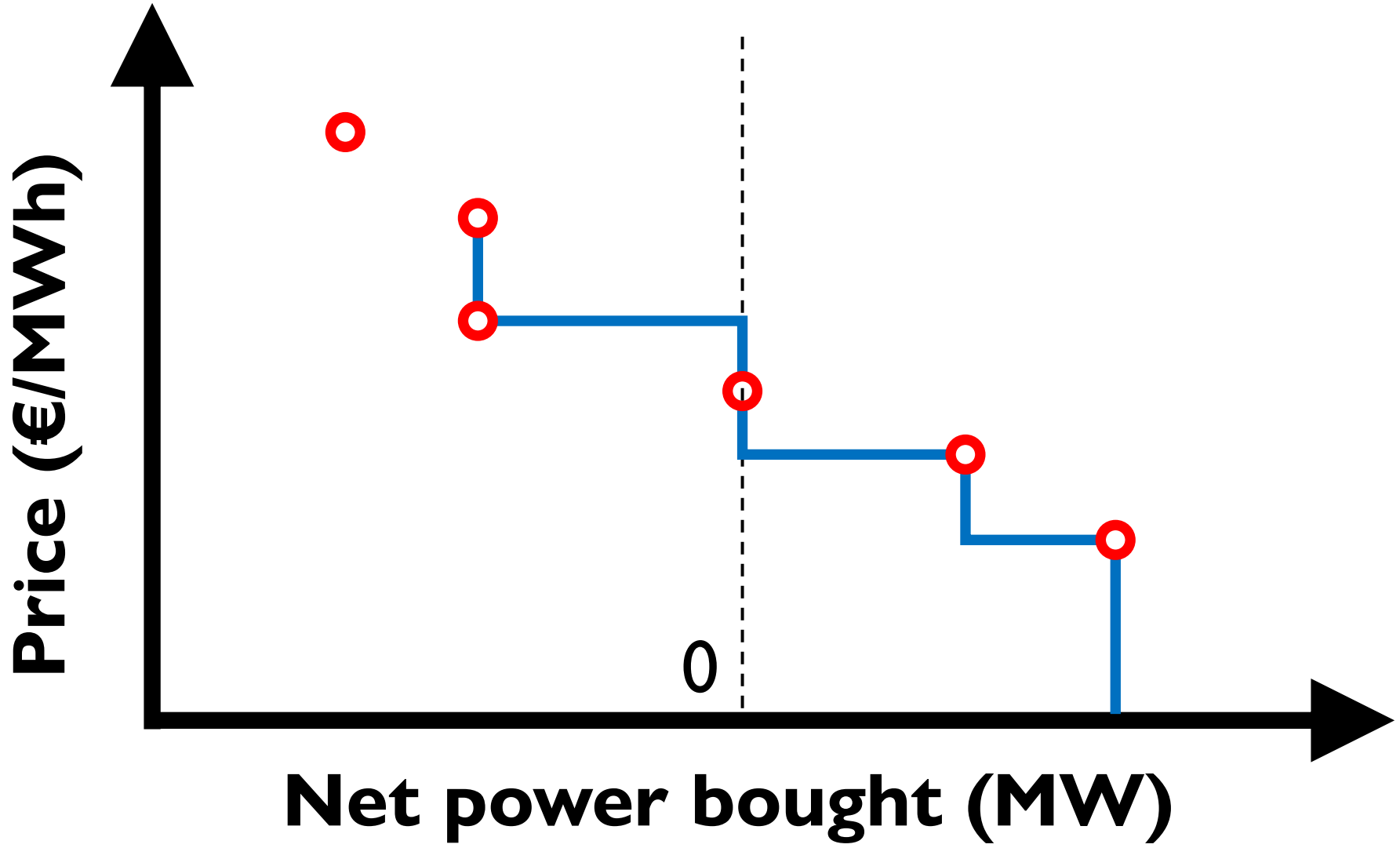
# SP – Building DA bidding and offering curves



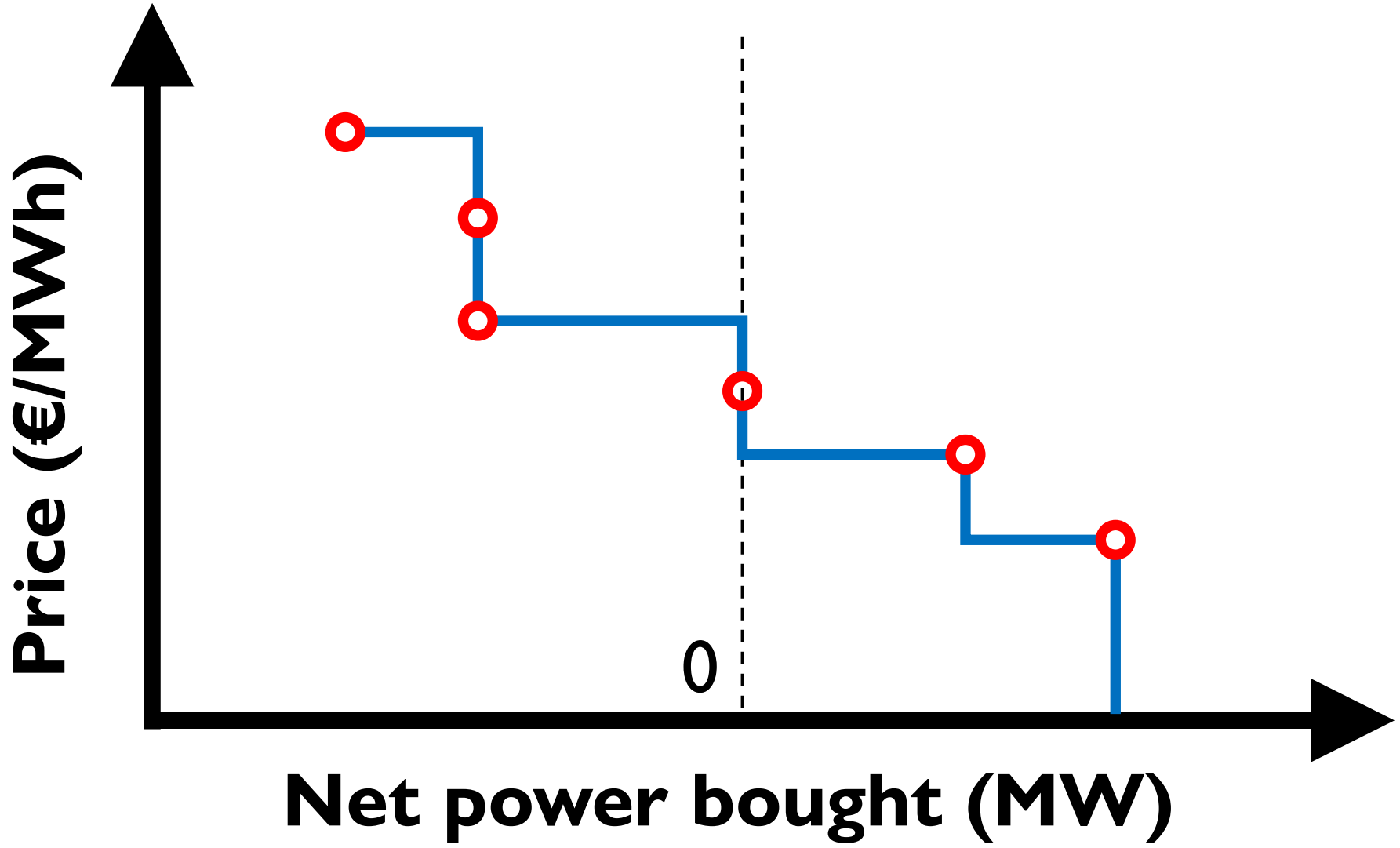
# SP – Building DA bidding and offering curves



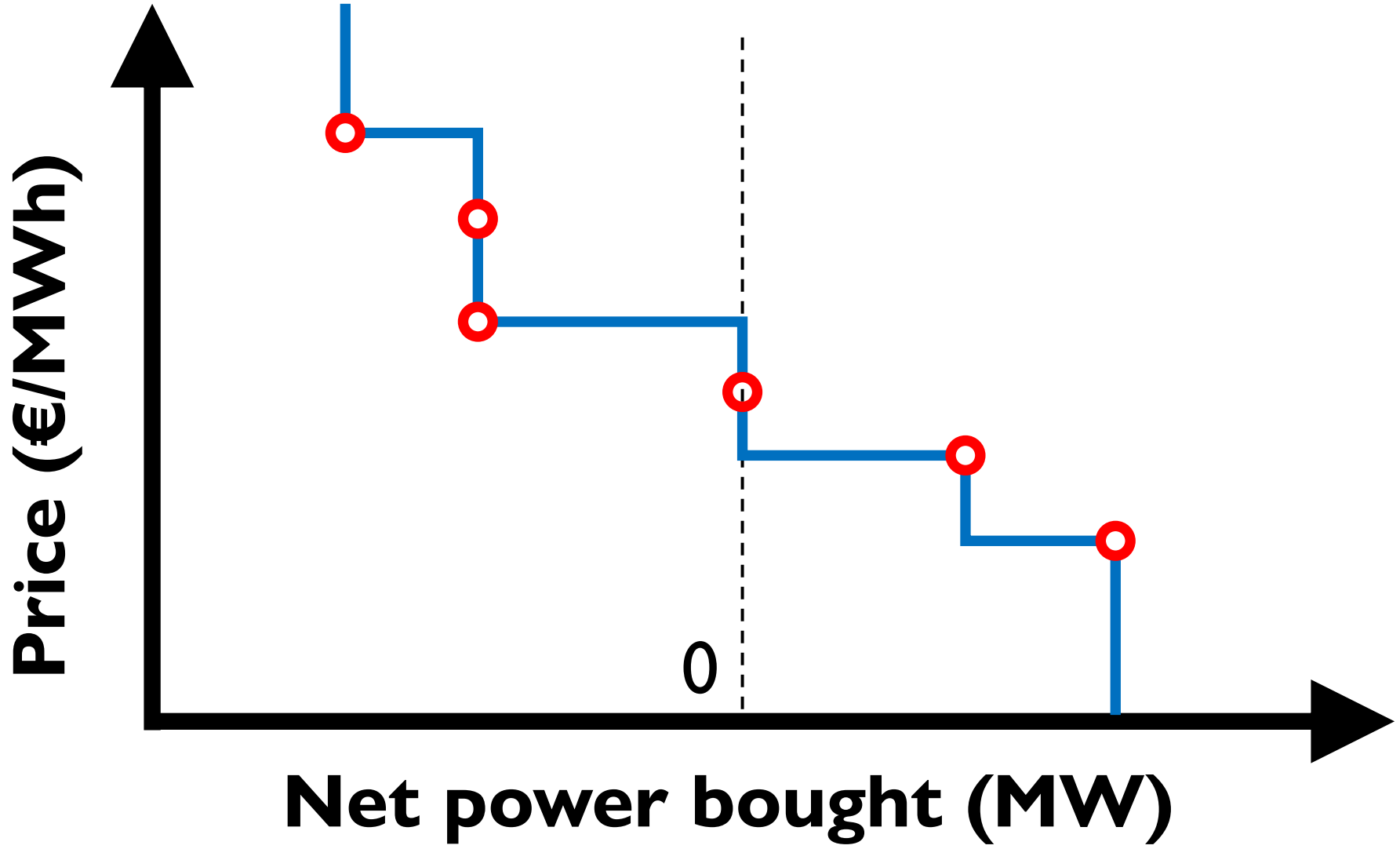
# SP – Building DA bidding and offering curves



# SP – Building DA bidding and offering curves



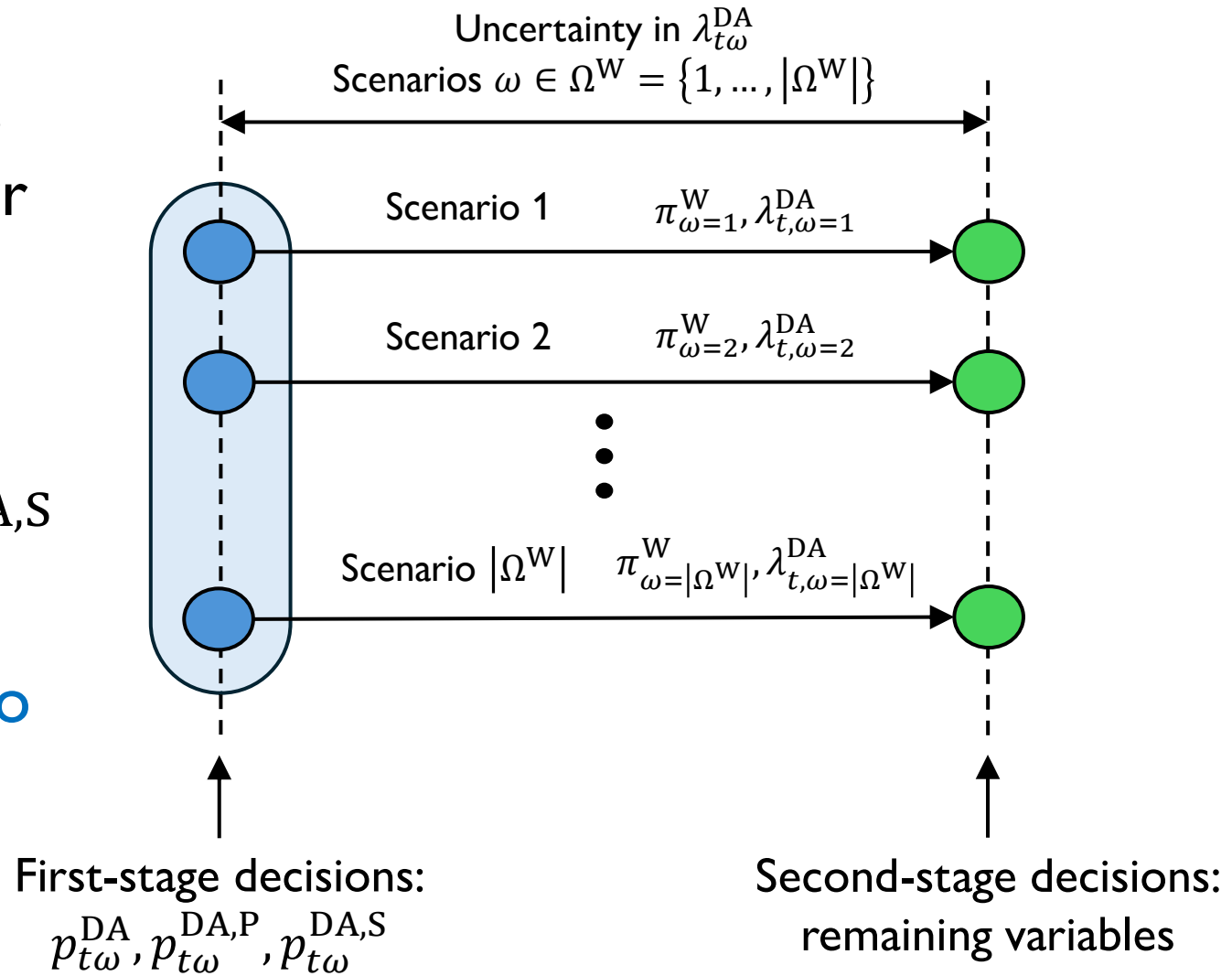
# SP – Building DA bidding and offering curves



# SP – Building DA bidding and offering curves

The bidding and offering curves are sent to the market operator before the uncertainty realization of $\lambda_{t\omega}^{\mathrm{DA}}$ is known.

This is why $p_{t\omega}^{\mathrm{DA}}, p_{t\omega}^{\mathrm{DA,P}}$, and $p_{t\omega}^{\mathrm{DA,S}}$ are first-stage decisions, even though they depend on scenario $\omega$.



# SP – Building DA bidding and offering curves

Monotonically decreasing curve:

$$p_{t\omega}^{\mathrm{DA}} \geq p_{t\tilde{\omega}}^{\mathrm{DA}} \text{ if } \lambda_{t\omega}^{\mathrm{DA}} \leq \lambda_{t\tilde{\omega}}^{\mathrm{DA}}, \forall h, \forall \omega, \tilde{\omega}$$

Variables $p_{t\omega}^{\mathrm{DA}}$ are associated with the first stage due to these constraints.

If these constraints were removed, then the resulting problem could be solved independently for each scenario $\omega$.

# Exercise 6 – VPP' self-scheduling – Curves

$$\max_{\Phi} \sum_{\omega \in \Omega^{\mathrm{W}}} \pi_{\omega}^{\mathrm{W}} \sum_{t \in \Omega^{\mathrm{T}}} \left( \lambda_{t\omega}^{\mathrm{DA}} p_{t\omega}^{\mathrm{DA,S}} - \left( \lambda_{t\omega}^{\mathrm{DA}} + \lambda_{t\omega}^{\mathrm{C}} \right) p_{t\omega}^{\mathrm{DA,P}} - C^{\mathrm{G,V}} p_{t\omega}^{\mathrm{G}} \right)$$

**s.t.:**

$$p_{t\omega}^{\mathrm{DA}} = p_{t\omega}^{\mathrm{DA,P}} - p_{t\omega}^{\mathrm{DA,S}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall t, \forall \omega$$

$$p_{t\omega}^{\mathrm{DA}} \geq p_{t\tilde{\omega}}^{\mathrm{DA}} \text{ if } \lambda_{t\omega}^{\mathrm{DA}} \leq \lambda_{t\tilde{\omega}}^{\mathrm{DA}}, \forall t, \forall \omega, \tilde{\omega}$$

$$p_{t\omega}^{\mathrm{DA,S}} - p_{t\omega}^{\mathrm{DA,P}} = p_{t\omega}^{\mathrm{G}} + p_{t\omega}^{\mathrm{PV}} + p_{t\omega}^{\mathrm{B,D}} - p_{t\omega}^{\mathrm{B,C}} - p_{t\omega}^{\mathrm{D}}, \forall t, \forall \omega$$

# Exercise 6 – VPP' self-scheduling – Curves

$$0 \leq p_{t\omega}^{\mathrm{G}} \leq \overline{P}^{\mathrm{G}}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{PV}} \leq \overline{P}^{\mathrm{PV}} A_t^{\mathrm{PV}}, \forall t, \forall \omega$$

$$e_{t\omega}^{\mathrm{B}} = e_{t-1,\omega}^{\mathrm{B}} + p_{t\omega}^{\mathrm{B,C}} \eta^{\mathrm{B,C}} - p_{t\omega}^{\mathrm{B,D}} / \eta^{\mathrm{B,D}}, \forall t, \forall \omega$$

$$\underline{E}^{\mathrm{B}} \leq e_{t\omega}^{\mathrm{B}} \leq \overline{E}^{\mathrm{B}}, \forall t, \forall \omega$$

$$e_{t\omega}^{\mathrm{B}} \geq E^{\mathrm{B,F}}, t = \left|\Omega^{\mathrm{T}}\right|, \forall \omega$$

$$u_{t\omega}^{\mathrm{B}} \in \{0,1\}, \forall t, \forall \omega$$

$$0 \leq p_{t\omega}^{\mathrm{B,C}} \leq \overline{P}^{\mathrm{B,C}} u_{t\omega}^{\mathrm{B}}, \forall t, \forall \omega$$

# Exercise 6 – VPP' self-scheduling – Curves

$$0 \leq p_{t\omega}^{\mathrm{B,D}} \leq \overline{P}^{\mathrm{B,D}}\left(1 - u_{t\omega}^{\mathrm{B}}\right), \forall t, \forall \omega$$

$$p_{t\omega}^{\mathrm{D}} = P_t^{\mathrm{D,I}} + p_{t\omega}^{\mathrm{D,F}}, \forall t, \forall \omega$$

$$\underline{P}_t^{\mathrm{D,F}} \leq p_{t\omega}^{\mathrm{D,F}} \leq \overline{P}_t^{\mathrm{D,F}}, \forall t, \forall \omega$$

$$\sum_{t \in \Omega^{\mathrm{T}}} p_{t\omega}^{\mathrm{D,F}} \geq \underline{E}^{\mathrm{D,F}}, \forall \omega$$

$$\Phi = \left\{e_{t\omega}^{\mathrm{B}}, p_{t\omega}^{\mathrm{B,C}}, p_{t\omega}^{\mathrm{B,D}}, p_{t\omega}^{\mathrm{D}}, p_{t\omega}^{\mathrm{DA}}, p_{t\omega}^{\mathrm{DA,P}}, p_{t\omega}^{\mathrm{DA,S}}, p_{t\omega}^{\mathrm{G}}, p_{t\omega}^{\mathrm{PV}}, u_{t\omega}^{\mathrm{B}}, \forall t, \forall \omega\right\}$$

# Exercise 6 – VPP' self-scheduling – Curves

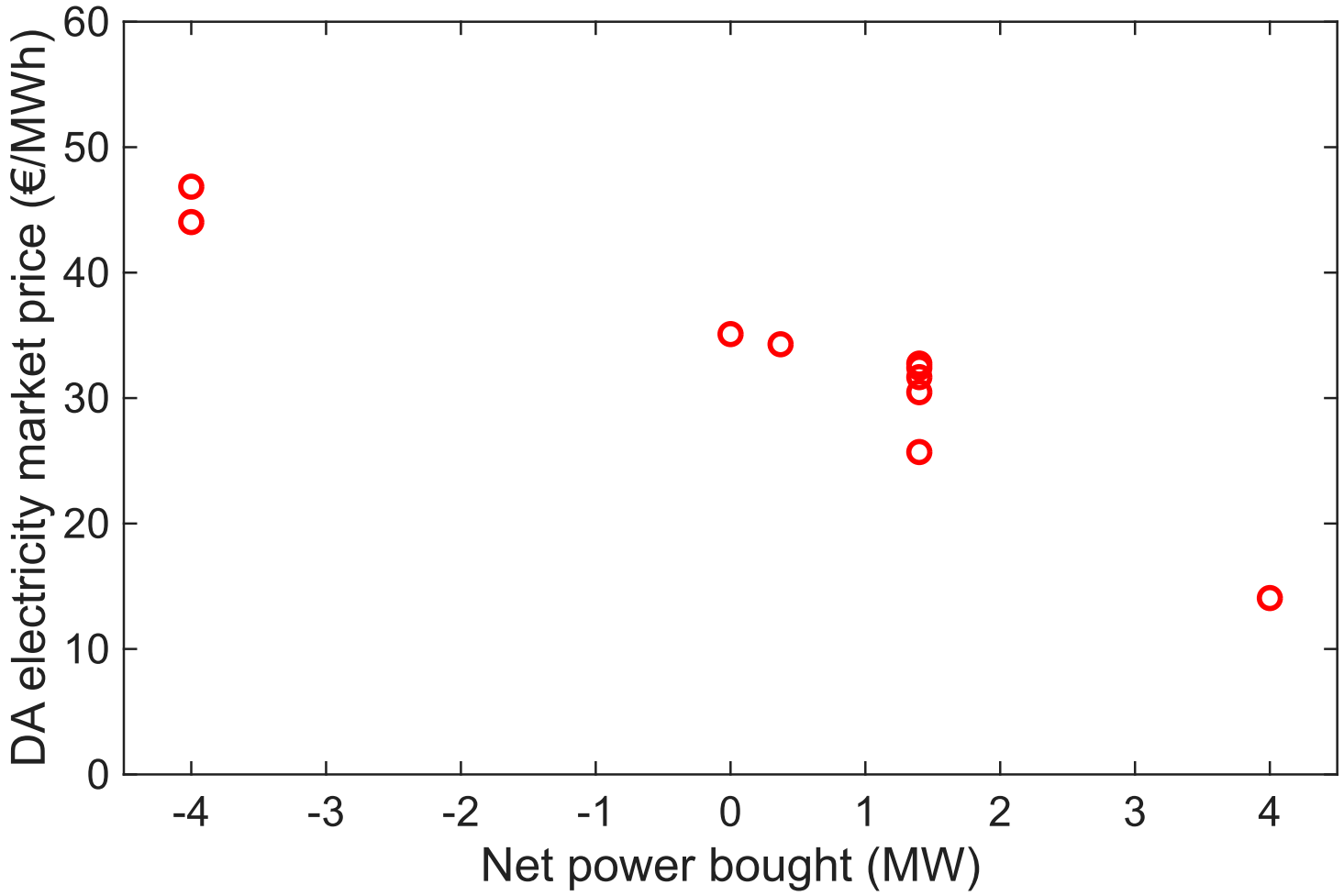
Solve the risk-neutral two-stage SP self-scheduling problem of a price-taker VPP using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_06.gms

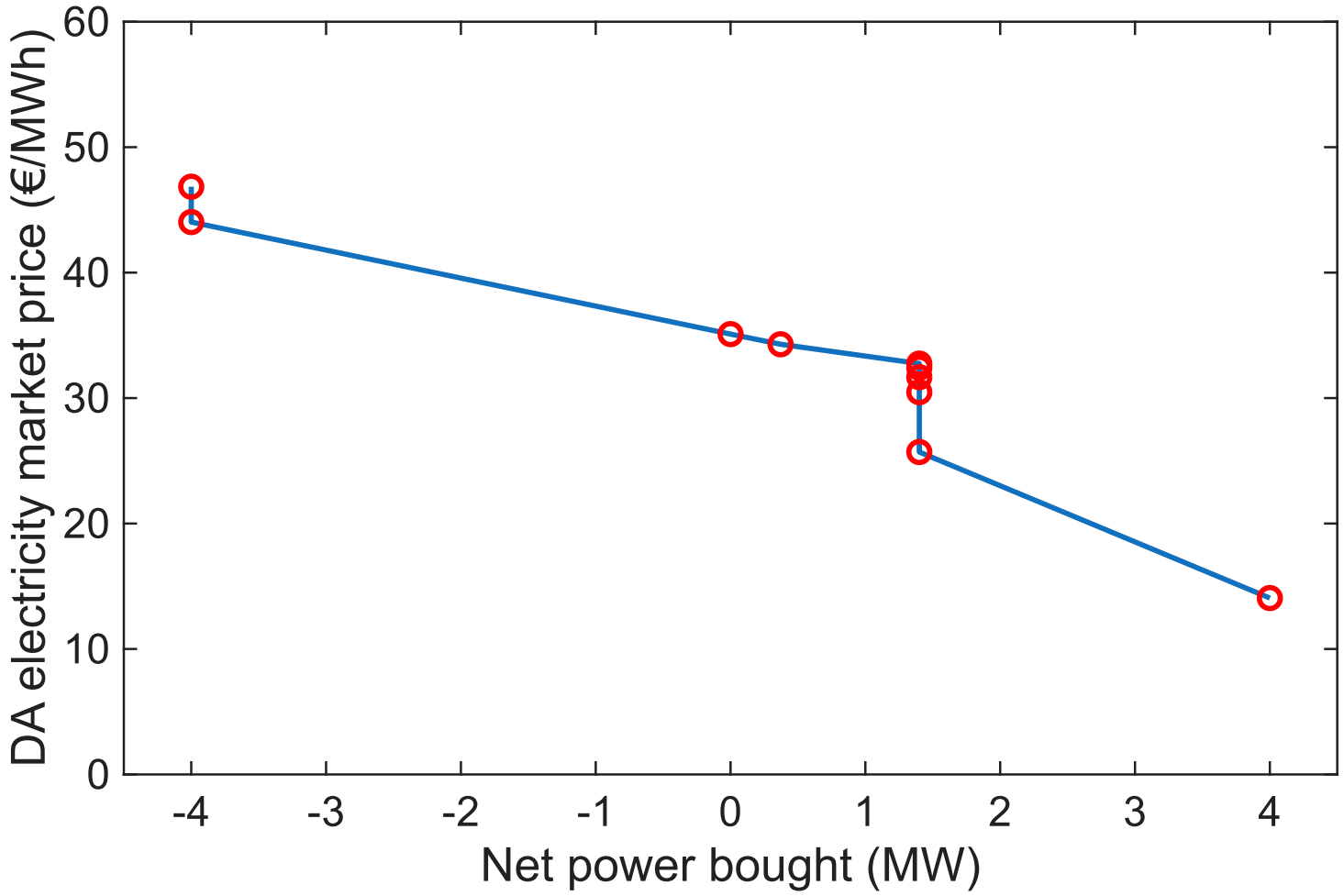
Solution:

Objective function value = − €619.15.

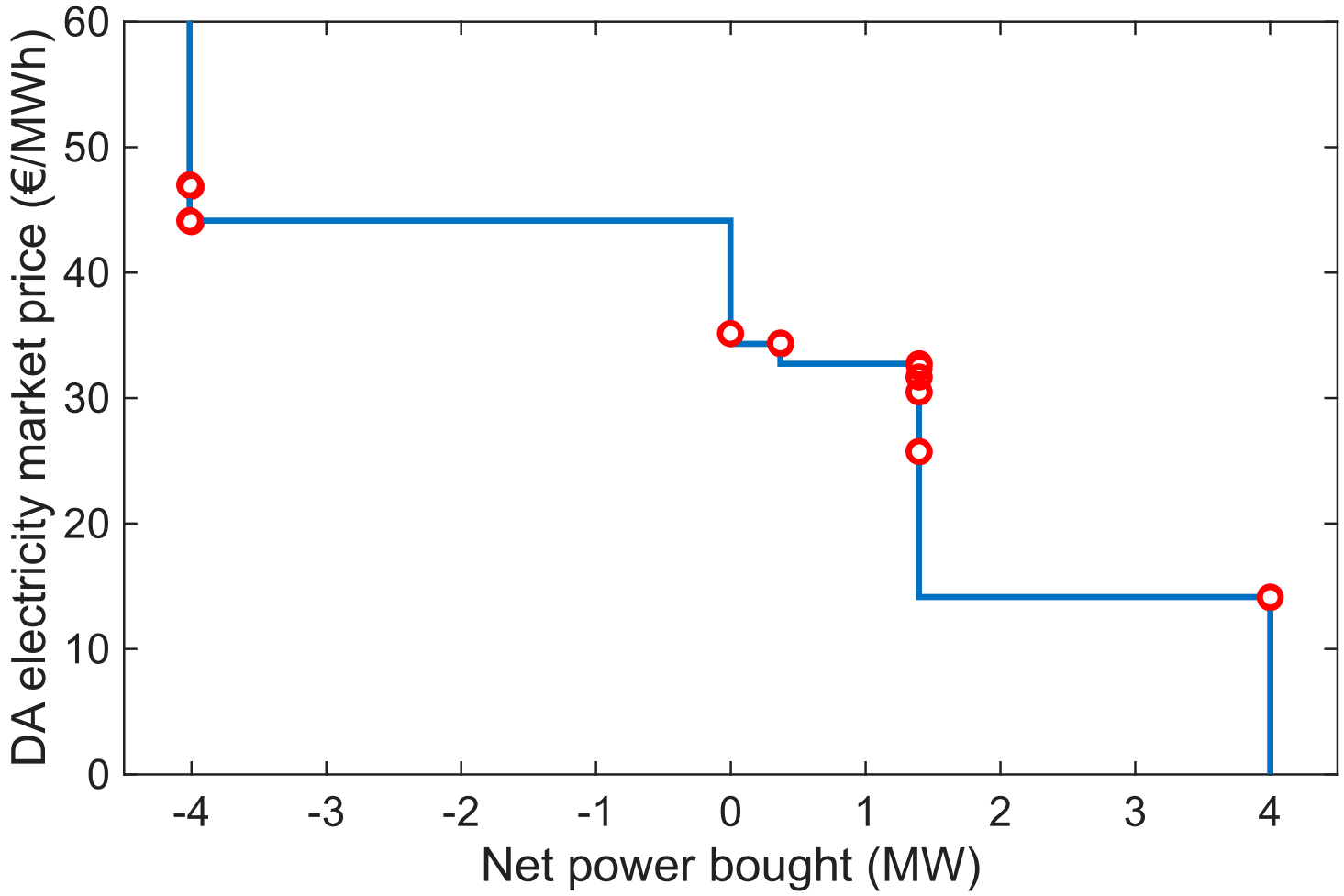
# Exercise 6 – VPP' self-scheduling – Curves



# Exercise 6 – VPP' self-scheduling – Curves

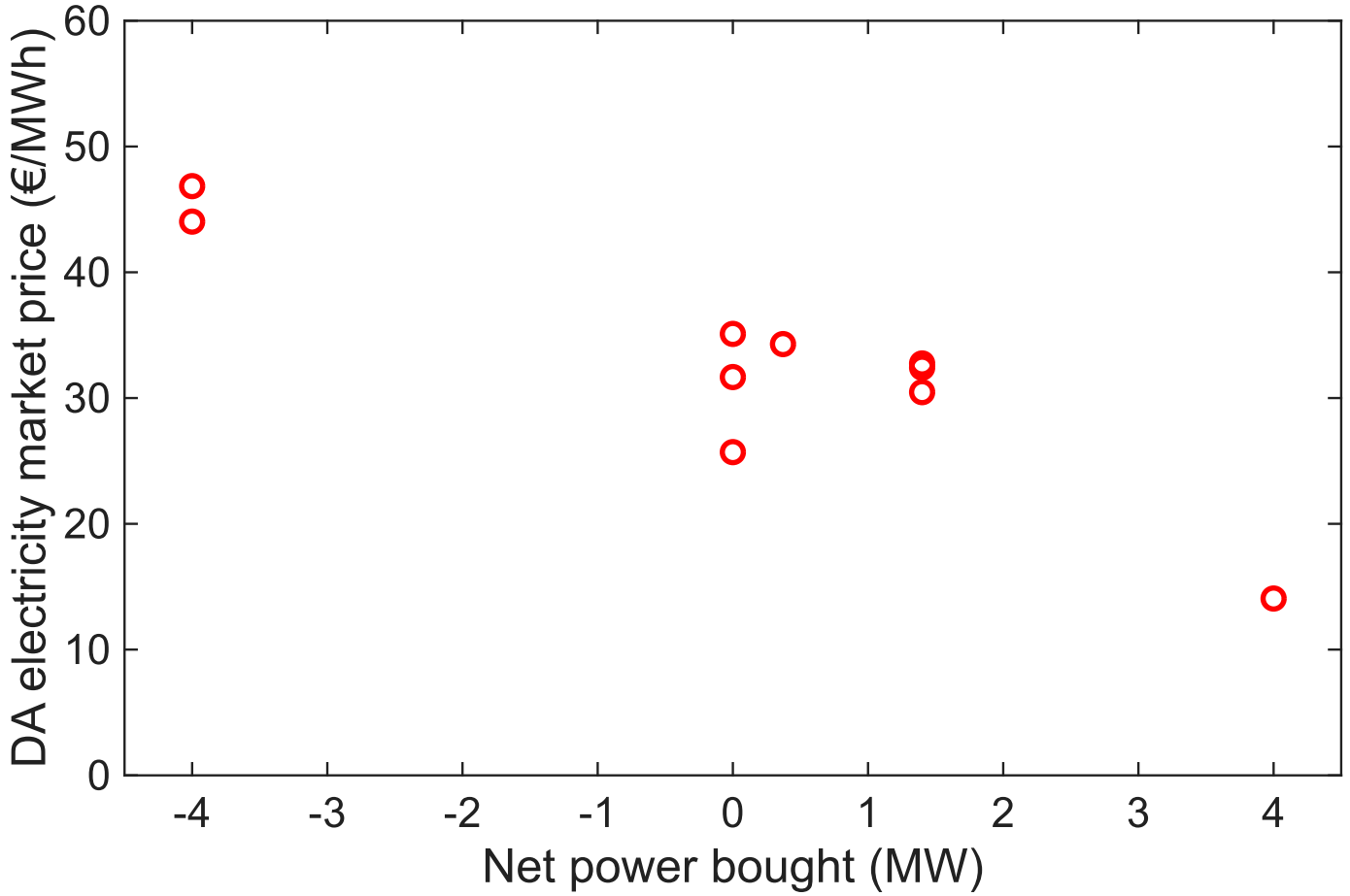


# Exercise 6 – VPP' self-scheduling – Curves



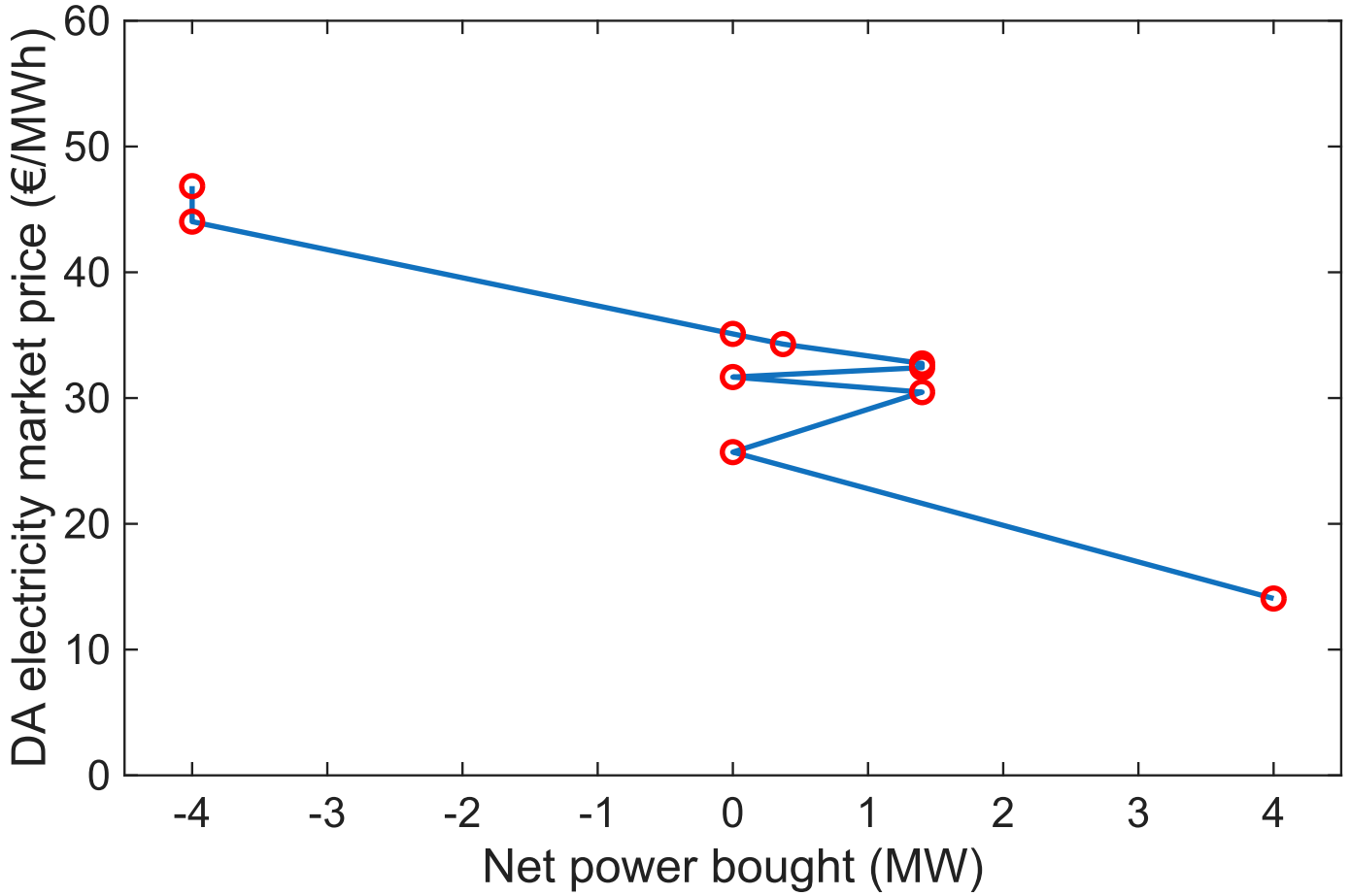
# Exercise 6 – VPP' self-scheduling – Curves

**Without the constraints that impose the monotonically decreasing curve**



# Exercise 6 – VPP' self-scheduling – Curves

**Without the constraints that impose the monotonically decreasing curve**



# Exercise 6 – VPP' self-scheduling – Curves

If the constraints of the monotonically decreasing curve are not considered, the objective function value is – €604.78, i.e., the profit is higher (lower net cost) than the profit obtained before, but the resulting bidding and offering curves are not monotonically decreasing.

This result highlights the need to consider these constraints.

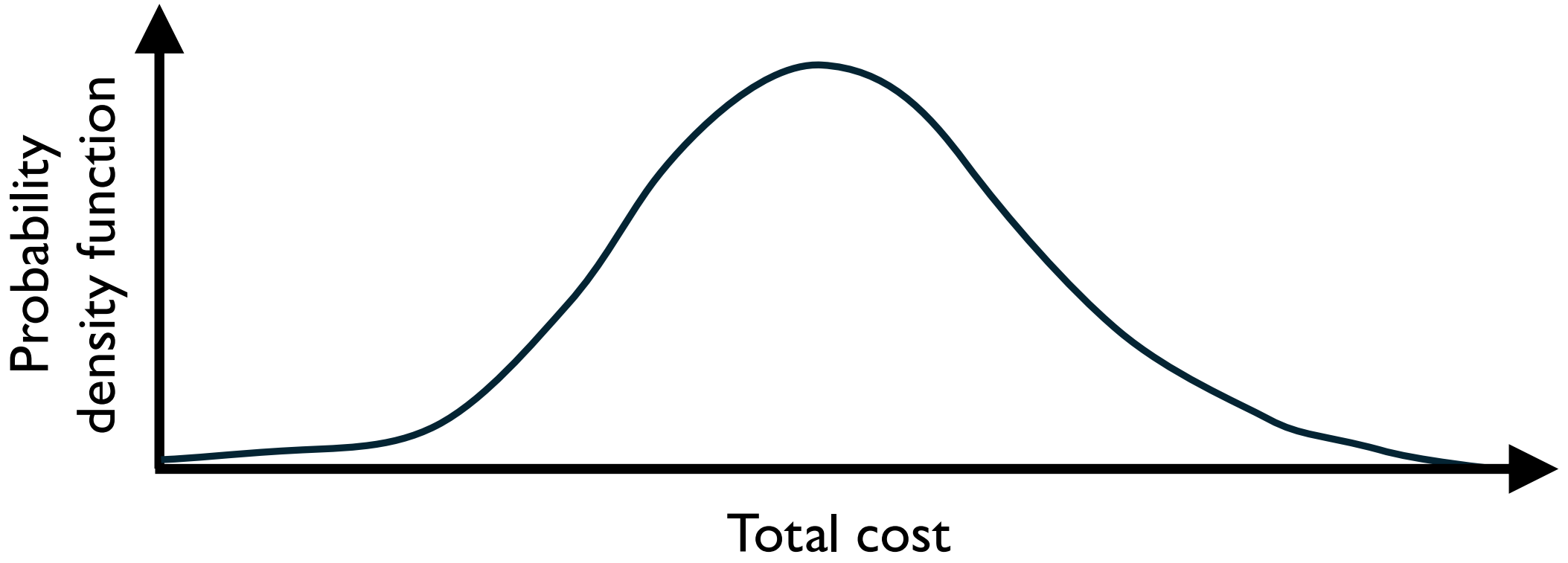
# Stochastic programming

- Risk-neutral approach.
- Quality metrics.
- Out-of-sample analysis.
- Building bidding and offering curves.
- **Risk-averse approach**.
- Decomposition techniques.

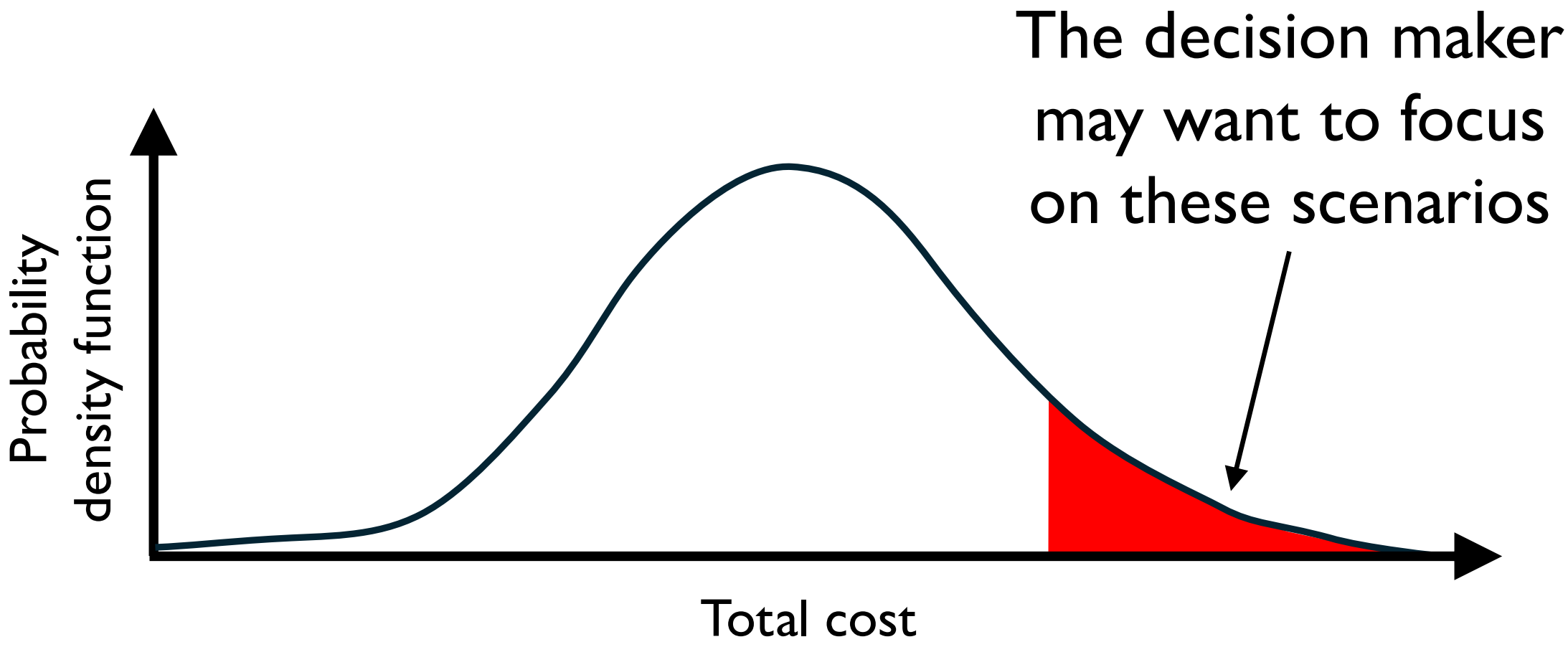
# SP – Risk-averse approach

Expected value → The risk aversion of the decision maker is not considered.

# SP – Risk-averse approach



# SP – Risk-averse approach



# SP – Risk-averse approach

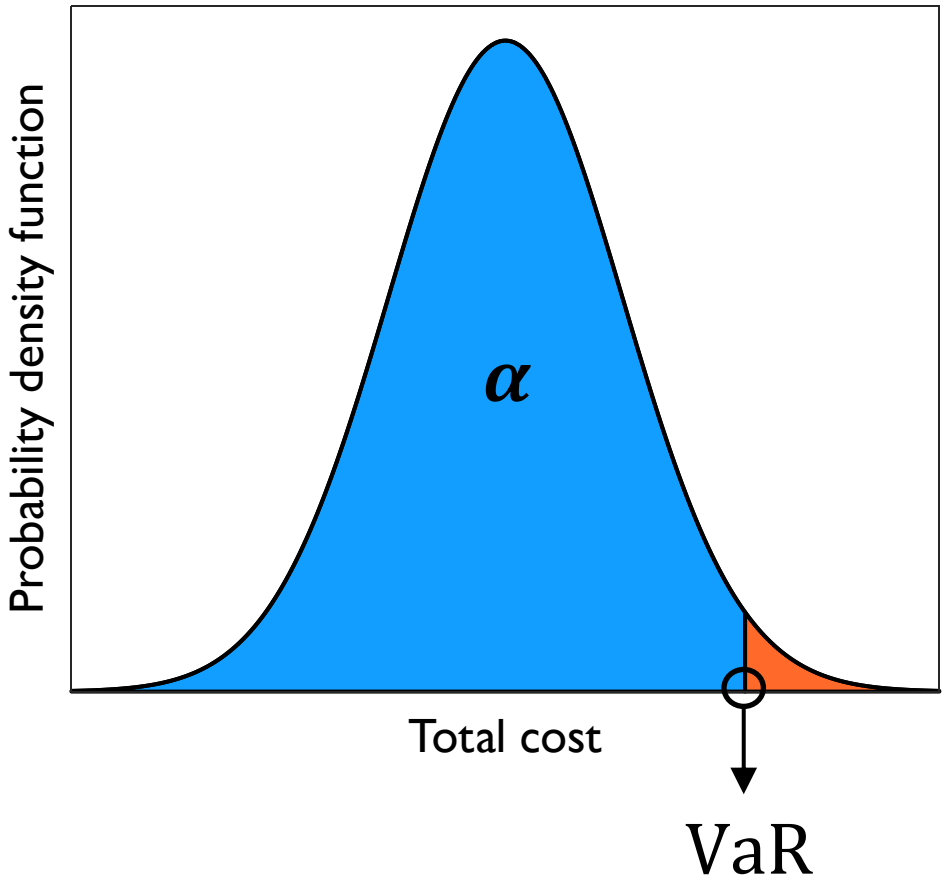
Expected value → The risk aversion of the decision maker is not considered.

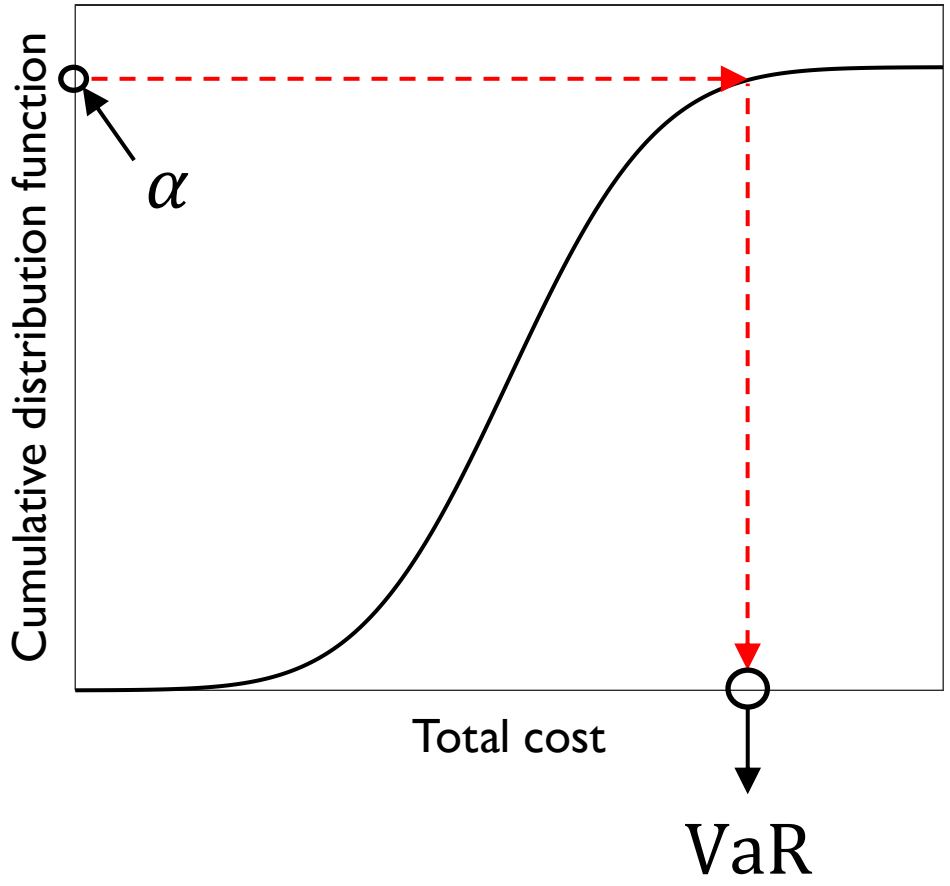
Risk measures:

- Value-at-risk (VaR).
- Conditional value-at-risk (CVaR).

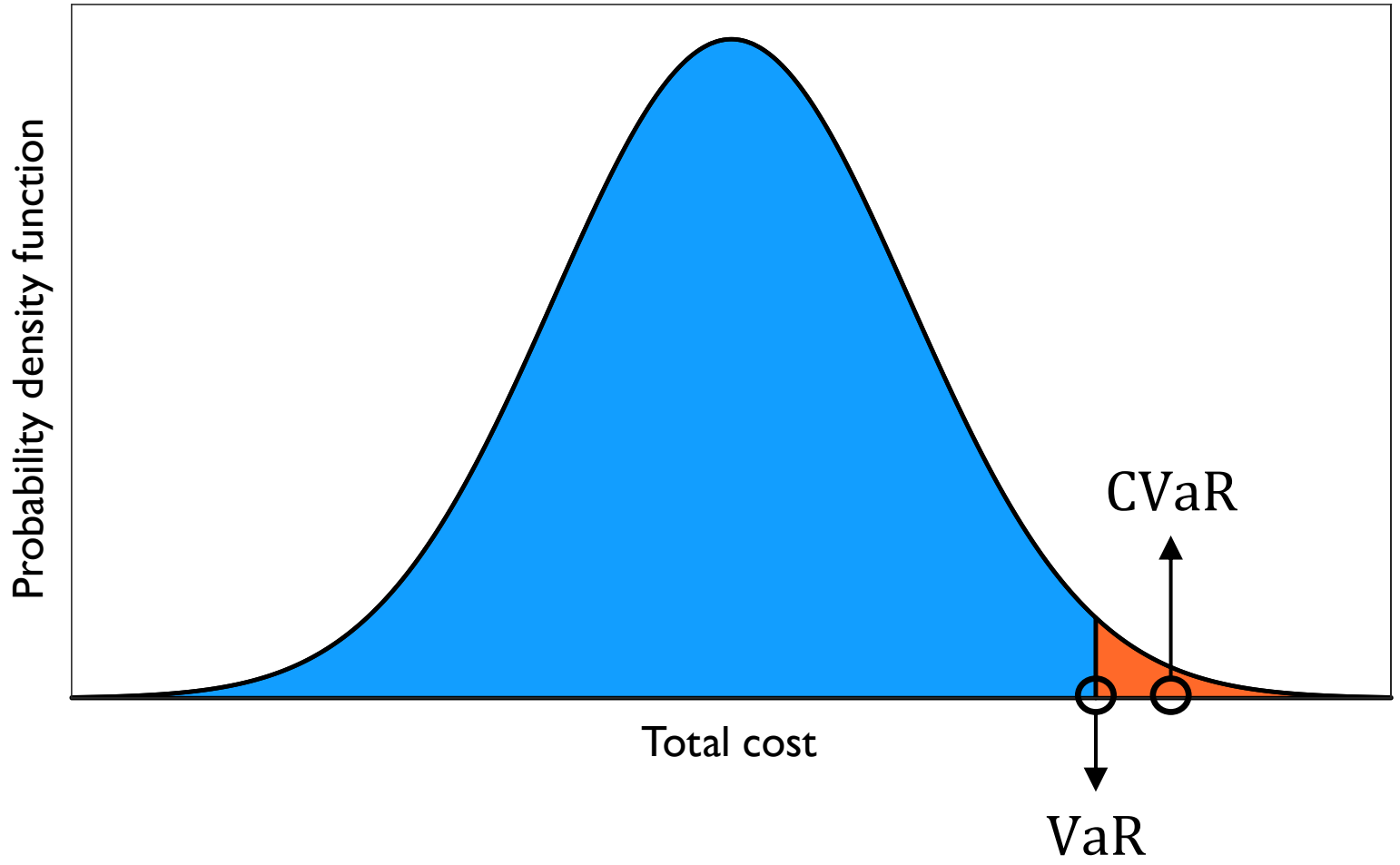
# SP – Risk-averse approach

Risk-aversion level → $\alpha \in [0, 1)$.



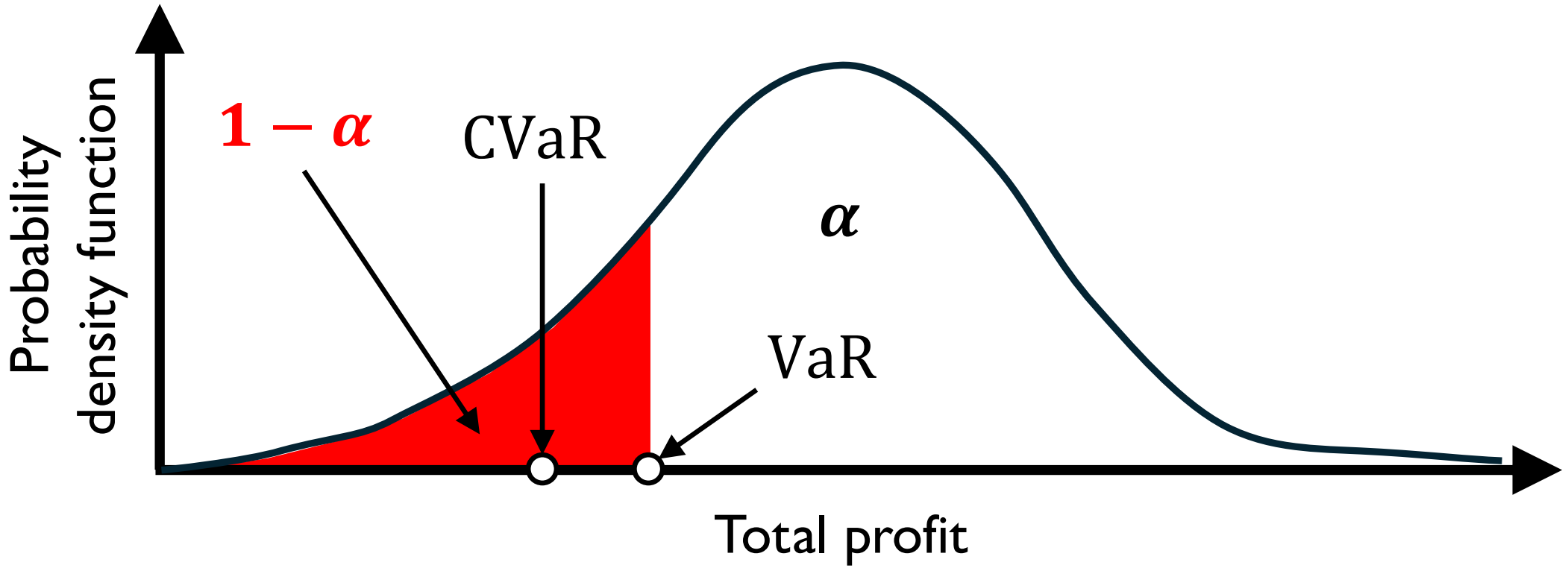



# SP – Risk-averse approach – CVaR



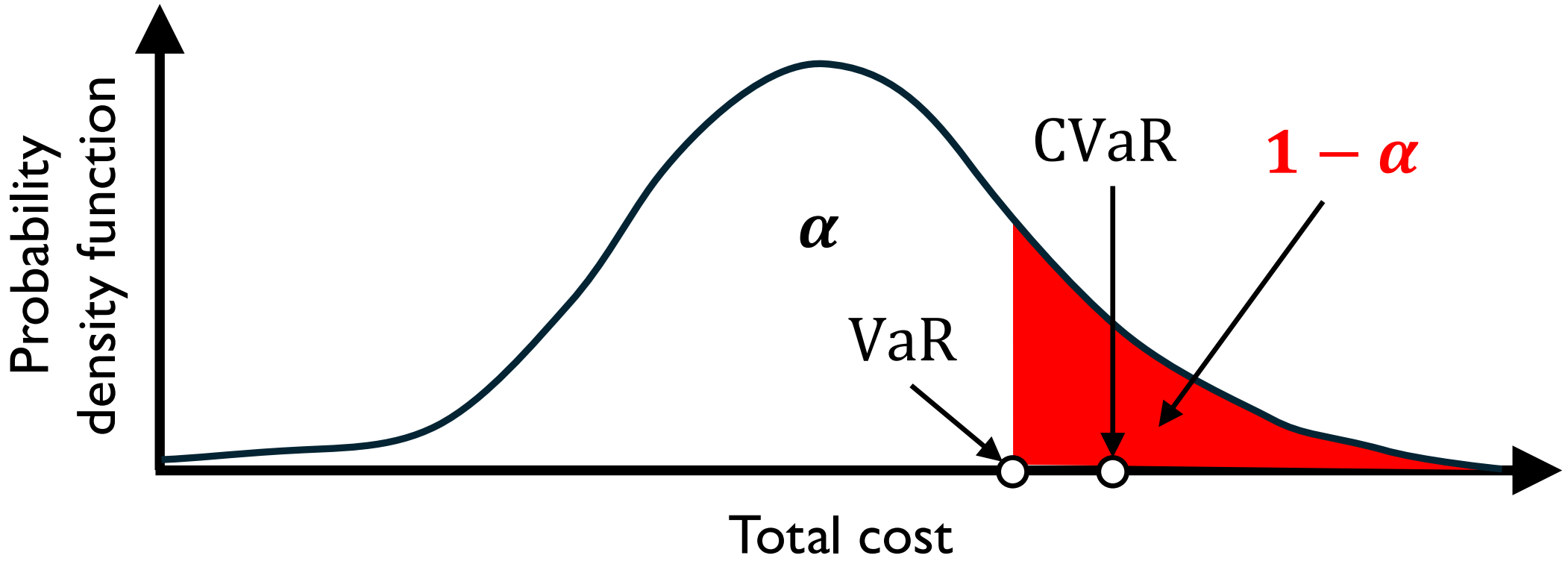
# SP – Risk-averse approach – CVaR

Risk-aversion level → $\alpha \in [0, 1)$.



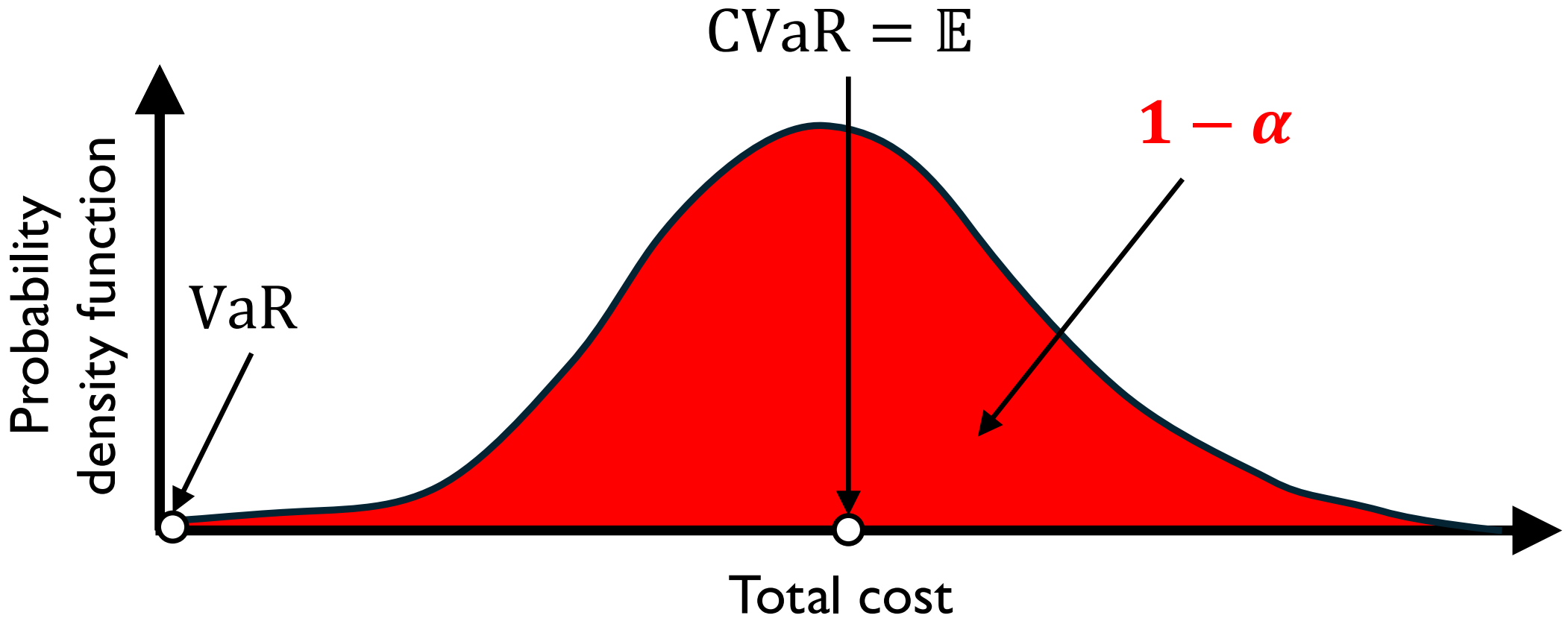
# SP – Risk-averse approach – CVaR

Risk-aversion level → $\alpha \in [0, 1)$.



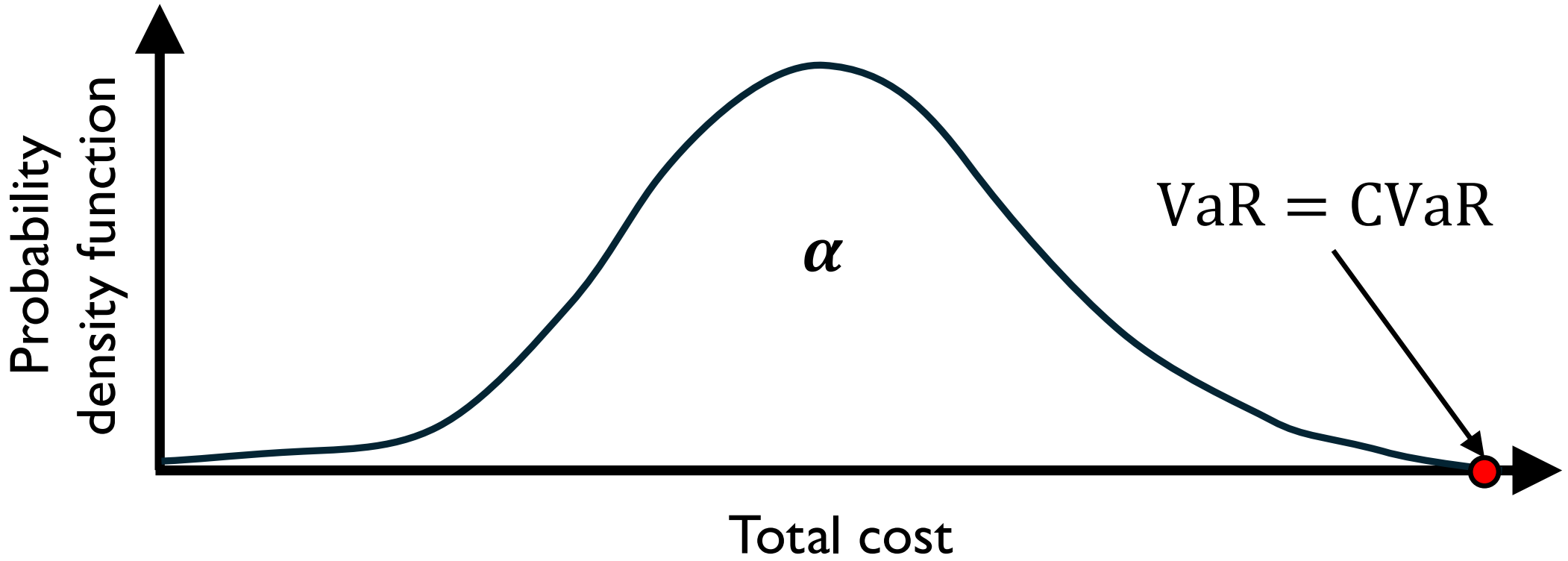
# SP – Risk-averse approach – CVaR

Risk-neutral approach $\rightarrow \alpha = 0$.



# SP – Risk-averse approach – CVaR

Risk-averse approach (worst-case) → $\alpha$ close to 1.



# SP – Risk-neutral approach

$$\min_{\boldsymbol{x};\boldsymbol{y}_\omega,\forall\omega} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \sum_{\omega\in\Omega^W} \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^W$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^W$$

# SP – Risk-averse approach – CVaR

$$\min_{\boldsymbol{x};\varphi;u_\omega,\boldsymbol{y}_\omega,\forall\omega} \quad z = \varphi + \frac{1}{1-\alpha}\sum_{\omega\in\Omega^{\mathrm{W}}} \pi_\omega\, u_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

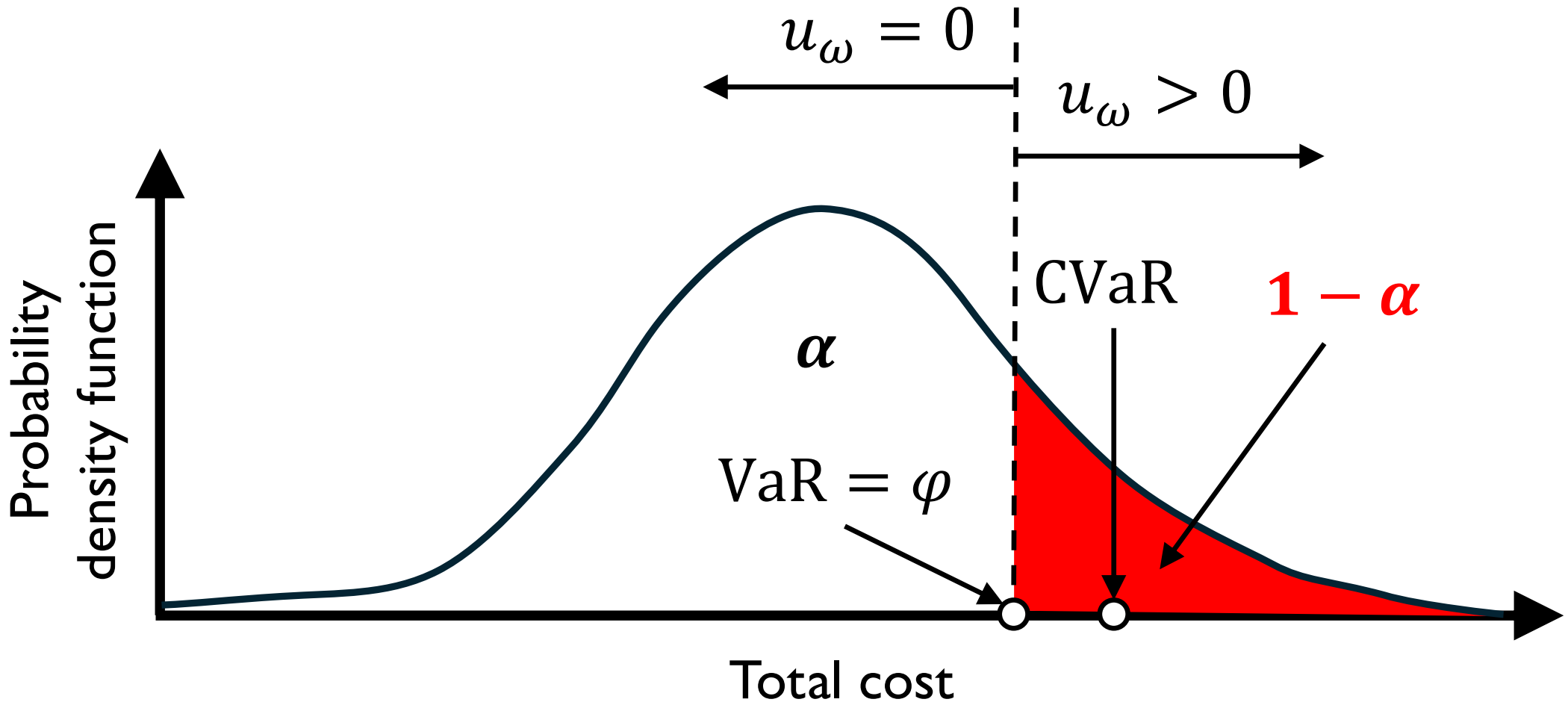
$$u_\omega \geq \boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega - \varphi, \forall\omega \in \Omega^{\mathrm{W}}$$

$$u_\omega \geq 0, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^{\mathrm{W}}$$

# SP – Risk-averse approach – CVaR



# SP – Risk-neutral approach

$$\max_{\boldsymbol{x};\boldsymbol{y}_\omega,\forall\omega} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \sum_{\omega\in\Omega^{\mathrm{W}}} \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^{\mathrm{W}}$$

# SP – Risk-averse approach – CVaR

$$\max_{\boldsymbol{x};\varphi;u_\omega,\boldsymbol{y}_\omega,\forall\omega} \quad z = \varphi - \frac{1}{1-\alpha}\sum_{\omega\in\Omega^{\mathrm{W}}} \pi_\omega\, u_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

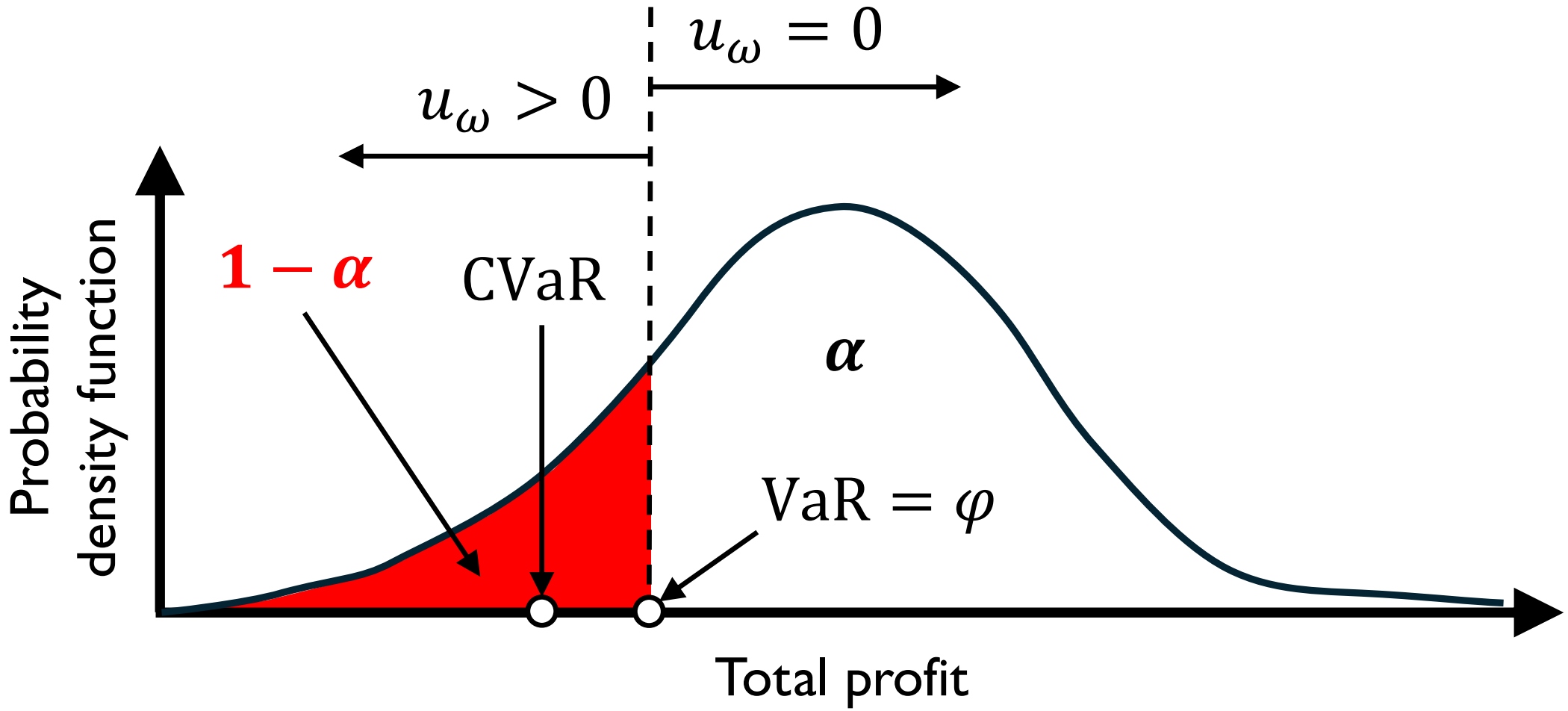
$$u_\omega \geq \varphi - (\boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega), \forall\omega \in \Omega^{\mathrm{W}}$$

$$u_\omega \geq 0, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^{\mathrm{W}}$$

# SP – Risk-averse approach – CVaR



# Exercise 7 – Centralized GEP – CVaR

$$\min_{\Phi} \quad \varphi + \frac{1}{1-\alpha} \sum_{\omega \in \Omega^{\mathrm{W}}} \pi_{\omega}^{\mathrm{W}} u_{\omega}^{\mathrm{W}}$$

**s.t.:**

$$u_{\omega}^{\mathrm{W}} \geq \sum_{t \in \Omega^{\mathrm{T}}} \rho_{t}^{\mathrm{T}} \left( C^{\mathrm{LS}} p_{t\omega}^{\mathrm{LS}} + \sum_{g \in \Omega^{\mathrm{G}}} C_{g}^{\mathrm{G,V}} p_{gt\omega}^{\mathrm{G}} \right) + \sum_{g \in \Omega^{\mathrm{G,New}}} C_{g}^{\mathrm{G,I}} CRF_{g} \overline{p}_{g}^{\mathrm{G}} - \varphi, \forall \omega$$

$$u_{\omega}^{\mathrm{W}} \geq 0, \forall \omega$$

# Exercise 7 – Centralized GEP – CVaR

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$0 \leq \overline{p}_g^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G,New}}, \forall g \in \Omega^{\mathrm{G,New}}$$

# Exercise 7 – Centralized GEP – CVaR

$$\sum_{g \in \Omega^{\mathrm{G}}} p^{\mathrm{G}}_{gt\omega} = P^{\mathrm{D}}_{t\omega} - p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{LS}}_{t\omega} \leq P^{\mathrm{D}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{P}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t, \forall \omega$$

$$\Phi = \left\{ \varphi; p^{\mathrm{G}}_{gt\omega}, \forall g, \forall t, \forall \omega; \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}; p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega; u^{\mathrm{W}}_{\omega}, \forall \omega \right\}$$

# Exercise 7 – Centralized GEP – CVaR

Solve the risk-averse two-stage SP centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_07.gms

# Exercise 7 – Centralized GEP – CVaR

The input data differ from those used in exercise 1:

- $C^{\mathrm{LS}}$ is reduced.
- $\pi^{\mathrm{W}}_{\omega_9}$ is increased and $\pi^{\mathrm{W}}_{\omega_{10}}$ is reduced.
- $P^{\mathrm{D}}_{t_4\omega_{10}}$ is increased.
- $\overline{C}^{\mathrm{I}}$ is increased.

These changes have been made to illustrate that first-stage variables $\overline{p}^{\mathrm{G}}_{g}$ may depend on the value of $\alpha$.

# Exercise 7 – Centralized GEP – CVaR

| **Risk-aversion level $\alpha$** | **0** | **0.50** | **0.99** |
|---|---|---|---|
| Objective function value (M€) | 45.17 | 48.45 | 50.93 |
| $\overline{p}^{\mathrm{G}}_{g_2}$ (MW) | 264.702 | 288.036 | 285.199 |
| $\overline{p}^{\mathrm{G}}_{3}$ (MW) | 415.298 | 401.964 | 403.585 |

# SP – Risk-averse approach – E + CVaR

$$\min_{x;\varphi;u_\omega,y_\omega,\forall\omega} \quad z = \beta\left(\boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega\right) + (1-\beta)\left(\varphi + \frac{1}{1-\alpha}\sum_{\omega\in\Omega^W}\pi_\omega\, u_\omega\right)$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$u_\omega \geq \boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega - \varphi, \forall\omega \in \Omega^W$$

$$u_\omega \geq 0, \forall\omega \in \Omega^W$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^W$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^W$$

# SP – Risk-averse approach – E + CVaR

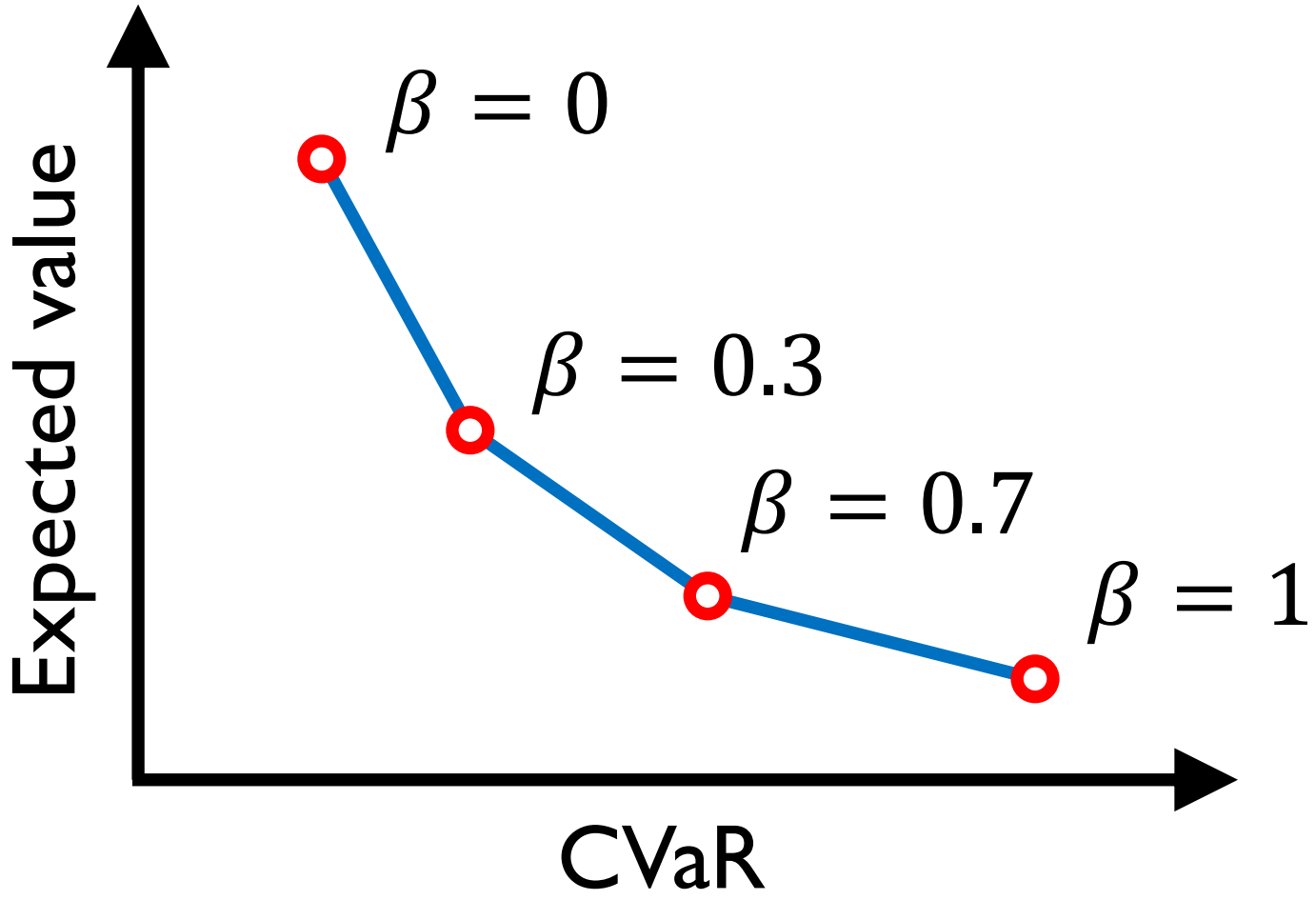
The risk-aversion level $\alpha$ is fixed, and the risk-weighting parameter $\beta$ is modified.

- $\beta = 1 \rightarrow$ The expected value is minimized.
- $\beta = 0 \rightarrow$ The CVaR is minimized.
- $\beta \in (0, 1) \rightarrow$ A combination of the expected value and the CVaR is minimized.

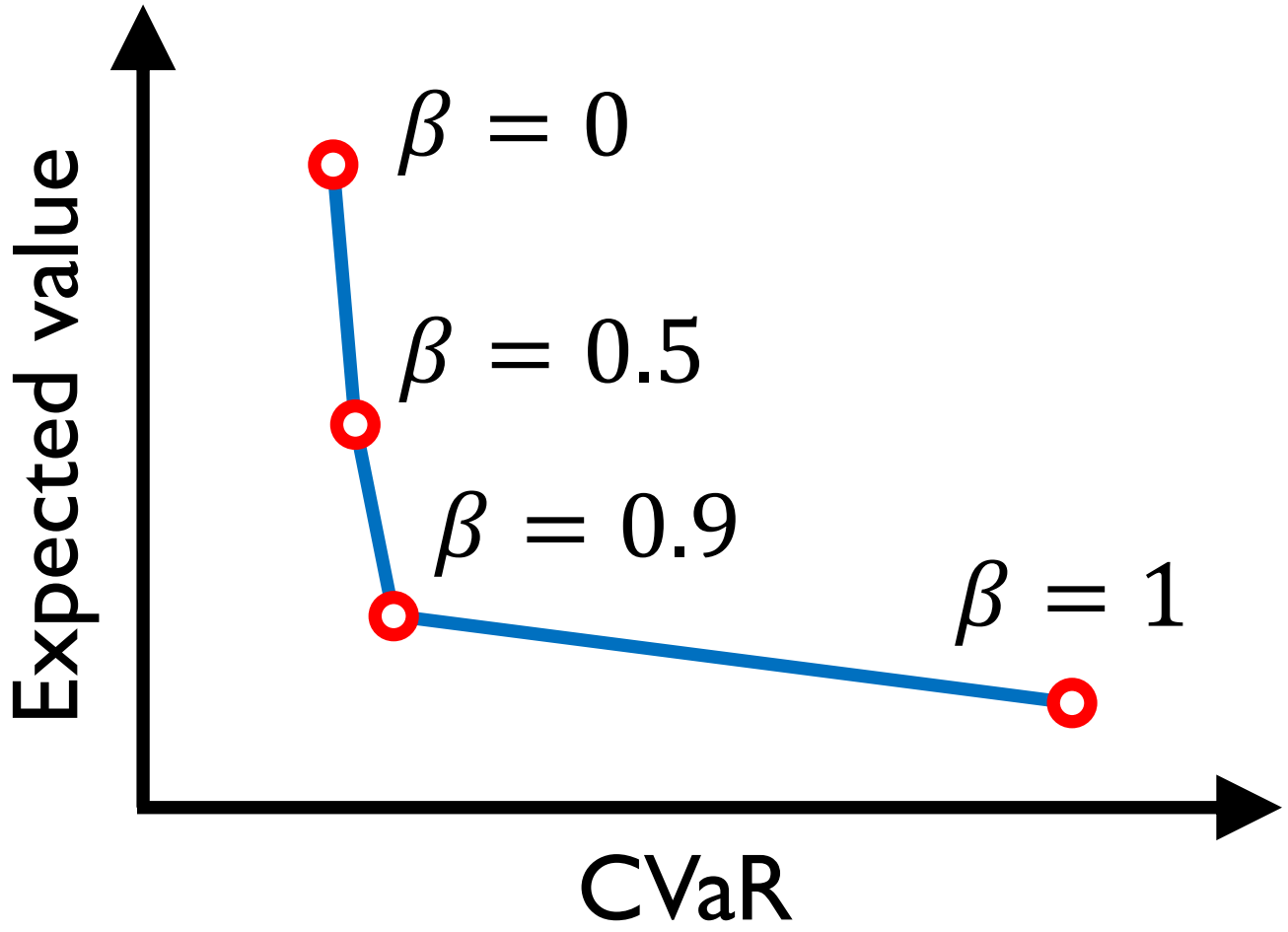
# SP – Risk-averse approach – E + CVaR

Efficient frontier → Useful tool that shows how the expected value and the CVaR change when the risk-weighting parameter $\beta$ is modified.

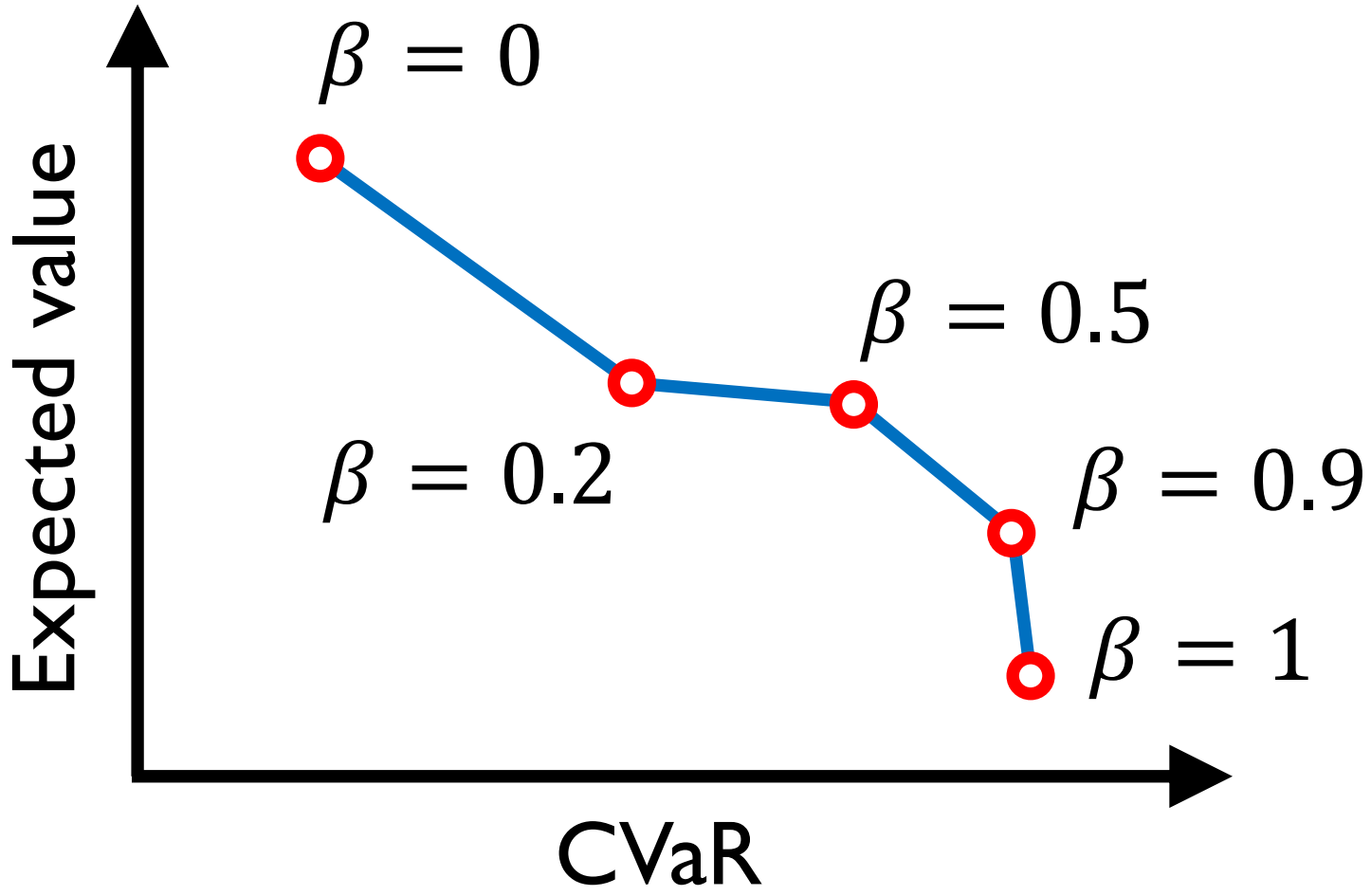
# SP – Risk-averse approach – E + CVaR



# SP – Risk-averse approach – E + CVaR



# SP – Risk-averse approach – E + CVaR



# Exercise 8 – Centralized GEP – E + CVaR

$$\min_{\Phi} \ \beta\left(\sum_{\omega\in\Omega^{\mathrm{W}}}\pi_{\omega}^{\mathrm{W}}\sum_{t\in\Omega^{\mathrm{T}}}\rho_{t}^{\mathrm{T}}\left(C^{\mathrm{LS}}p_{t\omega}^{\mathrm{LS}}+\sum_{g\in\Omega^{\mathrm{G}}}C_{g}^{\mathrm{G,V}}p_{gt\omega}^{\mathrm{G}}\right)+\sum_{g\in\Omega^{\mathrm{G,New}}}C_{g}^{\mathrm{G,I}}CRF_{g}\overline{p}_{g}^{\mathrm{G}}\right)$$

$$+(1-\beta)\left(\varphi+\frac{1}{1-\alpha}\sum_{\omega\in\Omega^{\mathrm{W}}}\pi_{\omega}^{\mathrm{W}}u_{\omega}^{\mathrm{W}}\right)$$

**s.t.:**

$$u_{\omega}^{\mathrm{W}}\geq\sum_{t\in\Omega^{\mathrm{T}}}\rho_{t}^{\mathrm{T}}\left(C^{\mathrm{LS}}p_{t\omega}^{\mathrm{LS}}+\sum_{g\in\Omega^{\mathrm{G}}}C_{g}^{\mathrm{G,V}}p_{gt\omega}^{\mathrm{G}}\right)+\sum_{g\in\Omega^{\mathrm{G,New}}}C_{g}^{\mathrm{G,I}}CRF_{g}\overline{p}_{g}^{\mathrm{G}}-\varphi,\forall\omega$$

$$u_{\omega}^{\mathrm{W}}\geq 0,\forall\omega$$

# Exercise 8 – Centralized GEP – E + CVaR

$$\sum_{g \in \Omega^{\mathrm{G,New}}} C_g^{\mathrm{G,I}} CRF_g \overline{p}_g^{\mathrm{G}} \leq \overline{C}^{\mathrm{I}}$$

$$0 \leq \overline{p}_g^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G,New}}, \forall g \in \Omega^{\mathrm{G,New}}$$

# Exercise 8 – Centralized GEP – E + CVaR

$$\sum_{g \in \Omega^{\mathrm{G}}} p^{\mathrm{G}}_{gt\omega} = P^{\mathrm{D}}_{t\omega} - p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{LS}}_{t\omega} \leq P^{\mathrm{D}}_{t\omega}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{P}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,Old}}, \forall t, \forall \omega$$

$$0 \leq p^{\mathrm{G}}_{gt\omega} \leq \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}, \forall t, \forall \omega$$

$$\Phi = \left\{ \varphi; p^{\mathrm{G}}_{gt\omega}, \forall g, \forall t, \forall \omega; \overline{p}^{\mathrm{G}}_{g}, \forall g \in \Omega^{\mathrm{G,New}}; p^{\mathrm{LS}}_{t\omega}, \forall t, \forall \omega; u^{\mathrm{W}}_{\omega}, \forall \omega \right\}$$

# Exercise 8 – Centralized GEP – E + CVaR

Solve the risk-averse two-stage SP centralized GEP problem using the data from the following link:

https://github.com/AlvaroGarciaCerezo/CourseOR/blob/main/Lecture%204/exercise4_08.gms

# Exercise 8 – Centralized GEP – E + CVaR

| **Risk-aversion level $\alpha$** | **0** | **0.50** | **1** |
|---|---|---|---|
| Objective function value (M€) | 50.71 | 47.98 | 45.17 |
| $\overline{p}^{\mathrm{G}}_{g_2}$ (MW) | 286.365 | 286.365 | 264.702 |
| $\overline{p}^{\mathrm{G}}_{3}$ (MW) | 402.919 | 402.919 | 415.298 |

# Exercise 8 – Centralized GEP – E + CVaR

| **Risk-weighting parameter $\beta$** | **0** | **1** |
|---|---|---|
| Expected value of the total cost (M€) | 48.99 | 45.17 |
| $\mathrm{CVaR}_{\alpha}$ of the total cost (M€) | 50.71 | 71.97 |

# Exercise 8 – Centralized GEP – E + CVaR

In this exercise the values of $\overline{p}_g^{\mathrm{G}}$ are the same for all the analyzed values of $\beta < 1$. Hence, the efficient frontier has only two points, i.e., those presented in the previous slide.

# SP – Risk-averse approach – Chance constraint

$$\mathbb{P}(A_\omega x \leq b_\omega) \geq P$$

Decision $x$ should fulfill that constraint $A_\omega x \leq b_\omega$ holds with a given probability $P$.

Types:

- Single chance constraint.
- Joint chance constraint.

# Stochastic programming

- Risk-neutral approach.
- Quality metrics.
- Out-of-sample analysis.
- Building bidding and offering curves.
- Risk-averse approach.
- **Decomposition techniques**.

# SP – Decomposition techniques

$$\min_{x;y_\omega,\forall\omega} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \sum_{\omega\in\Omega^W} \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^W$$

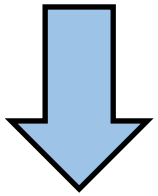
$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^W$$

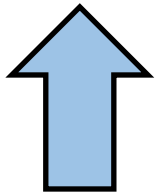
# SP – Decomposition techniques

**Master problem**

Relaxed version. Determine complicating decisions.

Complicating decisions





Additional constraints

**Subproblem**

Complicating decisions are fixed.
Determine the remaining decisions.

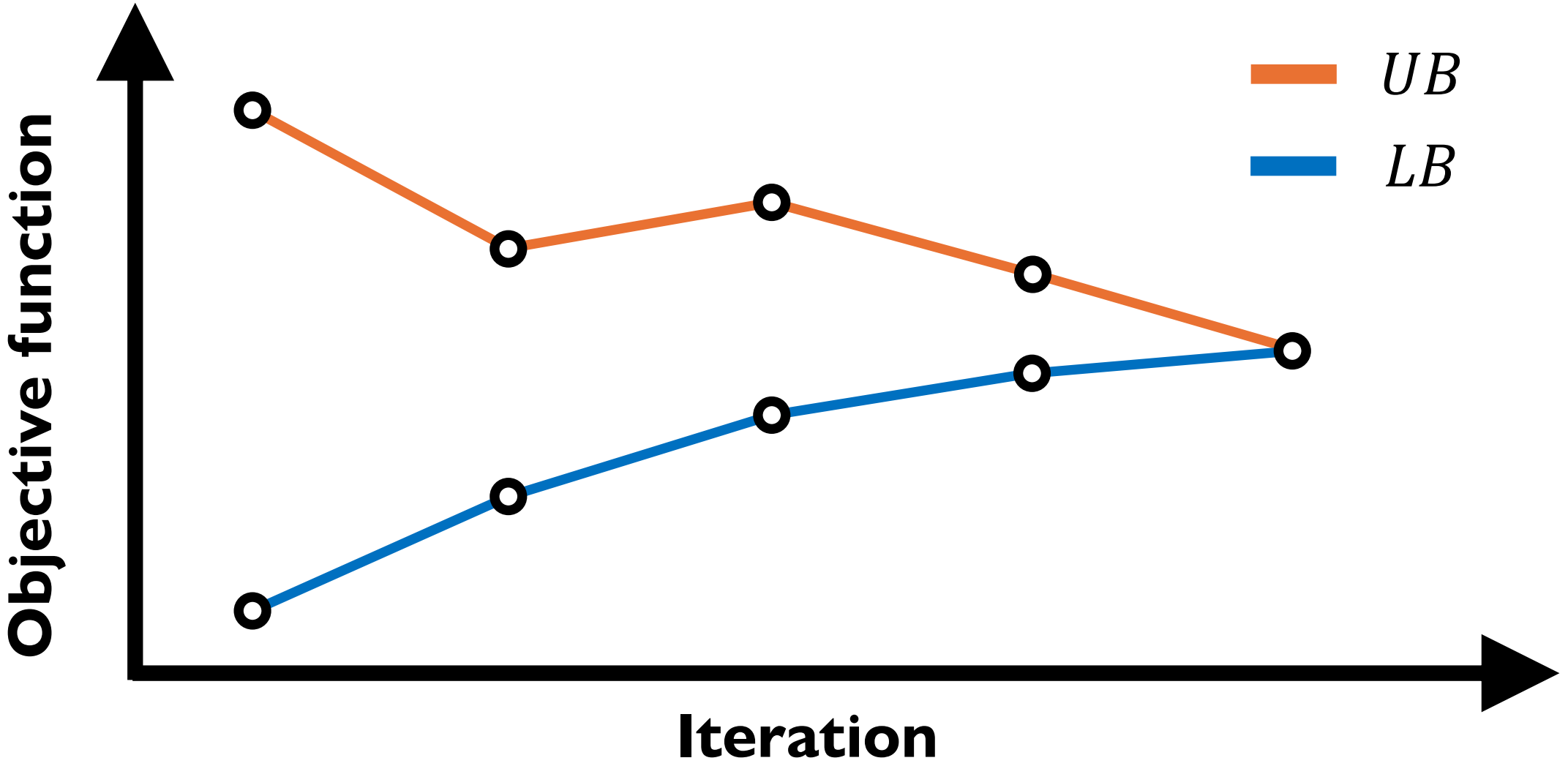
# SP – Decomposition techniques

Iterative procedure (iterations $j$)

Master problem $\rightarrow$ Lower bound $LB$

Subproblem $\rightarrow$ Upper bound $UB$

# SP – Decomposition techniques



# SP – Decomposition techniques

Decomposition techniques explained in this course:

- Benders' decomposition.
- Constraint generation-based algorithm.

# SP – Decomposition techniques

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- **Benders' decomposition**.
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# SP – Benders' decomposition

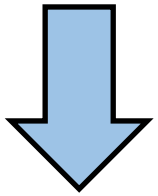
J. F. Benders, "Partitioning procedures for solving mixed-variables programming problems", *Numerische Mathematik*, vol. 4, pp. 238–252, Dec. 1962, DOI: 10.1007/BF01386316.

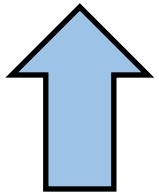
# SP – Benders' decomposition

**Master problem**

Relaxed version. Determine complicating decisions.

Complicating decisions





Additional constraints

**Subproblem**

Complicating decisions are fixed.
Determine the remaining decisions.

# SP – Benders' decomposition

For example, complicating variables may be first-stage decision variables in two-stage SP problems.

# SP – Benders' decomposition

$$\min_{\boldsymbol{x};\boldsymbol{y}_\omega,\forall\omega} \quad z = \boldsymbol{c}^T\boldsymbol{x} + \sum_{\omega\in\Omega^W} \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^W$$

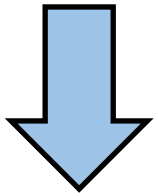
$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^W$$

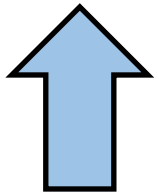
# SP – Benders' decomposition

**Master problem**

Relaxed version. Determine the first-stage decisions.

First-stage decisions





Additional constraints

**Subproblem**

First-stage decisions are fixed.

Determine the second-stage decisions.

# SP – Benders' decomposition

For given first-stage decision variables, the subproblem can be solved independently for each scenario $\omega$.

# SP – Benders' decomposition

Iterative procedure (iterations $j$)

Master problem $\rightarrow$ Lower bound $LB$

Subproblem $\rightarrow$ Upper bound $UB$

# SP – Benders' decomposition – Subproblem

$$\min_{\boldsymbol{y}_\omega} \quad z_\omega^{\text{Sub}} = \pi_\omega \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

s.t.:

$$\boldsymbol{T}_\omega \boldsymbol{x} + \boldsymbol{R}_\omega \boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}$$

$$\boldsymbol{x} = \boldsymbol{x}^{(j)} : \boldsymbol{\lambda}_\omega^{(j)}$$

# SP – Benders' decomposition – Subproblem

$$UB^{(j)} = \boldsymbol{c}^T \boldsymbol{x}^{(j)} + \sum_{\omega \in \Omega^{\mathrm{W}}} z_{\omega}^{\mathrm{Sub},(j)}$$

# SP – Benders' decomposition – Master problem

$$\min_{\boldsymbol{x};\alpha} \; z^{\text{Master}} = \boldsymbol{c}^T\boldsymbol{x} + \alpha$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

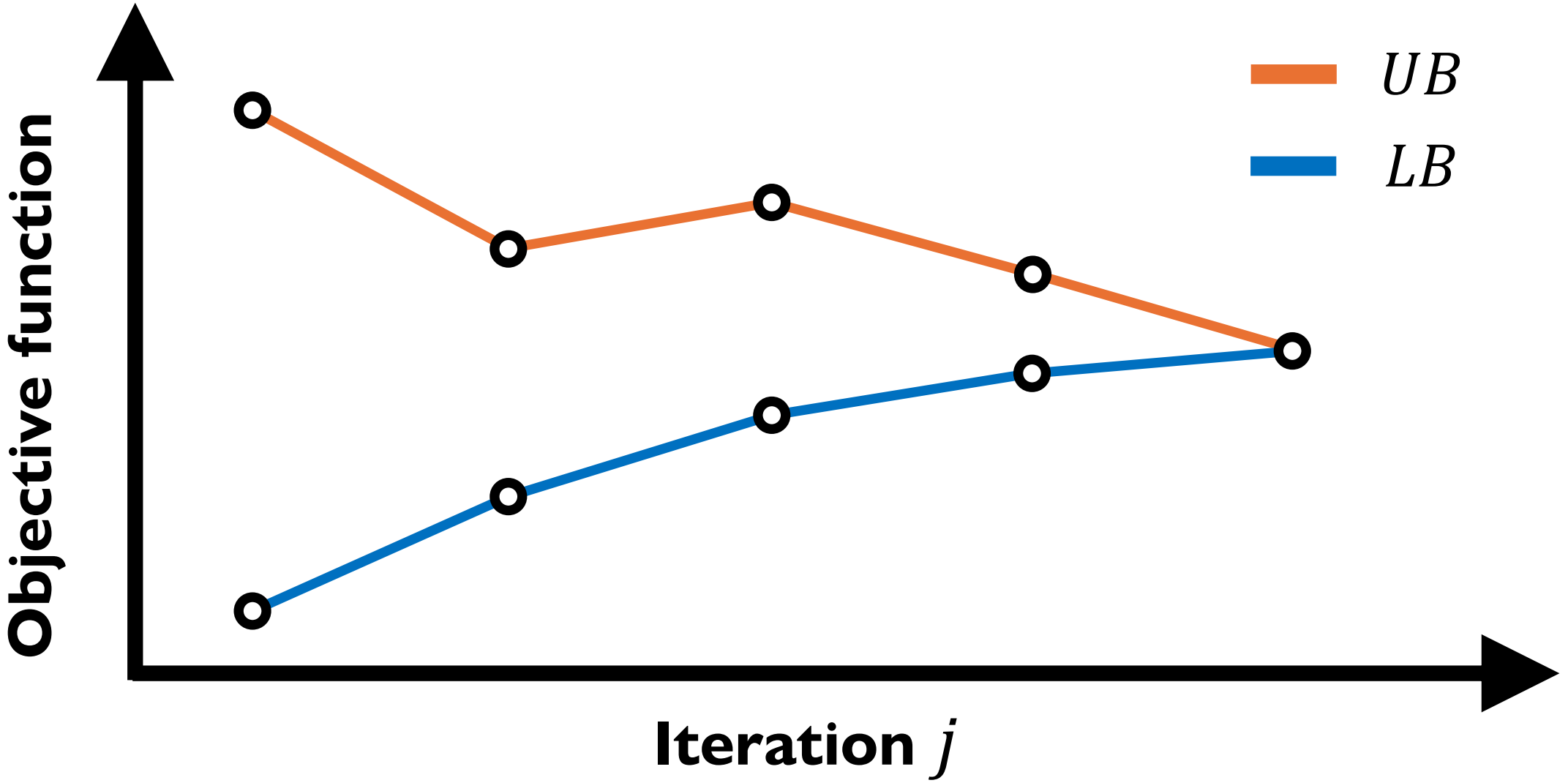
$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\alpha \geq \sum_{\omega \in \Omega^W} \left( z_\omega^{\text{SubP},(\nu)} + \boldsymbol{\lambda}_\omega^{(\nu)^T} \left( \boldsymbol{x} - \boldsymbol{x}^{(\nu)} \right) \right), \nu = 1, \ldots, j$$

# SP – Benders' decomposition – Master problem

$$LB^{(j)} = z^{\text{Master},(j)}$$

# SP – Benders' decomposition



# SP – Benders' decomposition

The master problem and the subproblem should include slack variables to prevent infeasible solutions, such as the load-shedding variables.

# SP – Decomposition techniques

Decomposition techniques explained in this course:

- Benders' decomposition.
- **Constraint generation-based algorithm**.

# SP – Constraint generation-based algorithm

R. Mínguez, W. van Ackooij, and R. García-Bertrand, "Constraint generation for risk averse two-stage stochastic programs", *European Journal of Operational Research*, vol. 288, no. 1, pp. 194–206, Jan. 2021, DOI: 10.1016/j.ejor.2020.05.064.

# SP – Risk-averse approach – CVaR

$$\min_{\boldsymbol{x};\varphi;u_\omega,\boldsymbol{y}_\omega,\forall\omega} \quad z = \varphi + \frac{1}{1-\alpha}\sum_{\omega\in\Omega^{\mathrm{W}}} \pi_\omega\, u_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$u_\omega \geq \boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega - \varphi, \forall\omega \in \Omega^{\mathrm{W}}$$

$$u_\omega \geq 0, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^{\mathrm{W}}$$

# SP – Risk-averse approach – CVaR

| Approach | Reasonable computational times | Exact solution procedure |
|---|---|---|

# SP – Risk-averse approach – CVaR

| Approach | Reasonable computational times | Exact solution procedure |
|---|---|---|
| Traditional solution procedure | × (only if $\left\|\Omega^{W}\right\|$ is not large) | ✓ |

# SP – Risk-averse approach – CVaR

| Approach | Reasonable computational times | Exact solution procedure |
|---|---|---|
| Traditional solution procedure | × (only if $\left\|\Omega^W\right\|$ is not large) | ✓ |
| Scenario reduction algorithms | ✓ | × |

# SP – Risk-averse approach – CVaR

| Approach | Reasonable computational times | Exact solution procedure |
|---|---|---|
| Traditional solution procedure | ✗ (only if $\left\|\Omega^W\right\|$ is not large) | ✓ |
| Scenario reduction algorithms | ✓ | ✗ |
| Constraint generation-based algorithm (CGA) | ✓ (based on previous works) | ✓ |

# SP – Risk-averse approach – CVaR

$$\min_{\boldsymbol{x};\varphi;u_\omega,\boldsymbol{y}_\omega,\forall\omega} \quad z = \varphi + \frac{1}{1-\alpha}\sum_{\omega\in\Omega^{\mathrm{W}}} \pi_\omega\, u_\omega$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

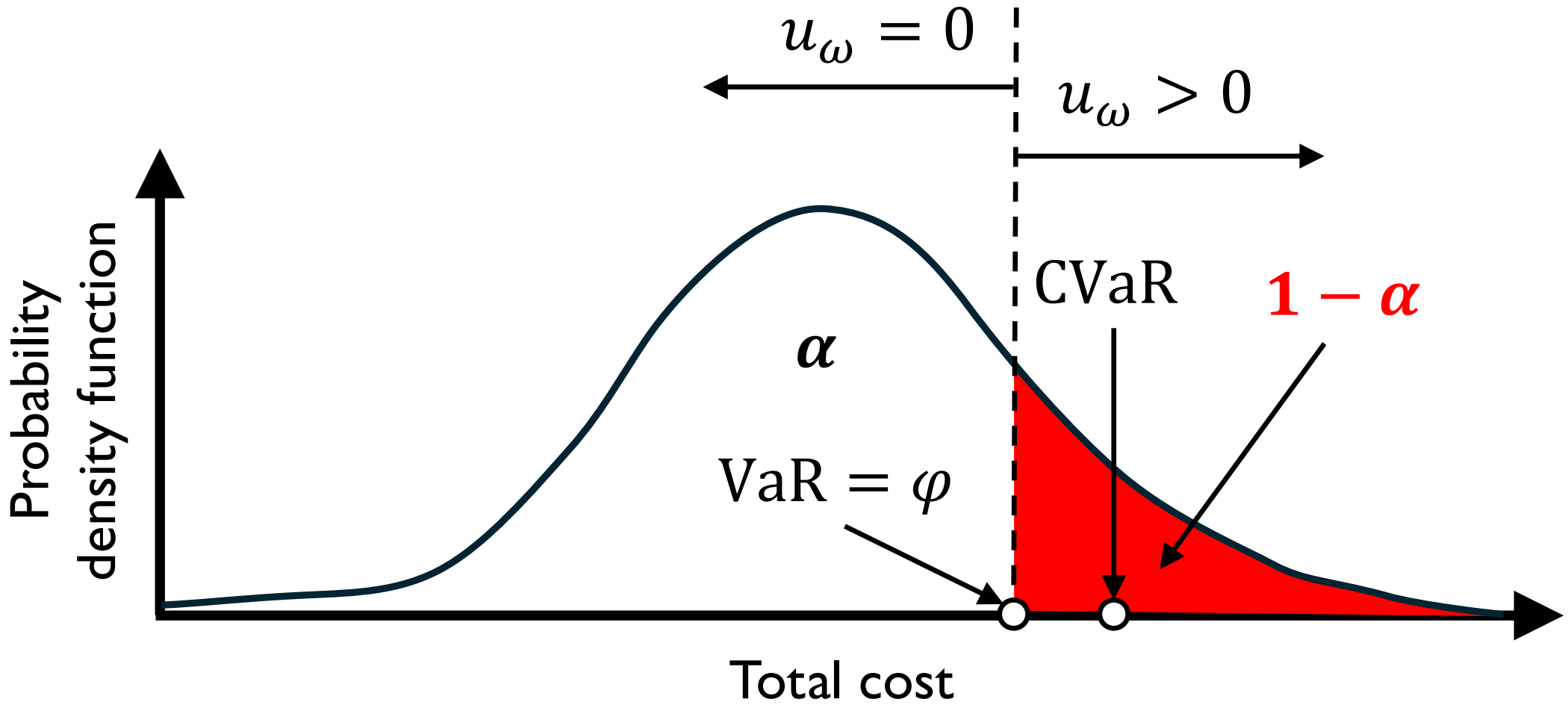
$$u_\omega \geq \boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{q}_\omega^T\boldsymbol{y}_\omega - \varphi, \forall\omega \in \Omega^{\mathrm{W}}$$

$$u_\omega \geq 0, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{T}_\omega\boldsymbol{x} + \boldsymbol{R}_\omega\boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega, \forall\omega \in \Omega^{\mathrm{W}}$$

$$\boldsymbol{y}_\omega \in \boldsymbol{Y}, \forall\omega \in \Omega^{\mathrm{W}}$$

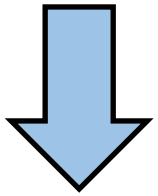
# SP – Risk-averse approach – CVaR



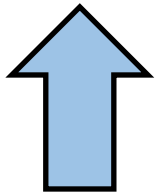
# SP – Constraint generation-based algorithm

**Master problem**

Relaxed version. Determine the first-stage decisions.



First-stage decisions

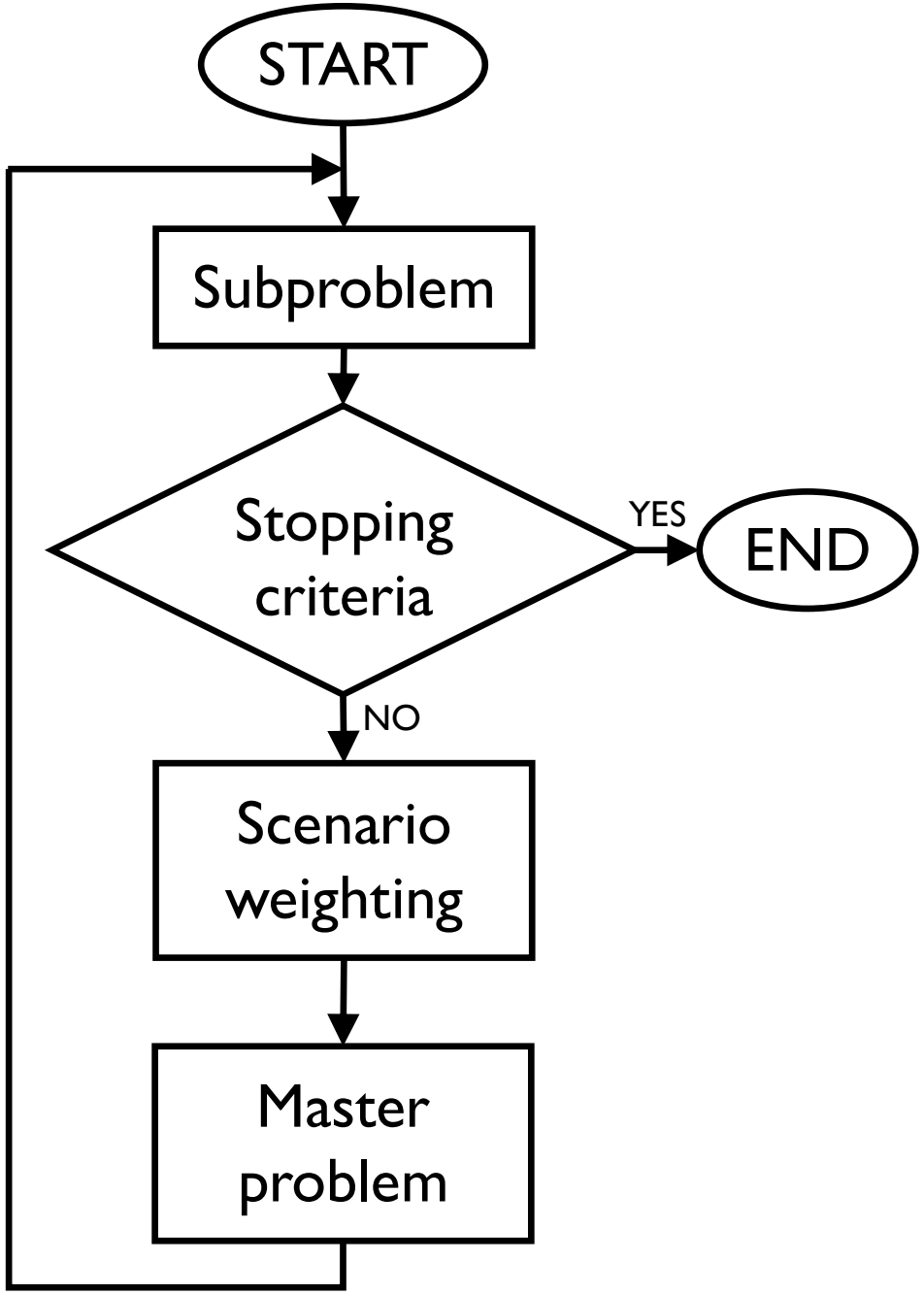


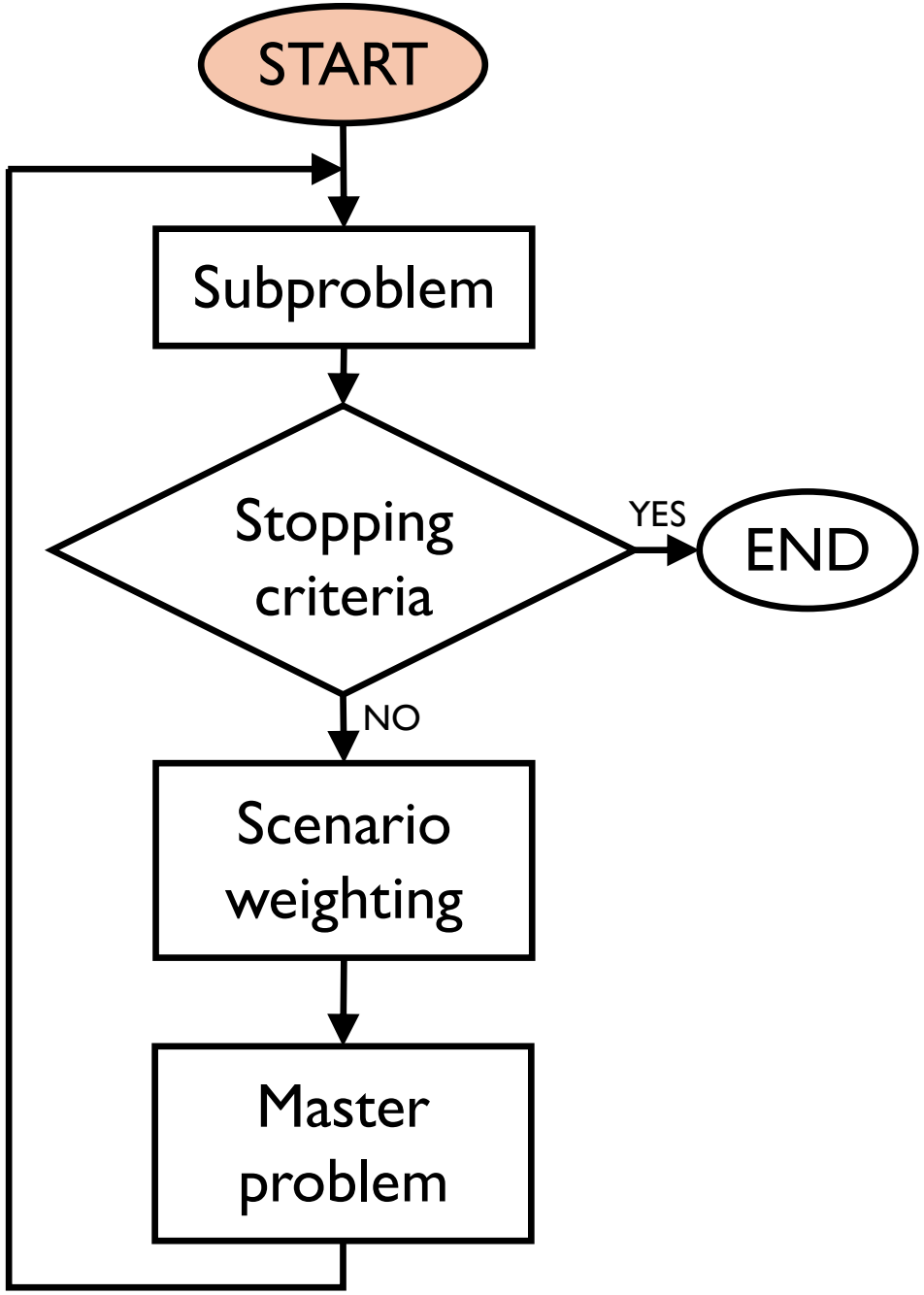
Active scenarios

**Subproblem**

First-stage decisions are fixed. Determine the second-stage decisions. Identify the active scenarios and compute their weights.

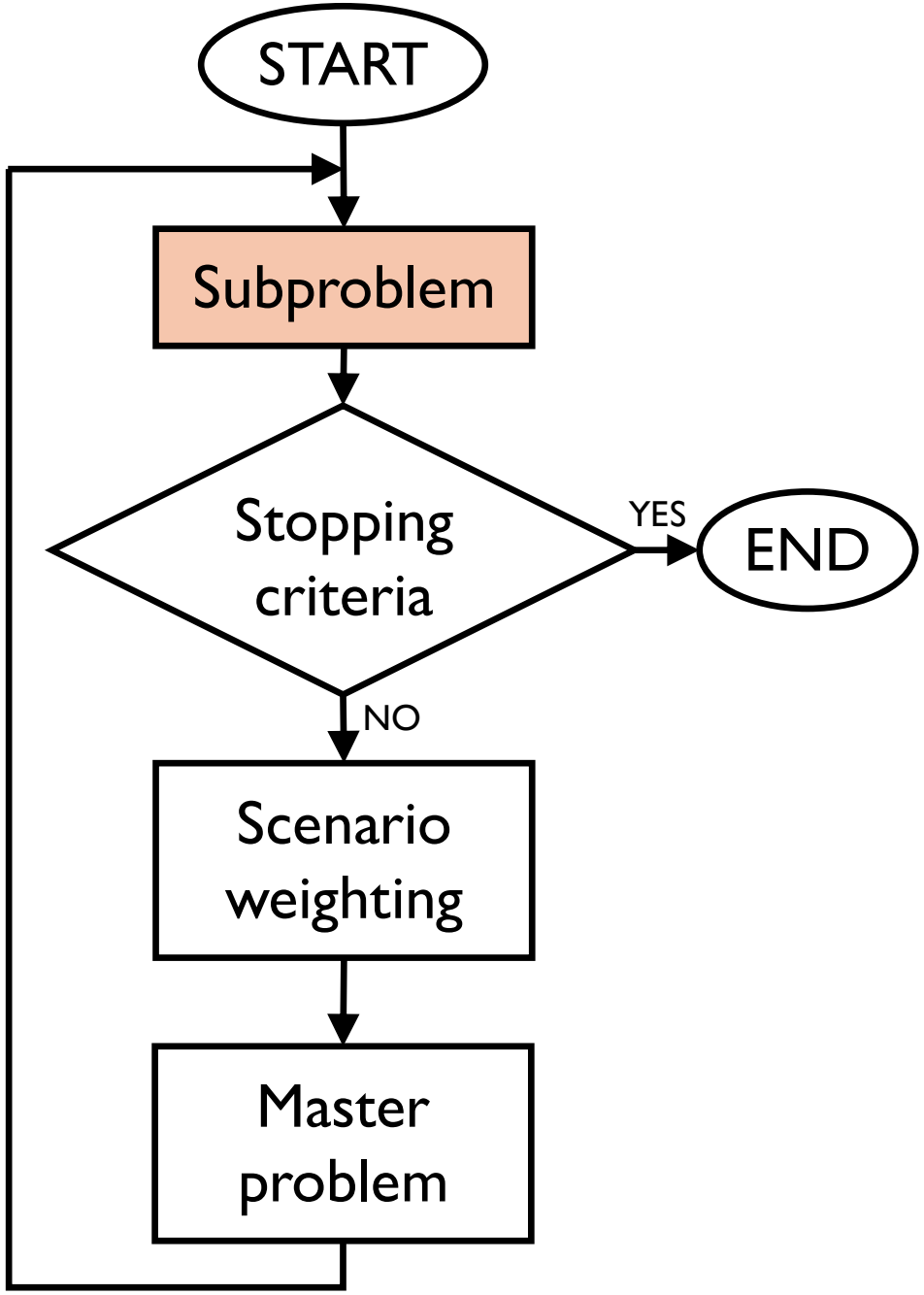
# SP – CGA


START
Subproblem
Stopping criteria
YES
END
NO
Scenario weighting
Master problem




# SP – CGA

- $UB = \infty, LB = -\infty.$
- Stopping tolerance $\varepsilon$.
- Iteration counter $j = 1$.
- Initialize $\boldsymbol{x}^{(1)}$.




# SP – CGA

- Solve the subproblem ($\forall \omega$):

$$\min_{\boldsymbol{y}_\omega} \quad z_\omega^{\text{Sub}} = \boldsymbol{c}^T \boldsymbol{x}^{(j)} + \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega$$

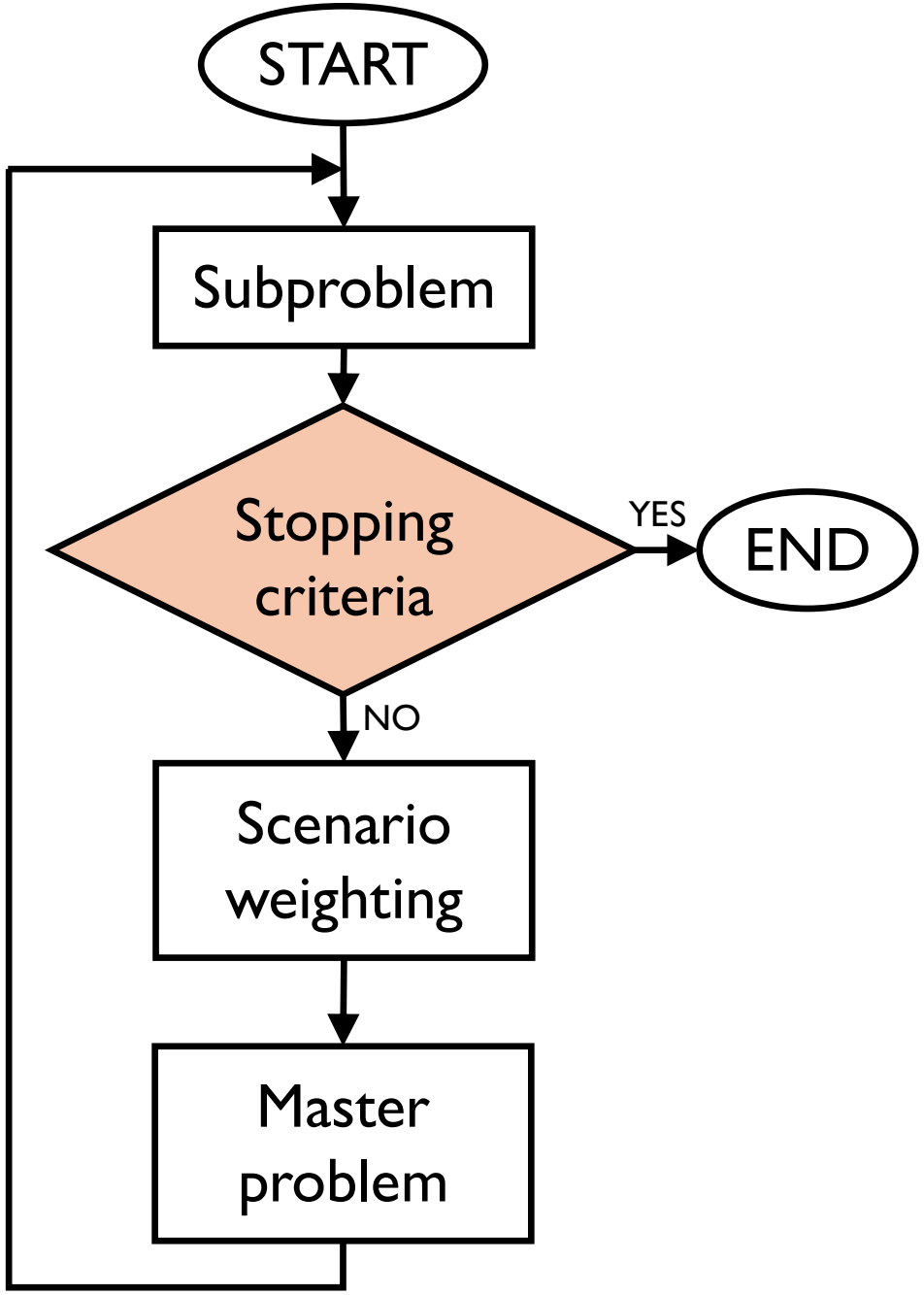
**s.t.:**

$$\boldsymbol{T}_\omega \boldsymbol{x}^{(j)} + \boldsymbol{R}_\omega \boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega$$

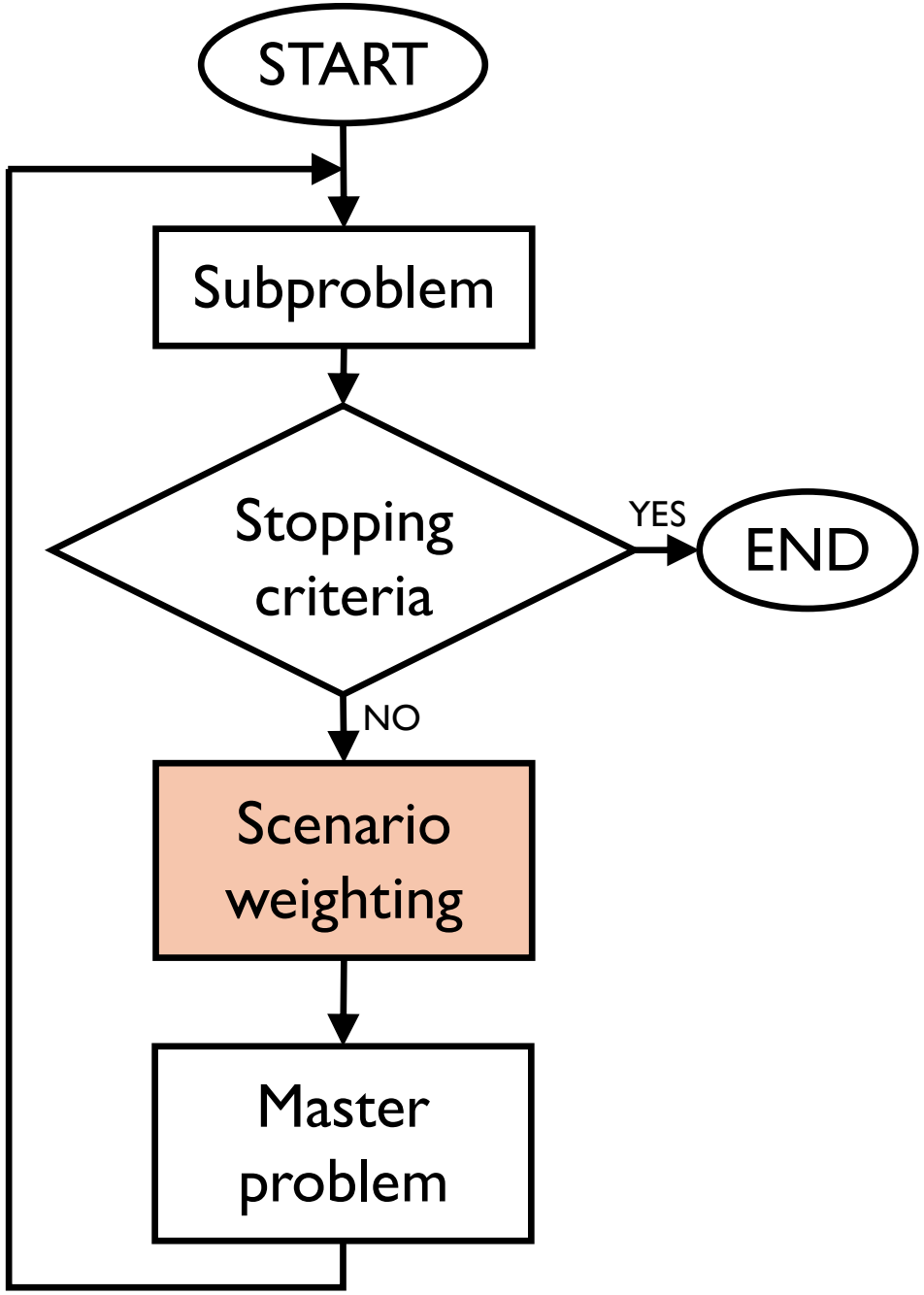
$$\boldsymbol{y}_\omega \in \boldsymbol{Y}$$

- $UB = \text{CVaR}_\alpha \left( z_\omega^{\text{Sub},(j)} \right)$.

# SP – CGA



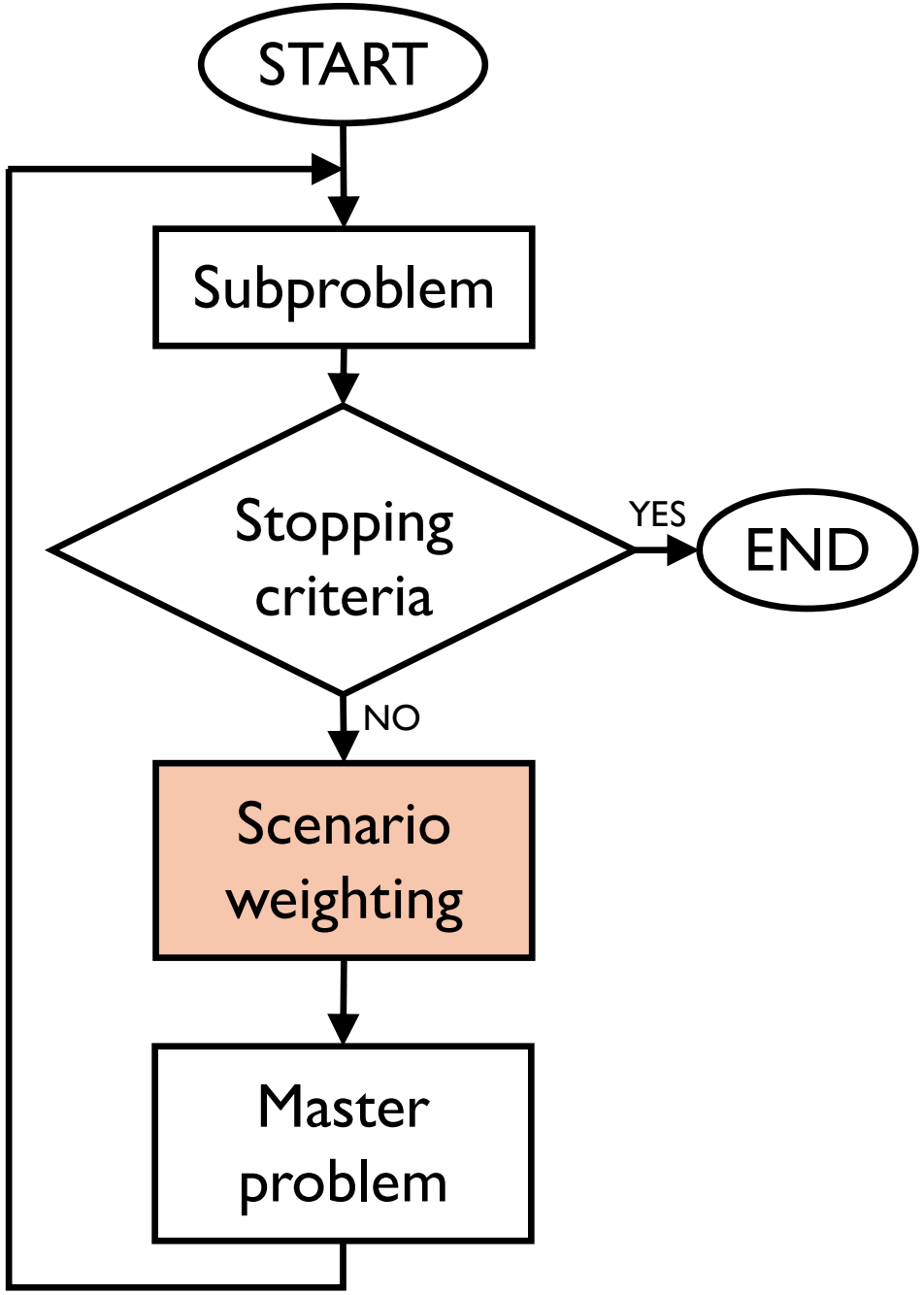

- If $\frac{|UB-LB|}{|UB|} \leq \varepsilon$, stop. Otherwise, go to the scenario weighting step.




# SP – CGA

- Sorting of the scenarios:

$\phi : \Omega^{W} \to \Omega^{W}$ such that $z^{\mathrm{Sub},(j)}_{\phi(1)} \leq \cdots \leq z^{\mathrm{Sub},(j)}_{\phi(|\Omega^{W}|)}$.




# SP – CGA

- Sorting of the scenarios:

$\phi : \Omega^{\mathrm{W}} \to \Omega^{\mathrm{W}}$ such that $z^{\mathrm{Sub},(j)}_{\phi(1)} \leq \cdots \leq z^{\mathrm{Sub},(j)}_{\phi(|\Omega^{\mathrm{W}}|)}$.

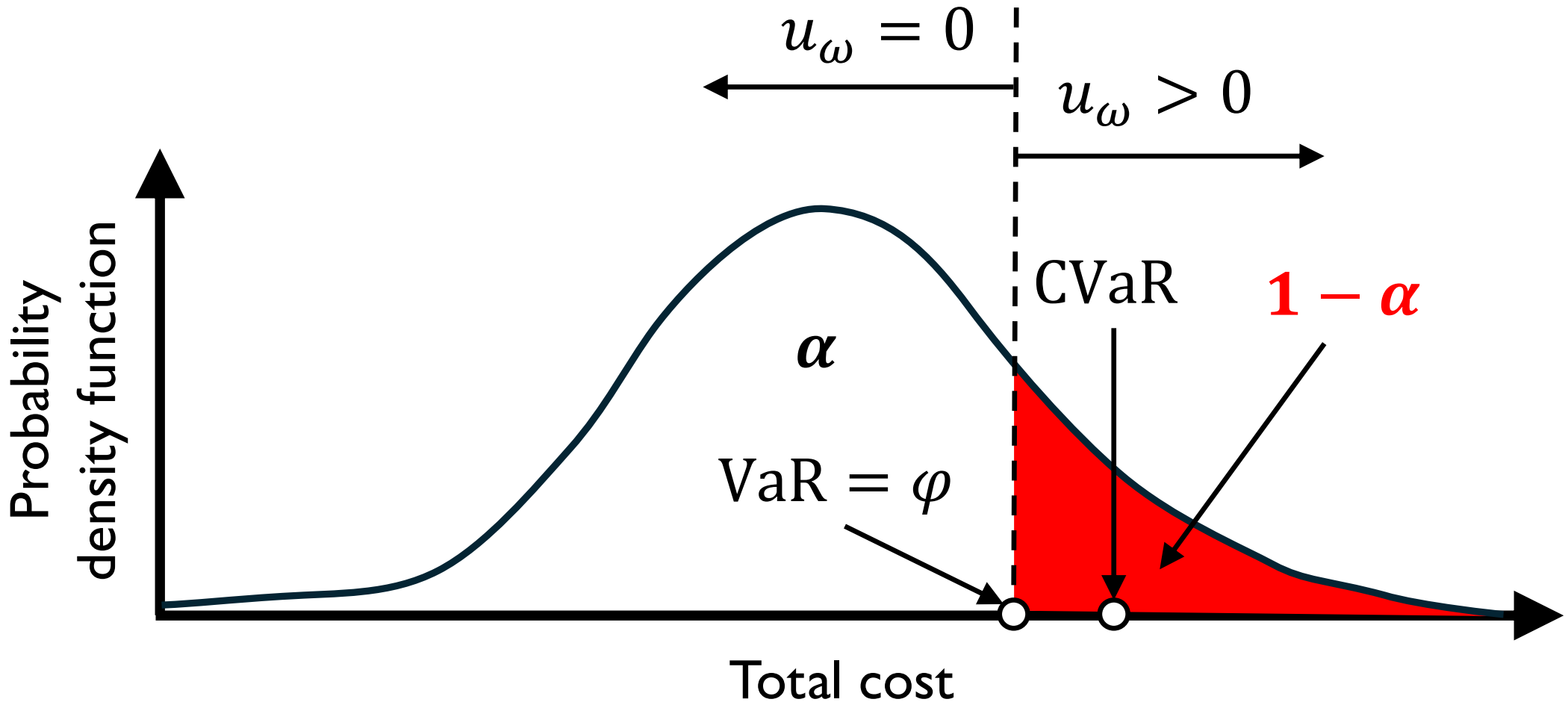
- Weighting of the scenarios:

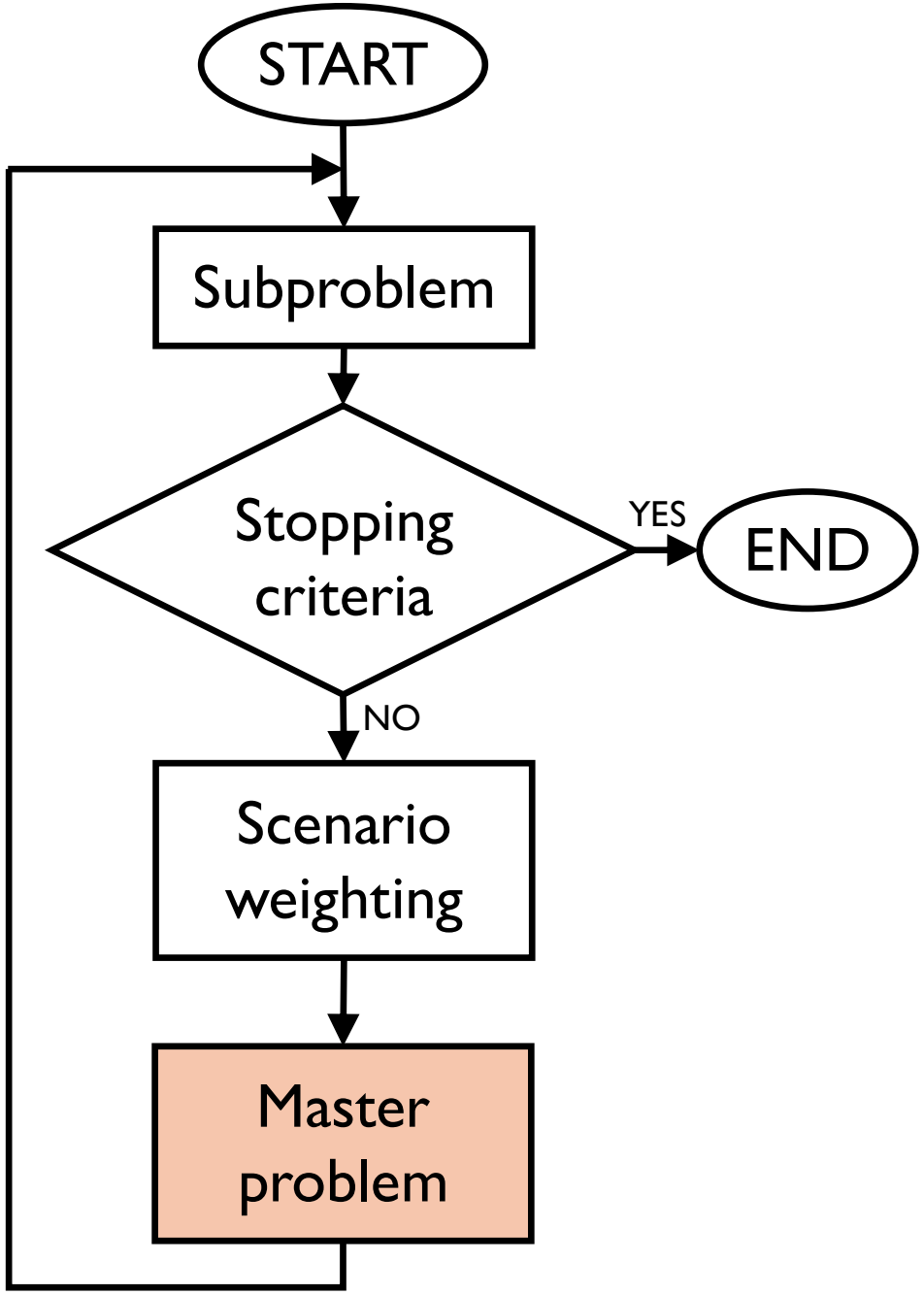
$$\rho^{(j)}_{\phi(\omega)} = \begin{cases} 0 & \text{for} \quad 1 \leq \omega < \hat{\omega} \\ \Pi_{\omega} - \alpha & \text{for} \quad \omega = \tilde{\omega} \\ \pi_{\omega} & \text{for} \quad \tilde{\omega} < \omega \leq |\Omega^{\mathrm{W}}| \end{cases},$$

where $\tilde{\omega} = \omega \in \Omega^{\mathrm{W}}$ such that

$$\sum_{\omega=1}^{\tilde{\omega}-1} \pi_{\phi(\omega)} < \alpha \,, \Pi_{\omega} := \sum_{\omega=1}^{\tilde{\omega}} \pi_{\phi(\omega)} \geq \alpha$$

# SP – Risk-averse approach – CVaR






# SP – CGA

- Solve the master problem:

$$\min_{\boldsymbol{x};\delta;\boldsymbol{y}_\omega} \; z^{\text{Master}} = \boldsymbol{c}^T\boldsymbol{x} + \delta$$

s.t.:

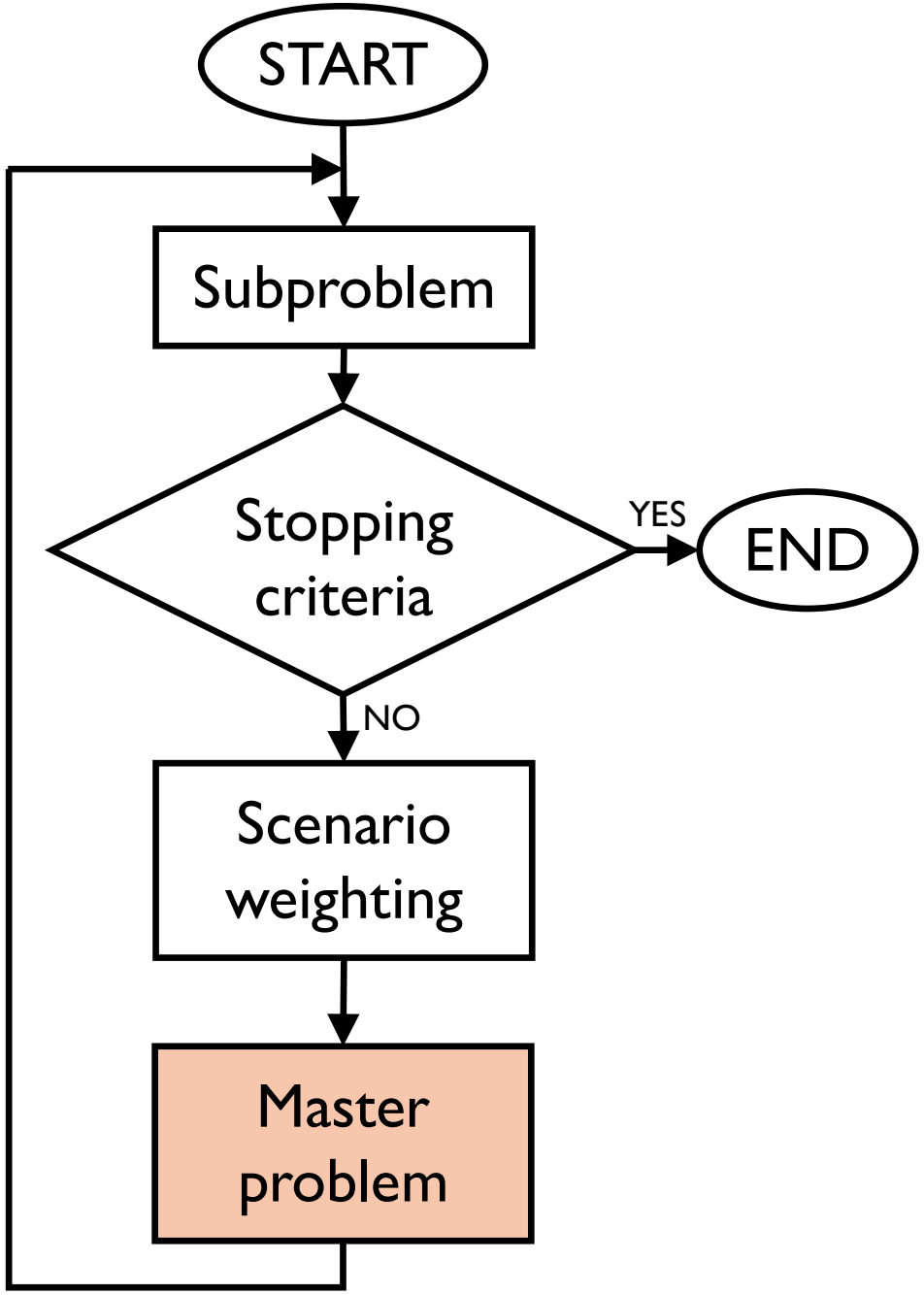
$$\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$$

$$\boldsymbol{x} \in \boldsymbol{X}$$

$$\delta \geq \frac{1}{1-\alpha} \sum_{\omega \in \Omega^{\text{W}}} \rho_\omega^{(\nu)} \boldsymbol{q}_\omega^T \boldsymbol{y}_\omega \,, \nu = 1, \dots, j$$

$$\left\{ \boldsymbol{T}_\omega \boldsymbol{x} + \boldsymbol{R}_\omega \boldsymbol{y}_\omega \leq \boldsymbol{h}_\omega \right.$$

$$\left. \boldsymbol{y}_\omega \in \boldsymbol{Y} \right\}, \text{s.t.: } \exists \nu = 1, \dots, j : \rho_\omega^{(\nu)} > 0$$




# SP – CGA

- $LB = z^{\text{Master},(j)}$.
- Increase the iteration counter:
$$j \leftarrow j + 1.$$
- Set $\boldsymbol{x}^{(j)} \leftarrow \boldsymbol{x}^{*}$.
- Go to the subproblem step.

# SP – Constraint generation-based algorithm

The master problem and the subproblem should include slack variables to prevent infeasible solutions, such as the load-shedding variables.

# SP – CGA – Illustrative example

| Scenario | Probability |
|---|---|
| $\omega_1$ | 0.2 |
| $\omega_2$ | 0.2 |
| $\omega_3$ | 0.2 |
| $\omega_4$ | 0.2 |
| $\omega_5$ | 0.2 |

# SP – CGA – Illustrative example

Iteration $j = 1$, $\alpha = 0.7$ (70%)

| Scenario | Total cost | Weight, $\rho_\omega^{(1)}$ |
|---|---|---|
| $\omega_3$ | 8 | 0 |
| $\omega_2$ | 10 | 0 |
| $\omega_5$ | 15 | 0 |
| $\omega_1$ | $\mathrm{VaR}_\alpha = 20$ | $0.8 - 0.7 = 0.1$ |
| $\omega_4$ | 25 | 0.2 |

Active scenarios $= \{\omega_1, \omega_4\}$

# SP – CGA – Illustrative example

Iteration $j = 2$, $\alpha = 0.7$ (70%)

| Scenario | Total cost | Weight, $\rho_\omega^{(2)}$ |
|---|---|---|
| $\omega_5$ | 4 | 0 |
| $\omega_4$ | 5 | 0 |
| $\omega_2$ | 6 | 0 |
| $\omega_3$ | $\mathrm{VaR}_\alpha = 7$ | $0.8 - 0.7 = 0.1$ |
| $\omega_1$ | 12 | 0.2 |

Active scenarios $= \{\omega_1, \omega_4\} \cup \{\omega_1, \omega_3\} = \{\omega_1, \omega_3, \omega_4\}$

# SP – CGA – Illustrative example

Iteration $j = 2$, $\alpha = 0.7$ (70%)

| | Weight, $\rho_\omega^{(j)}$ | |
|---|---|---|
| Scenario | Cutting plane $j = 1$ | Cutting plane $j = 2$ |
| $\omega_1$ | 0.1 | 0.2 |
| $\omega_2$ | 0 | 0 |
| $\omega_3$ | 0 | 0.1 |
| $\omega_4$ | 0.2 | 0 |
| $\omega_5$ | 0 | 0 |

Active scenarios $= \{\omega_1, \omega_4\} \cup \{\omega_1, \omega_3\} = \{\omega_1, \omega_3, \omega_4\}$

# SP – CGA – Computational issues

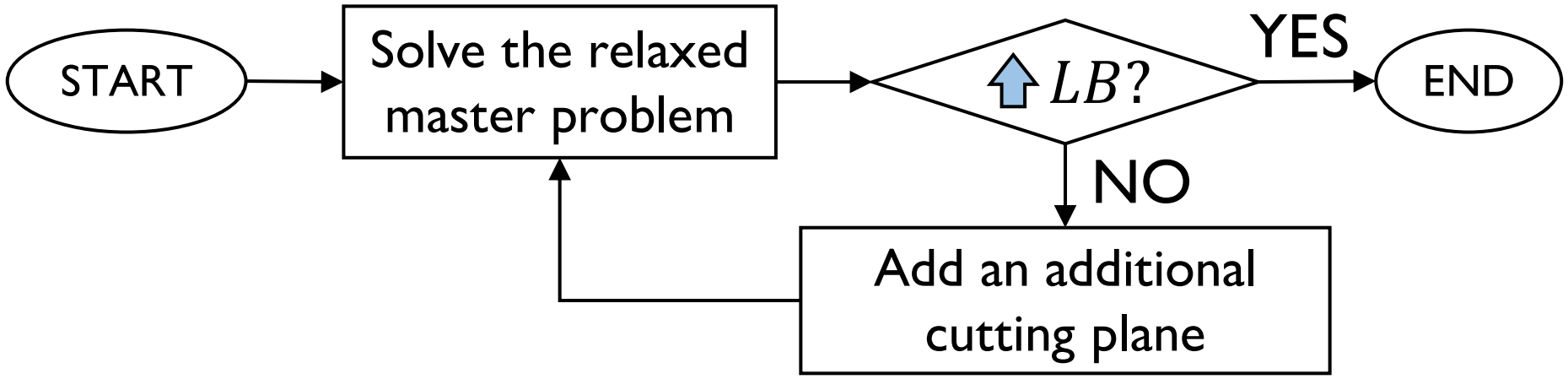
Computationally intractable problems may arise because more cutting planes related to new active scenarios are iteratively added to the master problem.

# SP – Relaxed CGA (RCGA)

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

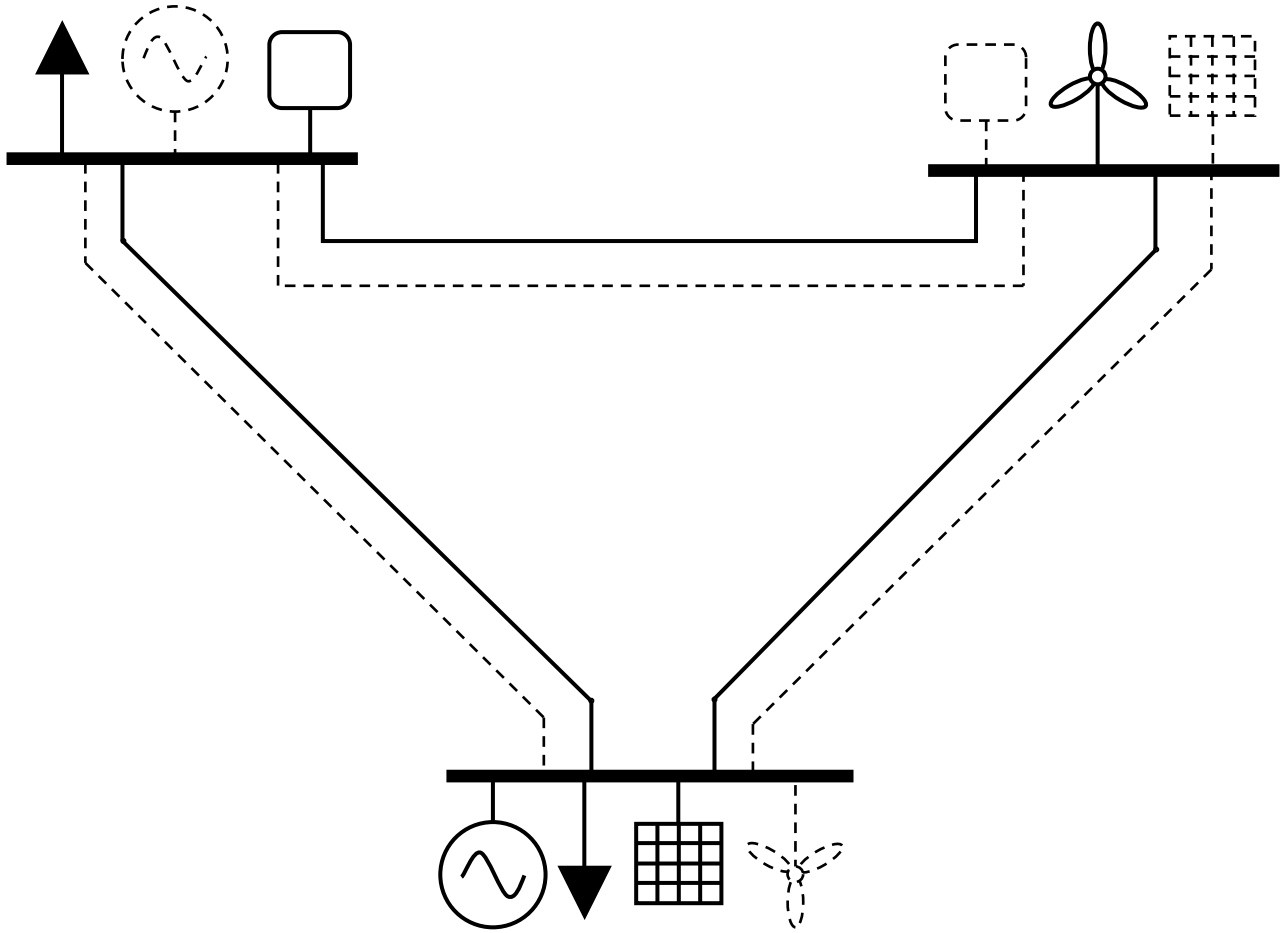
# SP – Relaxed CGA (RCGA)

Proposal → Relaxation of the master problem: latest active scenarios are considered, and more cutting planes are added until $LB$ increases.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

# SP – Relaxed CGA (RCGA)



Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

# SP – Relaxed CGA (RCGA)

Generation and transmission network expansion planning problem.

IEEE 24-bus Reliability Test System.

Uncertain parameters (100 scenarios):

- Peak power consumption of loads.
- Marginal production cost of conventional generating units.
- Capacity of conventional and renewable generating units.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

# SP – Relaxed CGA (RCGA)

MILP problem:

- Continuous variables: 1,544,135.
- Integer variables: 272,017.
- Constraints: 3,570,235.
- Computational time: not solved after 168 h (1 week).

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

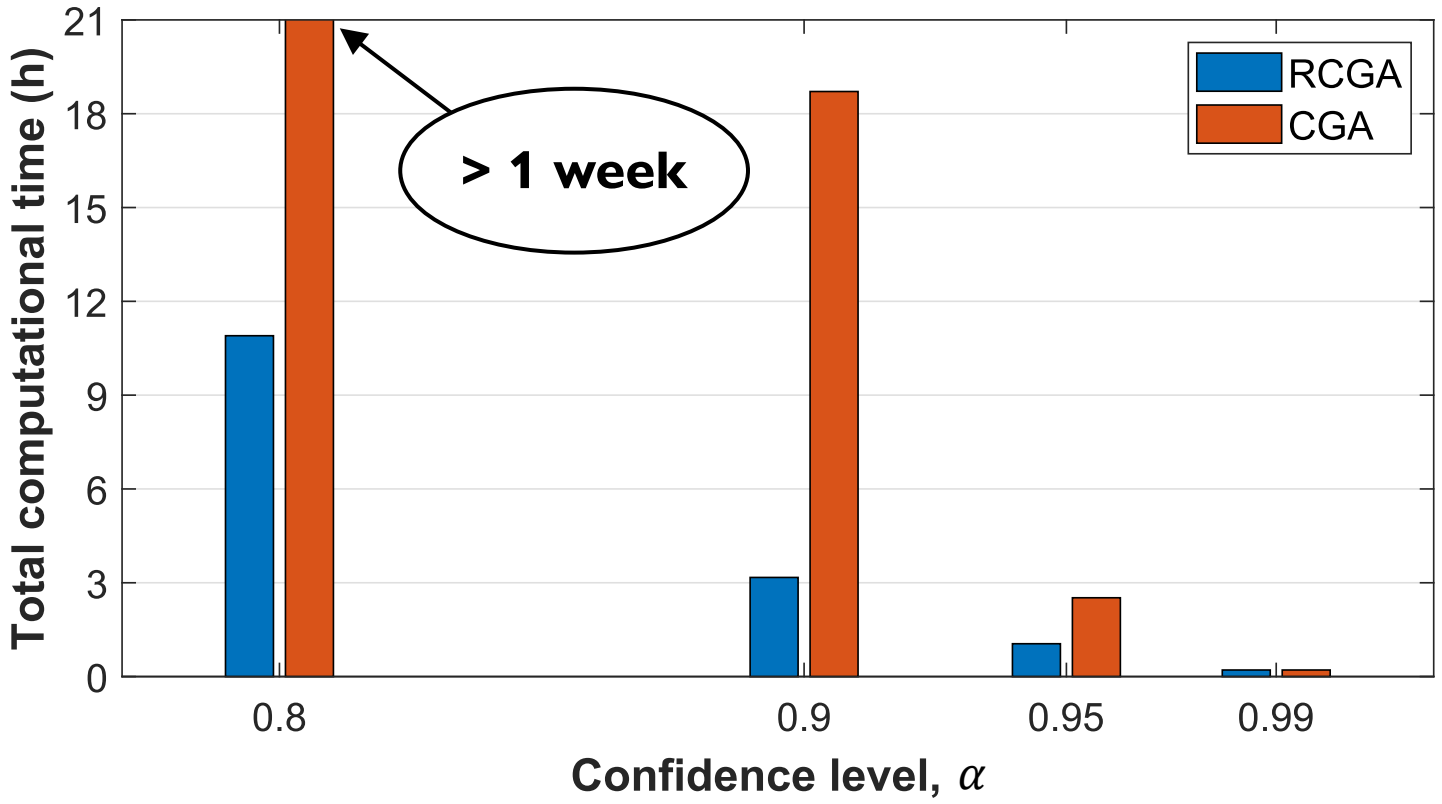
# SP – Relaxed CGA (RCGA)

| Confidence level, $\alpha$ | 0.80 | 0.90 | 0.95 | 0.99 |
|---|---|---|---|---|
| CVaR of the total annualized cost (M$) | 438.79 | 446.13 | 450.79 | 458.04 |
| Annualized investment costs (M$) | 354.99 | 358.65 | 360.38 | 369.10 |
| Computational time (h) | 10.90 | 3.37 | 1.05 | 0.21 |
| # outer-loop iterations | 3 | 3 | 3 | 2 |
| # inner-loop iterations | 1, 1 | 1, 1 | 1, 1 | 1 |

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

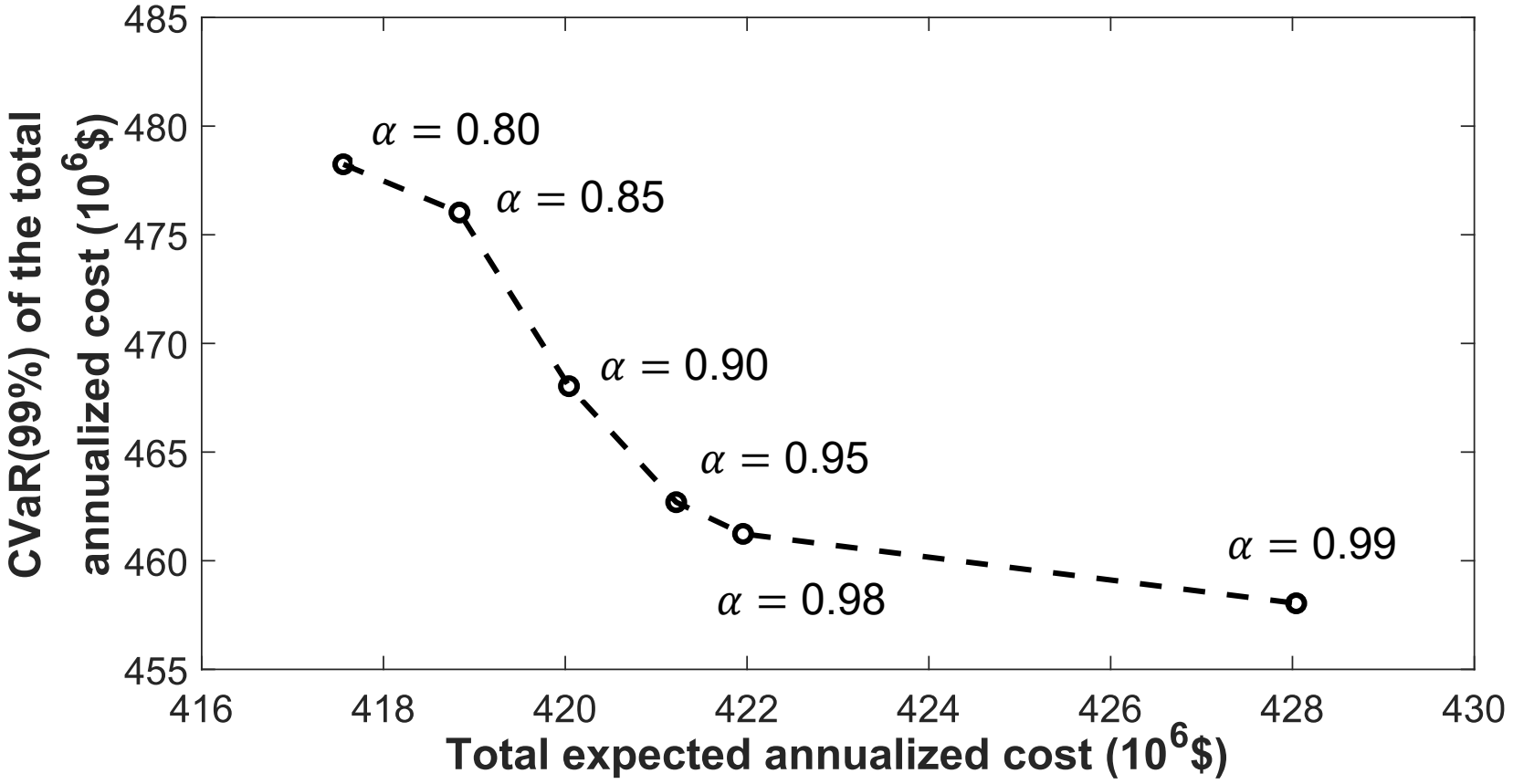
# SP – Relaxed CGA (RCGA)

Same solution, and the RCGA reduces the computational time.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

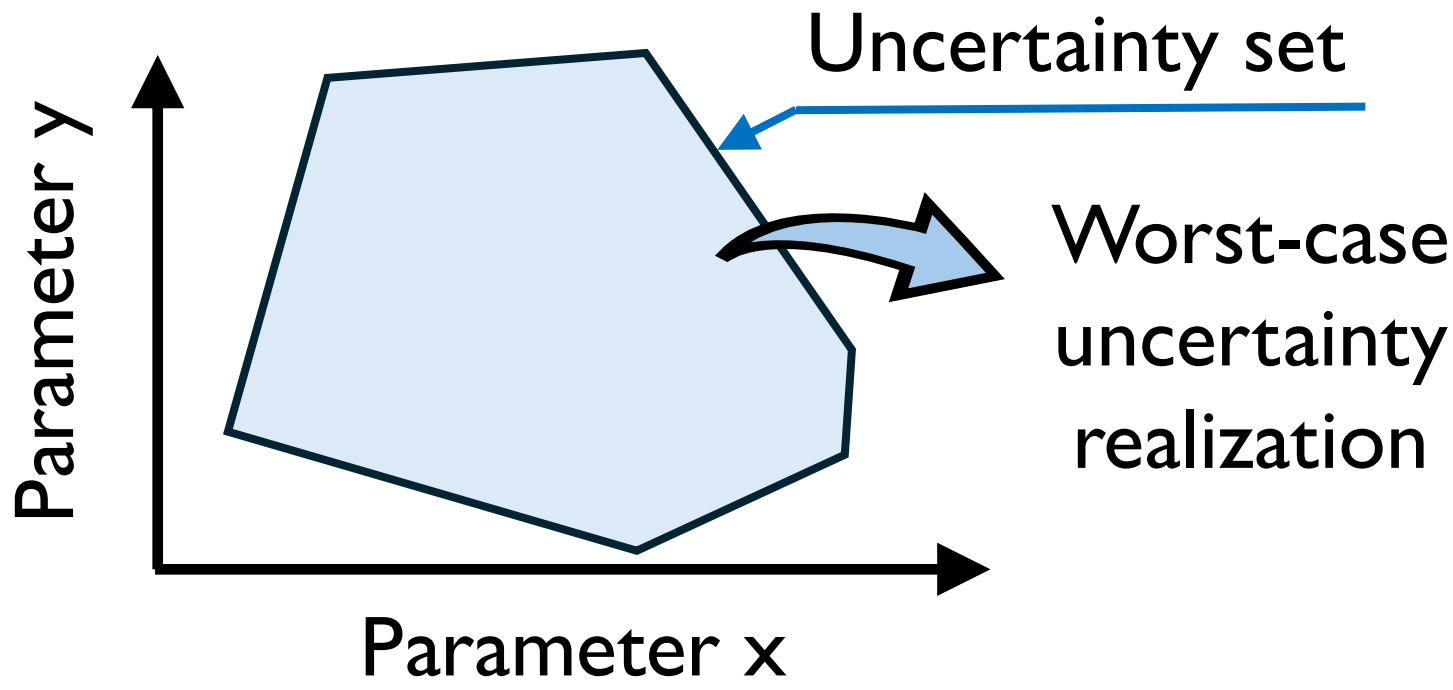
# SP – Relaxed CGA (RCGA)




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Computational performance enhancement strategies for risk-averse two-stage stochastic generation and transmission network expansion planning", *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 273–286, Jan. 2024. DOI: 10.1109/TPWRS.2023.3236397.

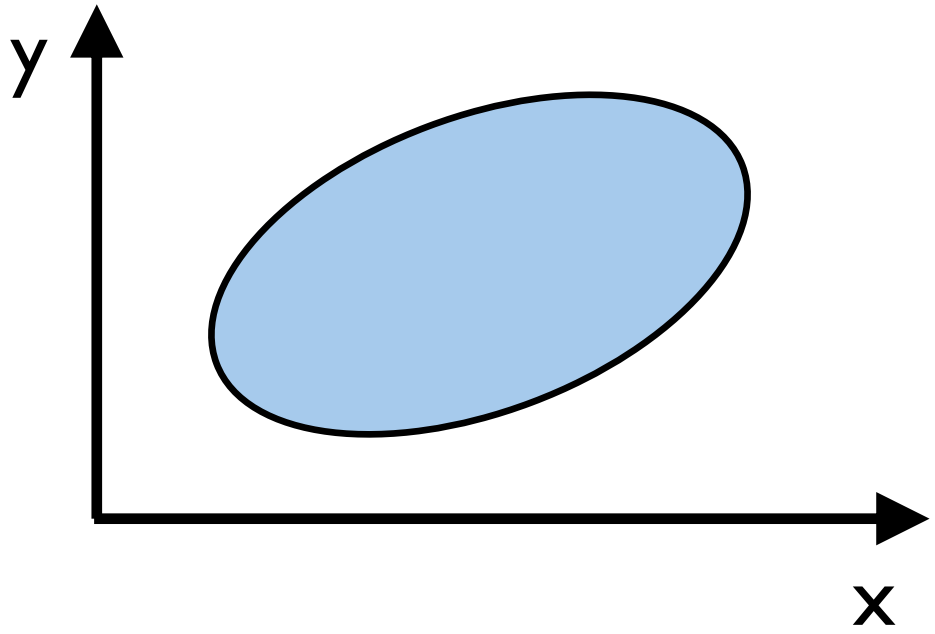
# 3. Robust Optimization (RO)

# Robust optimization

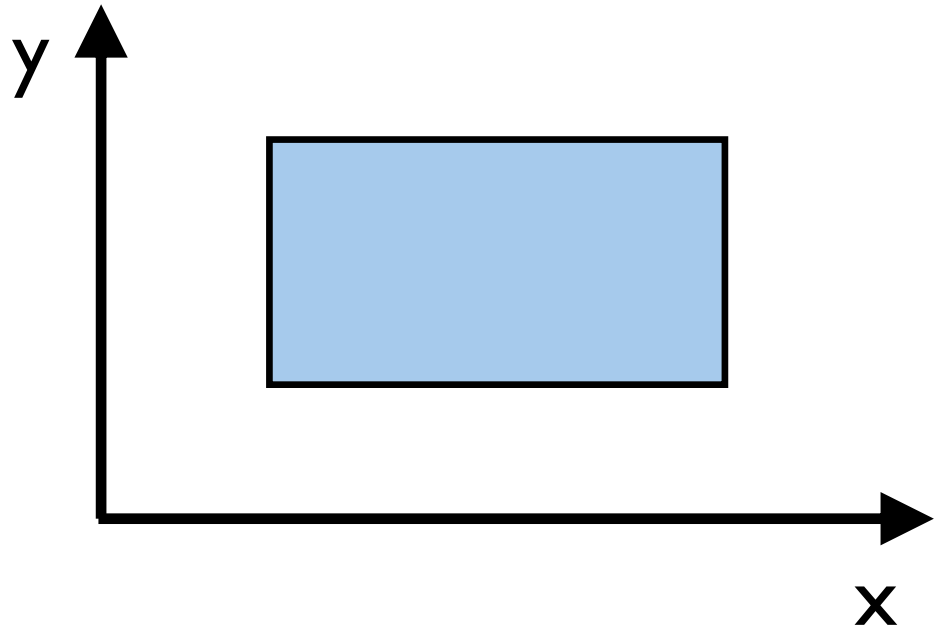


# Robust optimization – Uncertainty sets

Ellipsoidal uncertainty set

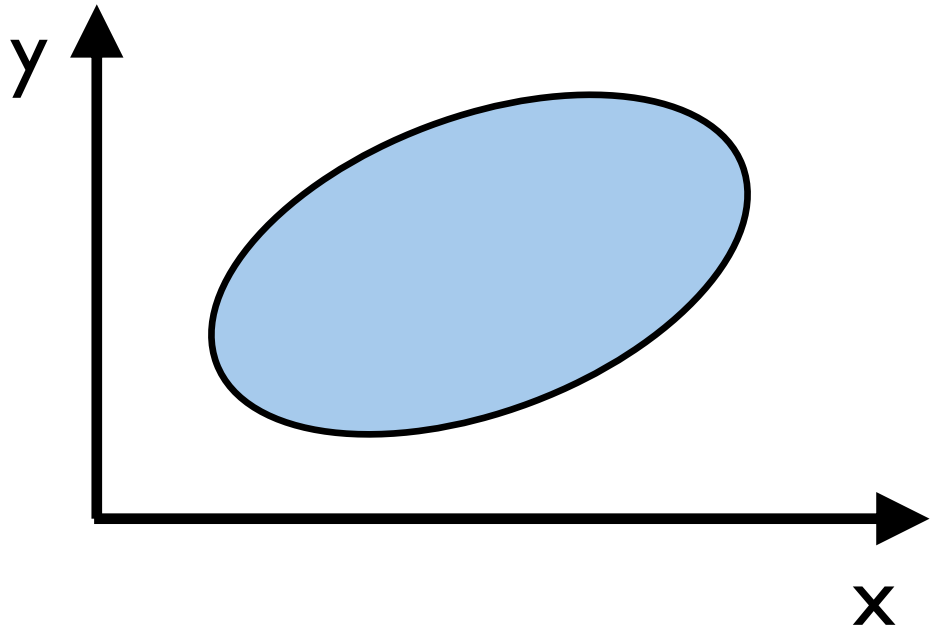



Polyhedral uncertainty set

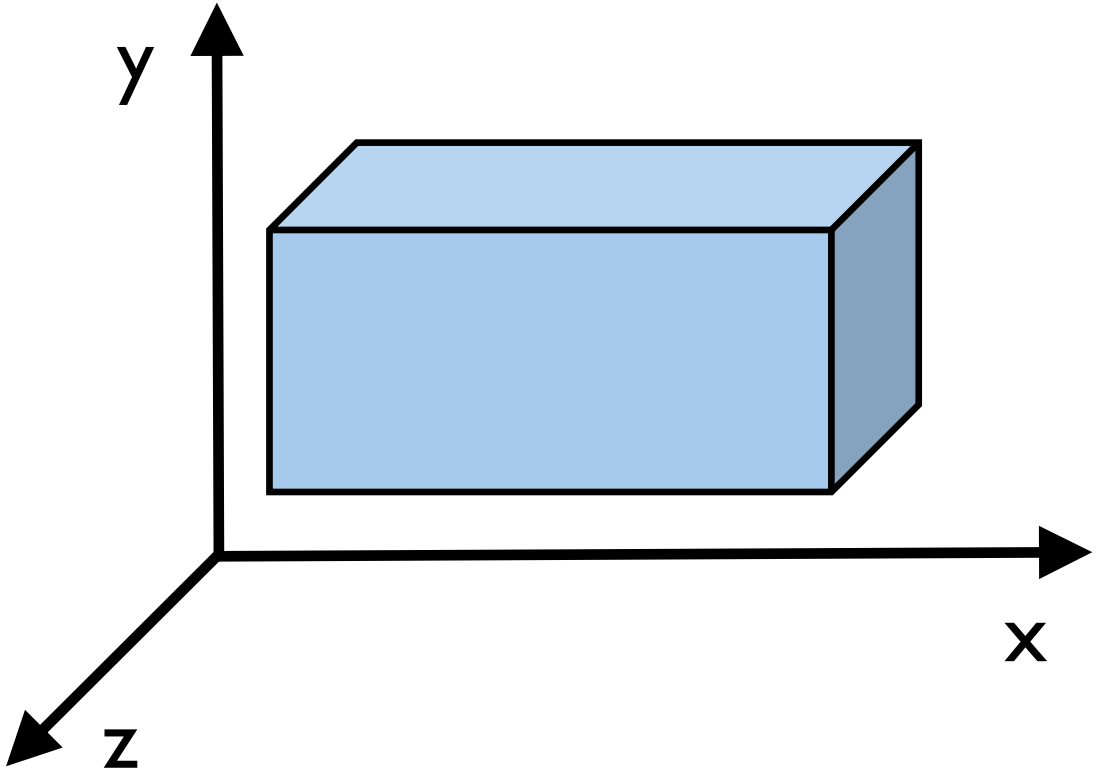


# Robust optimization – Uncertainty sets

Ellipsoidal uncertainty set




Polyhedral uncertainty set



# Robust optimization – Uncertainty sets

Control the conservativeness $\rightarrow$ Uncertainty budget $\Gamma$.

Sum of deviations with respect to the expected value $\leq \Gamma$.

$\Gamma = 0 \rightarrow$ Uncertainty realization $=$ Expected value.

The size of the uncertainty set is controlled through $\Gamma$.

# Robust optimization – Uncertainty sets

Cardinality-constrained uncertainty set with binary variables: the uncertainty budget $\Gamma$ limit the number of uncertain parameters that are allowed to vary from their expected values.

$$\mathcal{U} := \Big\{ u_i = \widetilde{U}_i + \widehat{U}_i^+ z_i^+ - \widehat{U}_i^- z_i^-, \forall i \in \Omega^{\mathrm{I}}$$

$$z_i^+, z_i^- \in \{0,1\}, \forall i \in \Omega^{\mathrm{I}}$$

$$z_i^+ + z_i^- \leq 1, \forall i \in \Omega^{\mathrm{I}}$$

$$\sum_{i \in \Omega^{\mathrm{I}}} (z_i^+ + z_i^-) \leq \Gamma \Big\}$$

# Robust optimization – Uncertainty sets

Cardinality-constrained uncertainty sets with binary variables were considered in the RO works presented next in this lecture.

# Robust optimization

- Robust optimization without recourse.
- Robust optimization with recourse.
  - Column-and-constraint generation algorithm.
  - Nested column-and-constraint generation algorithm.
  - Acceleration techniques.

# Robust optimization

- **Robust optimization without recourse**.

- Robust optimization with recourse.
  - Column-and-constraint generation algorithm.
  - Nested column-and-constraint generation algorithm.
  - Acceleration techniques.

# Robust optimization without recourse

All decision variables are identified before the uncertainty realization is known.

Also referred to as:

- Static robust optimization.
- Single-stage robust optimization.

# Robust optimization without recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}} \quad \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

# Robust optimization without recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T\boldsymbol{x} + c^{\mathrm{O,WC}}$$
$$\text{s. t. :}$$
$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$
$$\boldsymbol{x} \in \{0,1\}^n$$

Upper-level problem

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}} \quad \boldsymbol{g}^T\boldsymbol{u}$$
$$\text{s. t. :}$$
$$\boldsymbol{u} \in \mathcal{U}$$

# Robust optimization without recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}} \quad \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

Lower-level problem

# Robust optimization without recourse

Note that the lower-level objective function is part of the upper-level objective function.

Hence, if the lower-level problem is linear, it can be replaced with its corresponding dual problem, obtaining a single-level optimization problem.

$$\min \max \rightarrow \min \min \rightarrow \min$$

# Robust optimization without recourse

## Primal

min primal objective function (variables $x, y$)
subject to primal constraints (variables $x, y$) : $\lambda, \mu$

## Dual

max dual objective function (variables $\lambda, \mu$)
subject to dual constraints (variables $\lambda, \mu$) : $x, y$

# Robust optimization without recourse

## Primal

max primal objective function (variables $x, y$)
subject to primal constraints (variables $x, y$) : $\lambda, \mu$

## Dual

min dual objective function (variables $\lambda, \mu$)
subject to dual constraints (variables $\lambda, \mu$) : $x, y$

# Robust optimization without recourse

Alternative: to replace the lower-level problem with its Karush-Kuhn-Tucker (KKT) necessary optimality conditions.

The lower-level problem must be convex.

Main idea: the analytical solution of the KKT conditions is equivalent to the optimal solution of the optimization problem.

# Robust optimization without recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}} \quad \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

# Robust optimization without recourse

$$\min_{\boldsymbol{x},\boldsymbol{u}} \quad \boldsymbol{e}^T\boldsymbol{x} + \boldsymbol{g}^T\boldsymbol{u}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

KKT conditions of the lower-level problem

# Robust optimization without recourse

In my experience working with multi-level optimization problems, replacing an optimization problem with the corresponding dual problem results in lower computational times than using the KKT conditions due to the non-linear terms associated with the latter. These nonlinear terms can be handled using big-M linearizations.

# Robust optimization without recourse

The Karush–Kuhn–Tucker (KKT) Conditions and the Interior Point Method for Convex Optimization

https://www.youtube.com/watch?v=uh1Dk68cfWs

# Robust optimization without recourse

The main drawback of robust optimization without recourse is that it can give rise to solutions too conservative in practical applications that require making decisions in multiple stages based on the uncertainty realizations.

# Robust optimization

- Robust optimization without recourse.
- **Robust optimization with recourse**.
  - Column-and-constraint generation algorithm.
  - Nested column-and-constraint generation algorithm.
  - Acceleration techniques.

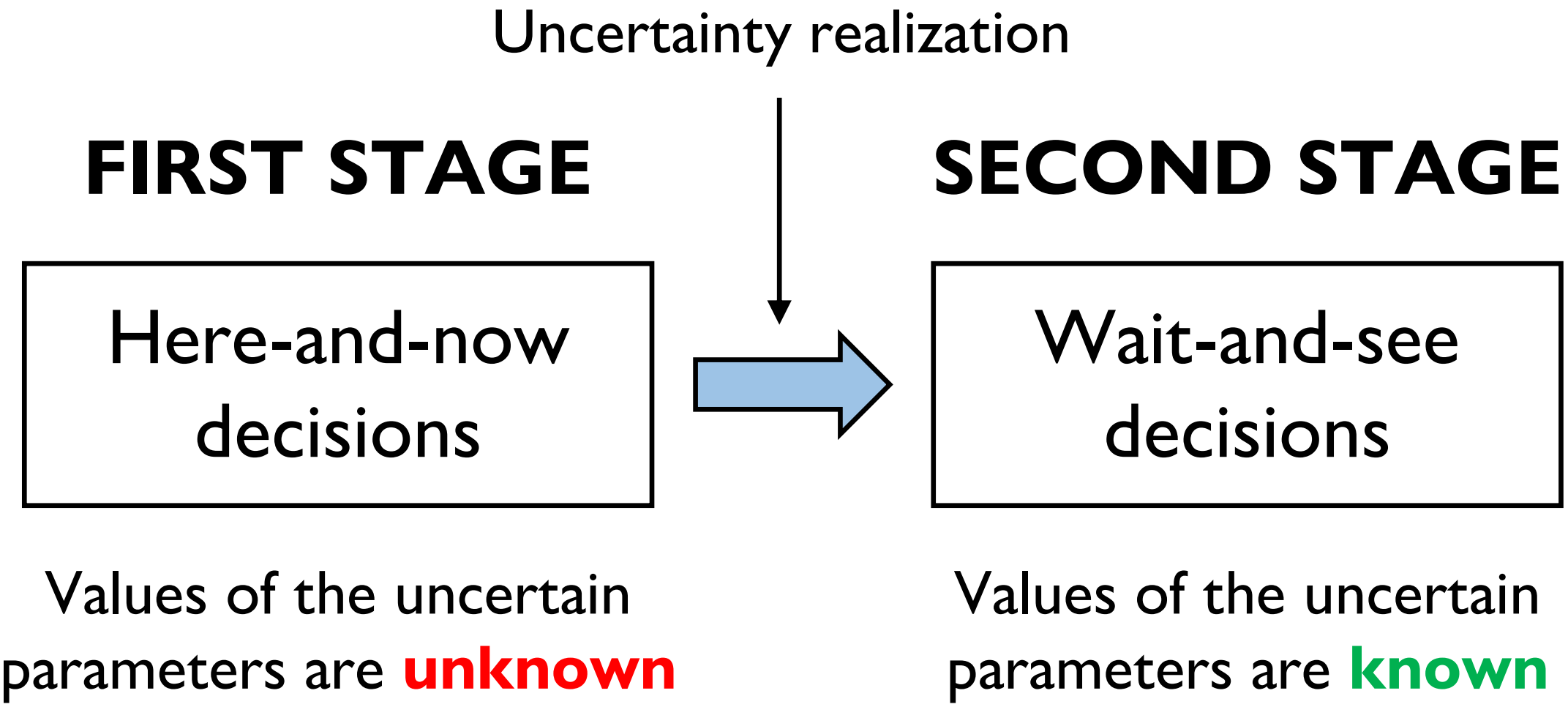
# Robust optimization with recourse

Decision variables are made before and after the uncertainty realization is known.

Also referred to as:

- Adaptive robust optimization (ARO).
- Multi-stage robust optimization.

# Robust optimization with recourse



# Robust optimization with recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}, c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

$$c^{\mathrm{O}} = \min_{\boldsymbol{y}} \quad \boldsymbol{b}^T \boldsymbol{y} + \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$

$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} \geq \boldsymbol{d}$$

# Robust optimization with recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$
$$\text{s.t.:}$$
$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$
$$\boldsymbol{x} \in \{0,1\}^n$$

Upper-level problem

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}, c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$
$$\text{s.t.:}$$
$$\boldsymbol{u} \in \mathcal{U}$$
$$c^{\mathrm{O}} = \min_{\boldsymbol{y}} \quad \boldsymbol{b}^T \boldsymbol{y} + \boldsymbol{g}^T \boldsymbol{u}$$
$$\text{s.t.:}$$
$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$
$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} \geq \boldsymbol{d}$$

# Robust optimization with recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}, c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

Middle-level problem

$$c^{\mathrm{O}} = \min_{\boldsymbol{y}} \quad \boldsymbol{b}^T \boldsymbol{y} + \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$

$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} \geq \boldsymbol{d}$$

# Robust optimization with recourse

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}, c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

$$c^{\mathrm{O}} = \min_{\boldsymbol{y}} \quad \boldsymbol{b}^T \boldsymbol{y} + \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$

$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} \geq \boldsymbol{d}$$

Lower-level problem

# Robust optimization with recourse

Two-stage robust optimization problems can be solved using the Benders' decomposition.

However, I will focus on the column-and-constraint generation algorithm (CCGA), which is the most common technique used in the literature.

# Robust optimization

- Robust optimization without recourse.
- Robust optimization with recourse.
  - **Column-and-constraint generation algorithm**.
  - Nested column-and-constraint generation algorithm.
  - Acceleration techniques.

# RO – Column-and-constraint generation algorithm

B. Zeng and L. Zhao, “Solving two-stage robust optimization problems using a column-and-constraint generation method”, *Operations Research Letters*, vol. 41, no. 5, pp. 457–461, May 2013, DOI: 10.1016/j.orl.2013.05.003.

# RO – Column-and-constraint generation algorithm

$$\min_{\boldsymbol{x}, c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u}, c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$

s. t. :

$$\boldsymbol{u} \in \mathcal{U}$$

$$c^{\mathrm{O}} = \min_{\boldsymbol{y}} \quad \boldsymbol{b}^T \boldsymbol{y} + \boldsymbol{g}^T \boldsymbol{u}$$

s. t. :

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$

$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} \geq \boldsymbol{d}$$

# RO – Column-and-constraint generation algorithm

$$\min_{x, c^{\mathrm{O,WC}}, y} \quad \boldsymbol{e}^T \boldsymbol{x} + c^{\mathrm{O,WC}}$$

s. t. :

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} \geq \boldsymbol{b}^T \boldsymbol{y}_i + \boldsymbol{g}^T \boldsymbol{u}^{(i)}, \forall \boldsymbol{i}$$

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y}_i + \boldsymbol{D}\boldsymbol{u}^{(i)} = \boldsymbol{a}, \forall \boldsymbol{i}$$

$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y}_i + \boldsymbol{L}\boldsymbol{u}^{(i)} \geq \boldsymbol{d}, \forall \boldsymbol{i}$$

Uncertainty realizations $i$

# RO – Column-and-constraint generation algorithm

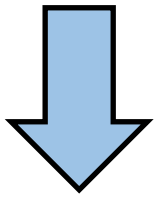
Computational issues are generally faced if all possible uncertainty realizations are considered.

Main idea of the CCGA: iteratively add uncertainty realizations until the convergence is reached.

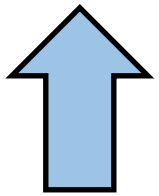
# RO – Column-and-constraint generation algorithm

**Master problem**

Relaxed version. Determine the first-stage decisions.



First-stage decisions



Worst-case uncertainty realization

**Subproblem**

First-stage decisions are fixed.

Identify the worst-case uncertainty realization and the corrective actions related to the second-stage decisions.

# RO – Column-and-constraint generation algorithm

The subproblem involves the two lowermost optimization levels, that can be recast using the duality theory or the KKT conditions.

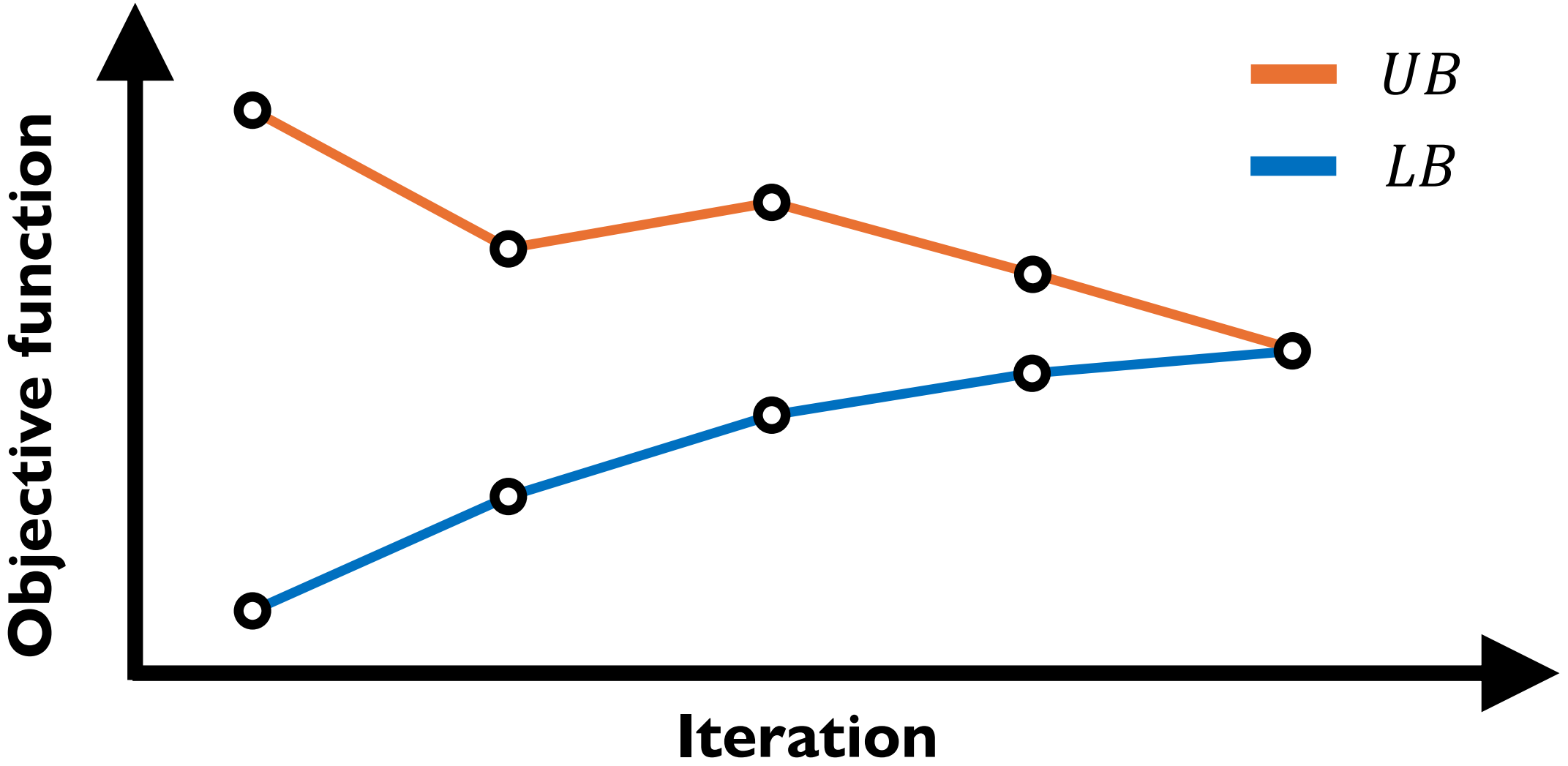
# RO – Column-and-constraint generation algorithm

Iterative procedure

Master problem $\rightarrow$ Lower bound $LB$

Subproblem $\rightarrow$ Upper bound $UB$

# RO – Column-and-constraint generation algorithm



# RO – Column-and-constraint generation algorithm

The master problem and the subproblem should include slack variables to prevent infeasible solutions, such as the load-shedding variables.

# Robust optimization

- Robust optimization without recourse.
- Robust optimization with recourse.
  - Column-and-constraint generation algorithm.
  - **Nested column-and-constraint generation algorithm**.
  - Acceleration techniques.

# RO – Nested CCGA

L. Zhao and B. Zeng, *An exact algorithm for two-stage robust optimization with mixed integer recourse problems*. University of South Florida, Tampa, FL, USA, Jan. 2012. Available: www.optimization-online.org/DB_FILE/2012/01/3310.pdf.

# RO – Nested CCGA

$$\min_{\boldsymbol{x},c^{\mathrm{O,WC}}} \quad \boldsymbol{e}^T\boldsymbol{x} + c^{\mathrm{O,WC}}$$

s.t.:

$$\boldsymbol{K}\boldsymbol{x} \leq \boldsymbol{f}$$

$$\boldsymbol{x} \in \{0,1\}^n$$

$$c^{\mathrm{O,WC}} = \max_{\boldsymbol{u},c^{\mathrm{O}}} \quad c^{\mathrm{O}}$$

s.t.:

$$\boldsymbol{u} \in \mathcal{U}$$

$$c^{\mathrm{O}} = \min_{\boldsymbol{y},\boldsymbol{z}} \quad \boldsymbol{b}^T\boldsymbol{y} + \boldsymbol{g}^T\boldsymbol{u}$$

s.t.:

$$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x})\boldsymbol{y} + \boldsymbol{D}\boldsymbol{u} = \boldsymbol{a}$$

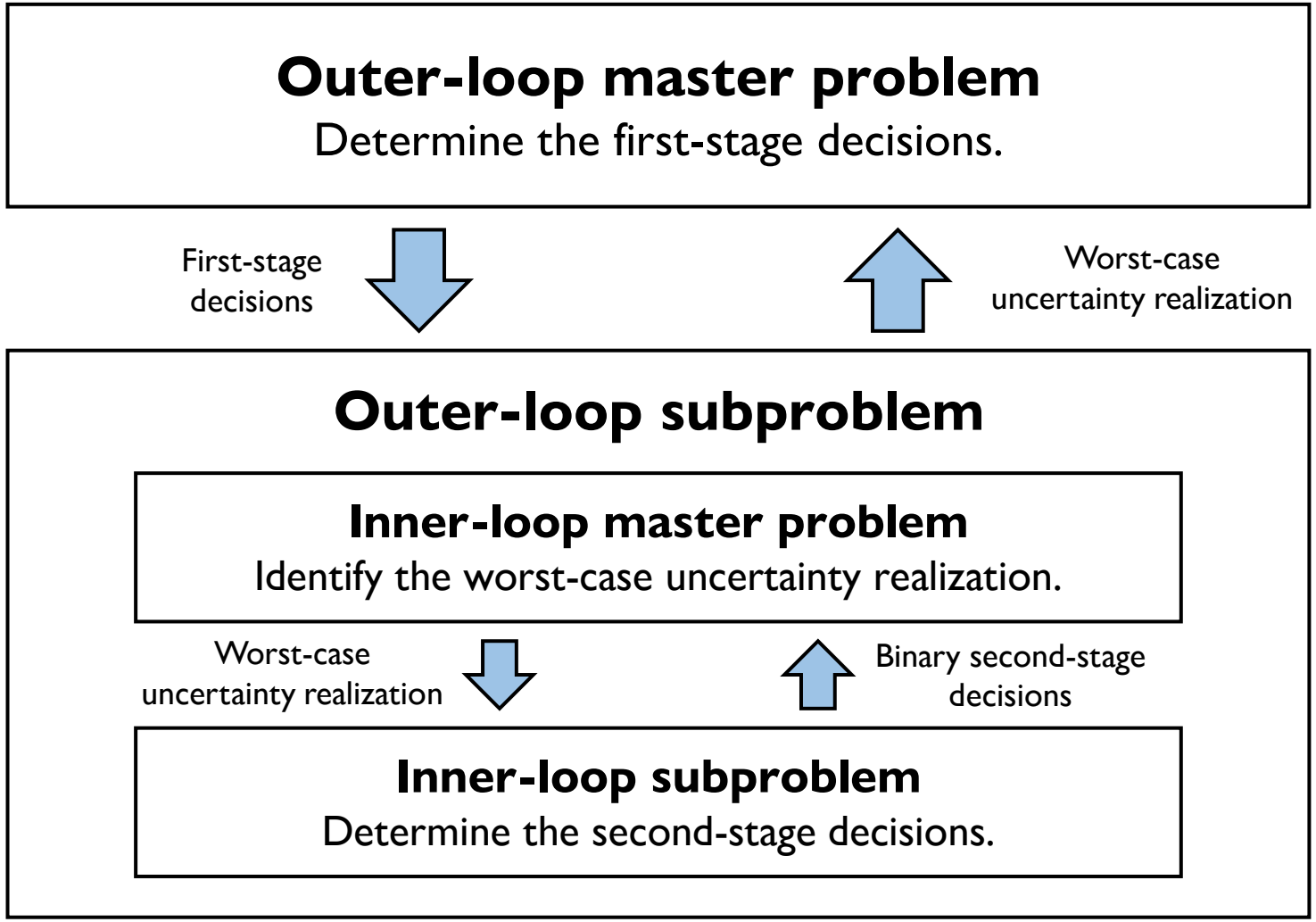
$$\boldsymbol{F}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{y} + \boldsymbol{L}\boldsymbol{u} + \boxed{\boldsymbol{M}\boldsymbol{z}} \geq \boldsymbol{d}$$

$$\boldsymbol{z} \in \{0,1\}^m$$

Binary lower-level variables ($\boldsymbol{z} \in \{0,1\}^m$) used to model:

- The commitment statuses of conventional generating units.
- The charging/discharging statuses of storage facilities.

# RO – Nested CCGA



# RO – Nested CCGA

Iterative procedure (outer and inner loops)

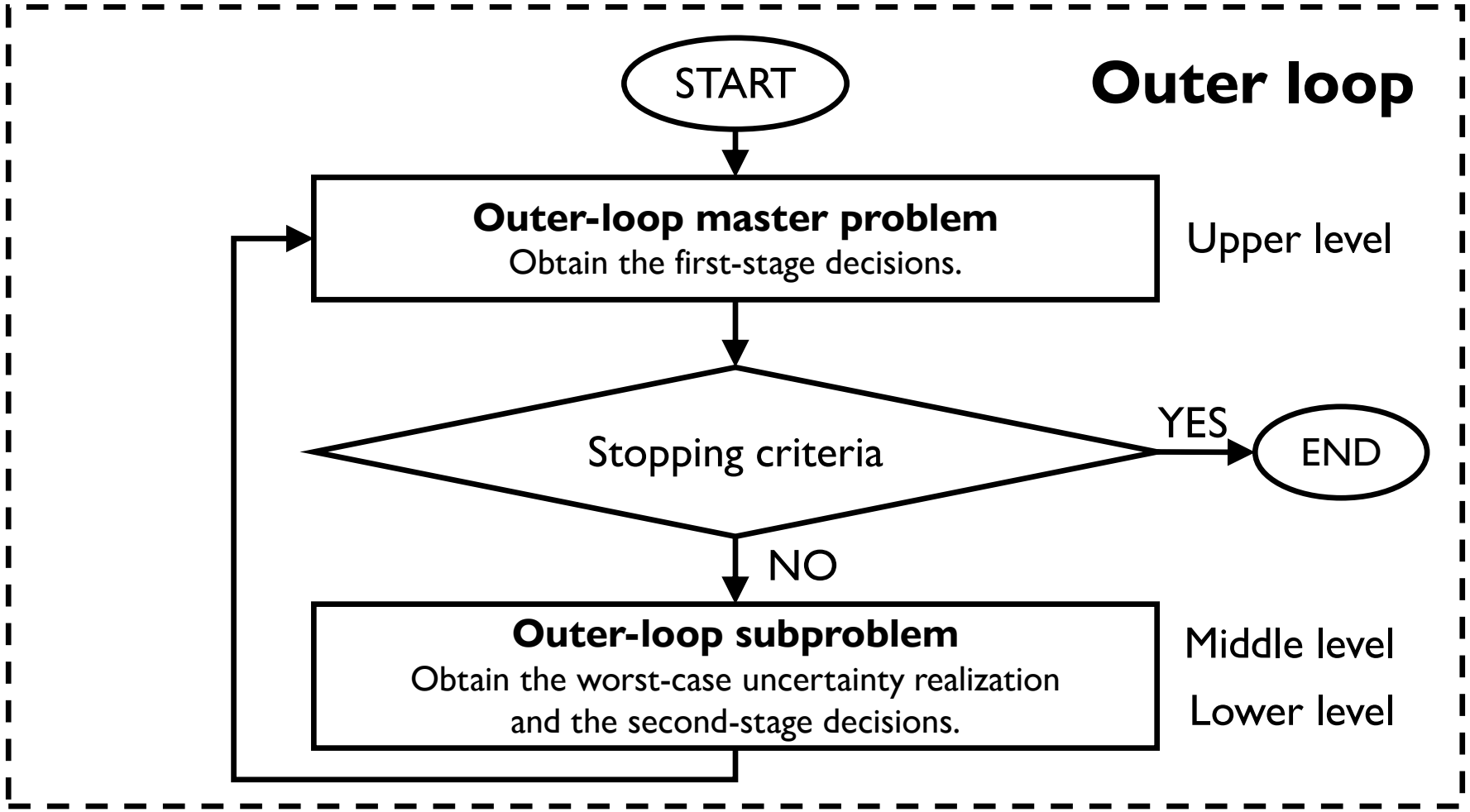
Outer-loop master problem → Outer-loop lower bound $LB^{\mathrm{O}}$

Outer-loop subproblem → Outer-loop upper bound $UB^{\mathrm{O}}$

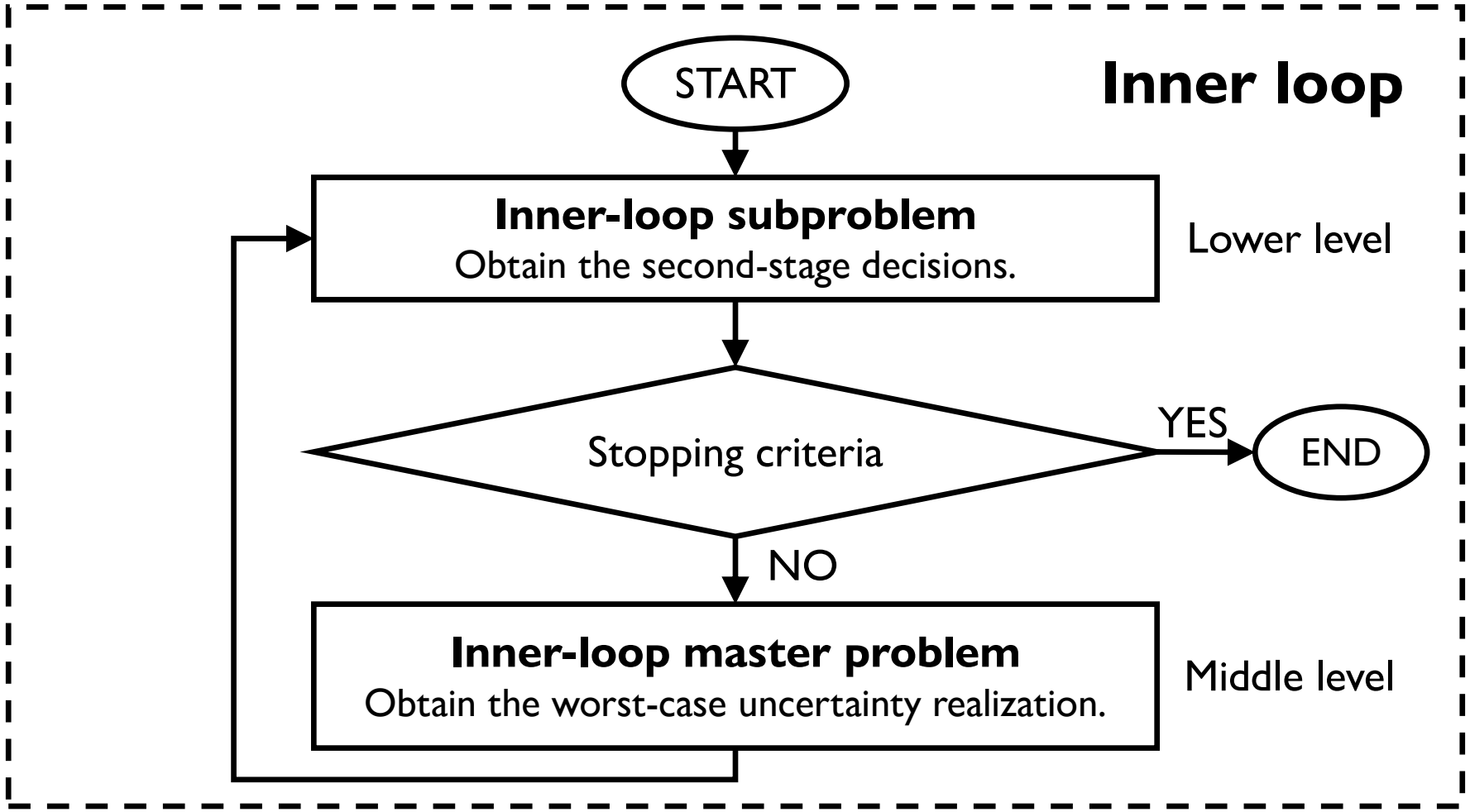
Inner-loop master problem → Inner-loop upper bound $UB^{\mathrm{I}}$

Inner-loop subproblem → Inner-loop lower bound $LB^{\mathrm{I}}$

# RO – Nested CCGA



# RO – Nested CCGA



# RO – Nested CCGA – Example

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

# RO – Nested CCGA – Example

Transmission network expansion planning problem.

IEEE 24-bus Reliability Test System.

Uncertain parameters:

- Peak power consumption of loads.
- Capacity of conventional and renewable generating units.

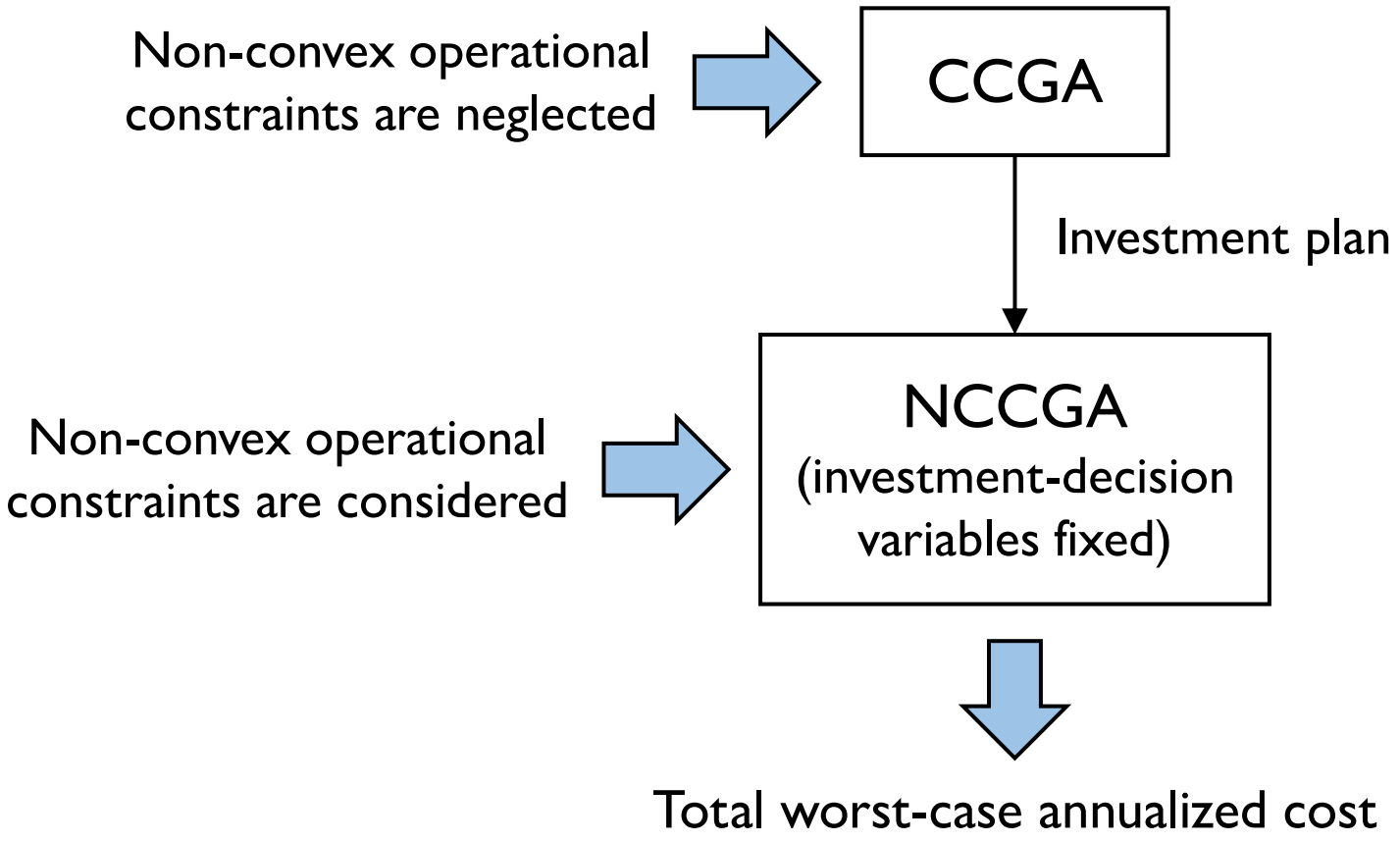
Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

# RO – Nested CCGA – Example

| Uncertainty level | Null | Low | Medium | High | Maximum |
|---|---|---|---|---|---|
| Total worst-case annualized cost (M$) | 130.28 | 149.15 | 166.35 | 177.09 | 182.84 |
| Investment costs (M$) | 9.75 | 50.63 | 76.88 | 91.88 | 93.00 |
| Transmission lines built | 21-22 | 7-8, 12-13, 15-16, 19-20, 21-22 | 3-9, 7-8, 8-10, 12-13, 15-16, 19-20, 19-20, 21-22 | 1-5, 3-9, 7-8, 8-10, 12-13, 15-16, 19-20, 19-20, 21-22 | 5-10, 7-8, 8-10, 10-12, 13-23, 15-16, 19-20, 19-20, 21-22 |

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

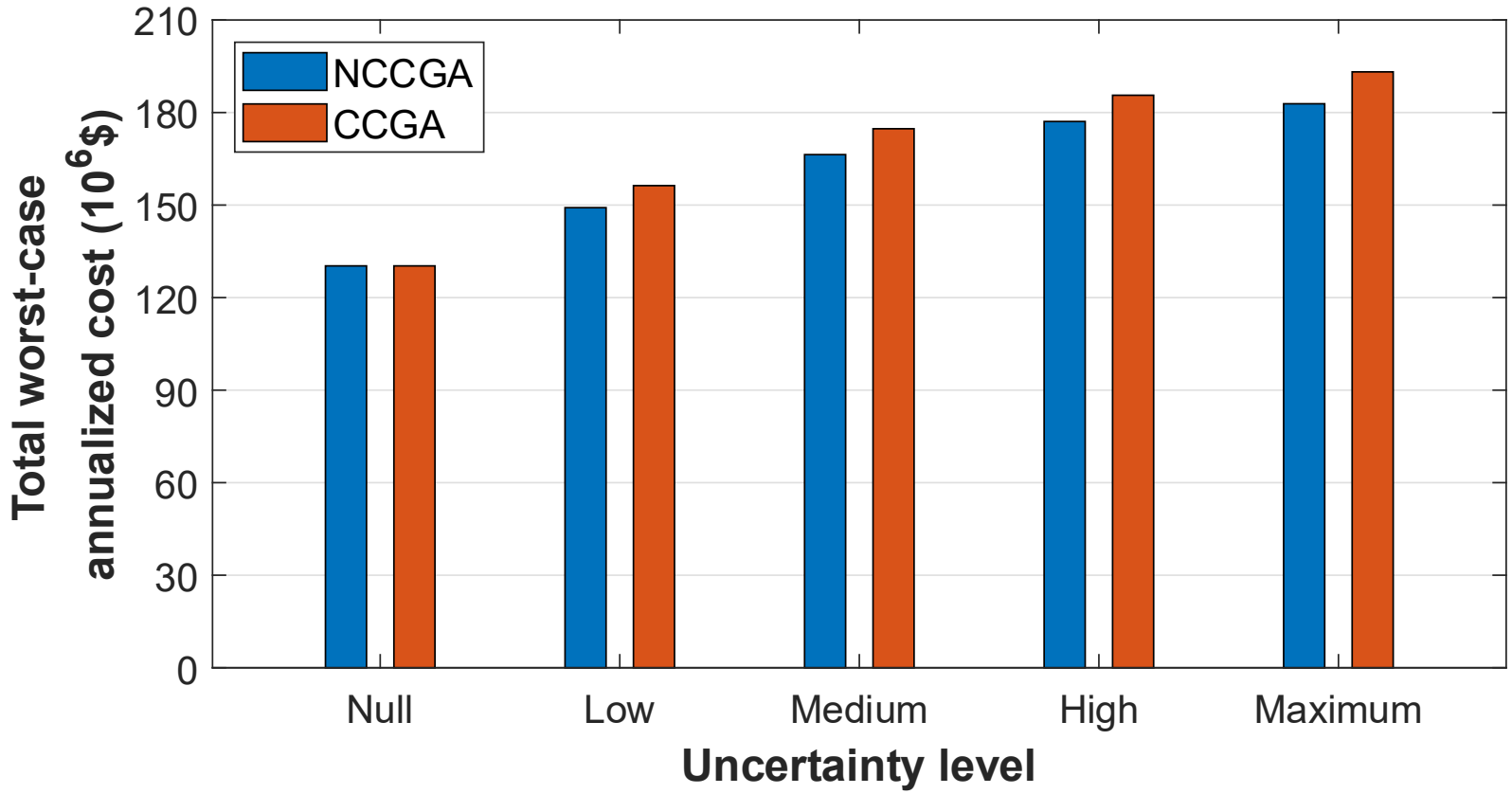
# RO – Nested CCGA – Example




Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

# RO – Nested CCGA – Example

CCGA → 5.66% increase in the total worst-case annualized cost.



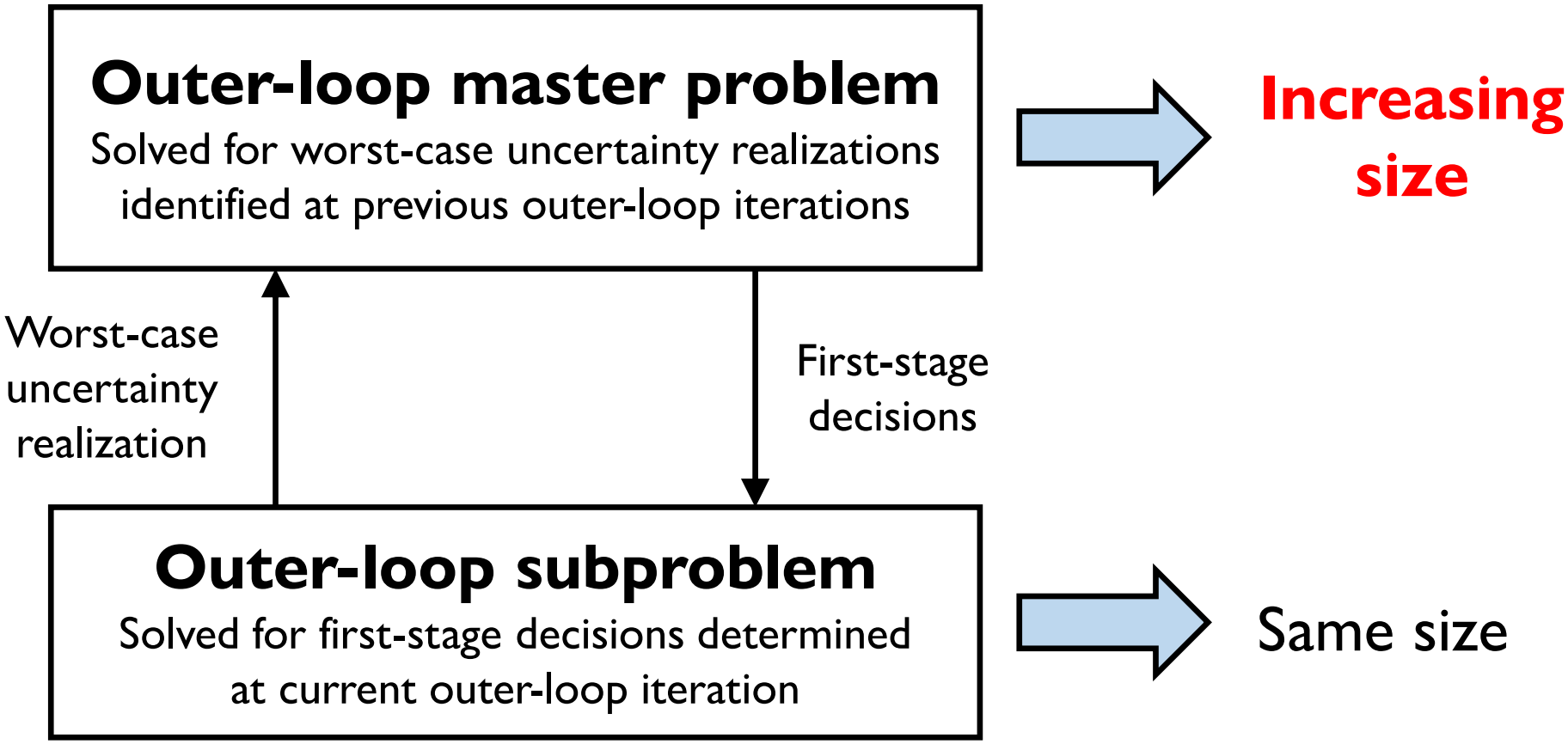

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

# RO – Nested CCGA – Example

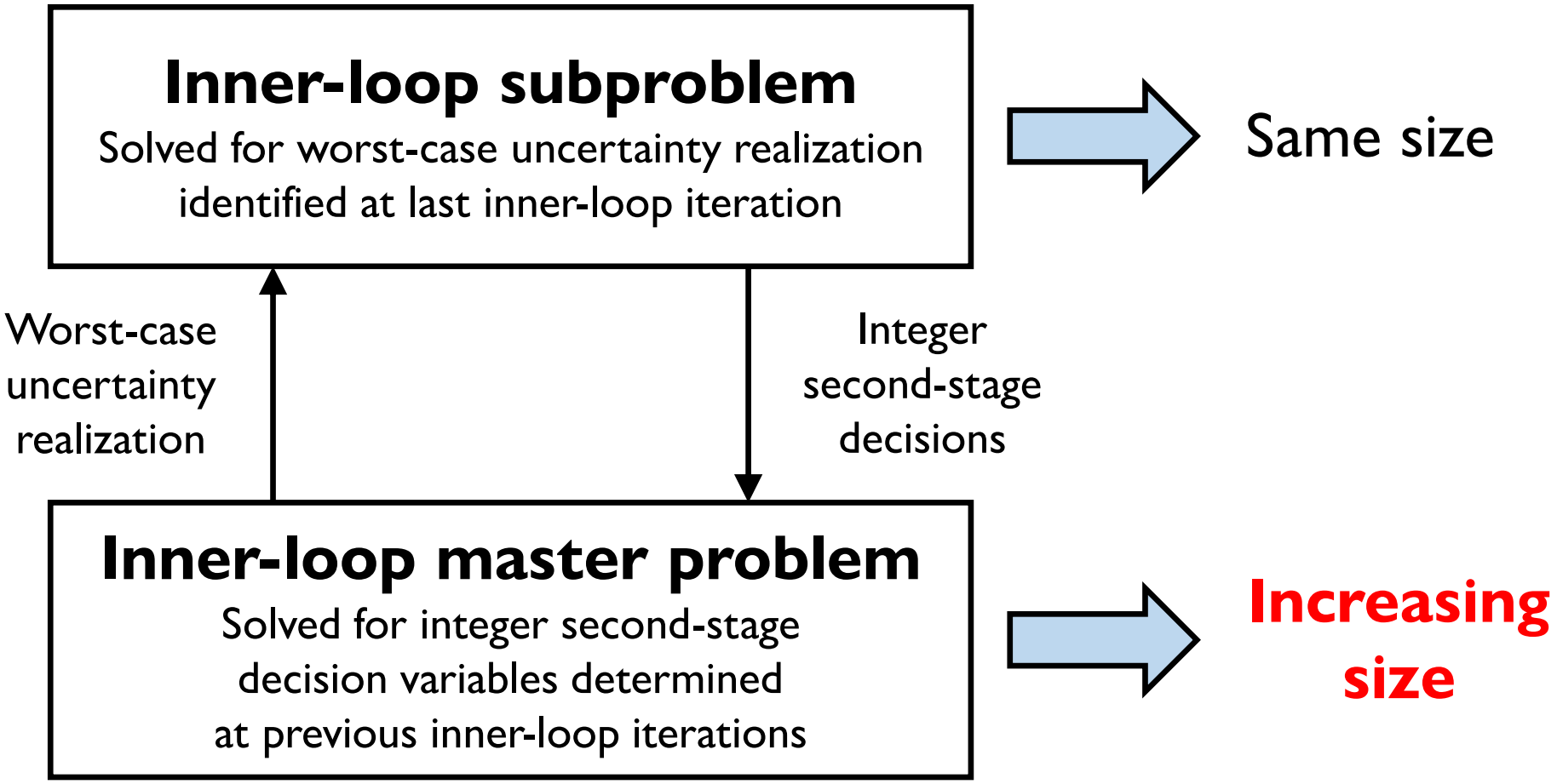
| Uncertainty level | Null | Low | Medium | High | Maximum |
|---|---|---|---|---|---|
| Computational time (h) | 0.01 | 17.24 | 53.69 | 4.42 | 1.69 |
| # outer-loop iterations | 1 | 5 | 5 | 3 | 2 |
| # inner-loop iterations | 1 | 3, 3, 4, 5, 0 | 3, 3, 3 ,3, 4 | 4, 3, 3 | 2, 3 |

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Robust transmission network expansion planning considering non-convex operational constraints", *Energy Economics*, vol. 98, art. no. 105246, Jun. 2021, DOI: 10.1016/j.eneco.2021.105246.

# RO – NCCGA – Computational issues



# RO – NCCGA – Computational issues



# Robust optimization

- Robust optimization without recourse.
- Robust optimization with recourse.
  - Column-and-constraint generation algorithm.
  - Nested column-and-constraint generation algorithm.
  - **Acceleration techniques**.

# RO – NCCGA – Acceleration techniques

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques

Transmission network expansion planning problem.

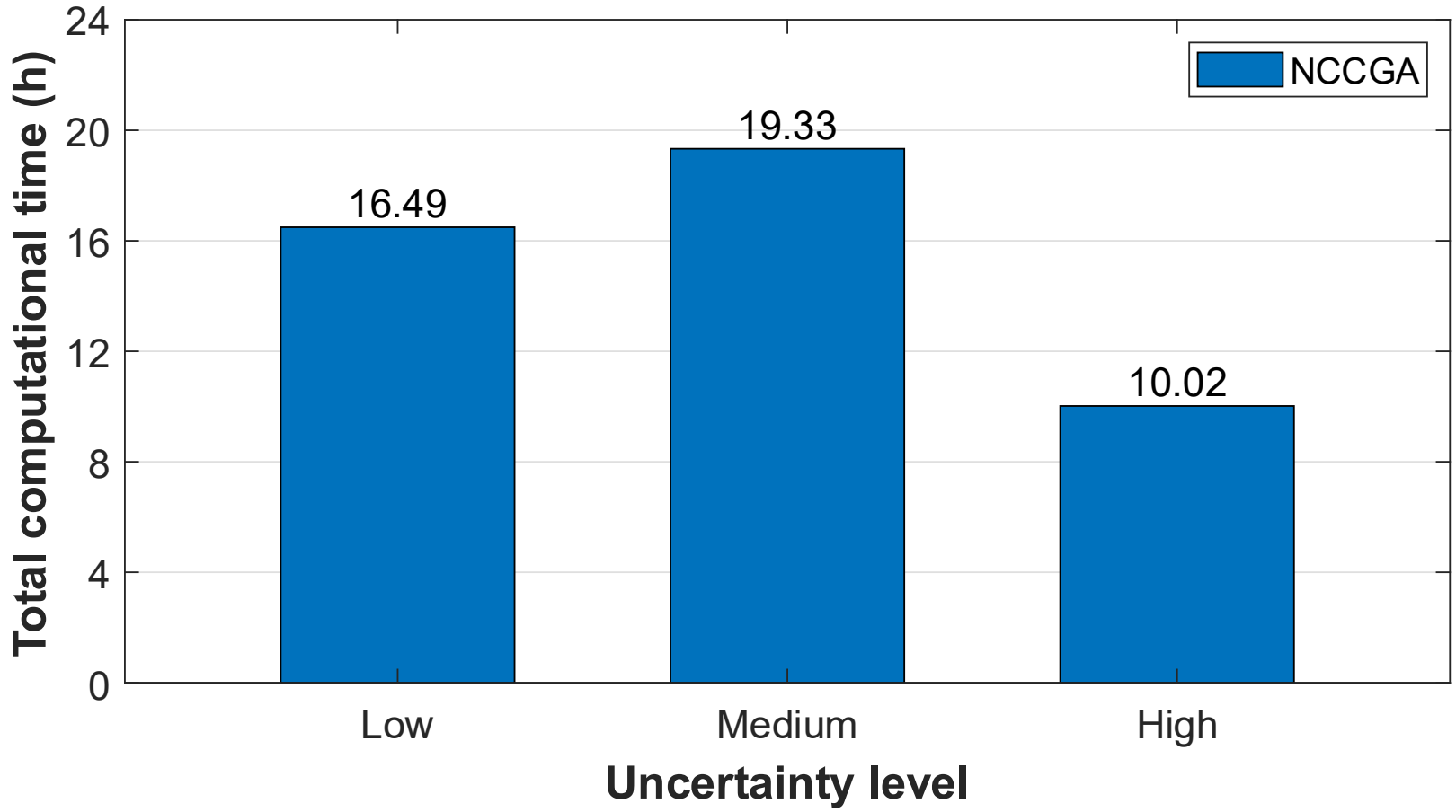
IEEE 24-bus Reliability Test System.

Uncertain parameters:

- Peak power consumption of loads.
- Marginal production cost of conventional generating units.
- Capacity of conventional and renewable generating units.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques

## Outer-loop master problem (medium uncertainty level)

| Outer-loop iteration | # constraints | # variables | Computational time (h) |
|---|---|---|---|
| 1 | 45,991 | 17,288 | 0.01 |
| 2 | 91,942 | 34,488 | 0.54 |
| 3 | 137,893 | 51,688 | 2.83 |
| 4 | 183,844 | 68,888 | 10.59 |

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

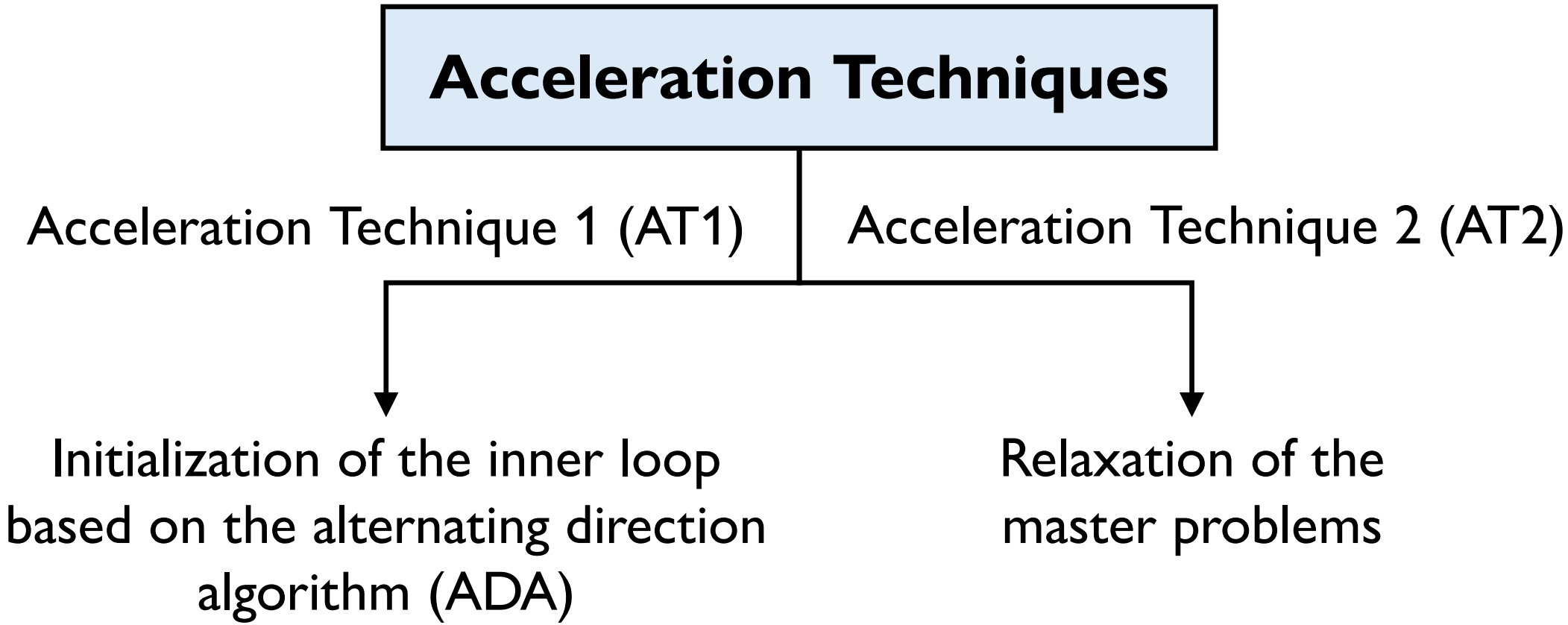
# RO – NCCGA – Acceleration techniques

Inner-loop master problem (high uncertainty level)

| Inner-loop iteration | # constraints | # variables | Computational time (h) |
|---|---|---|---|
| 1 | 37,807 | 42,299 | 0.01 |
| 2 | 75,613 | 84,549 | 0.15 |
| 3 | 113,419 | 126,799 | 0.39 |
| 4 | 151,225 | 169,049 | 0.83 |

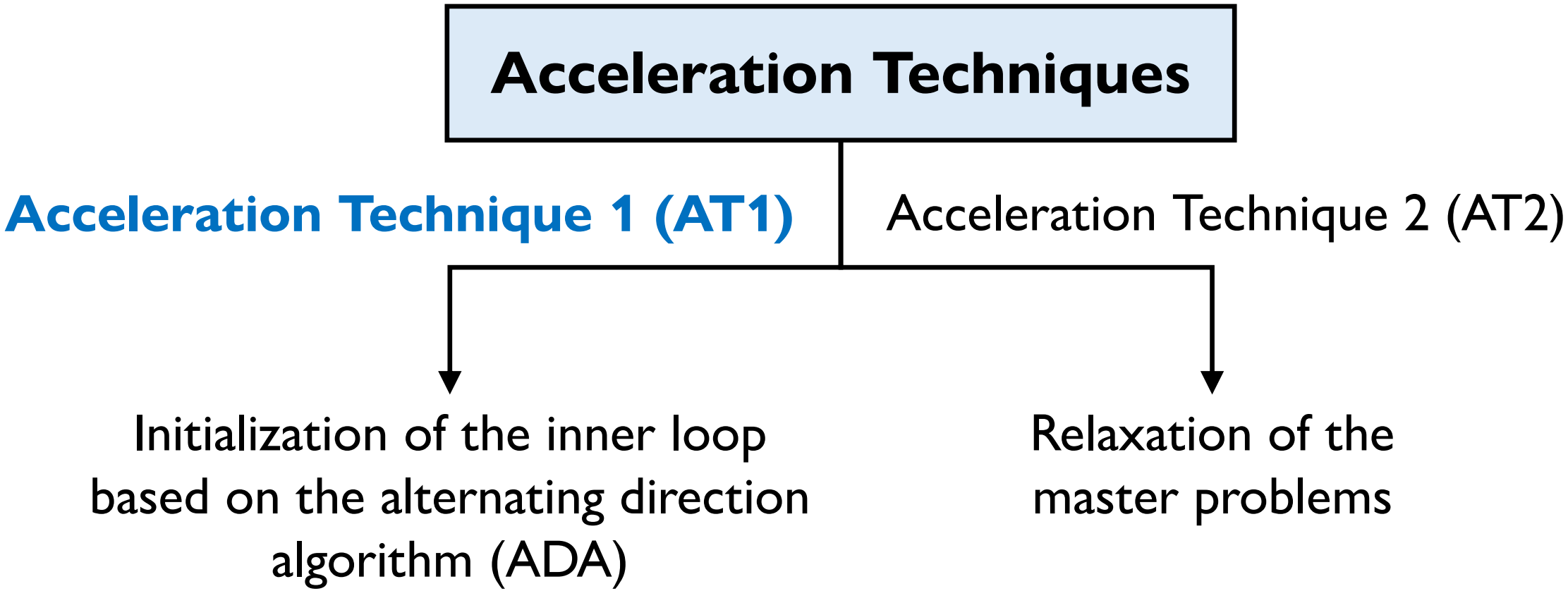
Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques



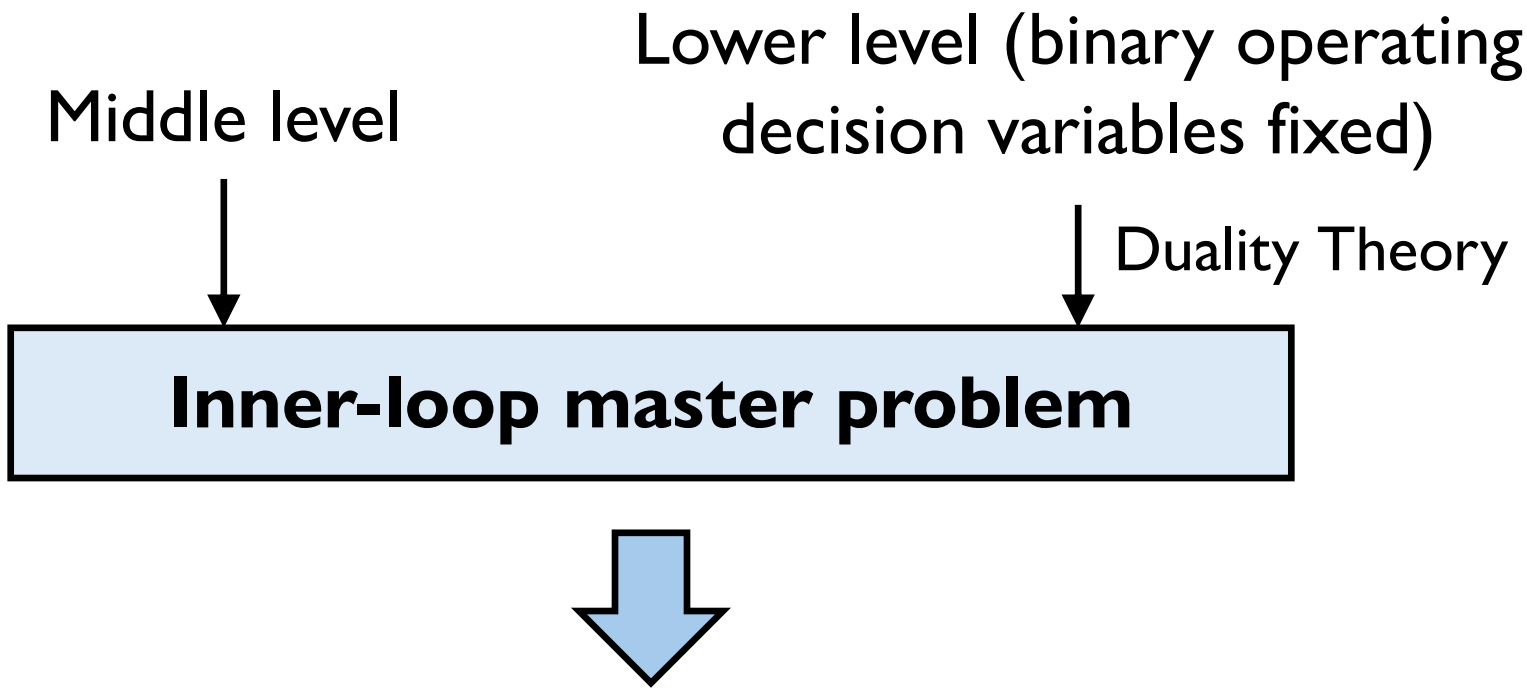

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

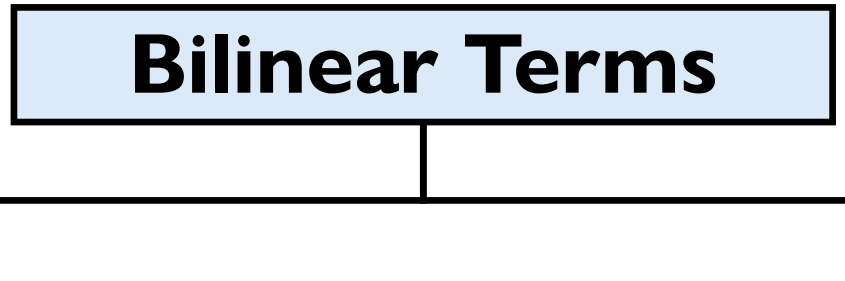
# RO – NCCGA & AT1




Bilinear terms (products of lower-level dual variables $\boldsymbol{\lambda}$ and variables that model the worst-case uncertainty realization $\boldsymbol{u}$).

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT1




Exact linearization procedure (based on the use of a cardinality-constrained uncertainty set with binary variables).

High computational times
Exact solution

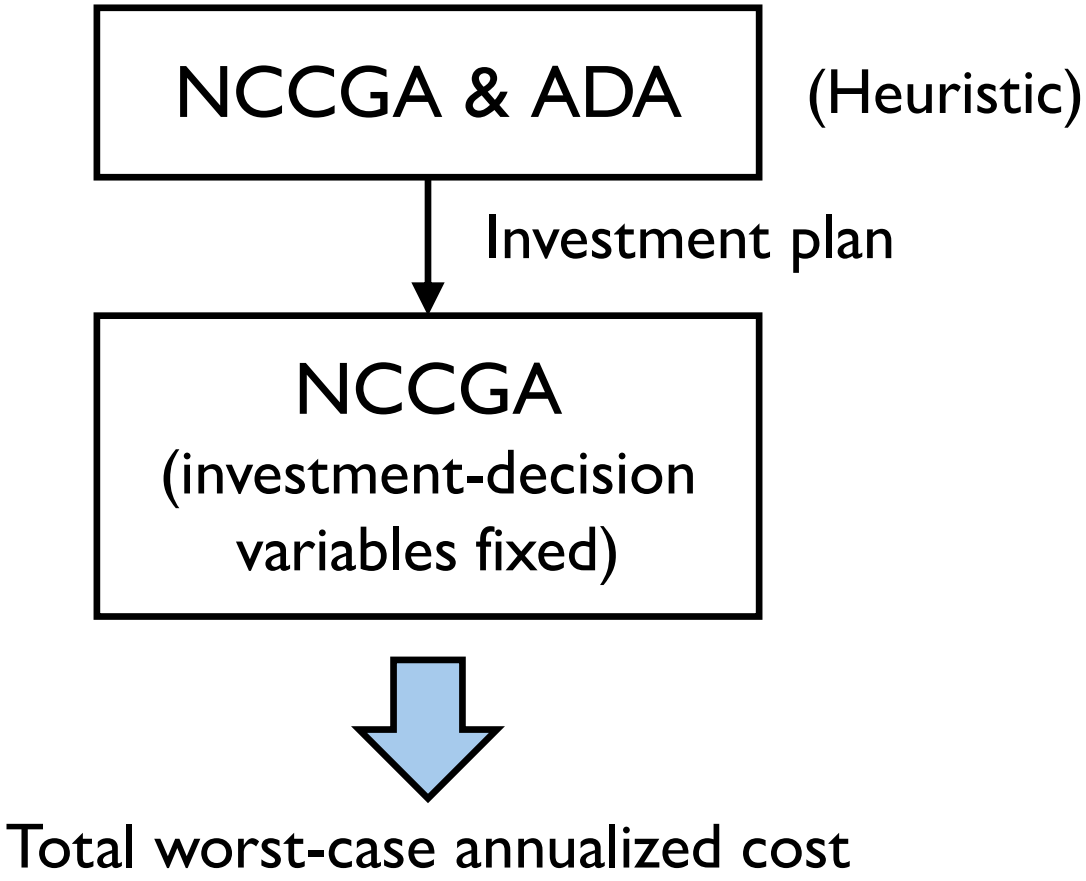
Alternating direction algorithm (ADA)

NCCGA & ADA
(used by other researchers)

Low computational times
Heuristic solution

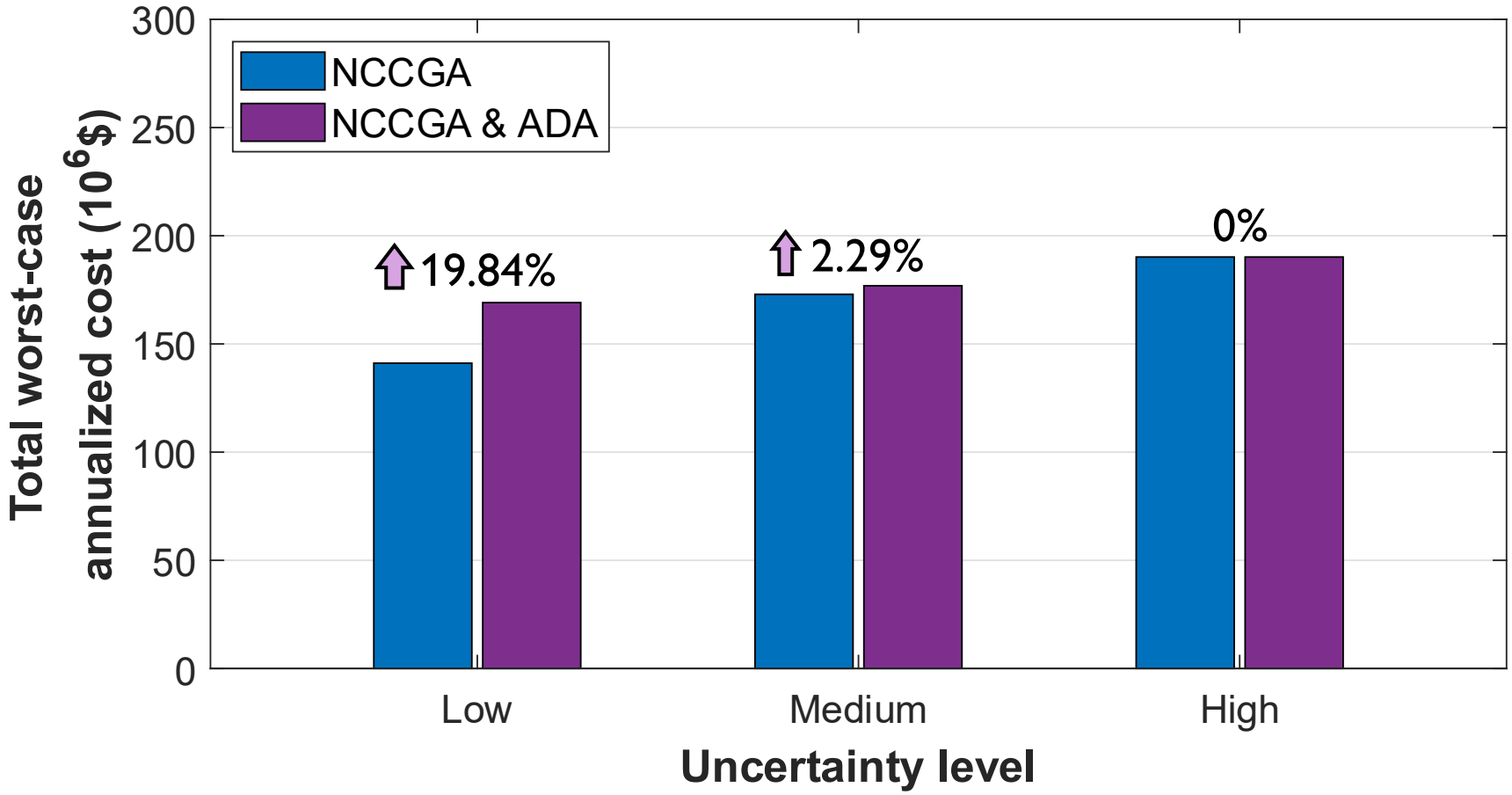
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# RO – NCCGA & AT1



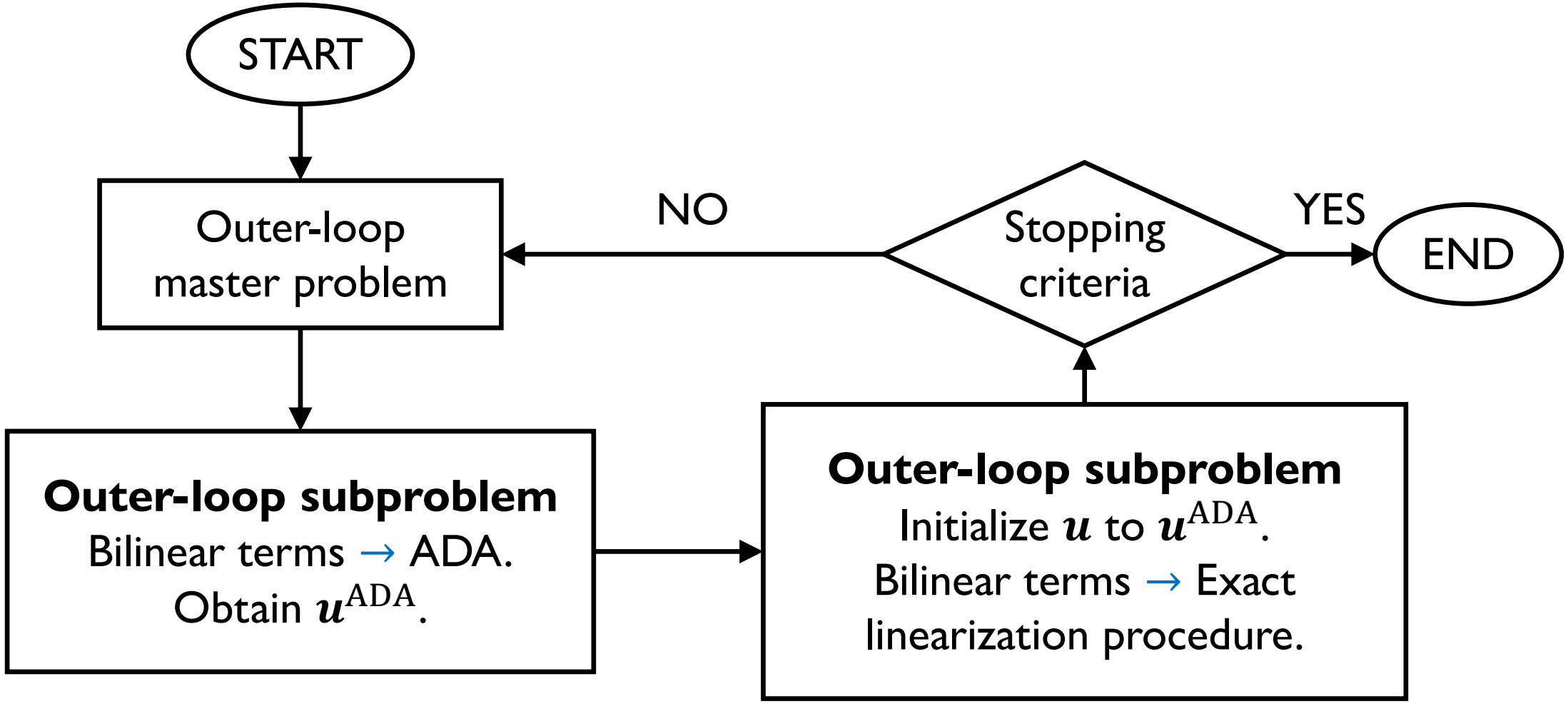

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT1




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT1




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

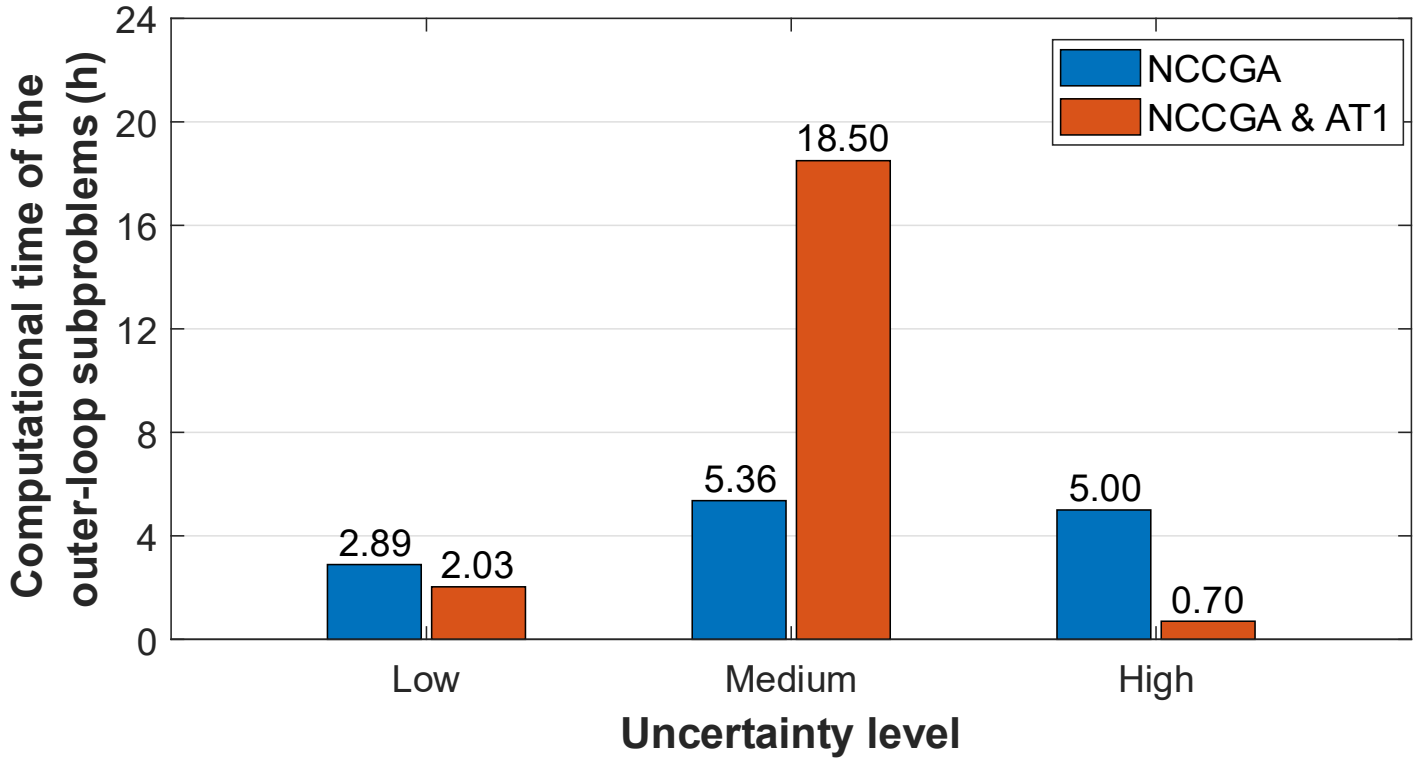
# RO – NCCGA & AT1

Main idea of the first acceleration technique: initializing the inner loop with a near-optimal uncertainty realization is expected to reduced the number of inner-loop iterations compared with the case without acceleration techniques.

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT1

Reduction in the computational time only when the number of inner-loop iterations decreases.



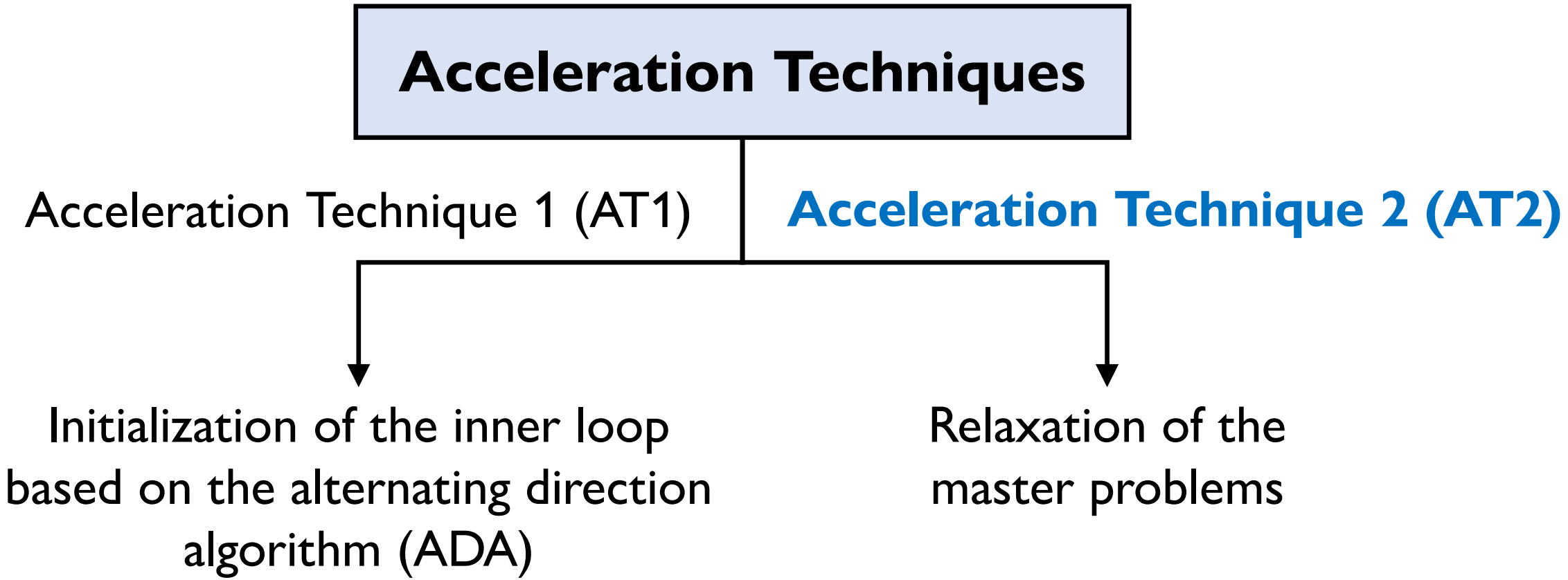

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT1

| Uncertainty level | Low | Medium | High |
|---|---|---|---|
| Reduction in the total number of inner-loop iterations | ✓ | ✗ | ✓ |
| # inner-loop iterations using NCCGA | 3, 3, 4 | 3, 3, 3, 3 | 5, 4 |
| # inner-loop iterations using NCCGA & AT1 | 3, 3, 3 | 2, 4, 4, 3 | 3, 3 |

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

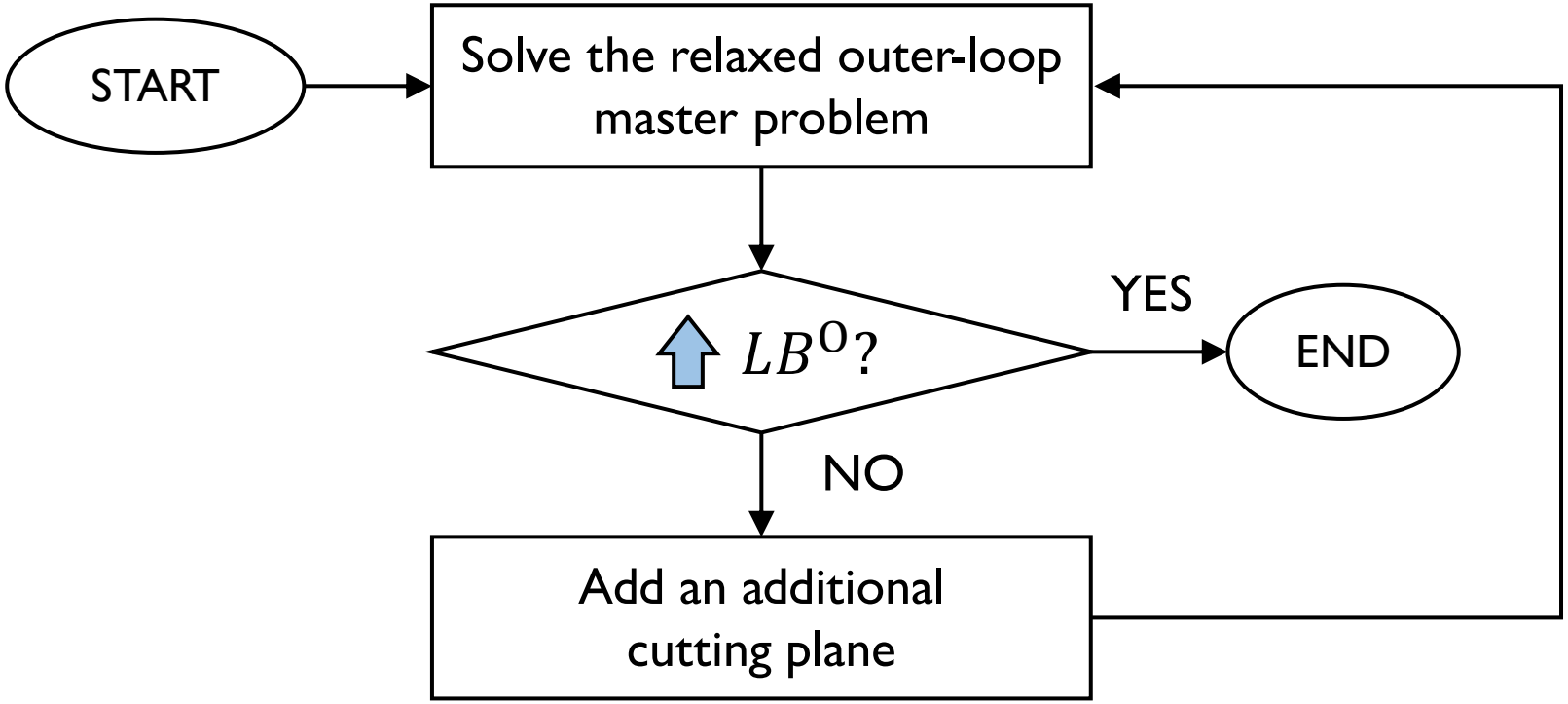
# RO – NCCGA – Acceleration techniques




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT2

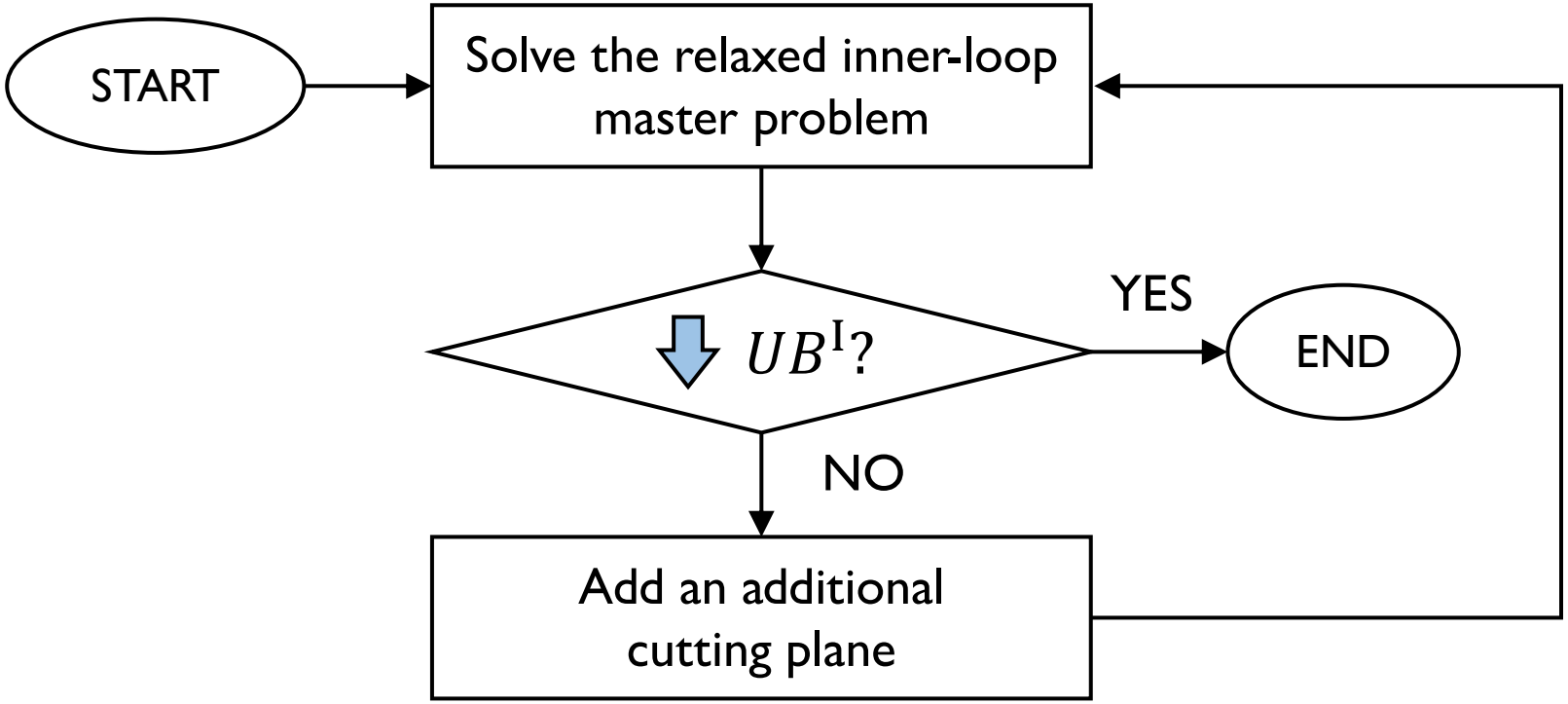
Relaxation of the outer-loop master problem → Latest $\boldsymbol{u}$ are considered and more cutting planes are added to the outer-loop master problem until $LB^{O}$ increases.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT2

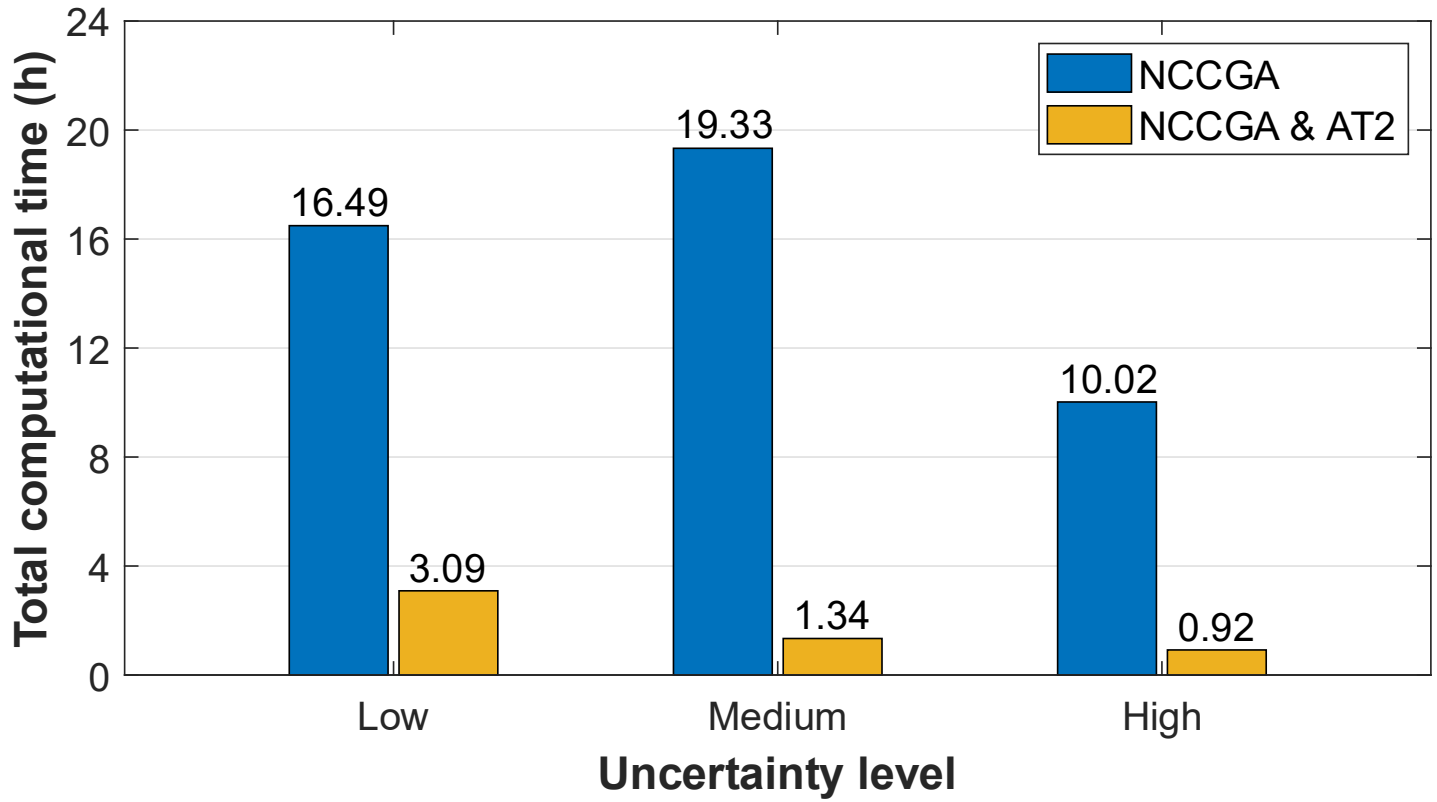
Relaxation of the inner-loop master problem → Latest $\boldsymbol{z}$ are considered and more cutting planes are added to the inner-loop master problem until $UB^{\mathrm{I}}$ decreases.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT2

Reductions in the total computational time greater than 81%.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT2

## Outer-loop master problem (medium uncertainty level)

| Outer-loop iteration | Relaxed outer-loop iteration | # constraints | # variables | Computational time (h) |
|---|---|---|---|---|
| 1 | 1 | 45,991 | 17,288 | 0.01 |
| 2 | 1 | 45,991 | 17,288 | 0.03 |
| 3 | 1 | 45,991 | 17,288 | 0.03 |
| 4 | 1 | 45,991 | 17,288 | 0.04 |
| 4 | 2 | 91,942 | 34,488 | 0.51 |

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA & AT2

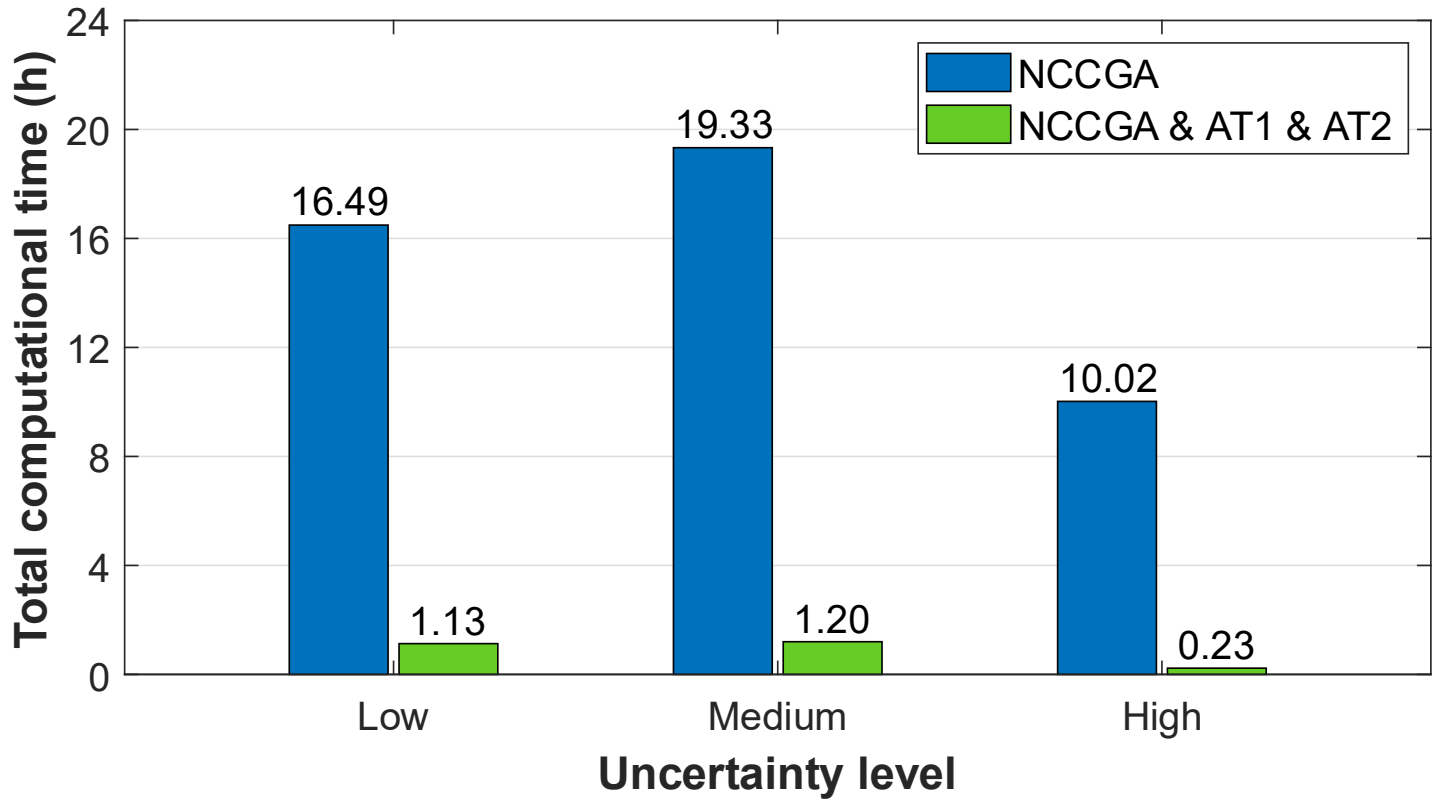
## Inner-loop master problem (high uncertainty level)

| Inner-loop iteration | Relaxed inner-loop iteration | # constraints | # variables | Computational time (h) |
|---|---|---|---|---|
| 1 | 1 | 37,807 | 42,299 | 0.03 |
| 2 | 1 | 37,807 | 42,299 | 0.01 |
| 3 | 1 | 37,807 | 42,299 | 0.01 |
| 4 | 1 | 37,807 | 42,299 | 0.01 |
| 4 | 2 | 75,613 | 84,549 | 0.14 |
| 5 | 1 | 37,807 | 42,299 | 0.01 |
| 5 | 2 | 75,613 | 84,549 | 0.14 |
| 5 | 3 | 113,419 | 126,799 | 0.33 |

Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

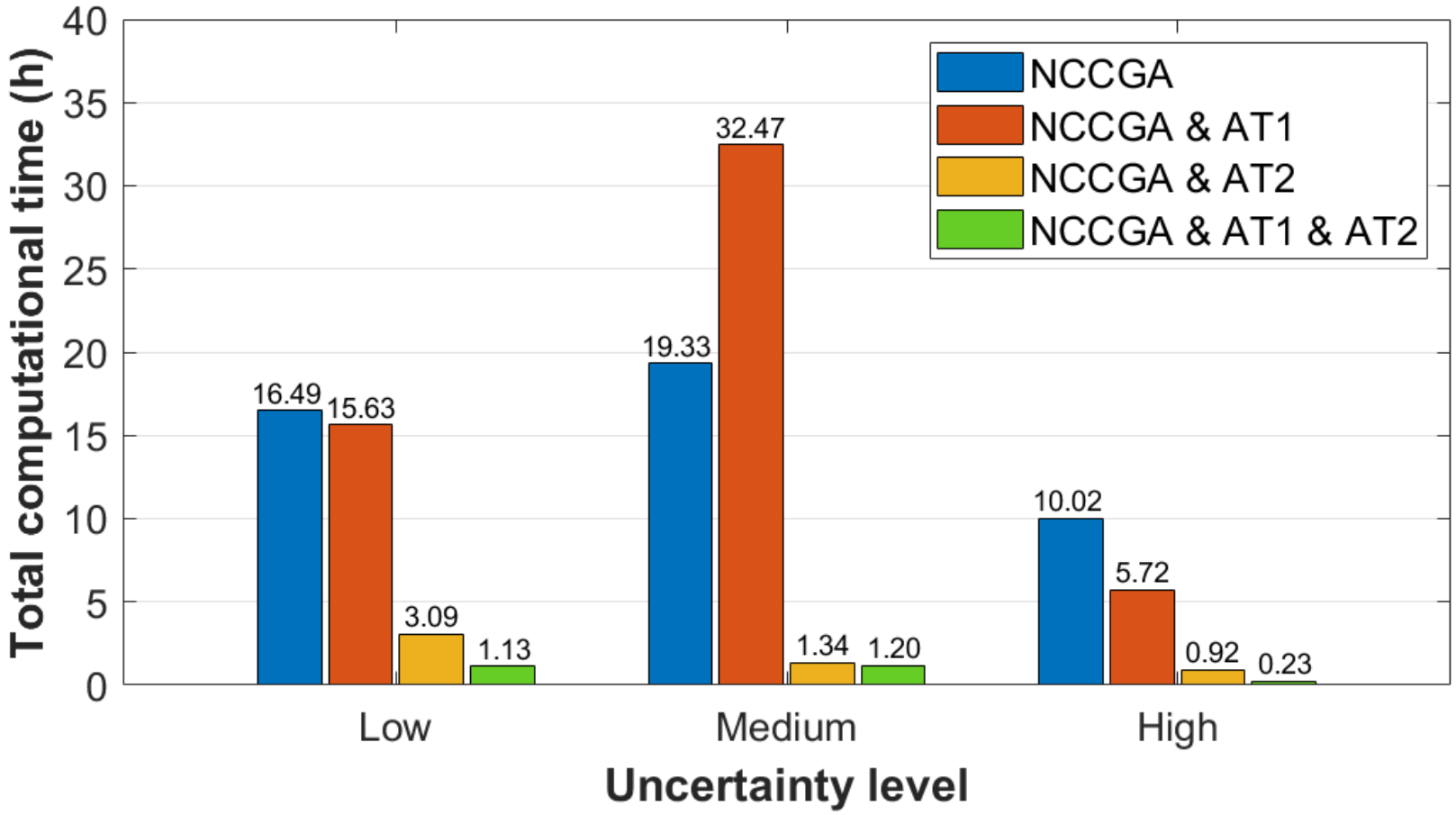
# RO – NCCGA & AT1 & AT2

Reductions in the total computational time greater than 93%.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

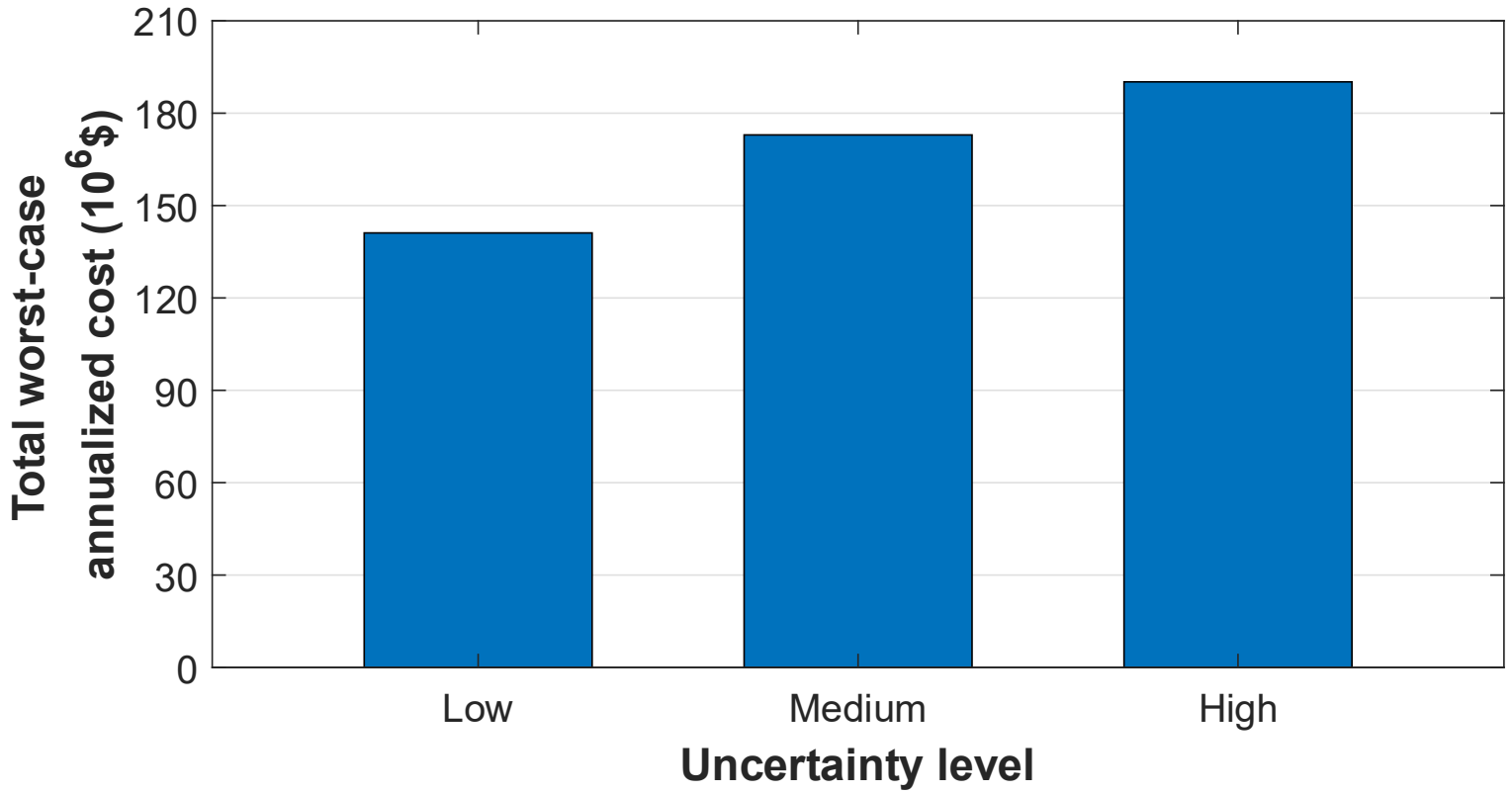
# RO – NCCGA – Acceleration techniques




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques

Proposed acceleration techniques → Same optimal solutions.




Á. García-Cerezo, R. García-Bertrand, and L. Baringo, "Acceleration techniques for adaptive robust optimization transmission network expansion planning problems", *International Journal of Electrical Power and Energy Systems*, vol. 148, art. no. 108985, Jun. 2023, DOI: 10.1016/j.ijepes.2023.108985.

# RO – NCCGA – Acceleration techniques

Multi-year approach:

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Expansion planning of the transmission network with high penetration of renewable generation: A multi-year two-stage adaptive robust optimization approach", *Applied Energy*, vol. 349, art. no. 121653, Nov. 2023, DOI: 10.1016/j.apenergy.2023.121653.

# RO – NCCGA – Acceleration techniques

Transmission network expansion planning problem.

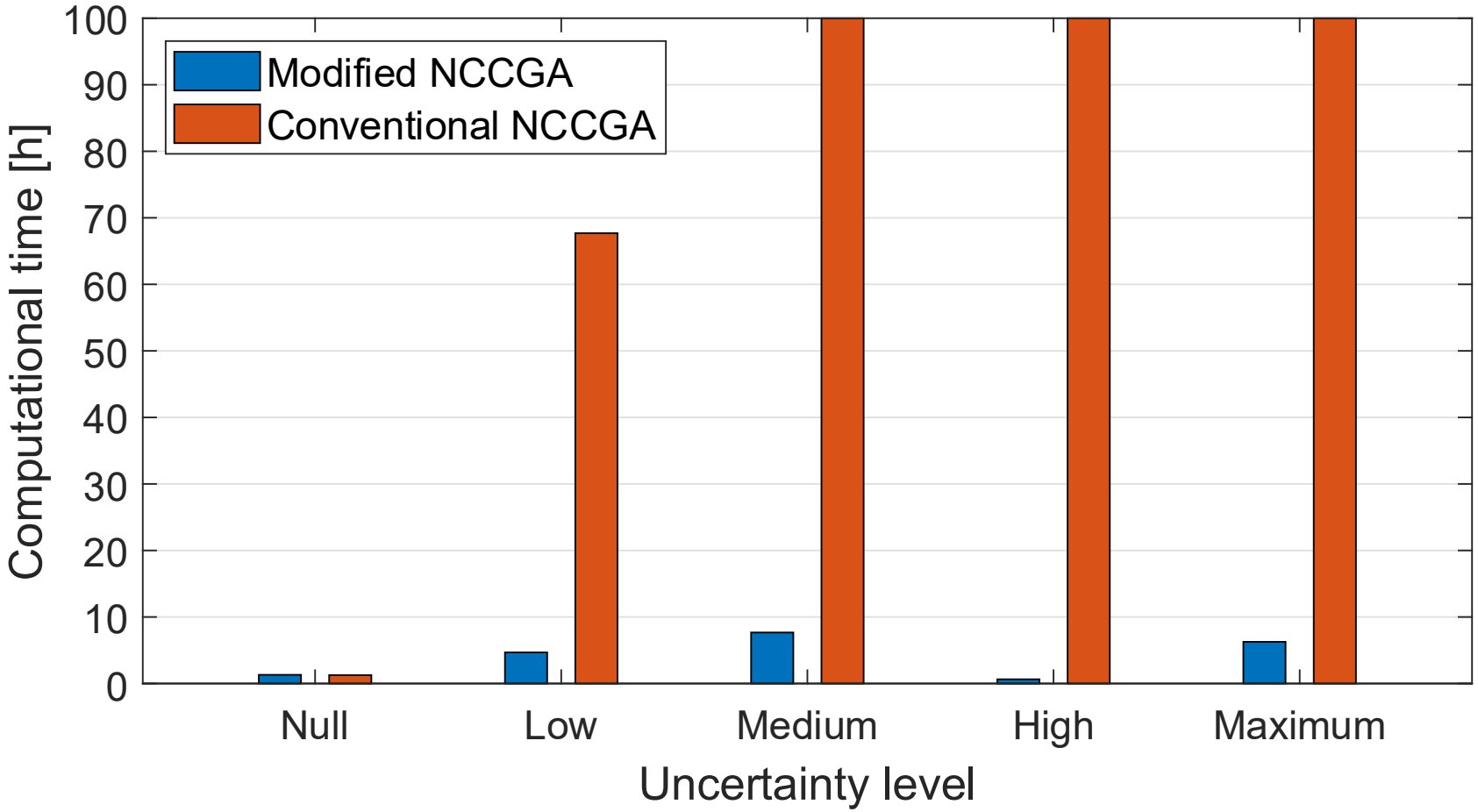
IEEE 118-bus Test System.

Uncertain parameters:

- Peak power consumption of loads.
- Marginal production cost of conventional generating units.
- Capacity of conventional and renewable generating units.

Á. García-Cerezo, L. Baringo, and R. García-Bertrand, "Expansion planning of the transmission network with high penetration of renewable generation: A multi-year two-stage adaptive robust optimization approach", *Applied Energy*, vol. 349, art. no. 121653, Nov. 2023, DOI: 10.1016/j.apenergy.2023.121653.

# RO – NCCGA – Acceleration techniques



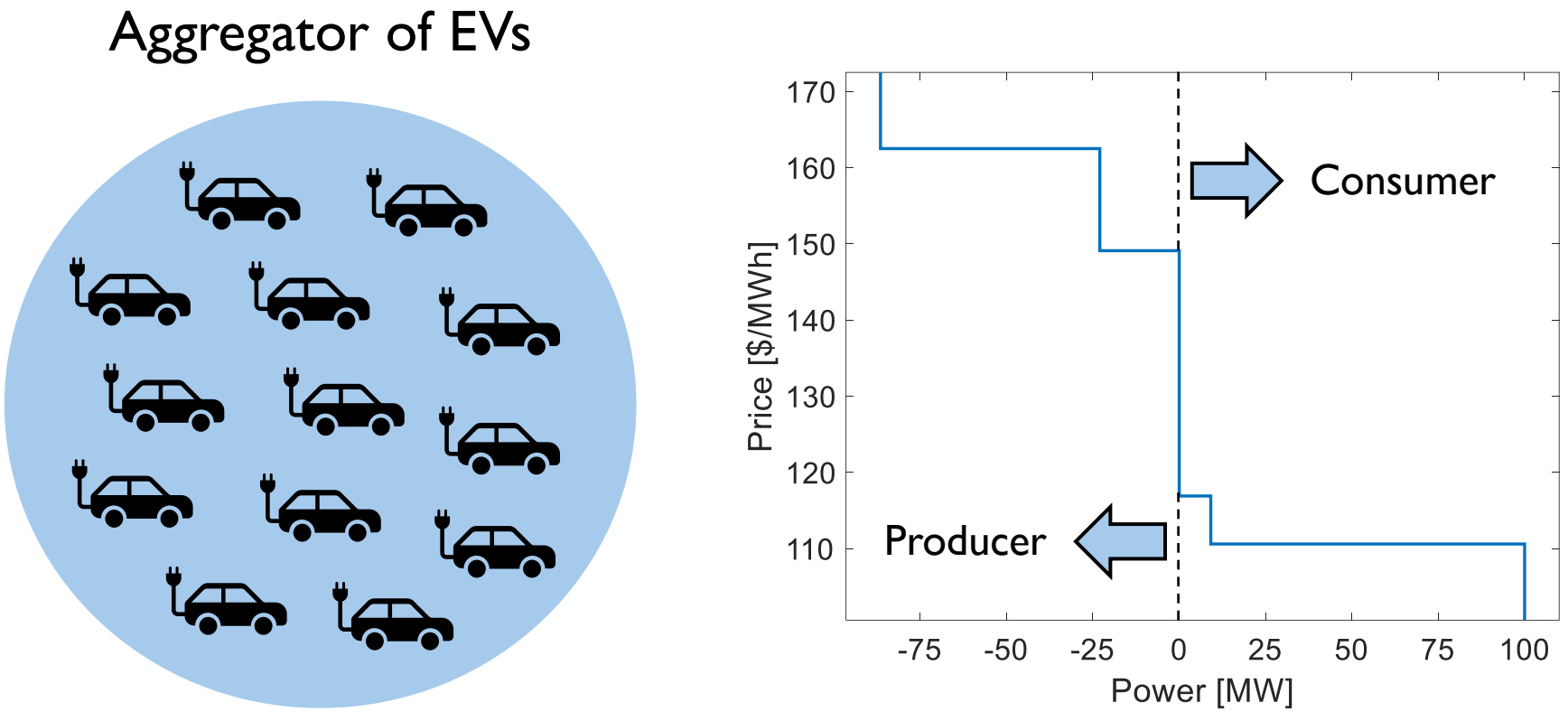

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# 4. SP-RO approach

# SP-RO approach

Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

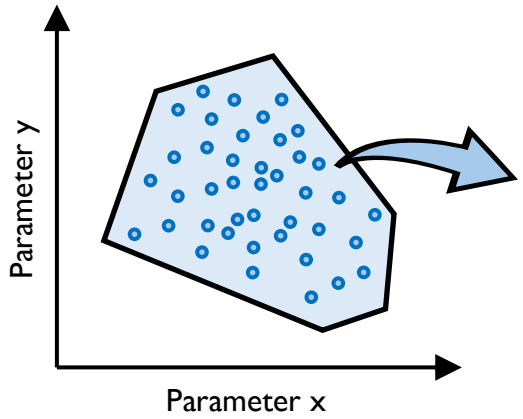



Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

Building of the DA bidding curves → subject to → Uncertainty: market prices and driving needs

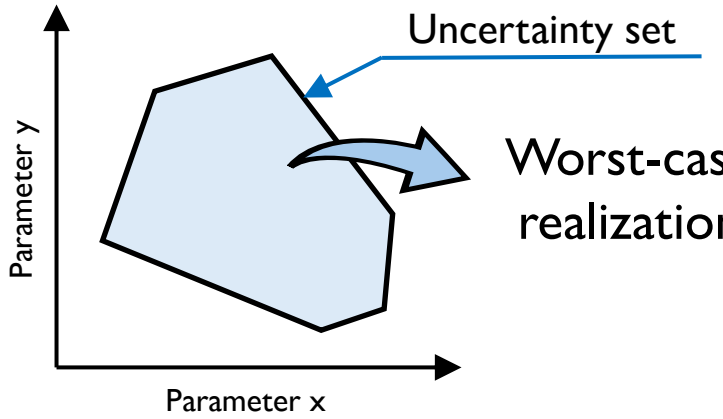
## Stochastic Programming




Discrete set of scenarios

Lack of information about the probability density function

## Robust Optimization



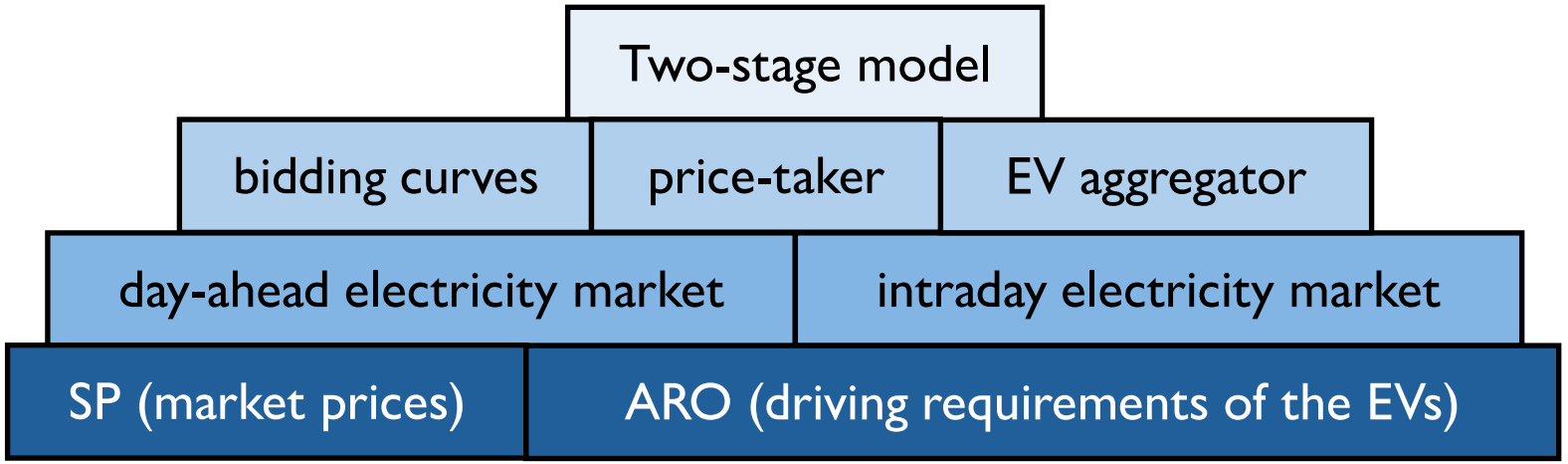

Worst-case realization

Too conservative solution

Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

Adaptive Robust Optimization (ARO) → Less conservative (corrective actions)




## Research question

Effect that the uncertainty level of the driving needs has on the bidding curves and the costs of the EV aggregator?

Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

min worst-case cost → Day-ahead market bidding decisions

max worst-case cost → Worst-case realization

min cost → Intraday market bidding decisions

Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

min worst-case cost → Day-ahead market bidding decisions

max worst-case cost → Worst-case realization

min cost → Intraday market bidding decisions

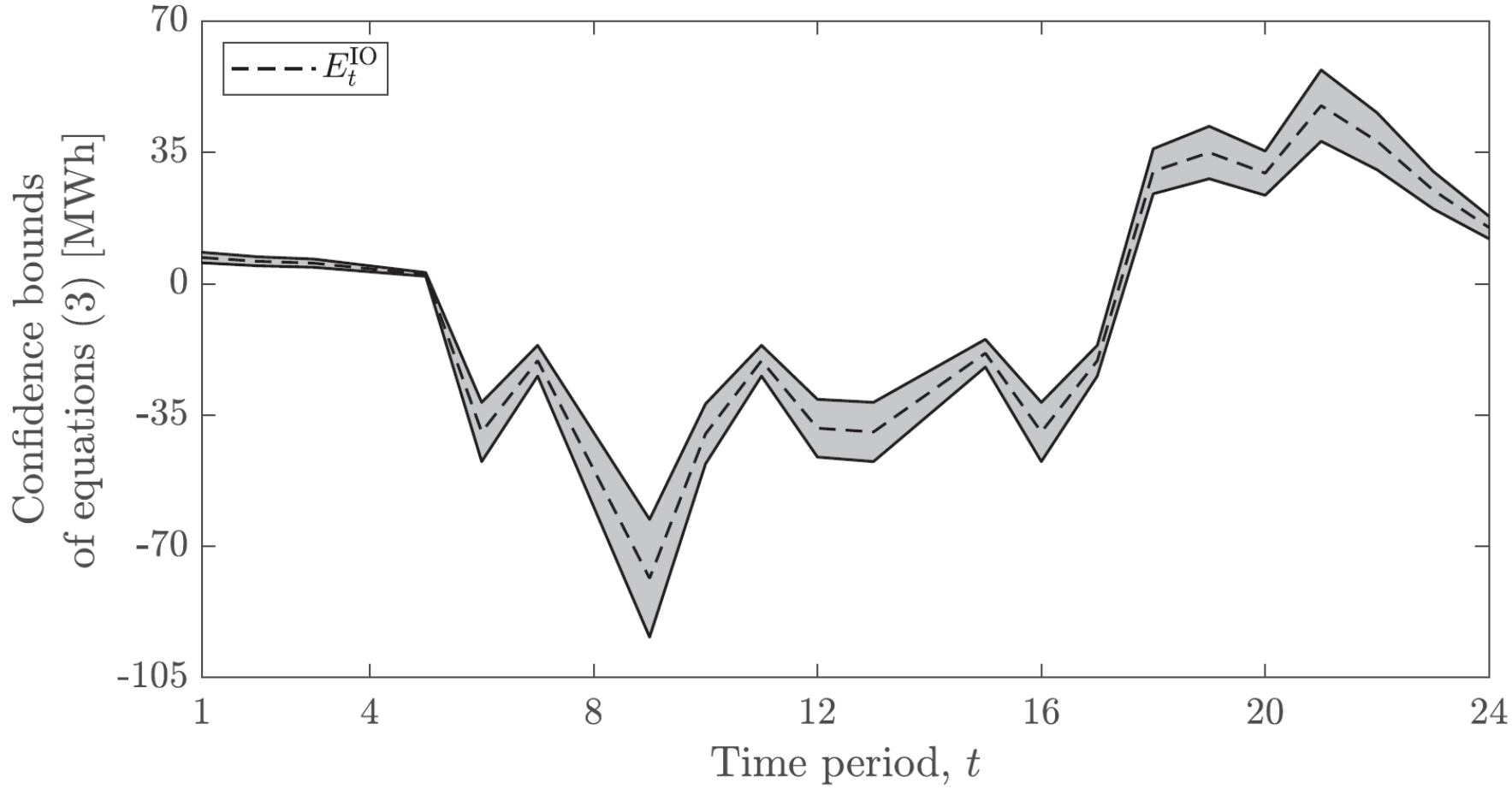
Market prices → SP.

Driving requirements → ARO.

- Net energy contribution (arrivals and departures).
- Minimum and maximum energy levels.
- Maximum charging and discharging power levels.

Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator




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# SP-RO approach – EV aggregator

min worst-case cost → Day-ahead market bidding decisions

max worst-case cost → Worst-case realization

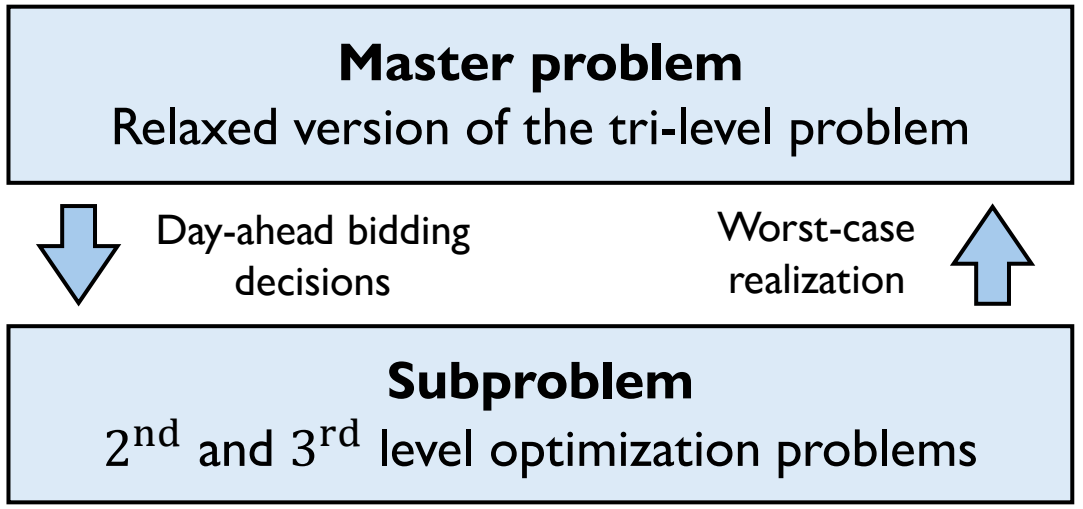
min cost → Intraday market bidding decisions

Market prices → SP.

Driving requirements → ARO.

- Net energy contribution (arrivals and departures).
- Minimum and maximum energy levels.
- Maximum charging and discharging power levels.

Column-and-constraint generation algorithm (CCGA)




Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator

Planning horizon → 24 hours.

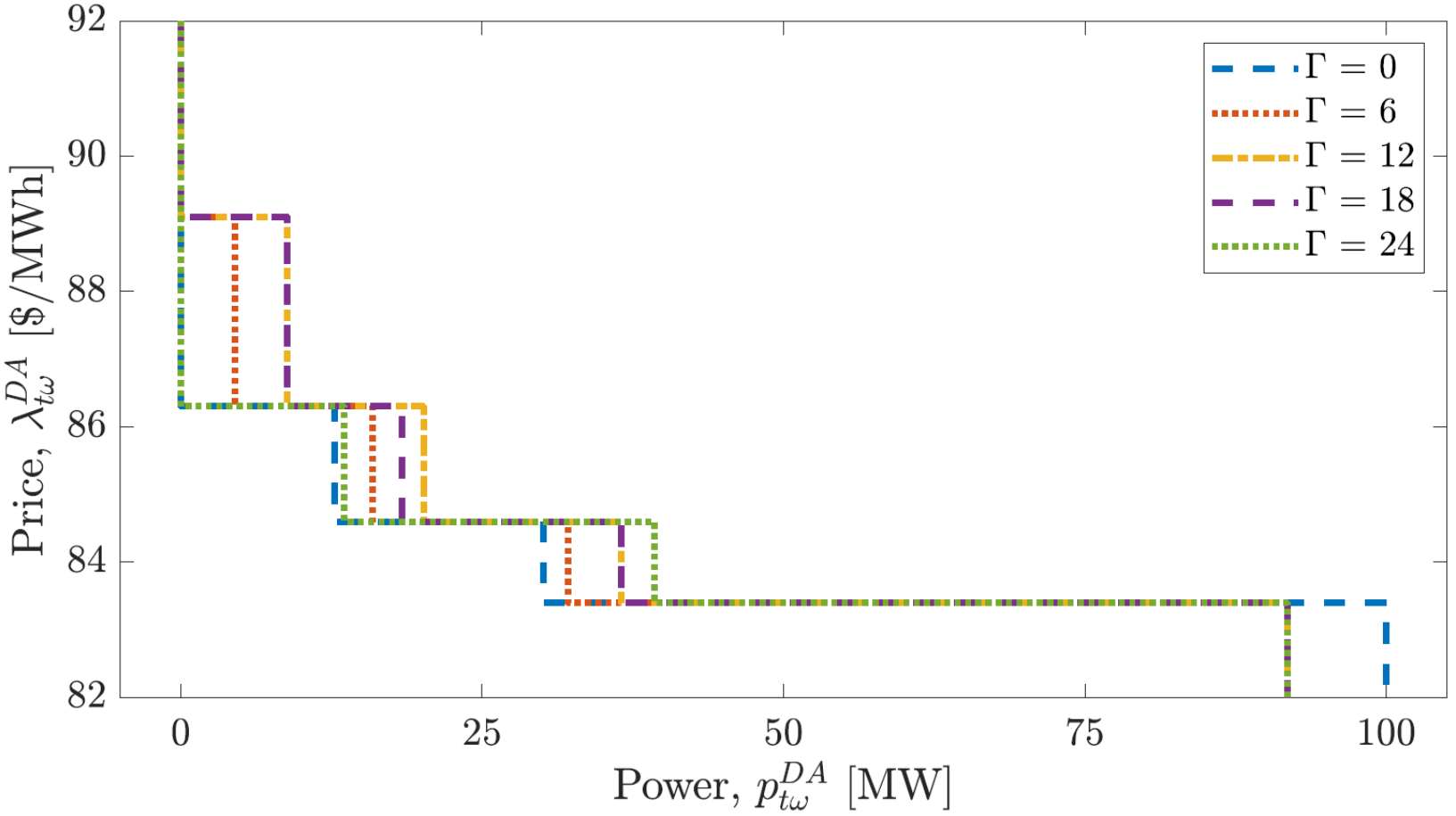
Risk-aversion → Uncertainty budget → $\Gamma \in [0, 24]$.

No uncertainty

Maximum uncertainty

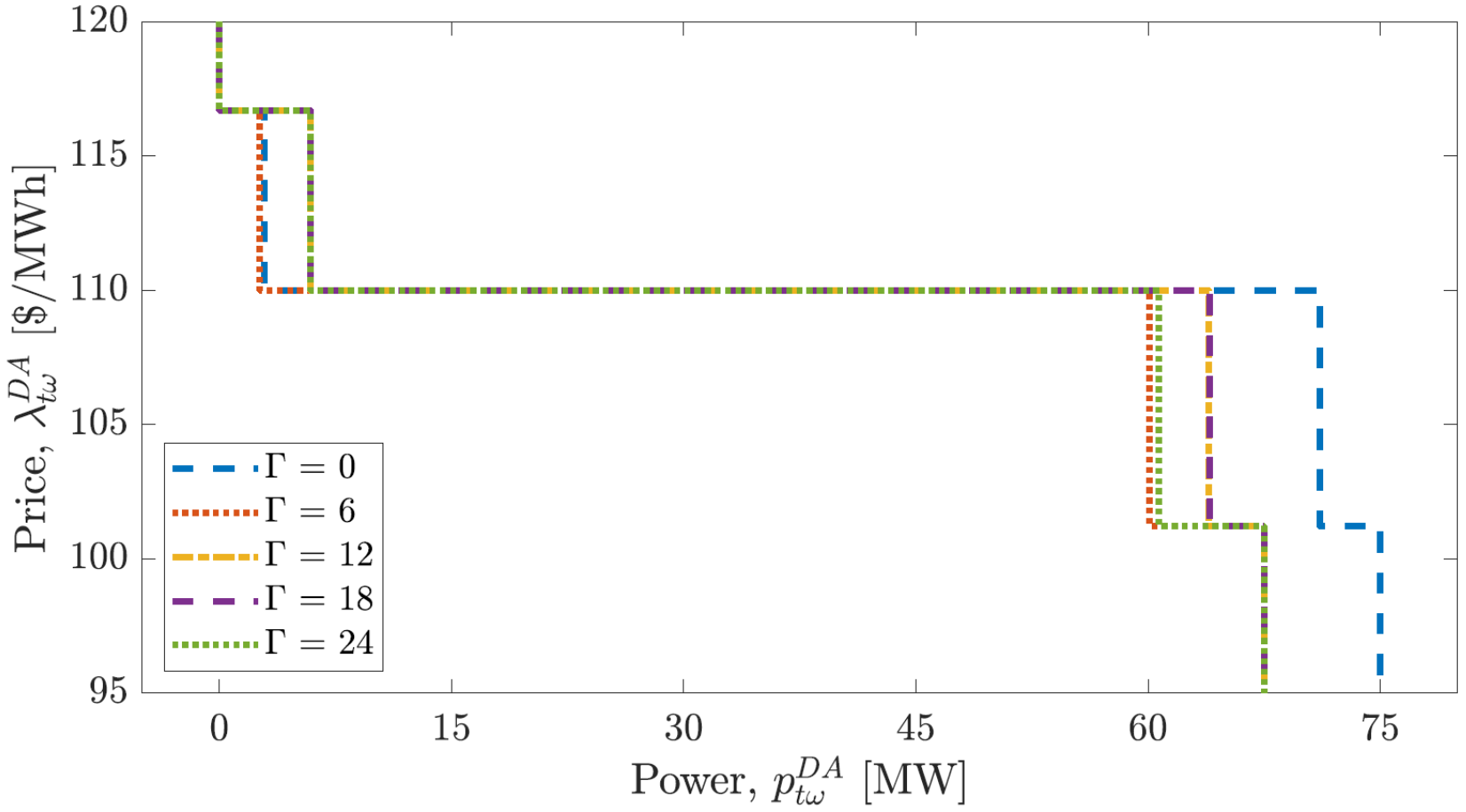
Á. García-Cerezo, D. Bonilla, L. Baringo, and J. García-González, "A stochastic adaptive robust optimization approach to build day-ahead bidding curves for an EV aggregator", *IEEE Transactions on Industry Applications*, vol. 62, no. 1, pp. 244–256, Jan.-Feb. 2026. DOI: 10.1109/TIA.2025.3587193.

# SP-RO approach – EV aggregator



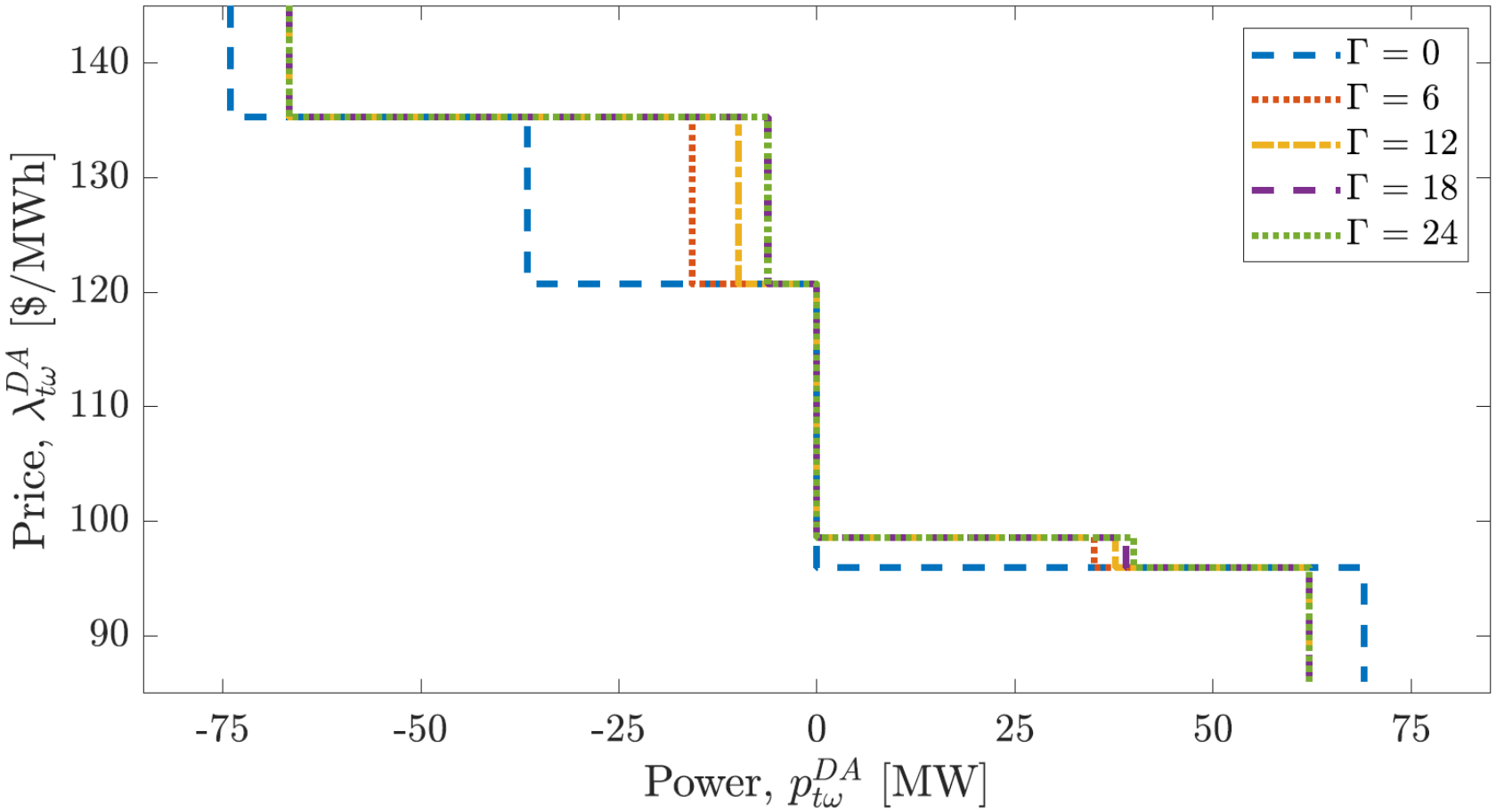

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# SP-RO approach – EV aggregator



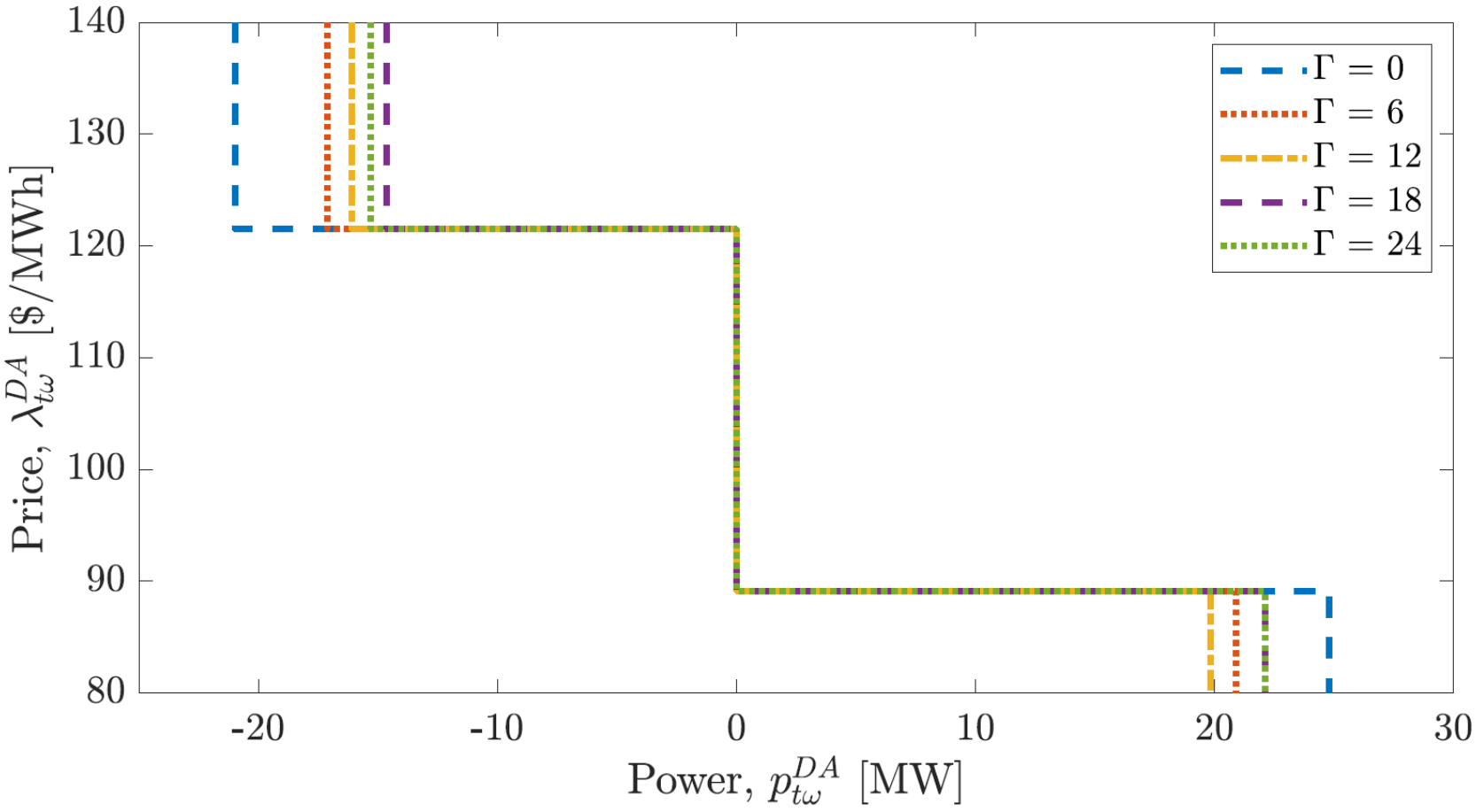

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# SP-RO approach – EV aggregator



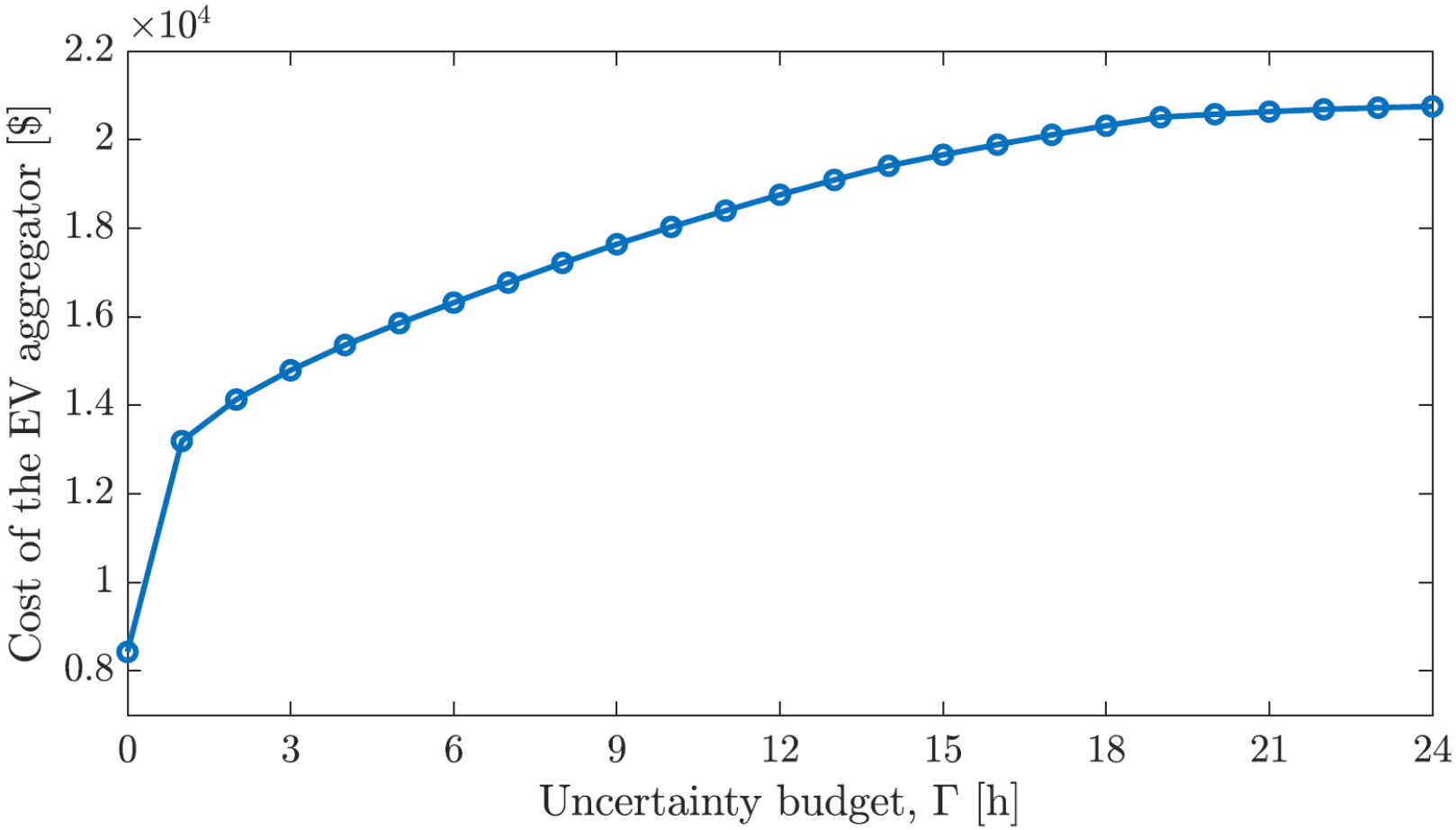

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# SP-RO approach – EV aggregator



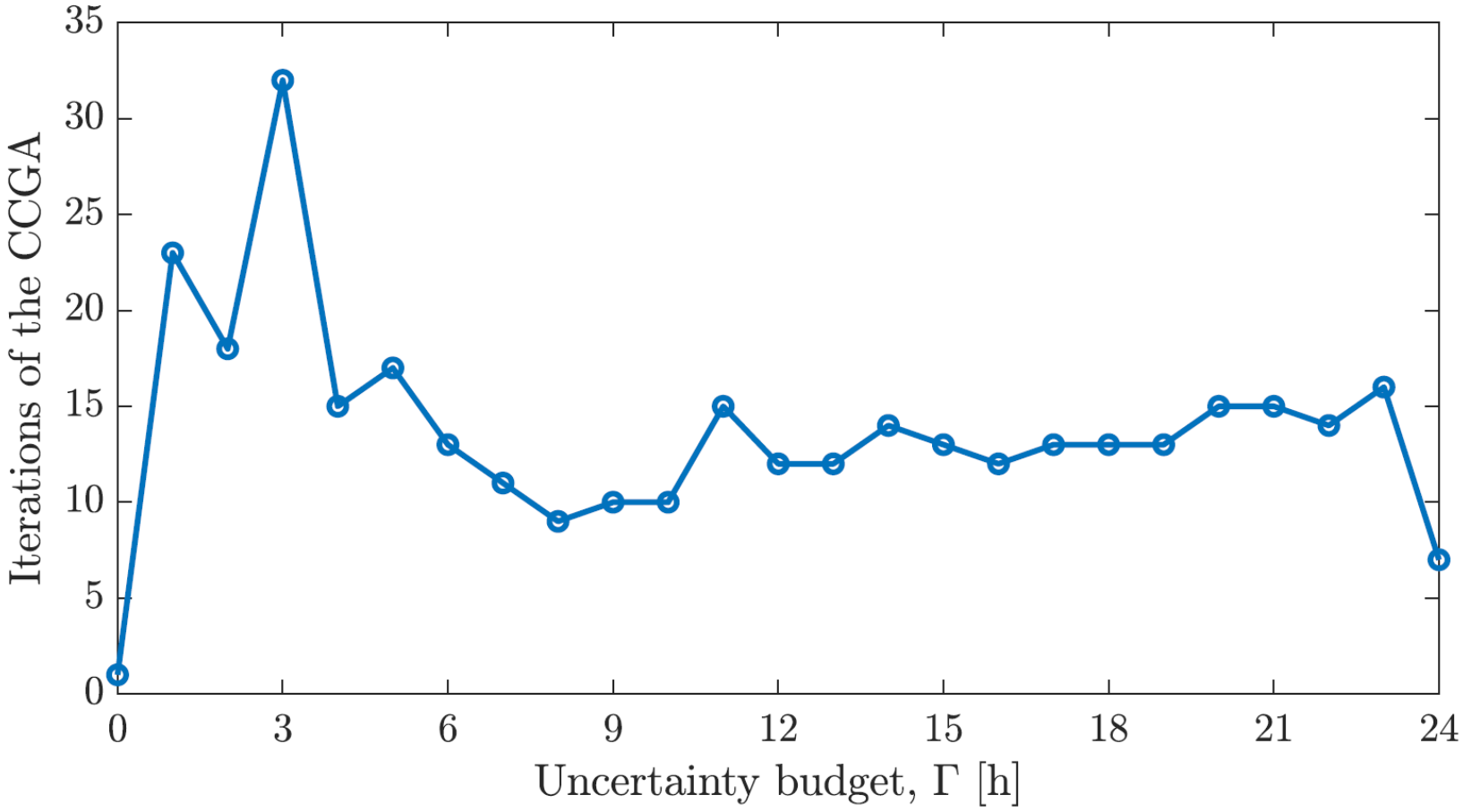

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# SP-RO approach – EV aggregator



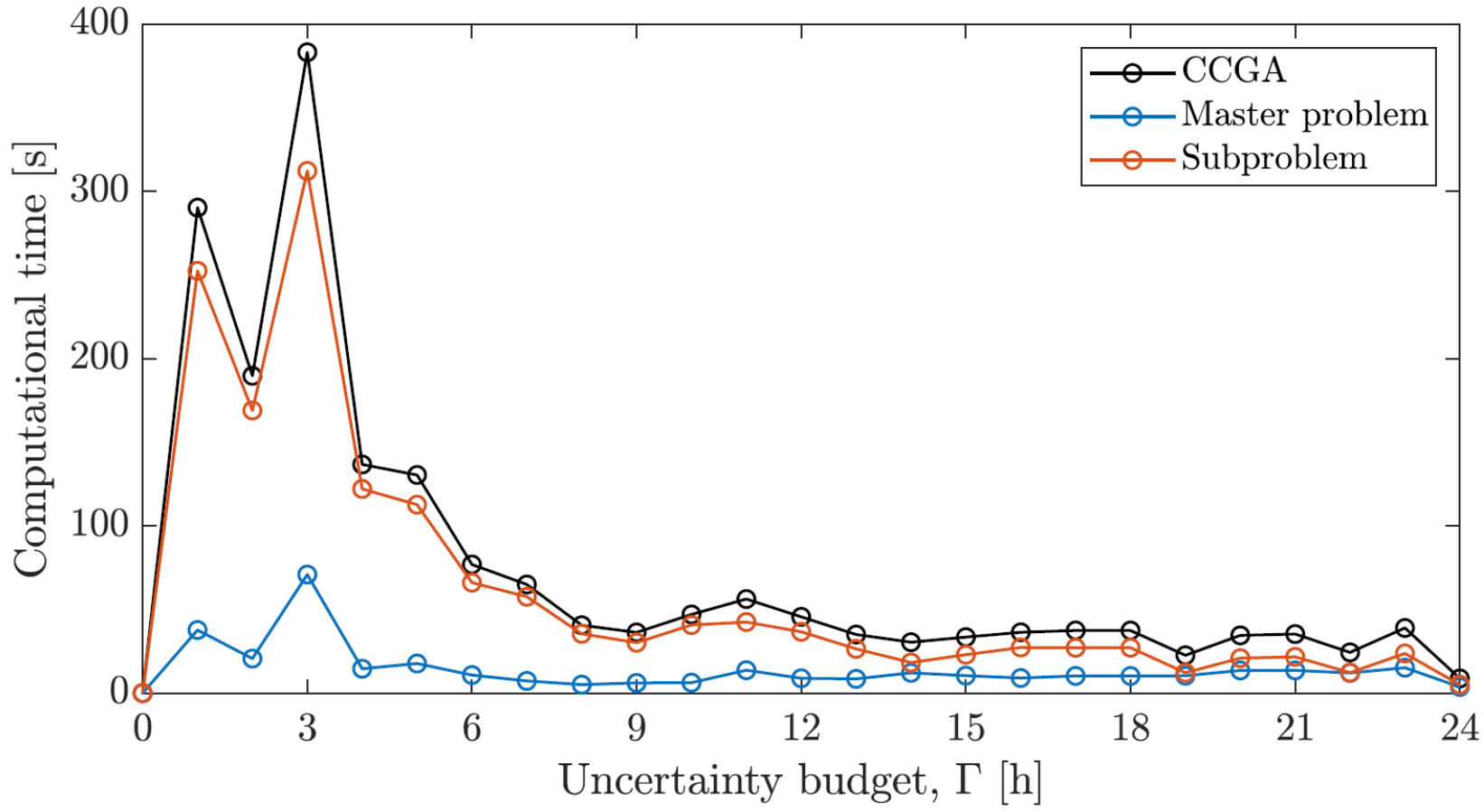

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# SP-RO approach – EV aggregator



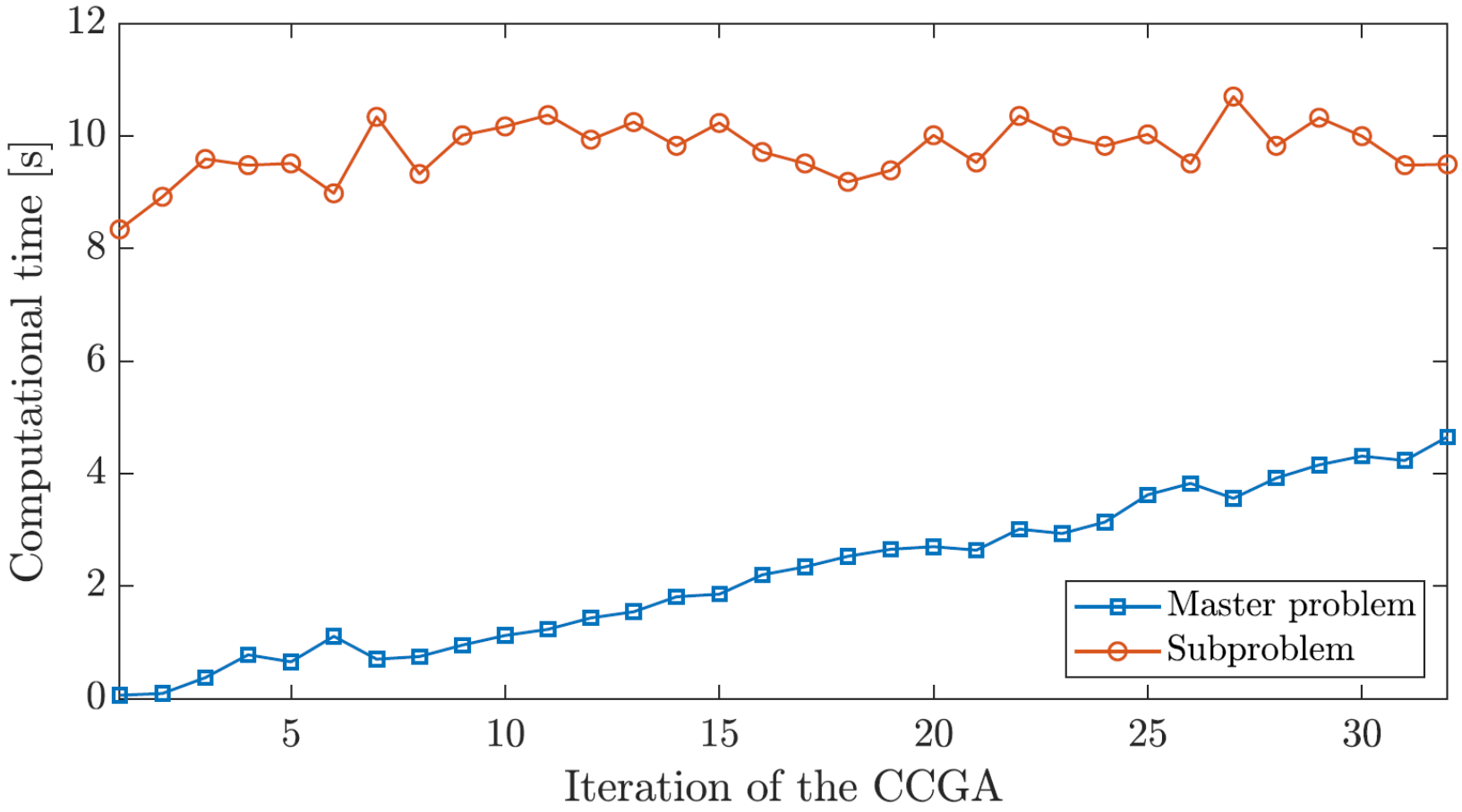

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# SP-RO approach – EV aggregator



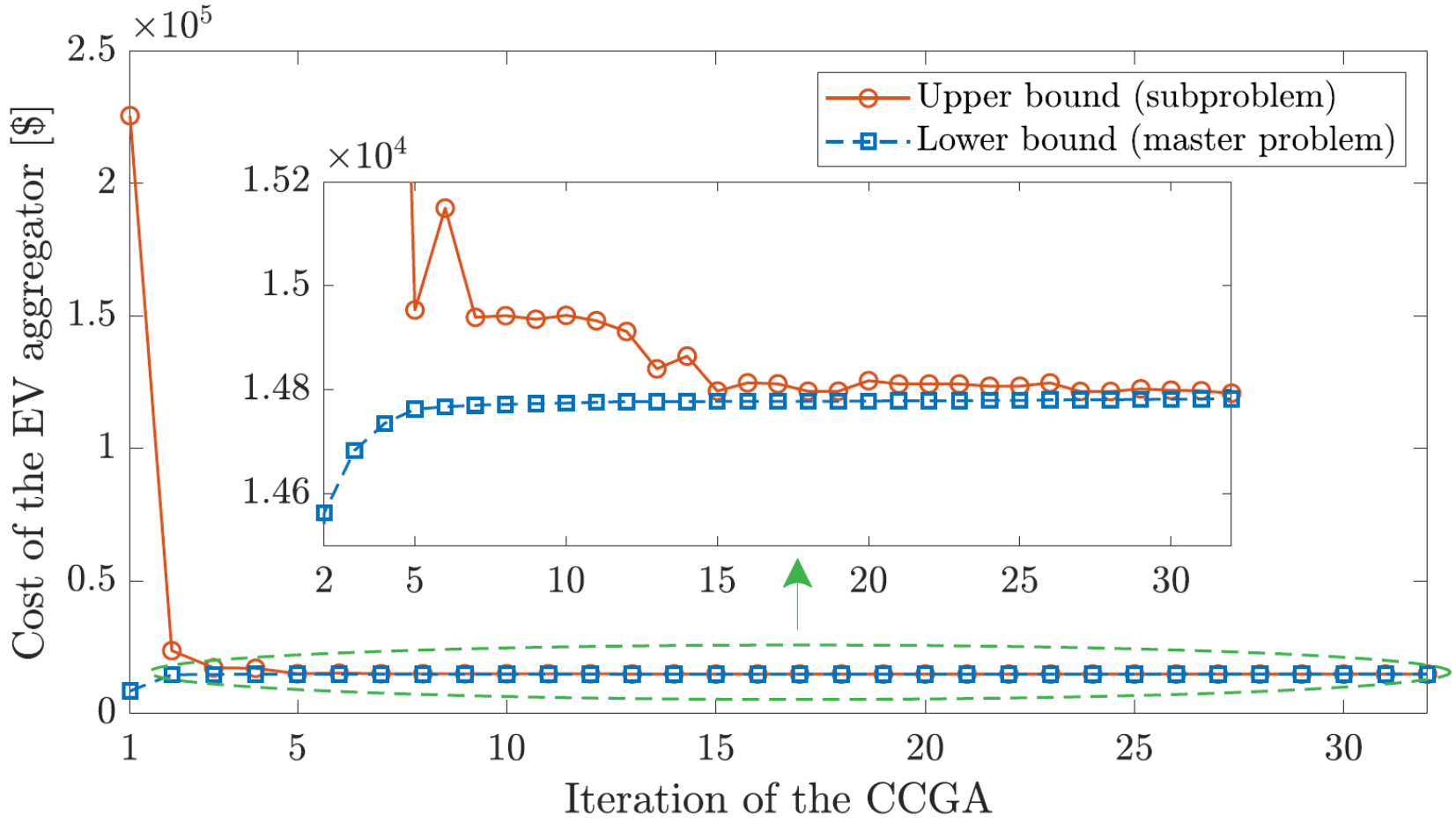

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# SP-RO approach – EV aggregator




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# SP-RO approach – EV aggregator




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# SP-RO approach

Is it possible to use SP and RO to model the uncertainty in the same parameter?

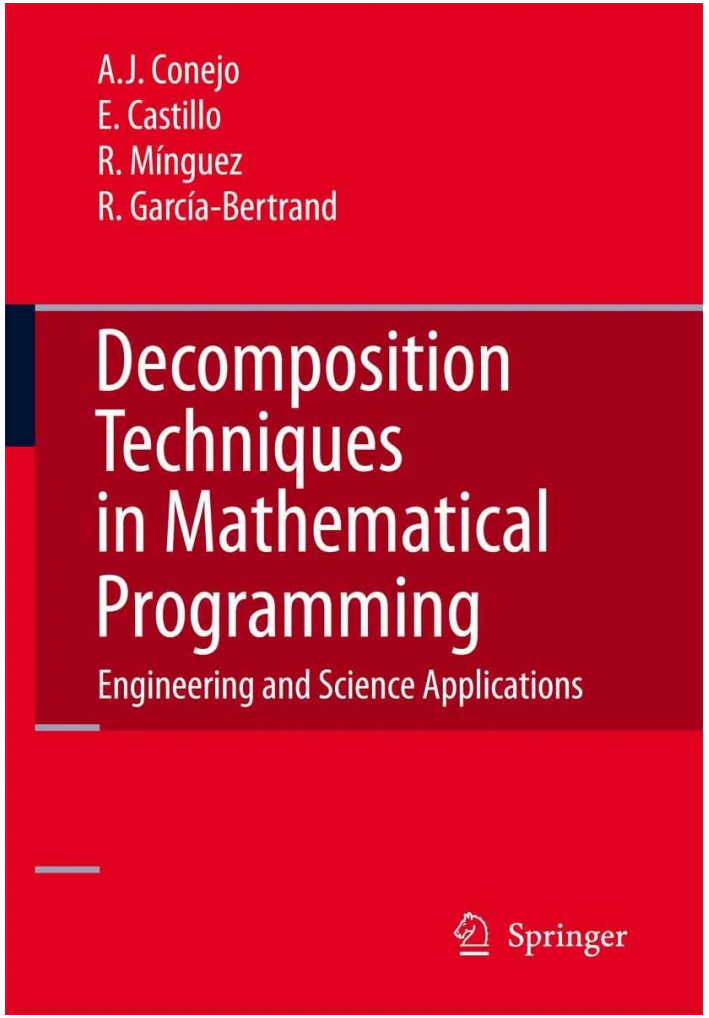
We will study this in lecture 5.

# Next lecture

- ~~Lecture 1: Introduction to operations research.~~
- ~~Lecture 2: Operation and planning problems.~~
- ~~Lecture 3: Operational variability.~~
- ~~Lecture 4: Uncertainty and risk.~~
- **Lecture 5: Further applications.**

# References

A. J. Conejo, E. Castillo, R. Mínguez and R. García-Bertrand, *Decomposition Techniques in Mathematical Programming*. Engineering and Science Applications. Springer-Verlag, Berlin, 2006. ISBN: 3-540-27685-8.



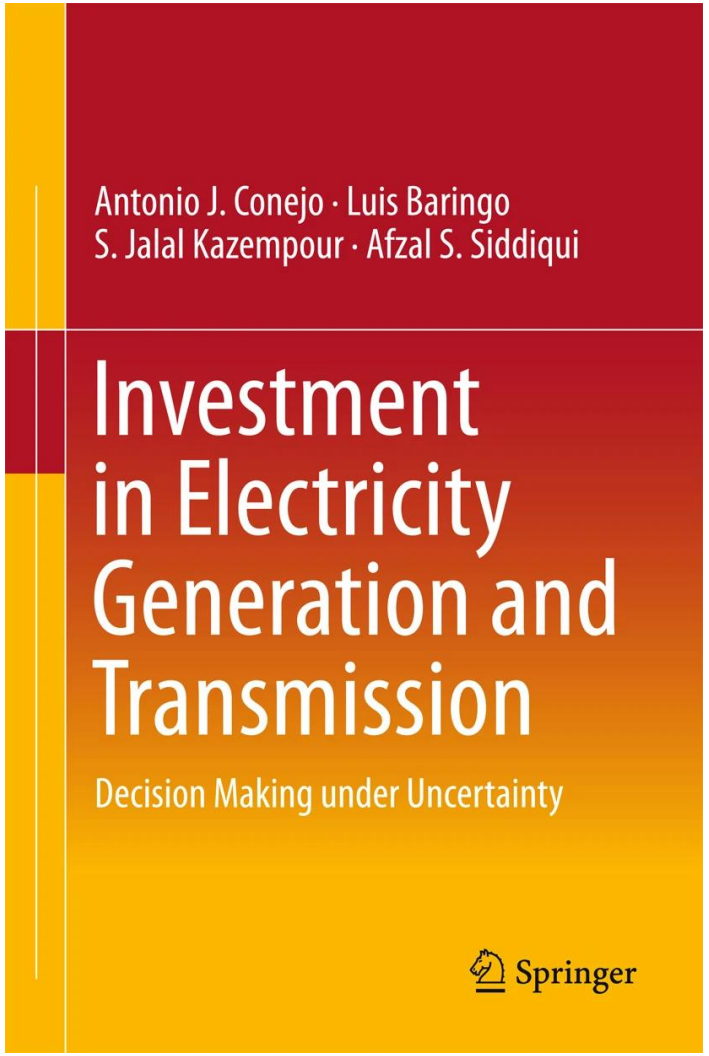
# References

A. J. Conejo, *ISE 5230 Decomposition Techniques in Mathematical Programming*. The Ohio State University.

https://u.osu.edu/conejo.1/courses/decomposition/

# References

A. J. Conejo, L. Baringo, S. J. Kazempour, and A. S. Siddiqui, *Investment in Electricity Generation and Transmission: Decision Making under Uncertainty*. Springer, Switzerland, 2016. ISBN: 978-3-319-29499-5. DOI: 10.1007/978-3-319-29501-5.



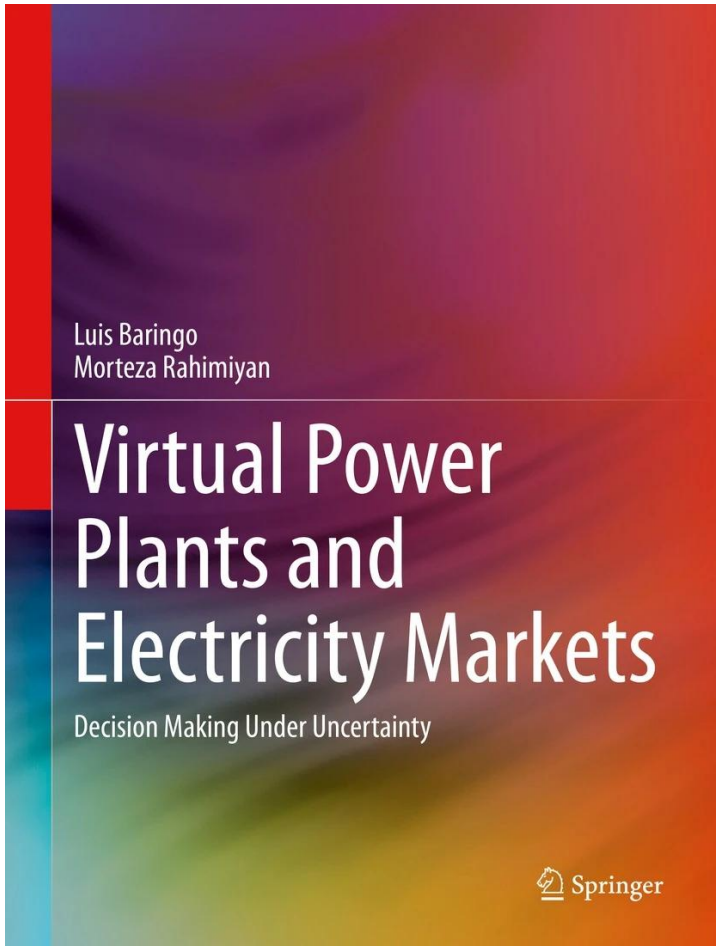
# References

N. González Cobos, *Robust generation scheduling in electricity markets*. PhD Thesis, Universidad de Castilla-La Mancha, Dec. 2018. Permanent link: https://hdl.handle.net/10578/20535.

See Chapter 2 (*Robust Optimization*).

# References

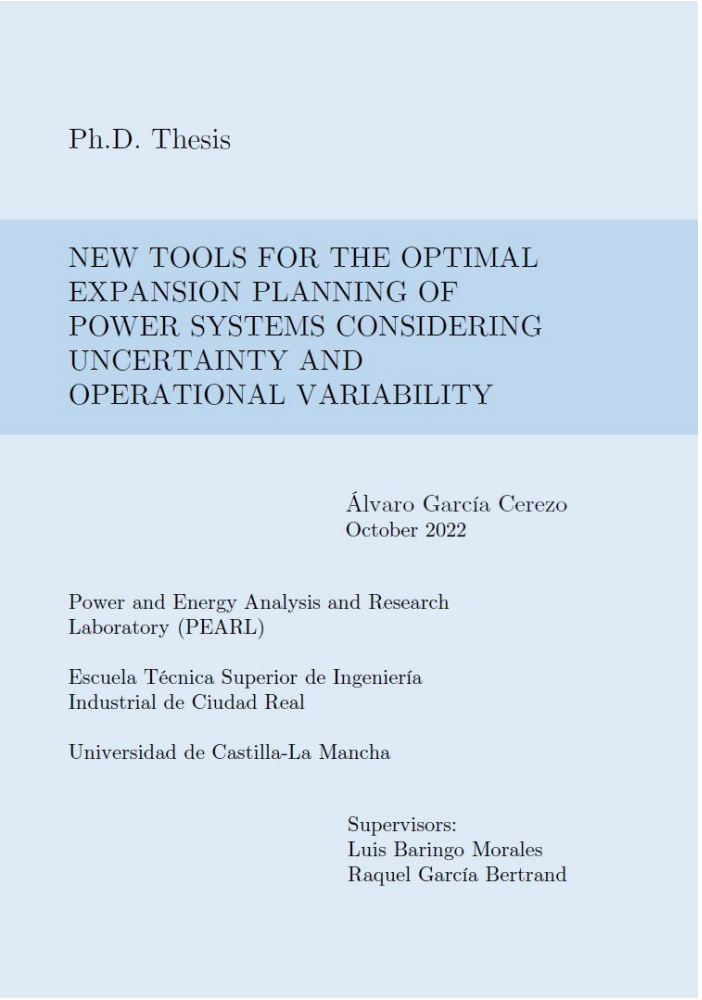
L. Baringo and M. Rahimiyan, *Virtual Power Plants and Electricity Markets: Decision Making under Uncertainty*. Springer, Cham, Switzerland, 2020. ISBN 978-3-030-47601-4. DOI: 10.1007/978-3-030-47602-1.



# References

Á. García Cerezo, *New tools for the optimal expansion planning of power systems considering uncertainty and operational variability*. PhD Thesis, Universidad de Castilla-La Mancha, Oct. 2022. Permanent link: https://hdl.handle.net/10578/30715.

See Sections 1.3.3 (*Modeling Uncertainty through Stochastic Programming*) and 1.3.4 (*Modeling Uncertainty through Robust Optimization*).



Ph.D. Thesis

NEW TOOLS FOR THE OPTIMAL EXPANSION PLANNING OF POWER SYSTEMS CONSIDERING UNCERTAINTY AND OPERATIONAL VARIABILITY

Álvaro García Cerezo
October 2022

Power and Energy Analysis and Research Laboratory (PEARL)

Escuela Técnica Superior de Ingeniería Industrial de Ciudad Real

Universidad de Castilla-La Mancha

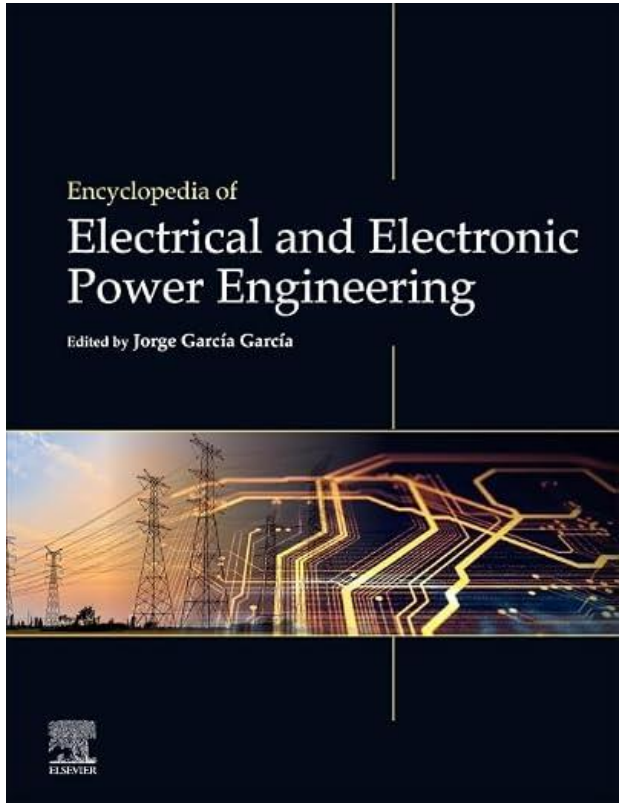
Supervisors:
Luis Baringo Morales
Raquel García Bertrand

# References


J. García, *Encyclopedia of Electrical and Electronic Power Engineering*. Elsevier, Oxford, UK, 2023. ISBN: 978-0-12-823211-8.

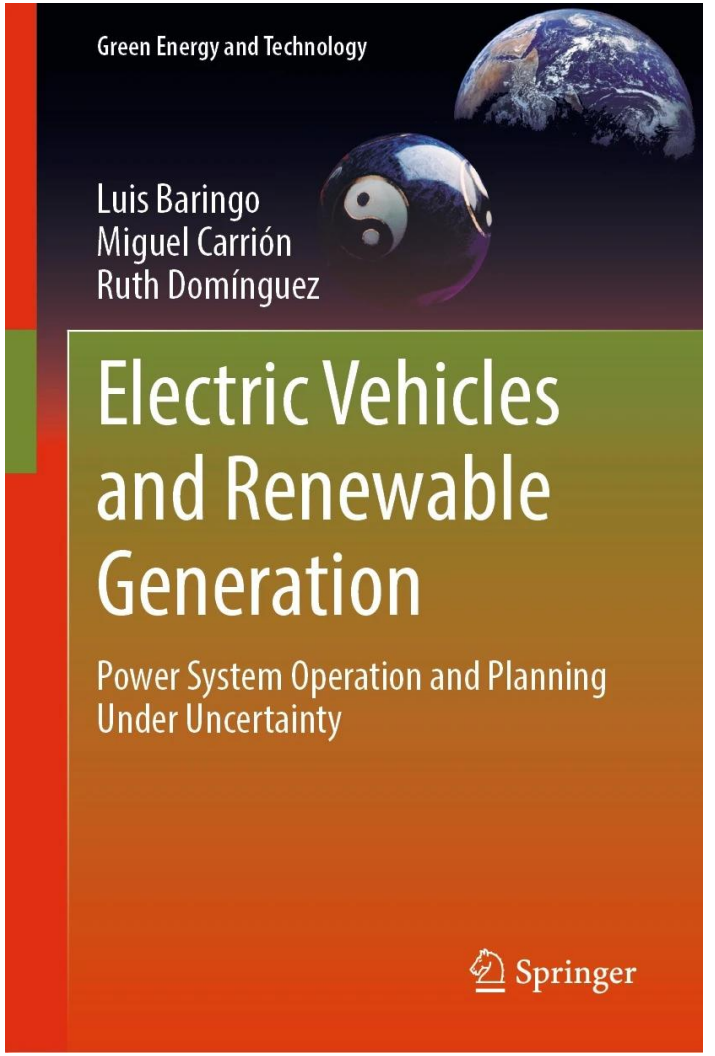
Chapters by R. García-Bertrand, L- Baringo, and Á. García-Cerezo:

- *Introduction to probability theory*. Pages 734–746. DOI: 10.1016/B978-0-12-821204-2.00016-7.
- *Uncertainty modeling in electrical power engineering: stochastic programming*. Pages 747–756. DOI: 10.1016/B978-0-12-821204-2.00090-8.
- *Uncertainty modeling in electrical power engineering: robust optimization*. Pages 757–764. DOI: 10.1016/B978-0-12-821204-2.00089-1.

# References

L. Baringo, M. Carrión, and R. Domínguez, *Electric Vehicles and Renewable Generation: Power System Operation and Planning under Uncertainty*. Springer, Cham, 2023. ISBN: 978-3-031-09078-3. DOI: 10.1007/978-3-031-09079-0.



# References

A. DeJeans Jr., *The Linearization Handbook for MILP Optimization: Modeling Tricks and Patterns for Practitioners*. BITBROS, 2025. ISBN: 979-8293657544.



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# Thank you very much for your attention!

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